The mass gap over the compact simple groups

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1 The label calculus

The section constructs the label data over the scalar-pair stage $\mathbb{P}$ (Definition 1); the classical reading is Remark 1's.

Definition 1 (The ground data)**.** The ground data are the positive naturals, generated from one by the successor and closed under the sum and the product, each injective in each argument ($a + c = b + c$ forces $a = b$, and $ac = bc$ forces $a = b$). A *read* of an object is an operation on its data given by formula in the displayed operations, returning ground data and decidable at every argument; an object *reads* the value the operation returns, and every ground object is natural data with decidable reads. Order is gap-existence: $q < p$ is the existence of the *gap*, the natural g at $q + g = p$, one value by the sum's injectivity, and of two naturals exactly one of three holds, a gap one way, the equality $q = p$, or a gap the other way, the trichotomy read, a sum exceeding its first summand by the shape of its data. A *count* reads how often a stated condition holds over a finite family: the naturals joined by the *vacant read*, the sum's unit, a summand at that unit fixing the other member and a factor at it reading it back, the sum injective in each argument with the product so at an occupied factor, the trichotomy running with the unit below every natural, a count *occupied* at a natural's read, and the *difference* of two counts reading the gap at the trichotomy's below outcome and the vacant read at the further two: the counts are a carrier with the displayed sum's unit. The *cofactor* c at $Y c = X$ is one value by the product's injectivity where it exists; order and factor statements are witness existence in this vocabulary, and a witness is a field of its statement: the gap a field of a stated order, the cofactor of a stated factorization, and a derived order's witness composed by formula through the sum and the product; a comparison is *priced* where its witness is so composed and displayed, the display its price. The scalar datum is a *pair*, written $[a:b]$ at $a, b \in \mathbb{N}$, the colon a separator of its data. The reads are displayed once and are the pair's whole interface:

$$[a:b] + [c:d] = [\,ad + cb \,:\, bd\,], \qquad [a:b]\,[c:d] = [\,ac \,:\, bd\,],$$

the order $[a:b] < [c:d]$ is the natural read $ad < cb$, and two pairs are one value exactly at $ad = cb$, the orientation read of two rays. A natural enters as $[a:1]$; a finite family sums by the fold over its index order, two fold orders equal by the ground identities. A later site may name a pair directly from two natural expressions; a naming that enters a value read maps every representative of its arguments to one value, the homogeneity principle's own test. On pairs the cofactor is total, the construction at the components' cross products, and the gap is the order's own witness at the cross-multiplied read. A comparison that runs on both sides of an equality is the *balance pair* $\langle u:v\rangle$'s, two ground data read jointly, the colon a separator of its data, the reads displayed once: sums add componentwise, the product reads cross-added, $\langle u:v\rangle\,\langle u':v'\rangle = \langle\, uu'+vv' \;:\; uv'+vu'\,\rangle$, the order $\langle u:v\rangle < \langle u':v'\rangle$ is the cross-added read $u+v' < u'+v$, and two balance pairs are one value exactly at the equality $u+v' = u'+v$, a summand on both members fixing the value; the equal-membered class is the sum's unit, that fixing its law, and a fold over a finite family seeds at the sum's unit and adds the members, total at every family, in this carrier and in every carrier with its displayed sum's unit; a family multiplies by the fold as well, seeded at a stated datum, in every carrier whose displayed product exchanges and regroups, two fold orders and seed placements one value at those reads, and at a key occupied once in the family the fold with that key's factor withdrawn, against the key's own factor, reads the whole fold, the factor rejoining across the exchanges; a ground datum enters at a one-member site, $\langle u:v\rangle + w = \langle u+w:v\rangle$. The pair's *side* is its two members' own trichotomy read, the equality among the outcomes, the exceeding member's gap its margin, and the gap of an ordered pair of balance pairs is the cross-added data's own, w at $(u+v')+w = u'+v$; matched lists of ground data pair componentwise, one balance pair per coordinate, the members equal exactly where every coordinate's are, and the *dot* of two matched lists, in this carrier and in every carrier with the displayed sum's unit and product, is the fold of their coordinate products, the coordinate reads' one contraction. The balance-pair displays read at every datum with the displayed sum, product and order reads, the scalar pairs among them, and a later site may name a pair at a balance-pair first datum: such a composite is the balance pair of its members' pairs at the one shared second datum, a natural expression,

$$[\,\langle u:v\rangle : c\,] \;=\; \Big\langle\, [\,u:c\,]:[\,v:c\,]\,\Big\rangle,$$

its reads the two displays' compositions, displayed once,

$$\begin{gathered}[\,\langle u:v\rangle : c\,] + [\,\langle u':v'\rangle : c'\,] = [\,\langle uc'{+}u'c : vc'{+}v'c\rangle : cc'\,],\\ [\,\langle u:v\rangle : c\,]\,[\,\langle u':v'\rangle : c'\,] = [\,\langle uu'{+}vv' : uv'{+}vu'\rangle : cc'\,],\end{gathered}$$

the order $[\,\langle u:v\rangle : c\,] < [\,\langle u':v'\rangle : c'\,]$ the cross-added read $uc'+v'c < u'c+vc'$ with one value exactly at its equality, the side the first datum's own at every second datum, and a scalar pair entering at a one-member site through the members' display. $\mathbb{P}$ denotes the scalar pairs with their reads, the stage glyph. Every statement reads pairs through the displayed reads alone, under the one *homogeneity principle*: every read is degree-known in each of its data, so a rescaling of any datum by ground data composes a comparison's witness by formula, one factor on each side, and the trichotomy reads the rescaled comparison back: a comparison holds exactly at its rescaling. Here it reads the representative freedom to one value, as it reads the form's scale and the family's: at $ad = cb$, rescaling both components of $[a:b]$ by d and of $[c:d]$ by b yields one pair, so two pairs at the identity read one value against every third. The harmonic numbers are $H_k = \sum_{j=1}^{k}[1:j]$; the *factorial* $k!$ is the product fold over the naturals from one to k, one at the vacant range; a pair of reads is componentwise data. Every later object reads the ground data through these operations.

Definition 2 (The ground polynomials)**.** A polynomial over the ground pairs, or over their balance pairs, is the recursion $P = c + z\,Q$, a constant read with a shifted family, finitely deep; equivalently its occupancy family of coefficients at the variable's monomials, the words $1, z, z^2, \ldots$ under the product with 1 their unit, the constant the unit monomial's read. The sum adds componentwise over the monomial keys, the product convolves, each coefficient one fold over the key pairs at its monomial, and the evaluation is the Horner read on the recursion. The *division* at a monic S, over the balance pairs, is the pair (Q,R), the *quotient* and the *remainder*, at the naming identity

$$P \;=\; S\,Q + R\;,$$

R's monomial keys below S's top, one value by the descent at the monic top, the identity descending the monomials from the top; at a linear factor $\langle z : r\rangle$, the variable against a value r, the remainder is the Horner read at r, and a *root* is a value at which the remainder's two members are equal, its multiplicity the count of iterated divisions there. The division reads exact at a stated factorization besides: a polynomial's *top* is its largest key at a coefficient off the sum's unit, one key by the trichotomy over the finitely many keys, and at factors of occupied tops the product's top sits at the summed key, the key's fold reading one summand at the two tops with every further summand reading a factor beyond its top, the sum's unit there, so the product's top coefficient is the tops' product, off the sum's unit by the entry carrier's product injectivity; a product reading the sum's unit at every key therefore puts one factor's every coefficient at that unit, and a shared factor of occupied top withdraws at two products of one value, the further factors' difference multiplying to the sum's unit and reading it key by key with the factors one value. The *exact division* at a divisor of occupied top and a dividend stated as the divisor's product descends at the tops: the quotient gains the entry carrier's cofactor at the two top coefficients on the cleared monomial at the keys' gap, the withdrawn dividend, the dividend joined to that multiple's balance partner, keeps its occupied keys below the prior top, and the descent closes at the dividend's top count with the quotient one value at the injectivity; the reads run over the ground pairs, over their balance pairs, and over the polynomials as the entry carrier in turn, each level's cofactor and injectivity the level below's own. The *derivative* P' is the successor-weighted shift, $P'_h = (h+1)\,P_{h+1}$ at every key, additive, with the Leibniz read $(PQ)' = P'Q + PQ'$ one convolution identity per key ($(i+j)\,P_iQ_j$ collected at either weighting over the key pairs joining to the successor). At a monic S the *remainder lists* are the coefficient lists at the monomial keys below S's top, sums componentwise and the product the convolution's remainder at the division by S.

Definition 3 (The ground eliminations)**.** A matrix over the ground data is its occupancy family of entries at row and column keys, total over stated finite key lists, the entries scalar pairs or their balance pairs (Definition 1). A *minor* at matched row and column lists is a formula in the entries, the first-row fold: at one row the entry, and at rows (i, I) and columns J the fold over J's splittings into a column j and a co-list K, the two joining to J, of the entry a_{ij} against the minor at $(I;\, K)$, the summand's side the count of J's columns before j, even at the first member and odd at the second. Iterating the splittings, a minor at matched lists collects to the *assignment fold*: one summand per assignment of a distinct column to each row, at the product of the assigned entries, the summand's side the assignment's swap grading (Construction 1), the steps' splitting sides joining to the count of the assignment's inversion list at the withdrawn co-lists. The Gauss elimination is the pivot descent: the first step's entries are the entries themselves, the second's the two-list minors at their naming $b^{(1)}_{ij} + a_{i1}a_{1j} = a_{11}a_{ij}$, and Sylvester's identity names each later step from the two before it, at every $k \geq 1$,

$$P_k\, b^{(k+1)}_{ij} \;+\; b^{(k)}_{i,k+1}\, b^{(k)}_{k+1,j} \;=\; b^{(k)}_{k+1,k+1}\, b^{(k)}_{ij}\ ,$$

$b^{(k)}_{ij}$ the pivot-bordered minor at the lists $(1..k,\, i\,;\, 1..k,\, j)$ and P_k the leading minor at $(1..k)$, one identity of minor formulas per step, the new entry the identity's cofactor: the descent closes on the sum and the product, each division exact at the prior pivot, the identity's read at every prior pivot outright: an equal-membered pivot reads the division's both members at the sum's unit, the identity collecting its join there with the cofactor's product beside it. The identity derives from the fold's key reads. A minor is additive and scale-reading in one row or one column, the assignment fold reading that key's entry once per summand, so a minor at a column read as a stated combination expands over the members, one summand per member at its coefficient; a minor at two row keys, or two column keys, of one value reads equal members, the keys' transposition pairing the assignments at exchanged sides (Construction 1); the exchanged row and column lists read one determinant, each assignment paired with its flip, the inverse permutation (Construction 1), at one term, the entries' product reindexed along the assignment, and at one class, the grading additive over composition with the identity even so the flip shares its assignment's class; and the assignment fold regroups at any row's column choice, one family per column of the entry against the withdrawn lists' minor at the sides' join. The *adjugate* of a square list is the cofactor family, the entry at (i, j) the minor at the withdrawn row j and column i on the side of the count $i+j$: a row's fold against its own cofactors collects the regrouped assignment fold, the determinant, and against a further row's cofactors it reads equal members, the fold a determinant at that row repeated, so the list against its adjugate reads the determinant's diagonal, the *adjugate identity*, and the adjugate against the list reads that diagonal as well: the exchanged row and column lists read one determinant at every struck pair, so a column's fold against its own cofactors collects the exchanged assignment fold and a further column's reads equal members. Sylvester's identity then reads at the shifted bordered list: the variable t joins the diagonal over the balance-pair polynomials, a carrier with the displayed reads (Definition 2), and the shifted bordered determinant is monic of degree $k + 2$ in t, the identity assignment's product reading the top at the unit with every further assignment at two or more entries off the diagonal, the moved places a pair at the least, two degrees below. Join the two *solved columns*, the shifted list's fold against its adjugate's two border columns, each the shifted bordered determinant at its own border key alone by the adjugate identity; the determinant at the border columns read to the solved ones expands twice, the two-column multilinear fold keeping the border pair's selections alone with every further selection a column repeated at equal members, against the direct fold along the two determinant-scaled unit columns: the two reads join the adjugate corner's two-by-two determinant against the shifted bordered determinant to that determinant's square against the shifted leading minor, the border cofactors' sides joining even at each pair, and the monic determinant withdraws at the division's one value (Definition 2). The evaluation at the equal-membered pair reads each shifted minor at the stated entries, the minor a formula in the displayed operations with the evaluation reading sums to sums and products to products (Definition 2): the identity holds at the entries themselves. A step pivots at an entry with unequal members, a row or column exchange swapping the members of the minors keyed at both. The descent reads the determinant with the rank: the terminal bordered minor at the last key pair is the minor at the full lists, the determinant, each step's entry the identity's own naming. At a leading minor off the sum's unit, the prior pivot's own read, a pivot column whose every bordered minor reads equal members reads the determinant at them as well: each bordered minor expands along its border row, one shared cofactor family over the leading columns with the off-unit leading minor the border key's own coefficient, a leading row's bordered minor a repeated-key read outright, so the column joins the leading columns' cleared combination at that off-unit coefficient, and the determinant, expanded at that column over the combination's members, reads a repeated column key per summand, equal members throughout with the off-unit coefficient withdrawing at the product's injectivity. The step keeps the span and the kernel: the update display reads the prior pivot's multiple of each new row as the pivot's multiple of the entry row against the entry's border multiple of the pivot row, and the entry rows read back at the divisions' witnesses, the two pivots off the unit, so the step's rows and the entry rows sit in one span at cleared combinations; a *kernel* member, a coordinate vector over the balance pairs pairing every row at the sum's unit, extends across the step at the pivot's clearing: the stepped member scales by the pivot and the pivot coordinate enters as the stepped row read's balance partner, the pivot row's own pairing collecting to the pivot's multiple of that read against its partner, the sum's unit, and the extension pairs the entry rows at the sum's unit exactly where the stepped member pairs the stepped rows, the update display's read at the cleared pivots, with the column exchanges permuting the coordinates as stated data. The descent names the *rank*, the pivot count, and the *kernel list* enters per pivot-free column by the back solve: at the terminal step every entry reads equal members and the unit family with one at the column's own key seeds the member, each earlier step extending at the pivot's clearing as above. The members pair every row at the sum's unit, the solves' reads; the list is independent, each member off the unit at its own pivot-free coordinate, the crossed pivots' product, and at the unit at the further pivot-free ones; and every kernel member joins the list's span, the pivot coordinates one value at the off-unit pivots by the product's injectivity with the cleared combination at the pivot-free coordinates reading the member whole: the list's length, the pivot-free count, is the *kernel dimension.* The *resultant* $\mathrm{Res}(p, q)$ of two polynomials at tops of unequal members, over the ground data or any carrier with their displayed reads (Definition 2), is the site-datum determinant of their *shift matrix*, rows the monomial shifts $x^a q$ at a below p's degree and $x^b p$ at b below q's, columns the monomial keys below the degrees' sum, one minor read; the adjugate row solves

$$A\,p + B\,q \;=\; \mathrm{Res}(p, q)$$

at polynomials of degrees below the other's, the solved witness's whole verification; a kernel vector reads the *cross pair* (A, B) at those degrees with $A\,p$ the balance partner of $B\,q$, so the resultant reads equal members exactly at an occupied cross pair, the descent's kernel read, and at a Bézout read $up + vq = 1$ it reads unequal members: at $\check{c}$ the memberwise swap of uB, the identity maps a cross pair's A to $(Av + \check{c}\,)\,q$ at q's degree beyond A's, every cross pair the sum's unit there.

Definition 4 (The deck families)**.** The *deck relation* at a variable y is the monic quadratic $y^2 + 1 = w\,y$, its coefficient the *chord* w, and its *shift read* is $y^{\,j} + y^{\,j+2} = w\,y^{\,j+1}$, the relation's $y^{\,j}$-multiple, one identity per key; a *root pair* of the relation is two data at product one joining to the chord. The *deck families* are the polynomials in the chord at the relation's recursion,

$$p_j + p_{j+2} = w\,p_{j+1}\,, \qquad Q_j + Q_{j+2} = w\,Q_{j+1}\,,$$

over the balance-pair carrier (Definition 2), each later member the join's datum, at the seeds $p_1 = w$, $p_2 + 2 = w^2$ and $Q_1 = 1$, $Q_2 = w$: p_b is monic of degree b and Q_b of degree e at $e + 1 = b$, each of its degree's parity in the chord, the recursion moving one parity per step from the seeds: the recursion's running pair reads its two degrees with the unit tops jointly, the join's degree the chord multiple's own at one beyond its factor's and the join's top that multiple's, the withdrawn member two below. The families' column reads at the relation close by induction on the shift read, one cross-multiplied identity per step:

$$y^{2b} + 1 \;=\; y^{\,b}\,p_b(w)\,, \qquad \langle y^{2b} : 1\rangle \;=\; \langle y^2 : 1\rangle\,y^{\,e}\,Q_b(w)\,,$$

the *plus read* and the *word read*, the word the geometric telescope's, $\sum_{c+d+1=b} y^{2d} = y^{\,e}\,Q_b(w)$ at $\langle y^{2b} : 1\rangle = \langle y^2 : 1\rangle \sum_{c+d+1=b} y^{2d}$; at a root pair the plus read collects to the matched powers' sum, $z^{\,b} + z'^{\,b} = p_b(w)$, the product's b-th power one.

Construction 1 (Places, shapes, and the column exhibit)**.** The declared scalar of the development is one natural, the *residue* r, unconstrained, declared here once with its successor name, the *fundamental count* $d_f := r + 1$ (Definition 1): d_f is read wherever a site counts the standard list below (a loop's closure, a trace, a row bound), r wherever a site reads the ordering data (the beta-sets' gaps, the fundamentals, a reduced shape's depth), the residue name the pairing's own read. $V := \mathbb{P}^{d_f}$ with its standard pairing and standard list $v_1, \ldots, v_{d_f}$; the power $V^{\otimes k}$, with the pairing of monomial tensors and the *place action* of S_k, $\sigma\,(x_1 \otimes \cdots \otimes x_k)$ the monomial reading x_j

at place $\sigma(j)$, each σ a permutation matrix on the monomial tensors, orthogonal at the transpose identity $\sigma^{\mathsf{T}}\sigma = 1$, the transpose the *inverse permutation*'s matrix, the assignment flipped, its value at j the witness k at $\sigma(k) = j$. A permutation's *inversion list* is the occupancy family of the place pairs $i < j$ at $\sigma(j) < \sigma(i)$; composing with an adjacent transposition moves the list by exactly one pair (the adjacent pair's read flips, and the pairs meeting the two places exchange theirs), so the descent at a reversed adjacent pair per step ends with every place pair sorted and writes σ as a word of adjacent transpositions, one letter per pair of the list. The *swap grading* is additive over composition, two words concatenating, *even* and *odd* its two classes: a word's letters flip one place pair's read each, the others exchanging theirs, so along a word from the sorted assignment the list moves by one pair per letter and the letter count is the witness sum $L = c + 2w$, c the inversion list's count and w the count of letters returning a pair: every word for σ shares the list's parity, one class per permutation, the identity's words at the paired letters alone, $L = 2w$, even outright; a transposition is odd, its inversion list its own place pair with two pairs per place between its two, one pair beyond an even family. The permutations are the words of unit content, one letter at every place once, one word per permutation at the assignment's own reads and at the factorial's count, the enumeration blocking at the head letter, one block per occupied letter at the withdrawn content's words, the unit content folding one letter out per step, the product fold's own read (Definition 1); and the *permuted display* of a list is its reads at a permutation's places, one entry per place. A permutation's *cycles* are its orbit words, from each least unwalked place the iterated images to the start's return: a walk's members are distinct until the return, two equal iterates reading equal predecessors down to the start at the assignment's own injectivity, so each walk closes within the place count, every place sits on one word, and the *cycle count* is the words' count; a permutation composed between a second permutation and its inverse permutation walks the composed orbits at the images, each iterate the second permutation's image of the walk's own at the flipped place, one orbit word per orbit either way, so the cycle count is one value across the composition. The *grading* is by content: the monomial tensor at indices $(i_1, \ldots, i_k)$ has *content* the occupancy family μ over the letters, a member letter at the count of places reading it; $V^{\otimes k}$ is the orthogonal sum of its content summands, a homogeneous member read at its coordinate family over its own content's monomials, the pairing the coordinate fold within one summand, and two members at distinct contents pairing at equal members outright, the orthogonality's read. A *shape* is its *column multiset*: finitely many columns, each of a positive length at most d_f, an occupancy family over the lengths; the *row list* is the derived display, λ_i the count of columns of length at least i, the occupancies the row list's consecutive gaps, one shape per display, the rows sorted at the occupied depths with the *degree* k the box total. The shape's *beta-set* X_λ is the fold over the column multiset's occupancies n_ℓ, columns of length exactly ℓ: descending from $x_{d_f} := n_{d_f} + 1$ by $x_i = x_{i+1} + n_i + 1$, a set of d_f distinct positive naturals with sorted display $x_1 > \cdots > x_{d_f}$, each consecutive gap x_i against x_{i+1} the occupancy successor $n_i + 1$, and the rows' read $x_i = \lambda_i + g + 1$ at the join $i + g = d_f$; the unit shape's is the *unit set* $U = \{1, \ldots, d_f\}$, a full column moves every member by one successor, so a shape's class at full columns is its gap word $(n_1, \ldots, n_r)$ alone, and a shape of c columns reads its *complement*'s set, the gap columns at $\ell + \ell' = d_f$, at the interval complement, $x_i(\lambda^*) + x_{i'}(\lambda) = c + d_f + 1$ at every index pair joining to $d_f + 1$. The *column exhibit* is the wedge tensor

$$t_\lambda \;:=\; w_{\lambda'_1} \otimes w_{\lambda'_2} \otimes \cdots\,, \qquad w_\ell \;:=\; \Big\langle \sum_{\sigma \text{ even}} v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(\ell)} \;:\; \sum_{\sigma \text{ odd}} v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(\ell)} \Big\rangle,$$

one factor per column at the sorted lengths $\lambda'_1 \geq \lambda'_2 \geq \cdots$, a column of one box reading its vector, $w_1 = v_1$, and a deeper column the balance pair of the two graded sums over S_ℓ, read componentwise on the monomial coordinates (Definition 1), each monomial on its permutation's side; the monomials are distinct, so the two members' supports are disjoint and the members differ, of content the columns' own read, letter i at the count of columns of length at least i: the row list λ; a wedge with a repeated vector has equal members, the repeat's transposition an odd move fixing every monomial and mapping either side's list onto the other's. The shapes compose by column union, the row lists adding, and the *unit shape* $\mathbf{1}$ is the union's unit; the powers compose by the tensor product with the scalars' line its unit, two spanning lists' tensors spanning the composite, a tensor of two combinations collecting to the pairs at the coefficients' products with a content summand of the composite reading its list at the pairs whose factor contents join to it, the grading's split; and $\mathbf{1}$'s exhibit is the scalar one on that line.

Lemma 1 (A transpose-closed action splits orthogonally)**.** *At a set of operators on a finite-dimensional space with positive pairing, closed under transpose, the orthogonal complement of an invariant subspace is invariant: at $u' \perp U$, g in the set and $u \in U$, $\langle gu', u\rangle = \langle u', g^{\mathsf{T}}u\rangle$ with $g^{\mathsf{T}}u \in U$, so the value is one of u''s pairings against U, of equal members by the complement's defining datum. The place action is such a set ($\sigma^{\mathsf{T}}\sigma = 1$, Construction 1), and so is every set of operators listed together with its transposes.*

Construction 2 (The unit action)**.** The matrix units act on V by $E_{ij}v_m = \delta_{jm}v_i$ and on a power by the Leibniz sum $\sum_{\text{places}} 1 \otimes \cdots \otimes E_{ij} \otimes \cdots \otimes 1$, written E_{ij} again; the pairing reads $E_{ij}^{\mathsf{T}} = E_{ji}$ place by place, and the commutators close on the table

$$[E_{ab}, E_{cd}] + \delta_{da}\, E_{cb} \;=\; \delta_{bc}\, E_{ad},$$

read on the standard list and extended to powers by the Leibniz rule, cross-place terms commuting. A *unit space* is a finite-dimensional space with positive pairing, operators E_{ij} obeying the table and the transpose identity, and the joint E_{ii}-grading by contents; the powers are unit spaces, and so are sums, tensor products (by the Leibniz rule), and invariant subspaces, a composite power's fold splitting at the arrangements' join, each factor's places collecting to its own fold read against the other factor's arrangement, the composition's letter actions the factors' own (Construction 1). The E_{ii} commute with integer grading, so the projection onto one content is a polynomial in them (one Lagrange interpolant per coordinate through the occupied integers), and an invariant subspace is graded: each of its elements' content components is its member. E_{ij} at $i < j$ is a *raising*, its content move the letter-pair move at (i, j), the count at i raised by one and the count at j lowered by one, strictly up the dominance order; $i > j$ is a *lowering*, strictly down; E_{ii} reads the grading.

Construction 3 (The generator table)**.** A *table space* is a finite-dimensional space with positive pairing and one interface of data, its fields. The *coordinates* are a finite key list, a *content* is a matched list over them (Definition 1), and the space is the orthogonal sum of its content summands, the occupied contents one *class*, every two joined by a balance-pair fold over the simple list. The *simple list* is finitely many contents $\alpha_1, \ldots, \alpha_s$; the *root list* is finitely many contents, each a natural fold $\sum_i c_i\, \alpha_i$ occupied off the unit fold; the *dominance order* is the fold's witness read, $\mu \preceq \lambda$ at the join $\mu + \sum_i c_i\, \alpha_i = \lambda$, the coefficients one solve at the simple folds' injectivity below. Per root the table reads a *raising* E_α at the content move $+\alpha$, its transpose the *lowering* $F_\alpha := E_\alpha^{\mathsf{T}}$ at the move's balance partner; per simple index, the *triple* (E_i, F_i, H_i) at the displays $[E_i, F_i] = H_i$ and $[F_i, H_i] = 2F_i$, H_i reading a content summand at its *coroot pair* $\mu(\alpha_i^\vee)$, one balance pair per content, additive over the contents' sum, at $\alpha_i(\alpha_i^\vee) = 2$; the simple folds' coroot reads are injective, a balance-pair fold over the simple list reading equal members at every coroot pair the unit fold, so two occupied contents at one value under every coroot pair are one content (their joining fold's reads equal members throughout), the joint H-grading is the content grading, and the projection onto one content is a polynomial in the H's, one Lagrange interpolant per coroot read through the occupied values; the *reflection* $s_i\mu$ is the content at the join $\mu = s_i\mu + \mu(\alpha_i^\vee)\,\alpha_i$, and a content is *dominant* where every coroot pair reads its upper side or equal members; the *closure*, the commutator of two field operators a span fold over the list at the moved content, a raising's or the H-span's at a root or the unit move and a lowering's below it; the *form*, one symmetric pair read per content pair, additive in each argument over the contents' sums, its doubled read against a root clearing at the root's length, the *root coroot pair* $\mu(\alpha^\vee)$ the cofactor at $\langle\alpha,\alpha\rangle\,\mu(\alpha^\vee) = 2\,\langle\mu,\alpha\rangle$ with the simple triples' reads its instances, the *root reflection* $s_\alpha\mu$ the content at the join $\mu = s_\alpha\mu + \mu(\alpha^\vee)\,\alpha$; the *root triples*, per positive root the display $[E_\alpha, F_\alpha] = H_\alpha$ at H_α reading a content summand at its root coroot pair, the further display $[F_\alpha, H_\alpha] = 2\,F_\alpha$ the grading's own read, the commutator moving by α at $\alpha(\alpha^\vee) = 2$; and the *Casimir*,

$$C \;:=\; Q + \sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\,\big(E_\alpha F_\alpha + F_\alpha E_\alpha\big)\ ,$$

the fold over the positive list at Q the diagonal read of the form's square $q(\mu)$ per content, commuting with every field operator: a content-λ vector whose every raising image is the sum's unit reads C at $q(\lambda) + \langle\lambda, 2\rho\rangle$, the fold's own evaluation, each $E_\alpha F_\alpha$ the commutator's read there and the weights collecting to the root fold's, $[\,\langle\alpha,\alpha\rangle : 2\,]\,\lambda(\alpha^\vee) = \langle\lambda,\alpha\rangle$. Sums of table spaces at one field set and one shared class are table spaces, and tensor products, the operators by the Leibniz rule, the coroot pairs additive and the classes adding; and so is an invariant subspace, graded by the projections. A member's table enters with every field a stated read, a series' tables polynomial pairs in the series' residue and a fixed member's finite data; the powers with the matrix units are the first table, the coordinates the letters, the roots the letter-pair moves with the consecutive moves simple, the triples the units' own displays, and the Casimir list the units themselves.

Construction 4 (The member weight tables)**.** A member's weight data instantiate the content fields of Construction 3: the coordinates, the simple and root lists with their folds, the coroot pairs, the form, and the root fold $2\rho := \sum_\alpha \alpha$ over the positive list. Two reads are the series' shared derivation, run on their displayed positive lists. *The reflection permutes the positive list*: the list is reflection-closed and a family member at one occupied key is its own simple root, every further member occupied off any one key, two reads per family of the displayed folds; $s_i\alpha_i$ is α_i's balance partner, and every further positive root keeps a positive fold coefficient off α_i (a one-key fold is its simple root's, the singleton check), kept by s_i, so s_i permutes the positive list off α_i with α_i's own key fixed under the image assignment, a further member's image at α_i's balance partner reading its root-fold dot on the lower side while every member's dot sits positive at its fold, $\langle\alpha, 2\rho\rangle = \sum_k c_k\,\langle\alpha_k,\alpha_k\rangle$, and its square positive at its family's displayed length, the form's fold read, and the fold reads $s_i(2\rho) + 2\,\alpha_i = 2\rho$: the root fold's coroot pairs read $(2\rho)(\alpha_i^\vee) = 2$ at every simple index. *The residue is one fold*: at the highest root's fold $\theta = \sum_i c_i\,\alpha_i$ and the form's read $\langle\theta,\theta\rangle = 2$,

$$\langle\theta, 2\rho\rangle \;=\; \sum_i c_i\,\langle\alpha_i,\alpha_i\rangle \;=\; 2\,r\ ,$$

the first equality the form's fold read at $\langle\alpha_i, 2\rho\rangle = \langle\alpha_i,\alpha_i\rangle$, the coroot read above cleared at the length, and the second the member's *derived residue* r, one fold per member. A third read joins them, the *reflection grading*, at any member whose positive list holds the permutation read, the first shared read's, s_i permuting the list off α_i. The member's *Weyl list* is the words' finite list of positive-list images: a word in the s_i acts on contents through the reflection joins, each letter an involution ($s_is_i\mu = \mu$, the moved coroot pair its own balance partner); a reflection keeps the form, the form's additivity read at the two joins,

$$\langle\mu,\nu\rangle = \langle s_i\mu,\, s_i\nu\rangle + \nu(\alpha_i^\vee)\,\langle s_i\mu,\,\alpha_i\rangle + \mu(\alpha_i^\vee)\,\langle\alpha_i,\, s_i\nu\rangle + \mu(\alpha_i^\vee)\,\nu(\alpha_i^\vee)\,\langle\alpha_i,\alpha_i\rangle\ ,$$

the middle terms collecting with the last to the sum's unit at the flipped coroot pairs, the halved products' read, so $\langle s_i\mu, s_i\nu\rangle = \langle \mu, \nu\rangle$; and a word fixing each member of the positive list keeps every content's coroot pairs, the kept form against the kept simples, so the member's coroot reads return the content, the displayed cofactors' solve with the fixed members' contents the coroot lists outright: two words at one image are one content map, and the list is finite at the images. A word's element reads its *inversion list*, the occupancy family of the positive list at the images of the positive list's balance partners; one further letter s_i moves the list by exactly one member, α_i's read flipping and the further members' exchanging along s_i's permutation of the positive list, so a word's letter count is the witness sum $L = c + 2w$ at c the inversion list's count, one parity per element, *even* and *odd* the two classes, the grading additive over composition, Construction 1's argument at the member's list with the place pairs' inversion lists the matrix units' instance; and the grading's fold reads ρ, $w\rho + \sum_{\alpha\in N}\alpha = \rho$ at N the inversion list, one letter per induction step, $s_i\rho + \alpha_i = \rho$ at the coroot read one, the halved $(2\rho)(\alpha_i^\vee) = 2$, the letter's permutation of the positive list mapping the further members across. The grading closes on four reads. *The transport*: at an image root $w\beta$ the reflection is the transported one, $s_{w\beta}\,\mu = w\,s_\beta\,w^{-1}\mu$, the coroot pair the kept form's own read, $\mu((w\beta)^\vee) = (w^{-1}\mu)(\beta^\vee)$; a reversed word's element shares its parity class at the one letter count. *The creation read*: along a word each letter moves the inversion list by one member, entering or leaving at the letter's own root and exchanging at the further members, so a list member names its *creating letter*, the member the later letters' image of that letter's simple root. *The deletion*: a word whose letter count exceeds its element's inversion list's count holds a leaving flip, the leaving member some earlier letter's created one, and the transport rewrites the word at two fewer letters, the leaving letter crossing to its creating one; so every word reduces to the list's count, and the element whose inversion list is the unit family is the identity, its reduced word letter-free. *The regular reads*: $w\rho = \rho$ forces w the identity, the ρ-fold's members at positive root-fold dots reading the list to the unit family; a dominant content κ at every simple coroot pair beyond the unit reads every positive pair beyond the unit, the fold's members positive, and its images are one per element with at most one dominant: at u off the identity the inversion list is occupied, a member γ the image of a partner, $u\beta = \check{\gamma}$ at a positive β, so $\langle u\kappa, \gamma\rangle$ is the balance partner of $\langle \kappa, \beta\rangle$, on its lower side, and $u\kappa$ sits off the dominant reads. A series member's list is the signed place permutations of its coordinates, every sign vector at B_ℓ and C_ℓ and the even flip counts at D_ℓ, the displayed list: the s_i generate it, the transpositions the place part with the last letter the flip at B_ℓ and C_ℓ and the flip pair at D_ℓ, a flip's transported images the further flips, and distinct signed permutations read distinct coordinate images, one list member each.

The series, at the rank ℓ, the coordinate count, the contents matched lists (Definition 1), the B and D lists at doubled coordinates, every read one value across the doubling by the homogeneity principle, their contents the coroot cofactors' lists at one parity each. Per series the coroot and form displays read the halved products, $2\,\langle \mu, \alpha_i\rangle = \mu(\alpha_i^\vee)\,\langle \alpha_i, \alpha_i\rangle$, one identity per simple class off the displayed data. B_ℓ at $\ell \geq 2$: the simple list α_i, 2 on key i against its balance partner on key $i{+}1$ at $i < \ell$, and α_ℓ, 2 on the last key; the positive list in three families at their folds, the difference family $c_k = 1$ on $i \leq k < j$, the sum family $c_k = 1$ on $i \leq k < j$ with 2 on $j \leq k \leq \ell$, and the short family $c_k = 1$ on $i \leq k \leq \ell$; the form $[\,1 : 4\,]$ at the coordinate dot, the lengths 2 at $i < \ell$ and 1 at the last; the coroot pairs $2\,\mu(\alpha_i^\vee) = \langle \mu_i : \mu_{i+1}\rangle$ at $i < \ell$ and $\mu(\alpha_\ell^\vee) = \mu_\ell$; the root fold's key-$i$ read $4g + 2$ at the join $i + g = \ell$; and θ the sum family's first member, 2 on keys one and two, its fold $c = (1, 2, \ldots, 2)$, the residue fold $2r = 2 + 4g' + 2$ at $g' + 2 = \ell$: $r = 2g$ at $g + 1 = \ell$. C_ℓ at $\ell \geq 3$: the simple list the letter-pair moves at $i < \ell$ and α_ℓ, 2 on the last key; the difference family $c_k = 1$ on $i \leq k < j$, the sum family $c_k = 1$ on $i \leq k < j$ with 2 on $j \leq k < \ell$ and 1 at the last, and the long family $2e_i$ at $c_k = 2$ on $i \leq k < \ell$ with 1 at the last; the form $[\,1 : 2\,]$ at the dot, the lengths 1 at $i < \ell$ and 2 at the last; the coroot pairs $\mu(\alpha_i^\vee) = \langle \mu_i : \mu_{i+1}\rangle$ and $\mu(\alpha_\ell^\vee) = \mu_\ell$; the root fold's key-$i$

read $2g$ at $i+g=\ell+1$; and θ the long family's first member, its fold $c=(2,\dots,2,1)$, the residue fold $2r=2g'+2$ at $g'+1=\ell$: $r=\ell$. D_ℓ at $\ell\ge 4$: the simple list the doubled letter-pair moves at $i<\ell$ and α_ℓ, 2 on keys $\ell-1$ and ℓ; the difference family $c_k=1$ on $i\le k<j$, the sum family $c_k=1$ on $i\le k<j$, 2 on $j\le k$ at the joins $k+2\le\ell$, and 1 on the last two keys, its $j=\ell$ member $c_k=1$ on $i\le k$ at $k+2\le\ell$ with 1 on the last key; the form $[\,1:4\,]$ at the dot, every length 2; the coroot pairs the cofactors $2\,\mu(\alpha_i^\vee)=\langle\mu_i:\mu_{i+1}\rangle$ at $i<\ell$ and $2\,\mu(\alpha_\ell^\vee)=\mu_{\ell-1}+\mu_\ell$; the root fold's key-$i$ read $4g$ at $i+g=\ell$ with the last key's read equal members; and θ the sum family's first member, the residue fold at its coefficient sum, $r=1+2g'+2$ at $g'+3=\ell$: $r=w$ at $w+3=2\ell$.

The fixed members enter at their simple keys, the contents the coroot lists, $\mu(\alpha_i^\vee)$ the key-i read, the Cartan reads $\alpha_j(\alpha_i^\vee)$ and the length list the member's displayed data with the form's simple reads the halved products $\langle\alpha_j,\alpha_i\rangle=[\,1:2\,]\,\alpha_j(\alpha_i^\vee)\,\langle\alpha_i,\alpha_i\rangle$, the root fold at its coroot list, 2 on every key, the doubled fundamental fold $2\sum_j\omega_j$ at the adjugate rows below, one content at those coroot pairs by the simple folds' injectivity, and the residue the one fold off the halved products: G_2, two keys at the edge read $\alpha_2(\alpha_1^\vee)$ the balance partner of 3 and $\alpha_1(\alpha_2^\vee)$ of 1, lengths $[\,2:3\,]$ and 2, θ's fold $(3,2)$, the residue fold $2r=3\,[\,2:3\,]+2\cdot 2$: $r=3$. F_4, four keys at the chain edges with the middle edge doubled toward the short pair, lengths $(2,2,1,1)$, θ's fold $(2,3,4,2)$, the residue fold $2r=4+6+4+2$: $r=8$. E_6, E_7, E_8, every length 2 at the simply-laced edge lists (the chain 1–3–4–5–6 with the branch 2–4; the chain grown by 7, and by 7–8), θ's folds $(1,2,2,3,2,1)$, $(2,2,3,4,3,2,1)$ and $(2,3,4,6,5,4,3,2)$, the residue folds twice the coefficient sums: $r=11$, $r=17$, $r=29$.

The adjugate rows. The fixed members' fundamental folds are the cleared displays

$$e\,\omega_i \;=\; \sum_k a_{ik}\,\alpha_k\ ,$$

at the *fold key* e, one at G_2, F_4 and E_8, three at E_6 and two at E_7, the rows a_i the five members' tables in their stated order,

$$\begin{pmatrix}2&1\\3&2\end{pmatrix},\qquad \begin{pmatrix}2&3&4&2\\3&6&8&4\\2&4&6&3\\1&2&3&2\end{pmatrix},\qquad \begin{pmatrix}4&3&5&6&4&2\\3&6&6&9&6&3\\5&6&10&12&8&4\\6&9&12&18&12&6\\4&6&8&12&10&5\\2&3&4&6&5&4\end{pmatrix},$$

$$\begin{pmatrix}4&4&6&8&6&4&2\\4&7&8&12&9&6&3\\6&8&12&16&12&8&4\\8&12&16&24&18&12&6\\6&9&12&18&15&10&5\\4&6&8&12&10&8&4\\2&3&4&6&5&4&3\end{pmatrix},\qquad \begin{pmatrix}4&5&7&10&8&6&4&2\\5&8&10&15&12&9&6&3\\7&10&14&20&16&12&8&4\\10&15&20&30&24&18&12&6\\8&12&16&24&20&15&10&5\\6&9&12&18&15&12&8&4\\4&6&8&12&10&8&6&3\\2&3&4&6&5&4&3&2\end{pmatrix},$$

each row's witness the coroot identities $\sum_k a_{ik}\,\alpha_k(\alpha_j^\vee)\;=\;e\,\delta_{ij}$ over the member's Cartan reads, one fold per key pair, and one row the θ-fold's e-multiple, the highest root the fundamental at that key. The rows' form reads close the fundamental data: the fold at the halved products $2\,\langle\alpha_k,\omega_j\rangle=\delta_{jk}\,\langle\alpha_k,\alpha_k\rangle$, the coroot displays' read, gives

$$2\,e\,\langle\omega_i,\omega_j\rangle=a_{ij}\,\langle\alpha_j,\alpha_j\rangle\ ,\qquad e\,\langle\omega_i,2\rho\rangle=\sum_k a_{ik}\,\langle\alpha_k,\alpha_k\rangle\ ,$$

the second at the root fold's all-twos coroot list, every read positive by its shape at the natural entries, and $C_2(\omega_i) = \langle \omega_i, \omega_i + 2\rho \rangle$ is the member's fundamental Casimir list, one fold per key; two contents are one class (Construction 3) exactly at one remainder family of their cleared folds at the key, the remainder families additive over the contents' sums, every content the unit's class at key one.

The fixed members' positive lists. A fixed member's root list enters at doubled coordinates, its members matched lists (Definition 1) with the coroot pairs the halved products at the displayed form scale, and the simple list's halved products read the member's displayed Cartan reads, one identity per key pair. The three E-members share eight coordinates at the form $[1 : 4]$ against the dot: the simple list is α_1, one on the first and eighth keys with its balance partner on the six between; α_2, two on the first two keys; and the letter-pair moves α_{j+1} at $2 \leq j \leq 7$, two on key j with its balance partner on the key before. The positive list in three families: the *difference family*, two on a key with its balance partner on an earlier key; the *sum family*, two on two keys; and the *half-sum family*, one on the eighth key and one or its balance partner on each earlier key, the partners even in count, at the counts 28, 28 and 64, the join $28 + 28 + 64 = 120$. E_8 reads the whole list; E_7 reads the members at the seventh and eighth entries' sum of equal members, the counts $16 + 15 + 32 = 63$; E_6 reads those with the sixth and eighth entries' sum of equal members as well, the counts $10 + 10 + 16 = 36$; a member letter's own selecting sums read equal members, so the member's letters keep its selection, and a selected member's families read at its own keys, the selection's fixed partners. The families close under the letters, the permutation read: a letter-pair move permutes its two coordinates, each family to itself off its own letter; the α_2 letter reads the first two entries to their swapped balance partners, exchanging the difference and sum members meeting the first pair and keeping the half-sums at two moved entries, the partner count moved by an even read; and the α_1 letter moves a member by α_1 at its coroot pair, the pair one, its balance partner or the unit read at every member off α_1, a difference or sum member at pair one to the half-sum at its moved entries and a half-sum back, the partner parity kept. Every member off the simple list holds a simple key at the coroot pair one: a difference member its top key's letter-pair move or, on the eighth key, α_1 at partners off the first key and the second key's letter-pair move at it; a sum member its lower key's letter-pair move off the first key, its top key's move on it, and α_1 at the first with the eighth; a half-sum member an ascent's letter-pair move, an earlier key's balance partner against its successor's one, α_2 at a leading plus pair, or α_1, the trichotomy complete by the sign list: off an ascent the list descends, off a leading plus pair at most the first entry is one, and the even partner count leaves α_1 alone. The *fold descent* moves at that key: the image is a member by the closure reads, its dot against the *height vector*

$$(0, 4, 8, 12, 16, 20, 24, 92)$$

falling by the letter's length at the pair, so the walk ends on the member's simple list and the fold witness collects along it, one natural multiple per step: the root list is the families at their natural folds (Construction 3's field). The height dots are positive per family, the entries increasing up the keys for the differences, two entries' sum at one positive member for the sums, and the last entry clearing the earlier seven's sum for the half-sums, the join $84 + 8 = 92$; and the vector is the root fold's, its coroot pairs two on every key, one content with the doubled fundamental fold $2\sum_j \omega_j$ at the simple folds' injectivity. F_4 enters at four coordinates, the form $[1 : 4]$ against the dot: the simple list is α_1, two on the second key with its balance partner on the third; α_2, two on the third with its balance partner on the fourth; α_3, two on the fourth key; and α_4, one on the first key with balance partners on the rest; the families are the *singles*, two on a key, the *difference and sum family*, two on a key with its balance partner or two on a later key, and the *half-sums*, one on the first key and one or its balance partner on each further key, the counts $4 + 12 + 8 = 24$, the singles and half-sums at length one and the pair family at two. The third letter flips the fourth entry, the letter-pair moves permute their coordinate pairs, and the α_4 letter moves a member by α_4 at its coroot pair: the families close, and every member off the simple list holds a key at a positive pair, twenty reads off the displayed families; the height vector is $(22, 10, 6, 2)$, the first entry clearing the rest's sum at the join $18 + 4 = 22$, its coroot pairs two on every key, and the fold descent reads the natural folds along it. G_2 enters at its simple keys outright, the six displayed folds

$$(1,0), \quad (0,1), \quad (1,1), \quad (2,1), \quad (3,1), \quad (3,2) \ ,$$

lengths $[\,2:3\,]$ at the first, third and fourth members and 2 at the rest, the form's fold reads; the closure is the displayed reflection reads, s_1 keeping the second key with the first key's image at the join $x + c_1 = 3\,c_2$, and s_2 keeping the first with the second's image at the join $y + c_2 = c_1$, ten reads off the flipped letters, and the six folds' sum $(10, 6)$ reads the coroot pairs two on every key; its alternant coordinates are three doubled keys at the form $[\,1:12\,]$ against the dot, α_1 two on the first key with its balance partner on the second and α_2 two on the second and third keys with the doubled partner on the first, each member's coordinates its fold's read, the root-fold vector twelve on the third key against the partners of four and eight on the first two, every member's dot positive. So every fixed member's positive list holds the permutation read, the reflection grading's scope: the member's Weyl list, its grading and its ρ-fold are the third read's at that list.

The defining tables. The series' defining table is the signed keys' space, each key i with its partner key $\bar\imath$, the contents $\pm e_i$ with the unit content at B_ℓ, the pairing diagonal at one with the B_ℓ null key at $[\,1:2\,]$: the root operators are the moved units at the form's twist, a difference move E_{ij} with $E_{\bar\jmath\bar\imath}$'s balance partner, a sum move $E_{i\bar\jmath}$ with $E_{j\bar\imath}$'s, a C_ℓ long move the single unit $E_{i\bar\imath}$, and a B_ℓ short move $E_{i0} + 2\,E_{0\bar\imath}$ at the weighted transpose. The *form* b and its *dual pair* c are the twist's two data: b pairs the partner keys at one on both sides at B_ℓ and D_ℓ, at one against its balance partner at C_ℓ, and reads the B_ℓ null square at the balance partner of $[\,1:2\,]$, the short move's own forcing ($b(Eu, v) + b(u, Ev)$ reads the sum's unit at every key pair, and at the null key against $\bar\imath$ it collects one against the doubled null square); c, the invariant line of the doubled defining power, pairs the partner keys at one on both sides at B_ℓ and D_ℓ, at one against its balance partner at C_ℓ, and reads the B_ℓ null square at the balance partner of 2, the annihilation's own forcing (the short move's Leibniz image collects the null coefficient against the paired coefficient's double at the sum's unit); and the two data read at the product the identity, one read at the displayed entries. Each triple display is one read, $[E, F] = H$ at the coroot pair with F the pairing's transpose, the further display $[F, H] = 2\,F$ the grading's own read, the consecutive and last letters' checks displayed reads.

The member alternants. A dominant content β at every simple coroot pair beyond the unit (a series member's sorted lists, the entries strictly decreasing) reads its *alternant pair*, the reflection-graded fold over the Weyl list,

$$a_\beta \ := \ \Big\langle \sum_{w \text{ even}} x^{\,w\beta} \ : \ \sum_{w \text{ odd}} x^{\,w\beta} \Big\rangle \ ,$$

monomials at matched-list exponents keyed by the contents, one key per content at the member's coroot solve (the displayed cofactors', Construction 3) with a coordinate display reading its exponent lists in by the displayed coroot pairs, the fixed members' keys the coroot lists outright; an odd element exchanges the members and a reflection-fixed list reads equal members (Construction 1's reads at the member's grading); the cleared presentation multiplies by the full monomial at the top entry, every exponent natural, and the grading reads the letters, $\operatorname{sgn}\sigma$ against the flip count's parity at B_ℓ and C_ℓ and $\operatorname{sgn}\sigma$ alone at D_ℓ, the flip pairs even at their letter pairs. Per coordinate the *chord* t_i enters at the deck relation $x_i^2 + 1 = t_i\, x_i$, and the deck families read the columns (Definition 4): the plus column the plus read $x^{2b}+1 = x^b\, p_b(t)$ and the word column the word read $\langle x^{2b}:1\rangle = \langle x^2:1\rangle\, x^e\, Q_b(t)$ at the join $e+1=b$, the families' degree and parity reads the definition's. The cleared alternant is the signed-column minor (Definition 3), the Weyl fold regrouping per permutation at the flip folds within columns, the minor's own splitting fold, entry (i,j) the pair $\langle x_i^{b_1+b_j}:x_i^{g_j}\rangle$ at the gaps $b_j+g_j=b_1$, and the rows factor at the word columns' reads, $\langle x^{\,b_1+b_j}:x^{\,g_j}\rangle = \langle x^2:1\rangle\, x^{\,h}\, Q_{b_j}(t)$ at the join $1+h=b_1$, the coordinate's power one read per row. At ρ, the root fold's half, the cofactor at two per key, the symbol degrees are consecutive and the minor column-reduces over the chord carrier to the chord alternant, the assembly's pair product at the chords (Theorem 1): $\prod_{i<j}\langle t_i:t_j\rangle$ at C_ℓ, and $\prod_{i<j}\langle t_i^2:t_j^2\rangle$ at B_ℓ and D_ℓ, the columns even, the D_ℓ read on the plus-column minor with the last column's two, the signed minor's last column at equal members and the doubled alternant's two clearing by the product's injectivity. The chords clear to the coordinates at the pair identities

$$x_ix_j\,\langle t_i:t_j\rangle = \langle x_i:x_j\rangle\,\langle x_ix_j:1\rangle\ , \qquad x_i^2x_j^2\,\langle t_i^2:t_j^2\rangle = \langle x_i^2:x_j^2\rangle\,\langle x_i^2x_j^2:1\rangle\ ,$$

each one cross-multiplied read at the deck relations, and the power bookkeeping closes exactly, the coordinate's shift the factor count, $g+1$ at C_ℓ, $2g+1$ at B_ℓ and $2g$ at D_ℓ over the pair count g at $g+1=\ell$: the *assembly identity* is

$$a_\rho \;=\; \prod_{\alpha}\,\big\langle\, x^{\alpha_+}:x^{\alpha_-}\,\big\rangle\ ,$$

the fold over the member's positive list at the moves' two sides, one factor per positive root, the difference family at $\langle x_i:x_j\rangle$ or its squares, the sum family at $\langle x_ix_j:1\rangle$ or its squares, and the per-coordinate family at $\langle x_i^2:1\rangle$, every identity one minor computation per series.

Lemma 2 (Tops)**.** *A top is a content vector whose every raising image is the sum's unit. (i) A vector off the sum's unit, of a content maximal among its invariant subspace's occupied contents, is a top: a raising's image is homogeneous at a content above the maximal occupied one, so its component in every occupied summand, and with them the image, is the sum's unit. (ii) A top's content is dominant: at a simple index i, the triple $(E,F,H):=(E_i,F_i,H_i)$ reads $[E,F]=H$ and $[F,H]=2F$ with H reading a content at its coroot pair $\langle p:q\rangle$ (Construction 3; at the matrix units the site datum at $E_{ii}=E_{i+1,i+1}+H$ over consecutive letters, the pair $\langle \mu_i:\mu_{i+1}\rangle$), and the commutator's Leibniz fold runs over the word's splittings,*

$$E\,F^m \;=\; F^mE + \sum_{a+b+1=m} F^a\,H\,F^b\ ;$$

a lowering strictly descends the finite content set, so at a least $m\ge 1$ the iterate F^mv is the sum's unit; reading the display at the top v of content μ, both outer reads are that unit and each summand reads its site's coroot pair $\langle p:q{+}2b\rangle$, two down per lowering at $\alpha_i(\alpha_i^\vee)=2$, on one shared iterate F^aF^bv, the splitting's word one letter short of F^m and its read off the sum's unit at the least m: the collected pair's members are equal, $m\,p+m = m\,q+m^2$ at the splitting count read $\sum_{a+b+1=m}(2b+1)=m^2$, and the product's injectivity at the factor m reads the coroot pair's gap, $q+m=p+1$: the pair on its upper side or equal members at every simple index, the content dominant. At a positive pairing the pair reads its side through the pairing as well, the crossed exchange's read: the lowered image's self-pairing walks across the transpose to the exchange at the top, $\langle Fv,Fv\rangle + q\,\langle v,v\rangle = p\,\langle v,v\rangle$ at the raising's unit, and at v off the sum's unit the positive pairing forces q at or below p, the consecutive letters' pair read at the matrix units. (iii) The column exhibit t_λ is a top at content λ: a raising E_{ij} reads each column by Leibniz, sending a v_j to v_i inside a wedge already holding v_i when $j\le\lambda'_c$, a repeat at equal members, and reading the sum's unit outright when $j>\lambda'_c$, the wedge free of v_j (Construction 1).

Lemma 3 (The lowering span)**.** *The invariant subspace generated by a top v is the span W of the lowering monomials read at v. Every raising and H-read maps W to itself, and every operator reading the grading with them, W graded by its words: at a lowering word Fw' with w' shorter, and X a raising or an H-read, $XFw' = FXw' + [X, F]w'$; the first term closes by induction on the word length, and the closure field reads $[X, F]$ as a span fold at the moved content (Construction 3; a unit or a difference of diagonals at the matrix units), closing by that induction where the move is a root or the unit move and by the span where it lowers. At the matrix units the letter pairs compose from the adjacent ones: a pair at a gap beyond one is the crossed composite at an interior letter, $E_{ij} = [E_{ik}, E_{kj}]$ at the table's shared-letter read with the far delta clear at the distinct ends (Construction 2), and the descent on the gap closes every pair from the adjacent ones, raisings and lowerings both; so the words run at the adjacent lowerings alone, a further lowering the descent's composite, and a vector reading the sum's unit at every adjacent raising reads it at every raising, the composite's two members each through an adjacent image: the raising kernel is the adjacent stack's. The words stabilize, a lowering descending the dominance order strictly over the space's occupied contents, finitely many at the finite dimension (length $k\,d_f$ at the matrix units). Consequently the generated subspace has every content at or below the top's, and its top-content summand is the line $\mathbb{P}\,v$, a lowering word of positive length reading strictly below. A span membership is one decidable read: at a list of members independent in the pairing, the Gram's adjugate column against its determinant solves the pairing data (Definition 3), and a vector joins the list's span exactly where the residual, the determinant-scaled vector against the solved combination, reads the sum's unit componentwise; the solve reads the residual against every list member at the sum's unit, a member vector puts the residual inside the span, where its self-pairing reads that unit at the positive pairing, and the determinant clears an independent list, a determinant of equal members reading its adjugate column as a dependency perpendicular to the list, the column's fold of the members the sum's unit at the pairing read, against the independence, the descent closing at the withdrawn last member's own determinant; and the determinant reads a dependency's coefficients back: a combination reading the unit family pairs against each cofactor column at its own coefficient's multiple of the determinant, the diagonal cofactor pairing the determinant's read with the off-column pairings the sum's unit, while the combination's side reads that unit through the members' own pairings, so at a determinant of unequal members every coefficient is the sum's unit by the product's injectivity: independence is the determinant's unequal-membered read at the Gram, one decidable read. At a stated list the membership runs at its collection, one member joined per refusal along the list: the collected members are independent, each join's residual refusing the span collected so far, every listed member sits inside the collection's span, a passing member at its residual's read and a joined member as the collection's own, and the collection's members are listed members, so a vector joins the list's span exactly where it joins the collection's, the two memberships one read at the collection's residual, the joined-collection read; and an independent list is its own collection: each leading part of an independent list is independent, the withdrawn tail's coefficients read at the unit family one member at a time, and a passing residual puts the next member inside its leading part's span, the membership's combination joined with the member at one off-unit coefficient reading the unit family against the extended part's own independence, so each residual refuses and every member joins. An independent list's count sits at or below the coordinate count: at an occupied first coordinate some member's first read is off the sum's unit, the pivot, and every further member joins the pivot at the scaled pair, the member at the pivot's first read against the pivot at the member's own first read's balance partner, every moved first read the sum's unit, so the moved list reads its tails at one coordinate withdrawn, independent at the pivot's clearing, and the descent closes on the count; at every first read the sum's unit the tails alone read the list. An independent list at the coordinate count spans every vector of the width: a vector off the list's span joins it at the membership's refusal, the joined list independent one past* *the count, against the bound. Two independent lists inside one span read one count, and the count comparison runs at the first list's independence alone: a member joins the second list's span at a clearing, a scalar off the sum's unit with the member's cleared read a combination of the second list, one coefficient row per member; a combination of the first list reads through the rows' combination at the clearings multiplied into the coefficients, the product's injectivity reading a coefficient back at its clearing, so the rows are independent at the second list's count, the width bound closes the comparison, and either count sits at or below the other at the two lists' memberships, the adjugate column against the determinant the membership's own decidable read at an independent second list. An operator's trace read, the operator at its matrix against the space's content coordinates (the monomial coordinates at the matrix units, Construction 1), at an orthogonal spanning list of the span, the members pairwise perpendicular with every self-pairing off the sum's unit, is the fold of the diagonal operator pairings, each against the further members' self-pairings, and two orthogonal spanning lists read one trace at the lists' cleared products, the first list's off-unit reads clearing the shared factor at the product's injectivity and the pairing's bilinearity the whole derivation: a span member expands over the list, the projection fold*

$$\Big(\prod_i \langle y_i, y_i\rangle\Big)\, v \;=\; \sum_j \langle v, y_j\rangle \Big(\prod_{i\neq j} \langle y_i, y_i\rangle\Big)\, y_j \ ,$$

the sides' gap perpendicular to every member at the orthogonality and a span member itself, its self-pairing the members' pairings' fold at the sum's unit and the gap that unit at the positive pairing, the display's read; pairing one list's projection fold against its own further member reads the crossed fold, $\langle x_i, x_{i'}\rangle \prod_j \langle y_j, y_j\rangle = \sum_j \langle x_i, y_j\rangle\, \langle y_j, x_{i'}\rangle \prod_{k\neq j} \langle y_k, y_k\rangle$; and the second list's diagonal operator pairings expand over the first, the double fold collecting at the crossed fold with the off-diagonal terms entering at the first list's own orthogonality's unit, so the two trace folds read one value at the cleared products. A pairwise perpendicular list of members off the sum's unit is independent: a combination at the unit family pairs against each member to the member's own coefficient at its self-pairing, the orthogonality clearing the rest and the positive pairing the coefficient. At a pairwise perpendicular list the Gram's determinant reads the self-pairings' product: every assignment off the identity reads an off-diagonal pairing's factor, at equal members by the perpendicularity, so its term enters the fold at the sum's unit and the identity assignment's product is the read. The determinant-scaled residual then reads one value entrywise with the product-scaled residual, the self-pairings' product against the vector at the projection fold's own sum withdrawn, at every vector of the list's width: each residual is perpendicular to every list member, each pairs the two residuals' gap at the determinant against the vector's read, so the gap's self-pairing reads the sum's unit and the positive pairing reads the gap at that unit entrywise, the two residuals one value at every coordinate. The graded dimensions are then counts at the read, one member joined per refusal.

Lemma 4 (Blocks)**.** *Write W_v for the subspace a top v generates (Lemma 3); the clauses read at the span calculus's stated data (Lemma 3). A subspace enters as a stated list of content-homogeneous members with its membership reads, each a decidable read at the list: the content projections read a general member's components as members (Construction 3; Construction 2 at the matrix units), so the homogeneous form covers every list, and the closure read is one membership read per raising or lowering and list member, the operator's image joining the list's span, an H-read scaling a homogeneous member inside it (Construction 3's grading). A map enters as its matrix family, one matrix per occupied content and per letter-pair image of one, against the space's content coordinates (the monomial coordinates at the matrix units, Construction 1), its value at a member the matrix read with the family's linearity the matrix action's own; the moved reads are the statement's data: per raising or lowering and span member, the operator's matrix at the member's value is the family's* *read of the operator's image. (i) W_v is irreducible. At a stated list L of members of W_v at the closure read, every W_v member joins L's span, or every member of L is the sum's unit; the top-content group splits the two, the group's span the line $\mathbb{P}\,v$ (Lemma 3). A top-content member of L off the sum's unit clears v into L's span, and every W_v member with it, a lowering word's read at v with the closure transporting the letters across L's combinations (Lemma 3). At every top-content member of L the sum's unit, L pairs v at equal members, distinct contents pairing at equal members outright, the grading orthogonal (Construction 1; Construction 3); a word's pairing walks across the transposes, $\langle x,\, E_{ij}\, y\rangle = \langle E_{ji}\, x,\, y\rangle$ with the transposed letter keeping L's span (Construction 2; the lowerings the raisings' transposes, Construction 3), so L pairs every W_v member at equal members; and a member's self-pairing, the fold of its own combination's pairings, reads equal members: the member is the sum's unit at the positive pairing. (ii) Endomorphisms are scalars. At a matrix family on W_v, the top's value a W_v member at the content λ with the moved reads at the words' spanning list (Lemma 3), the values read one cleared line,*

$$\langle v,\, v\rangle\ \varphi w \ =\ \langle \varphi v,\, v\rangle\ w \qquad \textit{at every member } w\,,$$

φw the family's value at the member w: the top's value sits in the top-content group's span, the line $\mathbb{P}\,v$ (Lemma 3), the display its residual read there; and the induction along the span's words passes the display down, a further member a lowering's image of an earlier one (Lemma 3), the moved read mapping the earlier member's display across the lowering with the matrix action clearing the scalars, the further values members by the display. (iii) Blocks at distinct top contents are disjoint. At a matrix family from W_v into W_u, values at the members' own contents with the moved reads and some value off the sum's unit, the top contents agree. (ii)'s induction reads every value as its member's own word at the top's value, so the top's value is off the sum's unit. The top u sits off the sum's unit: at a unit head every word's read and every settled value is the sum's unit, against the value off it. The value list joins (i)'s data inside W_u, its closure the moved reads across W_v's own (Lemma 3), and (i) at an off-unit value reads every W_u member into the values' span, whose occupied contents sit among the values', at or below the top's content λ (Lemma 3): u's content sits at or below λ, λ at or below u's at the top value's membership, and the two dominance witnesses join to the unit fold (Construction 3): the contents agree. (iv) One dimension per shape. Two tops v, u at one content λ, in powers of one degree $|\lambda|$ at the matrix units. The words' pairings read the top's content alone: two words at distinct moved contents pair at equal members outright, the grading orthogonal (Construction 1), and at lowering words m, m' at one moved content the pairing $\langle m\, v,\, m'\, v\rangle$ walks the reversed word across the transposes (Construction 2), the table folds each crossing into an H-read at the content with a shorter word's pairing (Construction 3), and a raising's read at the top is the sum's unit, so the pairing is the content's own coefficient against the self-pairing $\langle v,\, v\rangle$, one value per word pair at every top of content λ. The cross-content pairings are the grading's equal-membered reads on both lists (Construction 1). So, content group by content group, the word images at u of W_v's spanning list read the group's own Gram at the coefficient scale, the self-pairings off the sum's unit at the lists' independence, and each group's images are independent inside W_u's group at the moved content: the groups' counts sit at or below each other's, two independent lists inside one span reading one count (Lemma 3), the occupied contents' counts summing to the dimensions at the orthogonal grading, $\dim W_v = \dim W_u$, and $d_\lambda := \dim W_{t_\lambda}$ is one count per shape (Lemma 2(iii)).

Definition 5 (The block count)**.** For a table space M (Construction 3; the unit spaces among them, Construction 2) and a content λ, the shapes at the matrix units, the *block count* is

$$N_\lambda(M) \ :=\ \dim\{\text{content-}\lambda \text{ vectors of } M \text{ whose every raising image is the sum's unit}\}\ ,$$

a kernel dimension of the stacked raisings on the content summand (Definition 3), and the *fusion count* at contents a, b, c is $N_{ab}^c := N_c\big(W_a \otimes W_b\big)$ over blocks at those top contents, at the matrix units the shapes' blocks $W_\lambda := W_{t_\lambda}$ (Lemma 3; Lemma 2(iii)).

Lemma 5 (The block count, and the fusion data)**.** *(i) Exhaustion. A table space is an orthogonal sum of blocks: at occupied M take a top v at a maximal occupied content (Lemma 2(i)), split $M = W_v \perp W_v^\perp$ (Lemma 1), and descend the dimension. The complement enters as the stated members' residuals at the block's collected content groups, the joined-collection read's own residual (Lemma 3), and a joined-span member perpendicular to the block's rows sits in the complement's span alone: the block's part of its combination pairs every block row at the sum's unit, its self-pairing reads that unit through the rows' pairings, and the positive pairing withdraws it (Lemma 3). The descent closes at the stated members' count: each round's top is a member of its own block, its residual reading the unit through the solve (Lemma 3), so every round withdraws at least its own pick from the stated carrier and the count bounds the rounds. (ii) The count. $N_\lambda(M)$ (Definition 5) adds over orthogonal sums: the grading and the raising kernels split, a kernel member's two parts each a kernel member of its own stated summand, the parts' raising images inside their own summands' moved spans at the closure with the two images perpendicular and their join the sum's unit, the positive pairing withdrawing both (Lemma 3); and the kernel dimensions add at the kernel lists (Definition 3), the two sides' lists joining to an independent spanning list of the sum's kernel, the counts the graded dimensions' read (Lemma 3). A block reads $N_\lambda(W_u) = \delta_{\lambda,\mathrm{content}(u)}$ (Lemma 3, tops of a block sitting at its own top content alone by Lemma 4(iii)'s read). So $N_\lambda(M)$ is the number of λ-blocks of every exhaustion, and*

$$\dim M \;=\; \sum_\lambda N_\lambda(M)\, d_\lambda\ ,$$

the sum one summand per distinct top content of the exhaustion, each a shape's row list at the dominance read (Lemma 2(ii)) with d_λ its block's stated count (Lemma 4(iv)), and a shape off the tops' contents reading its count at the unit, the vacant collected group's kernel. The display refines to the contents: a content's dimension over M is the sum at $N_\lambda(M)$ against the shape's own occupancy at the content in place of d_λ, each block reading its shape's occupancy (Lemma 4(iv)'s group counts), the blocks pairwise perpendicular with each content summand split over their groups and the graded dimensions the counts at the read (Lemma 3). (iii) The fusion data. $N_{ab}^c = N_c\big(W_a \otimes W_b\big)$ (Definition 5): kernel-dimension counts by construction, the stacked raisings read against the stated pair list, the two blocks' spanning lists' tensors at complementary contents (Lemma 3), contents adding over the factors with the pairs spanning the summand (Construction 1), and the pair list is independent: two pairs pair at the factors' pairings' product, a concatenated arrangement reading its split at the first factor's degree with a block's contents at one degree, and two pairs at distinct first contents pair at equal members outright, each tensor's coordinates at the sum's unit off its own contents' splits (Construction 1), so a combination reading the unit family pairs every pair at the sum's unit and its coefficients read back through the two factors in turn, per first-factor member the contraction a combination of the second factor's group pairing every group member at that unit, its self-pairing reading that unit through the pairings' fold, the positive pairing withdrawing it and the group's independence reading its coefficients at the unit family (Lemma 3), the first factor's groups then reading the coefficients themselves: the kernel dimension is the display's own count, with

$$N_{ab}^c = N_{ba}^c, \qquad \sum_e N_{ab}^e N_{ec}^d = \sum_f N_{bc}^f N_{af}^d, \qquad N_{a\mathbf{1}}^c = \delta_{ac}, \qquad \sum_c N_{ab}^c\, d_c = d_a\, d_b,$$

the first by the flip of factors: the arrangements' exchange maps each pair to the exchanged pair at the commuted coordinate products, an orthogonal map at the place action's transpose identity *commuting with every raising, the Leibniz fold's places exchanged (Construction 2), so a kernel family reads across the flip at the exchanged coefficients and the two kernel dimensions read one count, each kernel list inside the other's span (Definition 3); the second by exhausting* $W_a \otimes W_b \otimes W_c$ *through either pairing, the composition associative at the arrangements' join (Construction 1): the parenthesized factor's exhaustion splits the triple orthogonally at its one degree, a block's contents at one degree with the exhausted carrier's members at the factors' degrees' sum, so two blocks' tensors against the further factor pair at the factors' pairings' products, the concatenated arrangements split at the one first-factor degree (Construction 1), the blocks perpendicular ((i)'s split), spanning the composite at the collected pairs (Construction 1), the counts adding over the split ((ii)); a block at a top content reads the fused count at the shape's own block: the block's members are its words' images at its top (Lemma 3), the one word family at the shape's own top spans the shape's block group by group, the images independent at the one-value word pairings with the occupancies agreeing (Lemma 4(iv)); a stacked combination of raised pairs reads the sum's unit exactly where its self-pairing does (the positive pairing), the raised tensors' pairings collecting to the factors' pairings' folds at the raisings' split over the concatenation (Construction 2), each fold the word pairings' coefficient against its top's self-pairing (Lemma 4(iv)), so the two carriers' unit reads agree at the off-unit self-pairings by the product's injectivity, the collections' refusals the pairings' own reads matched with them (Lemma 3), and the two counts are one value: either pairing's count is the display's sum, one summand per distinct top content of its exhaustion, the second pairing read across the flip; the third at the unit shape's line: the line's exhibit is the scalar one (Construction 1), a tensor against it reads the first factor's own coordinates at the vacant second arrangement, so the stated pair list is the block's own content group and the count is (ii)'s closing display, the Kronecker delta at the head's content; and the fourth by (ii) at* $M = W_a \otimes W_b$*, the fused carrier closed at the letters' split folds (Construction 2) with its dimension the pair list's count at every content, the two dimensions' product. Moreover* $N_{ab}^{a+b} \geq 1$*:* $t_a \otimes t_b$ *is a top off the sum's unit at content* $a+b$*, each raising's split reading a factor's top against the other factor tensored on (Lemma 2(iii); Construction 2), and its pair's own coefficient family is a kernel member off the unit family, the kernel list's span occupied (Definition 3).*

Lemma 6 (Strings, and the content symmetry)**.** *Fix a positive root* α *and its triple* $(E, F, H) := (E_\alpha, F_\alpha, H_\alpha)$ *on a table space* M *(Construction 3's root triples), the matrix units' at the letter pairs. The triple's span is transpose-closed (*$E^\mathsf{T} = F$*,* $H^\mathsf{T} = H$*), so* M *splits orthogonally into subspaces invariant for the triple and irreducible for it (Lemma 1, descending the dimension), each generated by a triple-top* v *at its* H*-eigenvalue* h*, at the join* $q+(h+1) = p+1$ *over the top's coroot pair* $\langle p\!:\!q\rangle$ *(Lemma 2(i),(ii) at the triple), and spanned by* $v, Fv, \ldots, F^h v$ *with* $F^{h+1}v$ *the sum's unit: the least* m *at which* $F^m v$ *is that unit has* $h+1 = m$ *(Lemma 2(ii)), so the string's* H*-eigenvalues descend from the top's* h *by two per lowering, the depth-*b *eigenvalue the pair* $\langle h\!:\!2b\rangle$ *at positive* $b \leq h$*, each once, two depths at the join* $b + b' = h$ *reading member swaps. At the matrix units the splitting is the residual walk at the ordered pair's height, the first letter's count against the second's: the span's contents run at the height descending, a raising moving the height key up by two, and at a content a group member off the collected members' span names a top, the membership test's residual itself (Lemma 3), perpendicular to the collected members by the solve's identity and inside the group's span; the top's raising image sits in the content above's group span, spanned by the walk's own members there, and perpendicular to them, so the image's self-pairing reads the sum's unit at the positive pairing and the image with it. The top's gap is the pairing's read, the lowered image's self-pairing joining the second count's multiple of the top's to the first's,* $\langle Fv, Fv\rangle + \nu_j \langle v, v\rangle = \nu_i \langle v, v\rangle$*, the crossed exchange's read at the letter pair (Lemma 2(ii)), the positive pairing forcing the second count at or below the first, the height their gap; each deeper member's self-pairing is the raise* *coefficients' telescoped multiple of the top's at the fold's read* $E\,F^{b+1}v = (b{+}1)(h{-}b)\,F^{b}v$*, off the sum's unit through the height and reading it at the further lowering, the coefficient's own unit there; and per content the members close at the exchange bound, one top per refusal, the walk's count the group's own (Lemma 3). The triple moves contents by* α *alone, fixing every reflection-fixed read, so the strings split each joint grading, and summing string by string,*

$$\dim M_\mu \;=\; \dim M_{s_\alpha\mu}$$

at the root reflection s_α *(Construction 3; the transposition of the two letters at the matrix units): content multiplicities are invariant under every root reflection and their compositions. At the matrix units a block's occupied contents are permutation-closed, sit at or below the top (Lemma 3), and so sit at or above its reversal* $w_0\lambda$ *in the dominance order, a sorted prefix sum falling as its tail rises; the reversal is occupied once, at the top's own multiplicity.*

Lemma 7 (The Casimir reads the shape)**.** $C := \sum_{p,q} E_{pq}E_{qp}$ *commutes with every unit, a diagonal unit at the grading's read, every fold term reading each content summand to itself with the unit at its scalar there (Construction 2), and an ordered letter pair at the derivation here: per factor the table's display reads the join at the factor's letters (Construction 2),* $E_{qp}E_{ab} + \delta_{bq}E_{ap} = E_{ab}E_{qp} + \delta_{pa}E_{qb}$ *and* $E_{pq}E_{ab} + \delta_{bp}E_{aq} = E_{ab}E_{pq} + \delta_{qa}E_{pb}$*, the second read inside the first's middle term chains the two; a letter off a content's support acts at the sum's unit there, the Leibniz fold over a vacant place index (Construction 2): at the unit's own letter off the source content's support the identity's two members read that unit together, and inside the fold the governing letter is the inner factor's lowered one, each member read at its own side's content, the source on the one side and the unit's image on the other, an image-side vacancy sitting at the unit's own lowered letter alone at its one source count, the term's surviving members closing at the display's own shorter instance, a factor written at a content pair off the action's own move reading the null map, the grading's read (Construction 2), and the vacant image letter's own pair closing at the diagonal reads' square, the one count twice; so the collected reads run at the letters each side supports, the unit's letter supported in the source throughout, the chained term's two move orders one content and the collected summands' contents one spelling at the round-trip and chain reads through the supported unit letter, the degenerate letters' members at the sum's unit on both sides; and the* (p,q)*-fold collects each side's delta families to the one join*

$$C\,E_{ab} + \sum_p E_{pb}E_{ap} + \sum_q E_{aq}E_{qb} \;=\; E_{ab}\,C + \sum_q E_{aq}E_{qb} + \sum_p E_{pb}E_{ap}\ ,$$

the two folds a summand on both members fixing the value (Definition 1): $C\,E_{ab} = E_{ab}\,C$*. On a block* W_v *at top content* λ *of degree* k *it is therefore one scalar (Lemma 4(ii)), read at the top, where* $E_{pq}E_{qp}v$ *at* $p<q$ *is* $[E_{pq},E_{qp}]v = \langle\lambda_p:\lambda_q\rangle\,v$*, the row pair's value, the commutator the site datum at* $E_{pp} = E_{qq} + [E_{pq},E_{qp}]$*, a vacant row* p *reading the term and the pair's value at the sum's unit together, the row list's descent putting the later row at that unit with it (Construction 1), and at* $p>q$ *is the sum's unit, the raising's value at the top:*

$$C\big|_{W_v} \;=\; q(\lambda) + \langle\lambda,2\rho\rangle, \qquad q(\lambda) := \sum_i \lambda_i^2, \quad \langle\lambda,2\rho\rangle := \sum_{p<q}\langle\lambda_p:\lambda_q\rangle\,,$$

the balance-pair fold over the place pairs, a natural at the sorted rows, collected per row as the balance-pair row weights $\sum_i \lambda_i\,\langle r+2:2i\rangle$*. The units instantiate the Casimir field (Construction 3): the diagonal read* $\sum_p E_{pp}^2$ *at the row squares, the pair terms* $E_{pq}E_{qp} + E_{qp}E_{pq}$ *at weight one, every length two, and the top read the display's. The trace recursion. On a table space's block at top* *content λ (Lemma 4), write* $\mathrm{mult}(\nu)$ *for the content summand's dimension and, per positive root at the root coroot pair* $\nu(\alpha^\vee) = \langle p : q\rangle$, $\nu^{(l)} := \nu + l\,\alpha$ *for the moved contents: tracing the Casimir fold over the* ν*-summand, the summand's trace the one value at every orthogonal spanning list (Lemma 3's trace read), the block scalar against the diagonal and the root terms' string traces (Lemma 6 at the root triples), reads*

$$q(\lambda{+}\rho)\,\mathrm{mult}(\nu) + \sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\, 2 \sum_{l\geq 1} q\,\mathrm{mult}(\nu^{(l)}) \\ = q(\nu{+}\rho)\,\mathrm{mult}(\nu) + \sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\, 2 \sum_{l\geq 1} (p{+}2l)\,\mathrm{mult}(\nu^{(l)}) \ ,$$

the ρ*-shift the form's bilinearity at the weight collection* $\langle \nu, 2\rho\rangle = \sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\,\nu(\alpha^\vee)$*, and per string one identity: an* α*-string through* ν *at depth* j *from its top reads the trace coefficient* $(j{+}1)\,d + j\,(d{+}1)$ *at the tie* $d + q = p + j$*, its moved reads collect at* $2\sum_{l\leq j}(p{+}2l) = 2j\,(d{+}1) + 2j\,q$*, the two folds one gap identity, and a string above* ν *sums its moved reads to equal members, the depth pairs at the join swapping members. The telescope. Per positive root, with* D_α *the coroot read per monomial,* $D_\alpha\, x^\nu = \nu(\alpha^\vee)\, x^\nu$*, and*

$$G_\alpha := \Big\langle \sum_\nu \Big(\sum_{l\geq 1} (p{+}2l)\,\mathrm{mult}(\nu^{(l)}) \Big) x^\nu \ : \ \sum_\nu \Big(\sum_{l\geq 1} q\,\mathrm{mult}(\nu^{(l)}) \Big) x^\nu \Big\rangle \ ,$$

each member finite at the occupied contents, one further move drops one l*, so at the move's two sides,* $\alpha = \langle \alpha_+ : \alpha_-\rangle$ *per coordinate,*

$$x^{\alpha_+}\, G_\alpha \ = \ x^{\alpha_-}\, \big(G_\alpha + D_\alpha\, \mathrm{ch}\big) \qquad \text{at} \quad \mathrm{ch} := \sum_\nu \mathrm{mult}(\nu)\, x^\nu \ ,$$

coefficientwise the drop's read, and the recursion collects to

$$q(\lambda{+}\rho)\, \mathrm{ch} \ = \ \mathcal{L}\, \mathrm{ch} + \sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\, 2\, G_\alpha \qquad \text{at} \quad \mathcal{L}\, x^\nu = q(\nu{+}\rho)\, x^\nu \ .$$

The generator list reads it through the Fierz sum (Definition 7): a matrix acts as $\sum_{pq} A_{pq} E_{pq}$*, so*

$$\Delta \ = \ \sum_a X_a^\dagger X_a \ , \qquad d_f\,\Delta + \xi\, \mathcal{Z}^2 \ = \ \xi\, d_f\, C, \qquad \mathcal{Z} := \sum_p E_{pp},$$

positive semidefinite by its shape, its split the fold of squares, and the block read is the additive bracket $d_f\,\Delta|_{W_v} + \xi\, k^2 = \xi\, d_f\, \big(q(\lambda) + \langle\lambda, 2\rho\rangle\big)$*, the cleared content read of the weight-lattice Casimir at the form's scale* ξ*. Checks: at the one-box shape* $\lambda = (1)$*,* $q = 1$*,* $\langle \lambda, 2\rho\rangle = r$*,* $k = 1$*, and the cleared bracket reads* $1 + r\,(r+2) = d_f\,(1 + r)$*, the one-value join* $r(r+2) + 1 = d_f^2$*: so* $\hat{c}_2(f) = [\, r(r+2) : 2d_f^2\,]$ *against the adjoint's below. At* $\lambda = (2, 1^c)$ *at* $c + 2 = d_f$*,* $k = d_f$*,* $q = r + 3$*,* $\langle\lambda, 2\rho\rangle = 2r$*, the cleared bracket reads* $k^2 + 2d_f \cdot d_f = d_f\, \big(q + \langle\lambda, 2\rho\rangle\big)$*, both sides* $3d_f^2$*, the traceless read* $2\,d_f$*: the adjoint Casimir is* $C_2(\mathrm{adj}) = d_f$ *at* $\xi = [\,1 : 2\,]$*, the Fierz weight of Definition 7.*

Lemma 8 (Duality, and the unit read)**.** *(i) The dual. A unit space* M *defines its dual action on the underlying paired space,* $E^\vee_{ij}$ *the balance partner of* E_{ji}*, the sum's every entry of equal members: a vector's partner is its scale by the unit's own partner (Definition 1's displayed reads), so a dual image is a scaled image, a matrix read of a scaled vector its own read's scale, two partners compose* *to the original, and a pairing reads the partner on either side, the coordinate fold's scale. The dual obeys the table,*

$$[E^\vee_{ab}, E^\vee_{cd}] + \delta_{da}\, E^\vee_{cb} \;=\; \delta_{bc}\, E^\vee_{ad} \ ,$$

the exchanged letters' read: the commutator's factors compose at the double partner to $[E_{ba}, E_{dc}]$*, its display at the letters* (b, a, d, c) *Construction 2's, and joining* $\delta_{cb}\, E_{da}$ *to both members reads each member as a unit against its own partner, the first through that display at* $\delta_{ad}\, E_{bc}$ *and the second at* $\delta_{cb}\, E_{da}$ *outright, both at the sum's unit, a summand on both members fixing the value (Definition 1); and the transpose identity,* $(E^\vee_{ij})^\mathsf{T} = E^\vee_{ji}$*, the partner read across the pairing over the original's transpose. Each content is the original's balance partner,* $\mu + \mu^\vee$ *of equal members coordinatewise, the dual diagonal reading a content summand at its coordinate's partner, the diagonal's read at the partner scalar. Invariant subspaces agree: a dual image is the exchanged letters' image at the partner's scale, a span member's scale inside the span, and the double partner reads the original images back; so duals of blocks are irreducible, Lemma 4(i) at the agreement, and* $(M^\vee)^\vee = M$*, the exchange twice at the partner twice, each letter's read the original's own. The pairing's evaluation* $\mathrm{ev} : M^\vee \otimes M \to \mathbb{P}$ *is equivariant to the unit: a letter pair's Leibniz image reads* $\langle E^\vee_{ij} f,\, x\rangle + \langle f,\, E_{ij} x\rangle$*, the first summand the partner of* $\langle E_{ji} f,\, x\rangle$*, the transpose identity reading that pairing as the second summand, and a value joined to its partner reads the sum's unit. Its Gram-dual coevaluation at a stated independent spanning list* y *is the determinant-scaled identity datum,*

$$\mathrm{coev} \;:=\; \sum_{j,k} A_{jk}\; y_j \otimes y_k \;\in\; M^\vee \otimes M \ ,$$

A *the Gram's adjugate against its determinant, the list against the adjugate reading the determinant's diagonal (Definition 3; the grading splits the Gram at one block per occupied content, the cross pairings of equal members, so the adjugate collects each block's own at the further blocks' determinants), and the coevaluation is equivariant to the unit as well: a letter pair's Leibniz image paired against a member pair* (y_a, y_b) *collects at the adjugate identity to the determinant against one crossed pairing per summand, the partner of* $\langle E_{ji} y_b,\, y_a\rangle$ *at the first slot's read and* $\langle E_{ij} y_a,\, y_b\rangle$ *at the second's, the transpose identity with the fold's exchange (Definition 1) joining the two at partners, so the moved coevaluation pairs every member pair at the sum's unit. And* $\mathrm{inv}(M^\vee \otimes W) \cong \mathrm{Hom}(M, W)$ *at the equivariant maps: an invariant of the dual tensor is a family* $\sum_i f_i \otimes w_i$ *at matched contents, the dual diagonals' unit read, whose every letter pair's Leibniz image, the dual action at the first factor, pairs every member pair of the two carriers' stated lists at the sum's unit. The invariant reads as the equivariant* $x \mapsto \sum_i \langle f_i, x\rangle\, w_i$*, the fold the argument's own content group at the matched contents with every cross-content pairing the sum's unit: the invariance read at the pair of a stated argument with a further member joins two crossed folds, the first the map's value at the moved argument through the evaluation's equivariance and the second the moved value's own pairing, so the map's two moved reads agree against every member and the moved images are one value at the pairing's definiteness, the map's data one matrix per occupied content with the moved reads (Lemma 4's matrix families); and back through the coevaluation, an equivariant* φ *reading the invariant* $\sum_{j,k} A_{jk}\; y_j \otimes \varphi(y_k)$*, the coevaluation's second slot at the map's value, the family at matched contents with each value at its member's own content: the invariance read at the pair of a member with a further vector collects at the adjugate identity twice, the member's own column reading the determinant against the map's moved value at the member's key with the equivariance transporting the letter across the slot, and the first slots walked across the transpose identity to the members' pairings against the member's letter image, the fold at the image's pairing data the adjugate solve (Lemma 3); the image's cleared combination over the list's span reads the solve at the determinant against the combination's coefficients, the map's value additive and scale-reading across the members, so the two reads join at partners and the clearing withdraws at the* *product's injectivity. The round trips read the determinant's scale: the composite at an equivariant φ reads $\det G \cdot \varphi$, the adjugate identity at the map's argument with the members' reads extending over the list's span, both composites additive and scale-reading in the argument, a span member's cleared combination reading them through the members with the clearing withdrawn at the product's injectivity (Lemma 3), and the composite at an invariant reads its $\det G$-multiple, the identity at each f_i over the list's span; the determinant sits off the sum's unit at the independence (Lemma 3), every read one value across the scale (Definition 1's homogeneity principle), and the two spaces' counts read one value at the correspondence's own carriers. An invariant enters at its coefficient family over the two stated lists' matched member pairs, the moved reads one stacked linear system in the coefficients, each row the moved form's fold at a letter pair and a member pair, with its count and kernel list the elimination's (Definition 3). A family whose map reads every listed member at the sum's unit is the unit family: per content the value is the second list's combination at the first list's weighted pairings, the second list's independence reads those weights back (Lemma 3), and per second member the weights pair the first list's group against its every member at the sum's unit, the perpendicular combination the unit family at the group's independence (Lemma 3). An equivariant map enters the grid through the second list's solve, each value's cleared combination over that list's span at the graded Gram's adjugate against its determinant (Lemma 3; the grading splits the Gram at one block per occupied content, so the adjugate collects each block's own at the further blocks' determinants, the coevaluation's read at the second list), the grid's datum one block per occupied content, the content group's rows at the content's own values' solves with each member's row read at the member's place in its group, the moved reads at the sum's unit by the equivariance above and the map the determinants' scale of its own by the round trip: the kernel list and a stated map list read one count, two independent lists inside one span (Lemma 3). (ii) The full-column line, and the dual block. The full wedge w_{d_f} spans the block $W_{(1^{d_f})}$, the full-column line: every unit at $i \neq j$ maps it to the sum's unit, a repeat at each Leibniz read, so $d_{(1^{d_f})} = 1$; tensoring by it maps blocks to blocks: two monomial tensors pair at the factors' pairings' product, a concatenated arrangement reading its split at the first factor's degree (Construction 1), a tensored member's coordinates the factor products at the split arrangements with the unit tail exactly at the factor's own, and a letter's action splits over the concatenation by the Leibniz read, the wedge factor's term the repeat's unit, a letter vacant in the further factor leaving the joint image at that term alone, the unit tail: a lowering word's image at $w_{d_f} \otimes t_\lambda$ is the wedge tensored onto its image at t_λ, the exhibits' composition at the column union reading $t_{\lambda+(1^{d_f})}$ as $w_{d_f} \otimes t_\lambda$ (Construction 1), the span memberships transport at the wedge tensored on: a combination's image is the image's combination, and two images at one value read their gap's tensor at the unit tail, the tensored coordinates' split above reading the gap at the further factor against the wedge's differing members (Construction 1), so a member joins its group's span exactly where its image joins the image group's (Lemma 3's membership at its cleared combination, the two directions the solve's own reads), so the block $W_{\lambda+(1^{d_f})}$'s span is W_λ's span at the wedge tensored on, member for member, $W_\lambda \otimes W_{(1^{d_f})} \cong W_{\lambda+(1^{d_f})}$ with $w_{d_f} \otimes t_\lambda$ its top, and the dimension, the occupancies at the shifted contents, and the fused counts read the mapped span's: the fused pool against a further block pairs the mapped members at the arrangements' join (Construction 1), so the pair list at a shifted content is the content's own list mapped; a stacked raising's image at a mapped pair is the wedge tensored onto its image at the pair, the wedge factor's term the repeat's unit, and a tensored image reads the unit tail exactly at the further factor's own read, the split coordinates at the wedge's differing members, so the stacked raisings annihilate a coefficient family's combination at the mapped list exactly where they annihilate it at the list, the kernel data one family at the two pair lists (Definition 5) with the collections two lists in one span at one count (Lemma 3), and the count at the shifted content is the content's own; and the line* *reads at every power by the descent: a power's exhibit is the wedge tensored onto the prior power's, the column union's read, a lowering's image reading the wedge factor's term at the repeat's unit and the further factor's at the prior power's own unit read, so every lowering image of every power's exhibit reads the unit tail and* $W_{(m^{d_f})} = W_{(1^{d_f})}^{\otimes m}$ *is one line at every* m*. The dual block reads by the content symmetry:* W_λ*'s occupied contents sit between* $w_0\lambda$ *and* λ *with the reversal occupied once (Lemma 6), so* $W_\lambda^\vee$ *reads its maximal occupied content, the reversal* $w_0\lambda$*'s balance partner, on one line, a top (Lemma 2(i)), and*

$$W_\lambda^\vee \otimes W_{(1^{d_f})}^{\otimes \lambda_1} \;\cong\; W_{\lambda^*}\,,$$

λ^* *the complement shape, every column of* λ *at a gap against the full length read to the gap's column at* $\ell + \ell' = d_f$*, a full column reading to the full-column line above: two irreducible unit spaces sharing the top content, the complement shape's row list at the join* $\lambda^* + w_0\lambda = \lambda_1(1^{d_f})$ *(Lemma 4(iv)). (iii) The unit read. For shapes* a, b *the full-column powers' count in* $W_a \otimes W_b$ *is a Kronecker delta: at the tie* $m\, d_f = |a| + |b|$ *where a natural* m *witnesses it,*

$$N_{(m^{d_f})}\big(W_a \otimes W_b\big) \;=\; \dim \operatorname{Hom}\big(W_a^\vee \otimes W_{(1^{d_f})}^{\otimes m},\; W_b\big)$$

the Kronecker delta, one exactly at b *and* a^* *differing by full columns. The count is the kernel dimension of the stacked raisings at the stated pair list, the two blocks' spanning lists' tensors at complementary contents (Definition 5; Lemma 5(iii)), and the display is (i)'s adjunction at the line* $W_{(1^{d_f})}^{\otimes m}$*: the kernel families and the equivariant maps read one count through the value map below, (i)'s count clause at the complementary grid. A kernel family reads as its value map, per argument the second slots' combination at the coefficients against the first slots' pairings, and the map is equivariant: the stacked adjacent raisings annihilate the family's combination, and the annihilation reaches every letter pair at distinct letters at the full-column content: the adjacent lowering's at (ii)'s line with the moved content's group vacant, and each further pair's at the crossed composite on the gap, the descent at raisings and lowerings both (Lemma 3), so the moved reads close at every such pair, the dual action at the argument against the action at the value, (i)'s moved reads at the exchanged letters. Off the join the kernel is the unit family. At a kernel family off the unit family some value sits off the sum's unit: per content a value is the second group's combination at the first group's weighted pairings, and a family whose every value reads the unit tail is the unit family, the two groups' independence reading the weights and the perpendicular combination back ((i)'s count clause at the complementary groups; Lemma 3). The value list at the first block's stated members is closed at the letters inside the second block's span, a letter's image of a value the value at the argument's dual move with the moved argument's combination reading through the fold, so at a value off the sum's unit every* W_b *member joins the value list's span (Lemma 4(i)), the head among them; the head's group in the value list holds a value off the sum's unit, a combination of unit tails reading the unit tail, and a value off that unit reads occupied slots, the pair guard's join exact:* b*'s row list joins an occupied content of* W_a *to the full columns, and the factors' exchange reads* a*'s row list so as well, the arrangements' exchange at the commuted coordinate products (Lemma 5(iii)). The occupied contents sit at or above the reversals (Lemma 6), and the two joins exchange the reads: at the full-column content a prefix of the complement joins the content's own prefix to the full columns' count, and at one degree a prefix of a reversal joins its list's complementary prefix to the degree, so the first factor's floor reads the occupied content at or above* a*'s reversal while the second factor's floor, read through the two exchanges, sits at or below it, and the two dominance witnesses join to the unit fold (Construction 3):* b*'s row list is the complement's read,* b *and* a^* *differing by full columns. At the join the kernel is one line. The first block's reversal member is occupied once (Lemma 6) and is the dual top, every dual raising's image at the unit tail ((ii)'s* *dual block), and its value's content, the complement of the reversal, is the join's read of the target's own top: the value at the dual top sits in the head's group, the top line (Lemma 3), a scale of the head. A kernel family off the unit family holds a value off the sum's unit at the reversal member: the head's group read above names a first-block member whose content joins b's row list to the full columns, as the reversal does at the join, and the sum's injectivity reads the two contents one (Definition 1), the reversal member occupied once; and the value map reads its coefficient family linearly, each coefficient entering its own weight once, so two families' cleared combination reads the two values' own at every argument (Definition 1): at that scale the unit the family is the unit family, two kernel families read one line, the cleared combination at the unit scale the unit family, and the count is at most one. And the kernel is occupied: the dual walk's H-read at a content is the partner pair, one value with the complement content's own at the full column's constant cross-added (Definition 1; (i)'s dual diagonal with (ii)'s line), a crossing of the walk reads the letter pair's site datum, two letters' reads at their gap (Lemma 7's display), so the constant withdraws at the gap's cross-add; the lowering moves the letter pair down and the dual moves it up, the joined content riding each letter's round trip at the occupied read, so the two walks' contents join to the full columns wherever both images sit off the unit tail, and every crossing of the dual walk reads the complement content's own crossing, and the dual words' pairings at the reversal member fold the walk's coefficients at those reads (Lemma 4(iv)): the dual word family at the reversal member and the word family at the target's top read one Gram at the two tops' self-pairings' scales, the source and the target one block at (ii)'s dual block, the dual twisted by the line reading the complement shape's carrier, and each dual word's partner word is the word at the exchanged letters, a dual move at a letter pair the complement's move at the exchanged pair ((i)'s dual action). The coevaluation at the two collected word families, the Gram's adjugate against its determinant weighting each dual word against its partner word, pairs every stacked raising's image against every pair at the sum's unit: a moved pairing is the walk's read at one further crossing, and the two slots' moved reads join at partners across the transpose identity at the exchanged letters ((i)'s coevaluation read at the two families); the coevaluation sits off the unit family at the determinant's diagonal, the collected families independent (Lemma 3): the count is one.*

Construction 5 (The label set)**.** Two shapes are one *label* when they differ by full columns, $W_{\lambda+(1^{d_f})} \cong W_\lambda \otimes W_{(1^{d_f})}$ (Lemma 8(ii)); each label reads one reduced shape, at most r rows deep. The data descend to labels: d is class-invariant (a line's tensor), and so is the fusion count at matched degrees, $N^{c+(1^{d_f})}_{a+(1^{d_f}),\,b} = N^c_{ab}$, the fused counts' read at the mapped span, the pair list at the shifted content the content's own mapped with the kernel data one family (Lemma 8(ii)); a full column crosses the factors at a fixed target, the display at the factors' exchange (Lemma 5(iii)) where the target holds a full column, and at a reduced target either side's count is the sum's unit, the mapped pool occupied at the last letter member for member (Lemma 8(ii)) with the target's group vacant (Lemma 5(ii)); the label count at two representatives reads at the matched-degree lift, the degree gap a full-column multiple at one class with the deficient side grown by the gap's columns, and the sum's unit at a gap off the multiples; the dual label $\bar{a}$ is the class of a^* (Lemma 8(ii)); the full columns' own class is the *unit label* **1**, the unit shape's, its multiplicity in a product one exactly at the dual label, the read of Lemma 8(iii); and the Casimir read descends, the bracket of Lemma 7 at $\lambda + (1^{d_f})$ gaining $2k + d_f$ from q and returning it to the degree term. The Cartan count sharpens to an equality on shapes, $N^{a+b}_{ab} = 1$: the content-$(a{+}b)$ summand of $W_a \otimes W_b$ is $(W_a)_a \otimes (W_b)_b$, one line, since a content ν of W_a with $a + b - \nu$ a content of W_b sits at and below a at once (Lemma 3, prefix sums).

Lemma 9 (The Casimir floor)**.** *Every nonunit label reads* $\hat{c}_2(\lambda) \geq \hat{c}_2(f) = [\, r(r+2) : 2d_f^2\,]$*, with* *equality exactly at f and $\bar{f}$; so a configuration of weight-free content at most Λ_C with L occupied links reads $L\,\hat{c}_2(f) \le \Lambda_C$, one pair comparison. The floor's descent is statement data: a box move joins two dominant shapes at one row pair $a < b$, the join $\lambda + e_b = \mu + e_a$; Q falls strictly along moves; every chain from a reduced shape ends at a fundamental or at the unit label, a step reaching the unit label passing θ's class; and the endpoints read $d_f\, Q(\omega_j) = j\, g_j\,(r+2)$ at the complement gaps $j + g_j = d_f$. At a member's table (Construction 4) the move is one positive root at dominant data, $\mu+\alpha = \lambda$, the box move the matrix units' instance at the letter-pair move, and the fall identity is the root arithmetic*

$$C_2(\mu) + \langle \lambda + \mu + 2\rho,\ \alpha\rangle = C_2(\lambda)\ ,$$

the join's two reads collected at the form's bilinearity, its read on its upper side at dominant data: a dominant content against a positive root reads $2\,\langle\lambda,\alpha\rangle = \sum_i c_i\,\lambda(\alpha_i^\vee)\,\langle\alpha_i,\alpha_i\rangle$ over the root's fold, every summand a product at the coroot reads, and $\langle 2\rho,\alpha\rangle = \sum_i c_i\,\langle\alpha_i,\alpha_i\rangle$ is positive. The member's end list reads its Casimirs off the tables at the end contents' cleared folds, the displays $e\,\mu = \sum_k b_k\,\alpha_k$ at a fold key e, the coroot identities $\sum_k b_k\,\alpha_k(\alpha_j^\vee) = e\,\mu(\alpha_j^\vee)$ each display's witness with the fixed members' folds the adjugate rows (Construction 4), and the Casimir the fold's form read at the halved products,

$$2\,e\,\langle\mu,\mu+2\rho\rangle \ = \ \sum_k b_k\,\mu(\alpha_k^\vee)\,\langle\alpha_k,\alpha_k\rangle \ + \ e\sum_\alpha\sum_k c_k\,\mu(\alpha_k^\vee)\,\langle\alpha_k,\alpha_k\rangle\ ,$$

the second fold over the positive list at each root's fold $\alpha = \sum_k c_k\,\alpha_k$, read against $C_2(\theta) = 2\,(r+1)$, the form and residue reads (Construction 4): at B_ℓ the vector, 2 on the first key, the fold key one at $b_k = 1$, at the reads 1 and $r+1$, so $\hat{c}_2 = [\,\ell : r+1\,]$, and the spinor, 1 on every key, the fold key two at the index list $b_k = k$, at the reads $[\,\ell : 4\,]$ and $[\,\ell^2 : 2\,]$, so $[\,\ell\,(2\ell+1) : 8\,(r+1)\,]$, the list's least member the trichotomy at $2\ell+1$ against 8; at C_ℓ the first fundamental, the fold key two at twos with one at the last key, at $[\,2r+1 : 4\,(r+1)\,]$, below the short dominant root's $[\,r : r+1\,]$, the root its own fold at key one; at D_ℓ the vector, the fold key two at twos with one on the last two keys, at $[\,r+2 : 2\,(r+1)\,]$, and the two spinors, the fold key two at the index list with the exchanged tail pairs $(c,\, c')$ at the joins $2c = \ell$ and $c'+1 = c$ at even rank and the fold key four at the doubled index list with the exchanged tails $(\ell,\, w)$ at $w+2 = \ell$ at odd, at $[\,\ell\,(r+2) : 8\,(r+1)\,]$, the trichotomy at ℓ against 4 with equal members at $\ell = 4$; and the fixed members' evaluations, $[\,1 : 2\,]$ at G_2's short dominant root ($C_2 = 4$ at the reads $[\,2 : 3\,]$ and $[\,10 : 3\,]$), $[\,2 : 3\,]$ at F_4's ($C_2 = 12$ at 1 and 11), $[\,13 : 18\,]$ at E_6's minuscule pair ($[\,4 : 3\,]$ and 16, the join $[\,52 : 3\,]$ against 24), $[\,19 : 24\,]$ at E_7's minuscule member ($[\,3 : 2\,]$ and 27 against 36), and at E_8 the highest root's own read, $\hat{c}_2(\theta) = 1$, the level one.

Proof. Write $d_f\,Q(\lambda)$ for the gap read of Definition 6 at degree k, $k^2 + d_f\,Q(\lambda) = d_f\,q(\lambda) + d_f\,\langle\lambda, 2\rho\rangle$, the cleared traceless bracket of Lemma 7, a label read (Construction 5: a full column adds $2k + d_f$ to q and returns it to the degree term), with $d_f\,Q(\mathrm{adj}) = 2d_f^2$; the claim is $d_f\,Q \ge r\,(r+2)$ on nonunit labels, one natural comparison. A *box move* joins two dominant shapes at one row pair $a < b$, the join $\lambda + e_b = \mu + e_a$, one box read at either row; it fixes the degree and lowers Q:

$$d_f\,Q(\lambda) = d_f\,Q(\mu) + d_f\,\langle \lambda_a + \mu_a \ :\ \lambda_b + \mu_b\rangle + 2\,d_f\,g,$$

the pair the summed rows' at the move's places, on its upper side or of equal members at the sorted rows' orders $\lambda_b \le \lambda_a$ and $\mu_b \le \mu_a$, and $2\,g$ the places' doubled gap at $a + g = b$: the display is the join's read at the fixed degree, the square fold moved by the summed rows' pair, each place's square difference the pair member's read at the one-box gap, and the row-weight fold by the places' doubled gap, the row weights $\langle r+2 : 2i\rangle$ at the two keys (Lemma 7's fold), the weights' gap the places' doubled own. Every reduced shape off the fundamentals and the unit moves: at some row of length at least 2, take a the last such row and b one past the last occupied row: the target μ stays sorted, $\mu_a \geq 1 \geq \mu_{a+1}$ at $\mu_a + 1 = \lambda_a$ and $\mu_b = 1 \leq \mu_{b'}$ at $b' + 1 = b$, a full column re-reducing with Q fixed; a one-column shape is the fundamental at its length, and every move off it breaks the sorting. Q falls strictly along moves, one natural measure descending, so every chain ends at a fundamental or at the unit label, and a step reaching the unit label starts at a constant row list moved by one box, dominance forcing the box from the first row to the last: θ's class, at $d_f\, Q(\theta) = 2d_f^2$. The floor is the least endpoint read, the row weights summing to $j\, g_j$ at the complement gap $j + g_j = d_f$, the cross-added read $j\,(j+1) + j\, g_j = j\,(r+2)$:

$$d_f\, Q(\omega_j) + j^2 = d_f\, j + d_f\, j\, g_j \ , \qquad d_f\, Q(\omega_j) = j\, g_j\, (r+2) \ ,$$

least at $j \in \{1, r\}$ with value $r\,(r+2)$, strictly larger between, and $2d_f^2 = r\,(r+2) + (r^2 + 2r + 2)$, the unit-reaching step strictly above the floor. At the series' tables the descent closes on coordinate moves, each one positive root at a stated family with the sorted order, the parity and the class kept. At B_ℓ: an even-parity content at two or more occupied keys takes the pair move, the sum root at the last two occupied keys, both reads dropping by two at the sorted tail's end; at one occupied key it takes the short root there, the chains through the vector, the short dominant root's content, to the unit; an odd-parity content off the spinor holds a read at three or beyond, the short root at the last such key moves it, the following read one at most, below the moved read; and the spinor is move-free, every root's subtraction breaking the last read or the order. At C_ℓ: two or more occupied keys take the pair move, one occupied key at two or beyond takes the long root, and the chains end at the first fundamental or reach the unit through the two dominant roots' contents. At D_ℓ: at the last key's read equal members, the pair move at the last two occupied keys at two or more of them, and the difference root at keys one and two at one occupied key reading four or beyond; at the last key's read off equal members, the pair move of the last key against key $\ell{-}1$ at the last read's magnitude two or beyond, the sign at the last read's side and the last read moved toward equal members, and against the rearmost key reading three or beyond at magnitude one, the sign flipping the last read at its magnitude; the vector and the two spinors are move-free, every root's subtraction breaking the order or the last pair's dominance. Each move's fall is the identity's read on its upper side, and the Casimirs descend strictly at the member's clearing factor, the form's displayed second members clearing every value to a natural, a strictly falling natural: every chain ends on the member's end list or at the unit through a dominant root's content, so every nonunit dominant content of a series member reads $\hat{c}_2$ at or above the least of the end list's and the dominant roots' reads, the statement's trichotomies, the member's floor. At a fixed member the descent runs in the coroot coordinates: a content with a coroot read at two or beyond moves by that simple root, the off-key Cartan reads balance partners of naturals (the member's displayed data), their subtraction raising the off-key coroot reads, so the move keeps dominance, one root with the class kept and the fall the identity's, terminating at the member's clearing factor; the contents at every coroot read at most one are finitely many, and each reads its Casimir at or above every occupied fundamental's,

$$C_2(\mu) \ = \ \sum_{i,j} \mu_i\, \mu_j\, \langle \omega_i, \omega_j \rangle + \sum_i \mu_i\, \langle \omega_i, 2\rho \rangle \ ,$$

every summand at the fundamental form reads' upper sides or equal members (the adjugate rows' form reads) with $\langle \omega_i, 2\rho \rangle = 2 \sum_j \langle \omega_i, \omega_j \rangle$ positive at the root fold's all-twos coroot list (Construction 4): the member's floor is the least fundamental read, one trichotomy fold over the member's fundamental Casimir list with the displayed minima the statement's, and at E_8 the fold reads the adjoint, the least nonunit label the highest root at level one. The link count follows, each occupied link contributing at least $\hat{c}_2(f)$ to the content. $\square$

Theorem 1 (The alternant identity)**.** *Write* $\mathrm{mult}_\lambda(\nu) := \dim(W_\lambda)_\nu$ *and* $\mathrm{ch}_\lambda := \sum_\nu \mathrm{mult}_\lambda(\nu)\, x^\nu$ *for a block's content generating polynomial, q the standard square, and, at a list β of naturals, a_β the alternant pair, the swap-graded sum over S_{d_f} (Construction 1), a balance pair of polynomials (Definition 1):*

$$a_\beta \;:=\; \Big\langle \sum_{w \text{ even}} x^{w\beta} \;:\; \sum_{w \text{ odd}} x^{w\beta} \Big\rangle,$$

each monomial on its permutation's side; an odd permutation of the exponent list exchanges the members and an even one fixes them (the grading additive, Construction 1), and a repeated exponent makes the members equal, the repeat's transposition mapping either side onto the other, the wedge's repeat read. Then, at the beta-sets of Construction 1, U the unit set with sorted display u and X_λ the shape's with sorted display $\lambda + u$,

$$a_u\ \mathrm{ch}_\lambda \;=\; a_{\lambda+u}\ .$$

Proof. The Casimir trace, string by string. Trace the Casimir over the content summand $(W_\lambda)_\nu$, the summand's trace the one value at every orthogonal spanning list (Lemma 3's trace read): the block scalar reads $\big(q(\lambda) + \langle\lambda, 2\rho\rangle\big)\,\mathrm{mult}(\nu)$ (Lemma 7), the diagonal part reads $q(\nu)\,\mathrm{mult}(\nu)$, and each pair $p < q$ traces $E_{pq}E_{qp} + E_{qp}E_{pq}$ over its strings (Lemma 6): a string through ν at depth j from its top, of top eigenvalue h, at the depth eigenvalue d, the pair $\langle h : j\rangle$, reads the ν-line coefficient $(j{+}1)\,d + j\,(d{+}1)$, reading Lemma 2(ii)'s fold twice, at the words $F^{\,j+1}$ and $F^{\,j}$, at the string's tie $d + \nu_q = \nu_p + j$ the coefficient the identity

$$(j{+}1)\,d + j\,(d{+}1) + \nu_q \;=\; 2j\,(d{+}1) + \nu_p\ ,$$

every read the balance pairs' own. The pair's value at a content is the pair $\langle \mu_p \,{:}\, \mu_q\rangle$ (Definition 1); writing $\nu^{(l)}$ for the pair's l-th move of ν, the occupied content at the joins $\nu_p^{(l)} = \nu_p + l$ and $\nu_q^{(l)} + l = \nu_q$ with every other coordinate equal (Construction 2's letter-pair moves iterated), the value at $\nu^{(l)}$ is the pair $\langle \nu_p{+}2l \,{:}\, \nu_q\rangle$; the string's values from its ν-line up to its top sum at the tie, $2\sum_{l\le j}(\nu_p{+}2l) = 2j\,(d{+}1) + 2j\,\nu_q$, while a string above ν sums its values to equal members: the value at depth m is the pair $\langle h : 2m\rangle$, the top's its one-member instance, two depths at the join $m + m' = h$ reading member swaps, the summed members equal. Collecting string by string, the string coefficients' ν-terms collecting to the two side weights, ν_i weighted per place by the gaps at $1{+}o = i$ and at $i{+}o' = d_f$, the equal-degree tie $\sum_i \lambda_i = \sum_i \nu_i$ with the shared reads $q(u)$ and twice the degree closing both sides, and the block's row read closing at $\langle\lambda, 2\rho\rangle + (d_f{+}1)\sum_i \lambda_i = 2\sum_i \lambda_i\, u_i$ (Lemma 7's fold), gives

$$q(\lambda{+}u)\,\mathrm{mult}(\nu) + 2\sum_{p<q}\sum_{l\ge 1} \nu_q\,\mathrm{mult}(\nu^{(l)}) \;=\; q(\nu{+}u)\,\mathrm{mult}(\nu) + 2\sum_{p<q}\sum_{l\ge 1}(\nu_p{+}2l)\,\mathrm{mult}(\nu^{(l)})\ ,$$

the pair sums over the places $p < q$ at their moves; the display runs at every content list of the block's width, the recursion family's keys the read's own finite carrier, and a key at the count's unit reads the display as the moved folds' one value, the above-string clause's read alone. *The generating telescope.* With $D_i := x_i\partial_i$ and the pairs' reads entering as balance pairs,

$$G_{pq} \;:=\; \Big\langle \sum_\nu \Big(\sum_{l\ge 1}(\nu_p{+}2l)\,\mathrm{mult}(\nu^{(l)})\Big)x^\nu \;:\; \sum_\nu \Big(\sum_{l\ge 1}\nu_q\,\mathrm{mult}(\nu^{(l)})\Big)x^\nu \Big\rangle,$$

the outer sum over the natural exponent lists, each member finite (an entry at ν reads occupied move targets alone, the occupied contents finitely many, the multiplicity the occupancy family's own count at the span's contents (Lemma 3's counts at the read), the recursion's keys the family joined to every pair move's sources, and a key off that join reading the count's unit on both sides): a balance pair of polynomials, and the inner sums telescope coefficientwise, one further move dropping one l, to

$$\langle x_p : x_q\rangle\, G_{pq} \;=\; x_q\,\langle D_p\,\mathrm{ch} : D_q\,\mathrm{ch}\rangle, \qquad \text{and the recursion reads} \qquad q(\lambda{+}u)\,\mathrm{ch} \;=\; \mathcal{L}\,\mathrm{ch} + 2\sum_{p<q} G_{pq},$$

$\mathcal{L} := \sum_i (D_i + u_i)^2$ reading $q(\nu{+}u)$ on x^ν. At a key with $\nu_p = 0$ the x_p-shift sits off the natural keys and the identity's line is the string's own palindrome: ch symmetric (Lemma 6) pairs the string's targets, two shifts at the join $l + l' = \nu_q$ reading one multiplicity, so the moved weights collect to the second exponent's count, $\sum_{l\ge 1} 2l\,\mathrm{mult}(\nu^{(l)}) = \nu_q \sum_{l\ge 0}\mathrm{mult}(\nu^{(l)})$. *The assembly.* $a_u = e\,V$ at the full monomial $e := \prod_k x_k$ and the pair product $V := \prod_{i<j}\langle x_i : x_j\rangle$ (Definition 1): the graded sum is graded, an odd variable swap exchanging its members (Construction 1), so at $x_i = x_j$ its members are equal, the swap fixing the evaluation while exchanging the members; the division at $\langle x_i : x_j\rangle$, monic in x_i (Definition 2), leaves that evaluation as remainder, members equal, so each factor's division is exact, the factors' in turn, a product of polynomial pairs of equal members exactly at a factor with equal members (the leading reads multiply), and e's cofactor reads monomial by monomial, every arrangement reading every variable at least once, the factors and the variables pairwise coprime (each factor monic linear in its higher variable, the variables prime); the two sides match at degree, $\sum_j u_j$ against $\binom{d_f}{2} + d_f$, and at the leading monomial x^u, on the even side at coefficient one: $a_u = e\,V$. So, at the cofactors Q_{ij} at $\langle x_i : x_j\rangle\, Q_{ij} = V$, one value each at the factor of unequal members with the index swap exchanging the members, the cofactor's coefficients are the monic letter's division descent (Definition 2), V's at the keys moved along the pair, and $D_i e = e$ with the derivative read $D_i V = \sum_{j\ne i} x_i\, Q_{ij}$ closing per key: a pair's moved line through a key reads V at its two arrangements alone, the two fillings of the line's open values at the exchanged grading; the monic descent opens at the key itself, one head read per variable beyond i, the swapped cofactors moving the letter strictly; at an arrangement with exponents v the collected reads close on the value's places split,

$$v_i + \#\{\, j > i :\; v_i < v_j \,\} \;=\; \#\{\, j > i \,\} + \#\{\, j < i :\; v_j < v_i \,\}\ ,$$

v_i the count of the values below the letter's over the places' trichotomy; and at every further key the lines' reads close in pairs: a line matched at both fillings joins its two arrangements' reads at the exchanged grading within its own fold, and at a one-sided hit the pair's value in the key occupies a second place, whose line keeps the hit's upper filling with its further filling at the receding value below the letter's, one one-sided hit per partner line, an involution off its own fixed points; the two hits' reads enter at opposite members, the places' transposition exchanging the grading at partner places on one side of the letter and the swapped cofactor's memberwise exchange reading the join at places the letter splits, so the collected reads join at equal members. Per pair the operator identity $\langle x_p D_p : x_q D_q\rangle = x_q\,\langle D_p : D_q\rangle + \langle x_p : x_q\rangle\, D_p$ reads

$$Q_{pq}\,\big\langle x_p D_p\,\mathrm{ch} : x_q D_q\,\mathrm{ch}\big\rangle \;=\; V\,G_{pq} + V\,D_p\,\mathrm{ch}, \qquad V\,G_{pq} = Q_{pq}\,x_q\,\big\langle D_p\,\mathrm{ch} : D_q\,\mathrm{ch}\big\rangle,$$

the second the cleared telescope at the cofactor, and summing pairs, a pair's two D_iV-terms collecting to the first display's left side, $\sum_i (D_iV)(D_i\,\mathrm{ch}) = V\sum_{p<q} G_{pq} + V\sum_{p<q} D_p\,\mathrm{ch}$. With the square's Leibniz rule $D_i^2(fg) = (D_i^2 f)g + 2(D_i f)(D_i g) + f(D_i^2 g)$, $\mathcal{L}_0 := \sum_i D_i^2$ reading q at

every arrangement, one value $q(u)$, on each monomial of a_u, the read $D_i a_u = a_u + e\, D_i V$, and the collection $\sum_{p<q} D_p + \sum_i D_i = \sum_i u_i D_i$,

$$\mathcal{L}_0\big(a_u\,\mathrm{ch}\big) = 2\,a_u\sum_{p<q} G_{pq} + a_u\Big(\mathcal{L}_0 + 2\sum_i u_i D_i + q(u)\Big)\mathrm{ch} = 2\,a_u\sum_{p<q} G_{pq} + a_u\,\mathcal{L}\,\mathrm{ch} = q(\lambda{+}u)\,a_u\,\mathrm{ch}\ ,$$

the first equality the Leibniz square's collection at those reads, the second $\mathcal{L}$'s display, the third the telescoped recursion. *The conclusion.* $a_u\,\mathrm{ch}_\lambda$ is graded, an odd variable swap exchanging its members (ch symmetric, Lemma 6; Construction 1), with natural coefficients on each member, hence a balance-pair combination $\sum_\mu c_\mu\, a_{\mu+u}$ over dominant μ: a repeated exponent's read has equal members, the repeat's transposition's, each strictly decreasing exponent orbit is one graded sum, and the walk to the representative is the exchanges' own: a key reading one value at two places reads the repeat's transposition, a distinct-valued key's first place pair off the descending order is an ascent whose exchange lowers the ascending place pairs' count by exactly one, and Construction 1's descent at the reversed order closes the walk; its coefficient c_μ is the pair of the two members' reads at $x^{\mu+u}$. The support sits at $\mu \preceq \lambda$: $\nu \preceq \lambda$ (Lemma 3) and $wu \preceq u$ add, the permuted display's prefix read. A leading prefix of wu joins the assignment's to $k\,(d_f{+}1)$, entry against entry at the display $u_{w_j} + w_j = d_f + 1$; an assignment's k leading values, distinct positive naturals at most d_f, sum at or beyond the count's triangle $\sum_{i\le k} i$, the top value withdrawn per step at the length capped by the bound; and the sorted display's prefix is the triangle's complement at $k\,(d_f{+}1)$, so it clears every permuted prefix. $c_\lambda = 1$: the top content sits on one line (Lemma 3) against $wu = u$ alone, every prefix equal forcing the assignment's prefixes to the triangle and each entry to the sorted order's own, the identity assignment. $\mathcal{L}_0$ reads each graded sum at $q(\mu{+}u)$, so the display forces the equality $c_\mu\, q(\lambda{+}u) = c_\mu\, q(\mu{+}u)$, and at dominant $\mu \preceq \lambda$, $\mu \neq \lambda$,

$$q(\lambda{+}u) \;=\; q(\mu{+}u) + \sum_{k<d_f} \delta_k\, g_k \qquad \text{at} \quad \sum_{i\le k} \mu_i + \delta_k = \sum_{i\le k} \lambda_i, \quad \lambda_{k+1}{+}\mu_{k+1}{+}g_k = \lambda_k{+}\mu_k{+}2,$$

the prefix gaps δ_k the dominance order's witnesses with some δ_k positive at $\mu \neq \lambda$, every g_k at least two at sorted shapes: so c_μ's two members are equal off λ, such a summand entering every value read equally on its two sides, and $a_u\,\mathrm{ch}_\lambda = a_{\lambda+u}$ is the one-value read. ☐

Theorem 2 (The member assembly)**.** *At a member's table with its positive list at the permutation read (Construction 4), the alternant pair at ρ assembles as the positive list's product: at the side fold h, the halved join $2h = \sum_\alpha(\alpha_+ + \alpha_-)$ over the displayed sides,*

$$x^h\, a_\rho \;=\; \prod_\alpha \big\langle\, x^{\alpha_+} : x^{\alpha_-} \,\big\rangle\ ,$$

monomials at the alternant's content keys (Construction 4), the positive-parts fold reading the shift, $\sum_\alpha \alpha_+ = \rho{+}h$. The series' chord-minor computations are the identity's displayed instances at their stated shifts, and the fixed members' side folds read 46*,* 11 *and* 6 *on every key at E_8, F_4 and G_2, with E_7's* $(26, 26, 26, 26, 26, 26, 17, 17)$ *and E_6's* $(16, 16, 16, 16, 16, 8, 8, 8)$ *at the shared coordinates.*

Proof. The subset fold. The product expands over the occupancy families S at the positive list, reads at most one per member: the S-term takes the second side at S's members and the first at the rest, its side S's count and its key h against the join's datum e_S at $e_S + \sum_{\alpha\in S}\alpha = \rho$, the shift the positive-parts fold, and the product collects factor by factor, each further factor reading every collected term twice, the term kept and the term at the key moved by the factor's root with the side exchanged, the collection a fold over the running families read per key at the two graded counts, the read the families' counts' sum at every key: the product is $x^h\,\Phi$ at the graded fold $\Phi := \big\langle \sum_{S \text{ even}} x^{e_S} : \sum_{S \text{ odd}} x^{e_S}\big\rangle$. *The grading.* Per simple index the pairing T_i joins α_i to the s_i-image of S at α_i off S, and reads the image of S's further members at α_i in S: an involution moving the count by one, with $s_i\, e_S = e_{T_i S}$ at the letter's read $s_i\rho + \alpha_i = \rho$ and the permutation read, so Φ is graded, an odd element exchanging its members. *The support.* Every image of a key sits at ρ against a natural fold: at an element w, the join $w\, e_S + \sum_{\alpha\in S} w\alpha = w\rho$ splits at the images, a member at a positive image contributing it and a member at a partner image withdrawing its partner from the inversion list, one member each (the list at the images of the balance partners), so the ρ-fold collects

$$w\, e_S \;+ \sum_{\beta\in N,\ \beta \text{ kept}} \beta \;+ \sum_{\alpha\in S,\ w\alpha \text{ positive}} w\alpha \;=\; \rho$$

at N the element's inversion list, the kept members those off the withdrawals: a natural fold of positives. *The pin.* A dominant key at a simple coroot pair the unit is the letter's own fix, $s_i\mu = \mu$ at one odd letter, so a graded fold's read there is its own member swap, equal members. A dominant key at every simple pair beyond the unit is $\nu + \rho$ at dominant ν, and its support read $\nu + \rho \preceq \rho$ solves ν against the unit fold: pairing the join $\nu + \sum_i c_i\,\alpha_i$ at the unit fold against ρ reads $\langle \nu, \rho\rangle + \sum_i c_i\,\langle \alpha_i, \rho\rangle$ at equal members; $\langle \nu, \rho\rangle$ sits at its upper side or equal members, twice its read the fold of $\langle \nu, \alpha\rangle$ over the positive list at 2ρ the root fold, each member's read the halved products' fold over its own natural fold, $2\,\langle \nu, \alpha\rangle = \sum_k c_k\,\nu(\alpha_k^\vee)\,\langle \alpha_k, \alpha_k\rangle$ (Lemma 9's read), every summand the length's natural multiple at a coroot pair on its upper side or equal members at dominant data, while every $\langle \alpha_i, \rho\rangle$ sits positive at its fold, so the join's fold is the unit family and ν with it: the key is ρ. *The conclusion.* Φ's read at a key is its dominant image's up to the walk's parity, the dominance walk raising at a lower-side simple pair, the raised key's root-fold dot the key's own joined to the pair's margin at the letter's positive dot, an occupied key's dot joining its family's or its witness fold's positive dots to ρ's own, and the raises closing at the gap's count; a key off both folds reads each graded fold at equal members, the two equal-membered reads one value. At the image ρ the read is one: $e_S = \rho$ forces $\sum_S \alpha$ the unit fold, every member's root-fold dot positive, so S is the unit family at side even; the element at each orbit key is one and ρ's images are one per element (the regular reads, Construction 4), so on ρ's images Φ reads the Weyl fold's own members, and at every further dominant image the pin reads both Φ and a_ρ at equal members: $\Phi = a_\rho$, key by key. □

Theorem 3 (The member character identity)**.** *At a member's table, on a table space's block at top content λ, with* ch_λ *the content generating polynomial (Lemma 7's* ch*),*

$$a_\rho\ \mathrm{ch}_\lambda \;=\; a_{\lambda+\rho}\ ,$$

the reads at cleared presentations with stated shifts, every display one value across the shifts (Definition 1's homogeneity principle), the A-series identity (Theorem 1) the first instantiation.

Proof. The eigen-read. Work at the coordinate dot, the form one scalar against it (Construction 4's form displays), every read one value across the rescaling: the trace recursion and telescope (Lemma 7) read $q_0(\lambda{+}\rho)\,\mathrm{ch} = \mathcal{L}_0^\rho\,\mathrm{ch} + \sum_\alpha \tilde w_\alpha\, 2\, G_\alpha$ at the dot weights $\tilde w_\alpha = [\,\mathrm{dot}(\alpha,\alpha) : 2\,]$, $\mathcal{L}_0^\rho$ reading $q_0(\nu{+}\rho)$ per monomial. Three reads close the product's recursion. The *cofactors*: at the assembly identity (Theorem 2), Q_α is the cofactor at $\langle x^{\alpha_+} \!:\! x^{\alpha_-}\rangle\, Q_\alpha = a_\rho$, and the telescope clears against it,

$$a_\rho\, G_\alpha \;=\; Q_\alpha\, x^{\alpha_-}\, D_\alpha\, \mathrm{ch}\ .$$

The *cross term*: a factor's Euler reads collect at the part-weighted folds D_{α_+} and D_{α_-}, their gap the dot read, $D_{\alpha_+} = \tilde w_\alpha D_\alpha + D_{\alpha_-}$ per monomial, so per factor $\sum_i (D_i F_\alpha)(D_i \mathrm{ch}) = x^{\alpha_+}\,\tilde w_\alpha D_\alpha \mathrm{ch} +$

$F_\alpha\, D_{\alpha_-}$ch, the first term clearing through the telescope at the cofactor, $Q_\alpha x^{\alpha_+} D_\alpha \mathrm{ch} = a_\rho\, G_\alpha + a_\rho\, D_\alpha \mathrm{ch}$, and the fold over the factors collects to

$$\sum_i (D_i\, a_\rho)\,(D_i\,\mathrm{ch}) \;=\; a_\rho\Big(\sum_\alpha \tilde{w}_\alpha\, G_\alpha + D_P\,\mathrm{ch}\Big) \qquad \text{at} \quad P := \sum_\alpha \alpha_+ \ ,$$

the positive-parts fold, $D_\alpha + D_{\alpha_-}\tilde{w}_\alpha$-collected to D_{α_+} per factor. The *fold reads the shift*: $P = \rho + h$ at the side fold (Theorem 2), the series' displayed checks per coordinate its instances at $h = b_1\,\mathbf{1}$: the positive-part counts $g_i + g + 2$ at C_ℓ, $2g_i + 2g + 2$ at B_ℓ and $2g_i + 2g$ at D_ℓ, over the joins $i + g_i = \ell$ and $g + 1 = \ell$, against ρ's key reads $g_i + 1$, $2g_i + 1$ and $2g_i$ with the shifts $g + 1$, $2g + 1$ and $2g$. So, at the shifted Euler reads $\tilde{D}_i$, the diagonal pairs $\langle \nu_i : h_i\rangle$ per monomial at the side fold's keys, and $\tilde{\mathcal{L}} := \sum_i \tilde{D}_i^2$: the cleared a_ρ's exponents read $w\rho + h$, the dot Weyl invariant, and the read rides the dominance walk at the graded fold: a key reading the fold off equal members raises along the walk to ρ, the walk and pin's read (Theorem 2), with each letter keeping the form (Construction 4), so q_0 reads one value at ρ and at every such key, $\tilde{\mathcal{L}}\, a_\rho = q_0(\rho)\, a_\rho$ at the graded counts key by key; and the square's Leibniz rule at the three reads collects to

$$\tilde{\mathcal{L}}\,\big(a_\rho\,\mathrm{ch}\big) \;=\; q_0(\lambda{+}\rho)\,\big(a_\rho\,\mathrm{ch}\big)\ .$$

The reads' searches close at the ρ-dot's own heights: the Gram's ρ-column reads each entry's double as the length's multiple of the key's coefficient count over the positive folds (Construction 4), every entry at its upper side or equal members against the unit fold, so a dominant key at every coordinate beyond the unit sits at or beyond ρ's own dot, the join at $\lambda + \rho$ capping its fold at the simple dots, the coefficients' fold against the simples' ρ-dots at or below $\langle\lambda, \rho\rangle$'s read, and a member on a root's line beyond a key sits within the two dots' gap at the root's positive dot. *The conclusion.* a_ρ ch is reflection-graded, ch invariant (Lemma 6) and a_ρ graded, so it is a balance-pair combination $\sum_\mu c_\mu\, a_{\mu+\rho}$ over dominant μ at the cleared shifts: an exponent key whose dominant image holds a unit simple pair reads equal members at that letter's own fix, the series' flips at an equal-membered coordinate and swaps at a repeated magnitude the displayed witnesses, and every further orbit reads one strictly dominant-shifted representative, the dominance walk's image at the regular reads (Construction 4). A shifted alternant is determined at four reads of a reflection-graded family: each letter joins a key to its image at the exchanged grading, every key sits at $\mu + \rho$ against a natural fold of simple roots, the read at $\mu + \rho$ is one at the even grading alone, and every key off $\mu + \rho$ at each simple coroot pair beyond the unit reads its two graded counts at one value (the regular reads, Construction 4): an occupied key raises along the walk to its dominant image, the raises closing at the fold's count and the counts mapped letter by letter, an image at a unit simple pair reads equal counts at its fixing letter, and the family reads $a_{\mu+\rho}$'s graded fold key by key. The support sits at $\mu \preceq \lambda$ ($\nu \preceq \lambda$ at the block, Lemma 3, and $w\rho \preceq \rho$ at the inversion list's fold, Construction 4, add), and $c_\lambda = 1$: a pair joining to $\lambda + \rho$ reads its two witness folds' ρ-dots joined at the sum's unit, every simple's ρ-dot positive at its fold (Theorem 2's pin), so both folds are the unit family and the pair is the two tops': the Weyl fold's one even read at ρ against the top content's one line (Lemma 3), the count one at the even grading alone. Each family is $\tilde{\mathcal{L}}$-diagonal at $q_0(\mu{+}\rho)$, so the eigen-read forces $c_\mu\, q_0(\lambda{+}\rho) = c_\mu\, q_0(\mu{+}\rho)$; at dominant $\mu \preceq \lambda$ off λ the two reads' gap is the dominance fold's,

$$q_0(\mu{+}\rho) + \mathrm{dot}\big(\lambda{+}\mu{+}2\rho,\ \textstyle\sum_i c_i\,\alpha_i\big) \;=\; q_0(\lambda{+}\rho) \qquad \text{at} \quad \mu + \textstyle\sum_i c_i\,\alpha_i = \lambda\ ,$$

positive at dominant data, the fall identity's own read (Lemma 9): c_μ's members are equal off λ, and the display is the one-value read. $\square$

Corollary 1 (The dimension is the Weyl product)**.** *$d_\lambda = \mathrm{ch}_\lambda(1,\ldots,1)$, and the substitution $x_j := t^{u_j}$ at the unit set's sorted display maps the character identity's two graded sums (Theorem 1) to full monomials against pair products, the assembly's one-variable read: the pair product expands over its occupancy families at the place pairs (Theorem 2's subset fold at the powers' gaps), the full monomial joining one display entry per place, so each family's key is the fold of the display against its taken counts' successors, per place the count of its pairs taken there with the joined entry's own one, and its side the family's own count; at two places of one taken count the family joins its exchange at its first tied place pair, the pairs enumerated at the later place with the earlier beneath it, one read of the family kept by the exchange: each further pair reads to its partner at the places' transposition, sorted at the trichotomy, its taken place the letter swap's image with the membership flipped exactly where the sorting crosses the taken position, and the shared pair's take flipped, one key at an odd move of the side, the joined graded reads entering at equal members; a family at pairwise distinct taken counts takes every pair at one place against each further place in turn, so its counts fill the degrees below the width and the family is an assignment's, its side the inversion list's parity (Construction 1): each graded sum reads its full monomial against its pair product, and the two full monomials' reads clear by the product's injectivity at the degree tie:*

$$\mathrm{ch}_\lambda\big(t^{\,d_f},\ldots,t\big)\prod_{j<k}\big\langle t^{\,u_j}\!:\!t^{\,u_k}\big\rangle \;=\; t^{\,|\lambda|}\prod_{j<k}\big\langle t^{\,l_j}\!:\!t^{\,l_k}\big\rangle \qquad (l:=\lambda+u).$$

Each factor clears by $\langle t^a\!:\!t^b\rangle = t^b\,\langle t\!:\!1\rangle \sum_{c+e+1=g} t^{\,e}$ at $b<a$ and its witness gap g at $b+g=a$, the geometric word one term per splitting, the two sides with one $\langle t\!:\!1\rangle$ factor per pair, which drop at the shared monic factors' withdrawal, two products of one value at the division's descent (Definition 2), and the polynomial identity read at $t=1$ gives

$$d_\lambda \prod_{j<k} g_{jk} \;=\; \prod_{j<k} g^{\lambda}_{jk} \qquad at \quad u_k+g_{jk}=u_j, \quad l_k+g^{\lambda}_{jk}=l_j\ ,$$

the unit set's member gaps against the shape's, every right factor at least its left partner. At a member the identity specializes at the principal point: the substitution x^ν to $t^{\,\mathrm{dot}(\nu,\rho)}$ maps a graded fold to the exchanged list's, $\mathrm{dot}(w\beta,\rho)=\mathrm{dot}(\beta,w^{-1}\rho)$ at the kept form with the reversed word's element at its parity class (Construction 4), so $a_{\lambda+\rho}$'s principal read is a_ρ's at the list $\lambda+\rho$'s own point, the series' role-symmetric signed and plus minors the displayed instances with the D_ℓ doubled alternant's two at both sides clearing by the product's injectivity; the assembly identity reads at either list (Theorem 2; Theorem 3), and each factor clears by the geometric word at its gap as above: the member dimension is the gap product

$$d_\lambda \prod_{\alpha} g_\alpha \;=\; \prod_{\alpha} g^{\lambda}_\alpha \qquad at \quad \mathrm{dot}(\rho,\alpha)=g_\alpha, \quad \mathrm{dot}(\lambda{+}\rho,\alpha)=g^{\lambda}_\alpha\ ,$$

the folds over the member's positive list, every right factor at or beyond its left partner.

Corollary 2 (The product's block counts, by the alternant pairs)**.** *For shapes a,b,c,*

$$N^{c}_{ab} \;+\; \sum_{\sigma\ \mathrm{odd}} \mathrm{mult}_a(\nu_\sigma) \;=\; \sum_{\sigma\ \mathrm{even}} \mathrm{mult}_a(\nu_\sigma) \qquad at \quad \nu_\sigma+\sigma(b+u)=c+u\ ,$$

σ over S_{d_f} at the swap grading (Construction 1), u the unit set's sorted display, ν_σ the tie's lattice solution and the multiplicity read the contents' occupancy read: $a_u\,\mathrm{ch}_a\mathrm{ch}_b = \mathrm{ch}_a\,a_{b+u}$ and $a_u\,\mathrm{ch}_a\mathrm{ch}_b = \sum_{c'} N^{c'}_{ab}\,a_{c'+u}$ (Theorem 1 twice, the second at the product's content data, the pair list's count at every content by Lemma 5(iii), read over the exhaustion's blocks at Lemma 5(i),(ii)'s

graded display), and the two reads match at the one monomial x^{c+u}: the product $\mathrm{ch}_a\, a_{b+u}$ reads it at the displayed ties, one summand per permutation at its side and its solution, while a graded sum $a_{c'+u}$ reads it at $\sigma(c'+u) = c+u$ alone, the identity assignment at $c'=c$, two strictly decreasing displays at distinct entries: the coefficient pair is the display's, the counts descending to labels at matched degree (Construction 5). At a member the identity runs at the member alternants: the content generating polynomials multiply over a tensor product, the contents' sums, and the tensor exhausts into blocks at the member counts (Lemma 5(i),(ii) at table spaces), so the product $a_\rho\, \mathrm{ch}_a\, \mathrm{ch}_b$ reads per key twice (Theorem 3 twice): as $\mathrm{ch}_a\, a_{b+\rho}$, a key's two counts the identity $a_\rho\, \mathrm{ch}_b = a_{b+\rho}$'s fold over the a-family, one instance at each member's withdrawn key, and as the channels' fold $\sum_{c'} N_{ab}^{c'}\, a_{c'+\rho}$, each channel's identity at the key itself. At the key $c+\rho$, dominant at every simple pair beyond the unit, a further channel's fold is vacant, the orbit's one dominant image its own shifted top, and c's channels read one at the even grading alone, the shifted key's top (the regular reads, Construction 4): at every member,

$$N_{ab}^{c} \;+\; \sum_{w \text{ odd}} \mathrm{mult}_a(\nu_w) \;=\; \sum_{w \text{ even}} \mathrm{mult}_a(\nu_w) \qquad \text{at} \quad \nu_w + w\,(b+\rho) = c+\rho \;,$$

w over the member's Weyl list at the reflection grading (Construction 4), the ties' solutions matched lists.

Lemma 10 (The member count and involution)**.** *At a member's table (Construction 3; Construction 4) the fusion count and the label involution are constructions of the table's data, one each.*

(i) The multiplicity family is the recursion's one solve. A block's family reads each occupied content at its summand's count, and the family closes key by key down the ρ-dots from the top. The top's summand is one line with the occupied contents at or below the top (Lemma 3); every count is its dominant image's (Lemma 6), the dominance walk raising at a lower-side simple pair with each raise's ρ-dot positive (Theorem 3's conclusion walk; Theorem 2's pin), so a dominant image's ρ-dot sits at or beyond its orbit's every member's; and at a dominant content below the top the trace recursion's display (Lemma 7) reads the count's multiple of the two Casimir reads' gap, positive at dominant data (Theorem 3; Lemma 9's fall identity), against the moved string folds' join, each moved count read at its dominant image of strictly larger ρ-dot: the product's injectivity reads the count off the higher keys' (Definition 1), the descent closes on the finite window at the dominant keys' ρ-dots capped by the top's own (Theorem 3), and the family is the recursion's one solution. The member's fusion count is the member display's gap read at the family, the two graded sums over the member's Weyl list at the first factor's family (Corollary 2); the display folds at the family: each occupied content's moved key, the target's shifted key joined to the content's balance partner, raises along the dominance walk to its dominant image (Theorem 3's conclusion walk), one letter per raise with the letter count reading the tie element's parity, the walk's word that element's reversal at one parity class (Construction 4's grading and transport), a walk ending at the shifted second factor naming the tie's one element at the regular reads (Construction 4) with every further end off the ties, the shifted factor its orbit's one dominant image, so the count is the family's fold at the walks' parities, one walk per occupied content; and the fusion row at two labels is the dominant targets at occupied counts, the targets at or below the labels' sum (the factors' contents adding over the tensor product, Construction 3, each at or below its own top, Lemma 3).

(ii) The involution is the shifted walk's read. The dual action is the transposes' partner read (Lemma 8(i)'s displays at Construction 3's fields, each commutator's transpose the exchanged pair's with two partners composing to the original), so a dual block's occupied contents are the block's at their balance partners. The reversal is the element at the full inversion list, grown one letter at a member off the list (Construction 4's creation read), its image of every positive member the

member's partner, the list's own read, and its image of ρ is ρ's partner, the root fold's image at the partner list; its negated image of a dominant content is dominant, the coroot pairs read across the partner list. The reversal's image of a dominance join is the exchanged join at the partnered folds, so the image order flips and a block's occupied contents sit at or above the reversal image of the top, occupied once at the top's line (Lemma 6; Lemma 3); the dual block's top is that image's partner, maximal among the dual's occupied contents and off the sum's unit (Lemma 2(i)). So the dual label reads at the shifted walk: the dual's top joined to ρ is the dominant image of the top-with-ρ's balance partner, one image at the regular reads (Construction 4), and the involution's construction is that walk, uniqueness by construction.

Lemma 11 (One box fuses up the shapes)**.** *$N^c_{\mu f} = 1$ at every dominant $c = \mu + e_i$, the row's members exactly those targets, f the one-box shape: Corollary 2 at $a = f$ reads the ties $\nu_\sigma + \sigma(\mu+u) = c+u$ at u the unit set's sorted display, the one-box contents being the e_i once each, so a contribution needs $c+u = \sigma(\mu+u)+e_p$ at one place p: the display $c+u$ sorts $\mu+u$ with one value raised by one, a raise at an occupied successor reading a repeat ($\mu_h = \mu_{h+1}$ at $h+1 = i$ puts the raised value at its neighbor's), which the strictly decreasing $c+u$ misses at every σ, and a raise at a vacant successor keeping its own place ($\sigma = \mathrm{id}$, one count, on the even side), the target $\mu+e_i$ dominant exactly there. At the complement's one-column shape f^*, the depth at the gap g at $g+1 = d_f$ (Lemma 8(ii)'s complement at the one-box shape), the contents are the full column's withdrawn letters, one letter withdrawn per content at one count each: a count read transports across the letters' exchanges through the shared sorted representative, so each withdrawn letter's content reads the top line's count, one, and a content holding a letter twice reads its sorted representative's count, vacant at the dominance cap (Lemma 3; Lemma 6), and the tie analysis reads $N^c_{\mu f^*} = 1$ at every dominant c at the join $c+e_i = \mu+(1^{d_f})$: a contribution needs $c+u+e_p = \sigma(\mu+u)+(1^{d_f})$ at one place p, the tie read at the inverse permutation with the displays transported across it, a lowering of the sorted display the flipped assignment's raise (Construction 1's inverse permutation): the display $c+u$ sorting the raised display $\mu+u+(1^{d_f})$ with one value lowered by one, a lowering at a vacant gap reading a repeat ($\mu_p = \mu_{p+1}$ puts the lowered value at its follower's), which the strictly decreasing $c+u$ misses at every σ, and a lowering at an occupied gap keeping its own place ($\sigma = \mathrm{id}$, one count, on the even side), the target dominant exactly there.*

Lemma 12 (The path count's square sum)**.** *The path count f^λ of a shape (Construction 1) is the count of its descents to the unit shape by one box removed at a time, the recursion $f^\lambda = \sum_{\mu\to\lambda} f^\mu$ over the shapes with one box removed, the unit shape's count one; at $k \le d_f$, over the shapes of k boxes,*

$$\sum_{|\lambda|=k} (f^\lambda)^2 \;=\; k!\ .$$

Proof. On the free $\mathbb{P}$-space of the shapes of at most k boxes, the unit shape among them, let U add a box over the covers and D remove one, transposes for the shape pairing, so $f^\lambda = \langle \lambda, U^k\mathbf{1}\rangle$, the recursion read at D's transpose. At a shape of fewer than d_f rows, every shape below the degree d_f among them, $DU = UD + \mathrm{id}$: two distinct shapes of one size share a cover exactly where they share a covered shape (their union and their intersection, one each, where they differ by one moved box), and one shape's addable corners exceed its removable ones by one (each is read off the distinct row lengths, the additions one more at the rows' end). So $D\,U^{j+1}\mathbf{1} = (j+1)\,U^j\mathbf{1}$ at every $j < d_f$ by induction, $D\,\mathbf{1}$ the sum's unit and $U^j\mathbf{1}$ at the degree j, and

$$\sum_{|\lambda|=k} (f^\lambda)^2 = \langle U^k\mathbf{1}, U^k\mathbf{1}\rangle = \langle U^{k-1}\mathbf{1}, D\,U^k\mathbf{1}\rangle = k\,\langle U^{k-1}\mathbf{1}, U^{k-1}\mathbf{1}\rangle = k!\ .$$

$\square$

Lemma 13 (The mixed power's invariants are the permutation span)**.** *At $k \le d_f$ the invariants of $V^{\otimes k} \otimes (V^{\otimes k})^\vee$ are spanned by the $k!$ place permutations, whose Gram is the cycle count $d_f^{\#\mathrm{cycles}(\sigma\tau^\mathsf{T})}$, the transpose the inverse permutation's matrix (Construction 1), the pairing's index sums closing each cycle to a factor d_f.*

Proof. The block count is the path count. $M \otimes V$ exhausts as $\bigoplus_\mu N_\mu(M)\,(W_\mu \otimes V)$ (Lemma 5), and $W_\mu \otimes V$ by one boxes (Lemma 11), so at every power $N_\lambda(V^{\otimes k+1}) = \sum_{\mu\to\lambda} N_\mu(V^{\otimes k})$, the unit power the scalars' line, over the shapes with one box removed: $N_\lambda(V^{\otimes k}) = f^\lambda$, the path count (Lemma 12), every shape of k boxes at most $k \le d_f$ rows deep. *The invariant count.* An invariant of $M \otimes M^\vee$ is the dual tensor's family at matched contents with every letter pair's Leibniz image null (Lemma 8(i)), its data one matrix per occupied content with the moved reads (Lemma 4's matrix families), and its form coordinates the matrices' entries at the contents' monomials; the families combine at the entries, the action's row fold reading each entry once (Definition 3), so a member's value, the moved reads and the coordinates are additive and scale-reading over the members. At an exhaustion of the power's carrier (Lemma 5(i)) every top's value under the family is a top at the top's own content, a raising image of the value the moved read at the top's null image; and a family whose every top value is the sum's unit is null: Lemma 4(ii)'s induction reads every value as its member's own word at the top's value, the blocks span the carrier, and a null map's matrices are null at the carrier's own coordinate vectors. So the stacked top values map an independent list of invariants to an independent list inside the raising kernels' joined span, a null image's family null with the coefficients read back at the independence, and the invariants' count sits at or below the kernels' count (Lemma 3's exchange clause), N_λ tops per content (Lemma 5(ii)) at kernel dimension N_λ each (Definition 5):

$$\sum_\lambda N_\lambda^2 \;=\; \sum_{|\lambda|=k} (f^\lambda)^2 \;=\; k!$$

(Lemma 12 at the path-count clause above). *Independence, the span, and the Gram.* A place permutation is equivariant (the units' Leibniz sum is place-symmetric, Construction 1), its family the permutation matrices per content, and at $k \le d_f$ the $k!$ of them are independent, reading distinct monomials on $v_1 \otimes \cdots \otimes v_k$ at distinct indices; an invariant off their span would extend the independent list one past $k!$, against the count above: the permutations span. Two permutations' forms pair at the count of the monomials their two reads agree on, and the reads agree exactly at a monomial constant along each cycle of $\sigma\tau^\mathsf{T}$, one free letter per cycle: the Gram is $d_f^{\#\mathrm{cycles}(\sigma\tau^\mathsf{T})}$. $\square$

2 The form, the carrier, and the setting

Definition 6 (Labels, the Casimir, and its weight-free read)**.** Lab is the label set of the calculus, one reduced shape per label (Construction 5) and one block at each (Lemma 4(iv)); and $C_2(\lambda) = \langle\lambda, \lambda + 2\rho\rangle$ the quadratic Casimir, the block read of Lemma 7 at the invariant form's scale (Proposition 1), with $\langle\cdot,\cdot\rangle$ the induced weight-lattice form, the reads $q(\lambda)$ and $\langle\lambda, 2\rho\rangle$ Lemma 7's display's. θ denotes the highest root, and

$$\hat{c}_2(\lambda) \;:=\; \left[\, d_f\, Q(\lambda) : 2d_f^2 \,\right] \quad \text{at} \quad k^2 + d_f\, Q(\lambda) = d_f\, q(\lambda) + d_f\,\langle\lambda, 2\rho\rangle \;,$$
$$d_f\, Q(\lambda) = \sum_{p<q} \langle\lambda_p : \lambda_q\rangle^2 + d_f\,\langle\lambda, 2\rho\rangle \;,$$

the second display the join's constructed witness: the square identity $d_f\, q(\lambda) = k^2 + \sum_{p<q} \langle \lambda_p : \lambda_q \rangle^2$ collects the row pairs' squares against the degree (k^2 reading $q(\lambda)$ with the place pairs' cross products twice, the pairs' squares the complement), every summand a natural at the sorted rows (Lemma 7's fold), and at every nonunit label two coordinates differ (equal coordinates are full columns, the unit label's class), their pair's square positive by its shape; the dual label reads one value, $\hat{c}_2(\bar{a}) = \hat{c}_2(a)$, the complement's pair gaps the label's own at reversed places, one fold of squares, and the complement's row weights the balance partners there, the full weight list's fold of equal members (Construction 5; Lemma 8(ii)); the definition reads at the nonunit labels, the unit label's line Δ's kernel, the kernel point's read; the clearing factor the fundamental count (Construction 1), the cleared traceless bracket at degree k against the adjoint's $Q(\theta) = 2d_f$ (Lemma 7): one quadratic form on the weight lattice read against itself, the form's components $\langle \lambda, \lambda + 2\rho \rangle$ and $\langle \theta, \theta + 2\rho \rangle$ rescaled by the one factor at every scale of the form, the pair one value across them (Definition 1; Proposition 1). The development runs on one object defined on the label index alone, $\Delta \chi_\lambda = C_2(\lambda)\, \chi_\lambda$.

Proposition 1 (One value on the form line)**.** *The pairings of the θ-block against its dual are one line, the form line: θ is self-dual ($\theta^* = \theta$, Construction 5) and an equivariant map between blocks is the unit of its map space at distinct labels and a scalar at one (Lemma 4(ii),(iii)). The invariant form is a point ξ of that line, a scalar pair, and rescaling the point rescales C_2 with it while fixing every character. Hence a pair of two reads, each of degree one in the form, is one value at every point of the form line. Every energy-type read below (E, ε_0, the coupling weights, the constants c_i) accordingly denotes its read against $C_2(\mathrm{adj})$, and every surviving constant is a count.*

Proof. Rescaling the form's point rescales Δ's block read (Lemma 7), hence $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$ at every λ, and a pair of two such reads is one value at every point, its two components rescaled by the one factor. A character is a label-index generator, defined without the form (Proposition 8). Δ, content and E are each of degree one in the form and χ_θ, V, M are form-free, so each displayed pair of two such reads is one value at every form point. $\square$

Definition 7 (Generators, where a proof reaches for them)**.** X_a are anti-Hermitian, the trace pairing reading $\mathrm{tr}(X_a X_b) + \xi\, \delta_{ab}$ at the sum's unit and the Fierz read cleared at the count,

$$d_f \sum_a (X_a)_{pq} (X_a)_{rs} + \xi\, d_f\, \delta_{ps} \delta_{rq} \;=\; \xi\, \delta_{pq} \delta_{rs}\,,$$

at the form's point ξ; the Hermitian companions T_a at $X_a = i\, T_a$ read ξ throughout; then $\Delta = \sum_a X_a^\dagger X_a$. A generator list is displayed at entries pairs over the ground at the formal i, the remainder lists at z^2+1 (Definition 2): per place pair $p<q$ the swap combination, entry one at (p,q) with its balance partner at (q,p), and the i-weighted sum, entry i at both places; per consecutive place pair the diagonal combination, entry i at the place's diagonal with i's balance partner at the successor's; each anti-Hermitian by its shape, the list independent (the off-diagonal families by their disjoint place pairs and components, the diagonal telescope by its leading place), its count at the join $2\binom{d_f}{2} + r + 1 = d_f^2$. The sums run over the list against its metric dual, $\sum_a X_a(\cdot) X^a(\cdot)$, pair data at the Gram's adjugate against its determinant; the orthonormal spelling is that sum's evaluation.

Construction 6 (The fusion interface)**.** The calculus's outputs the chain reads are one interface, its fields: the *label data*, a set with its equality read, involution and unit, each label equal to itself at that read; the *count laws*, the fusion counts at commutativity, associativity and the unit, with the unit read $N_{ab}^{\mathbf{1}} = \delta_{a\bar{b}}$ and the Cartan strictness $N_{ab}^{a+b} = 1$; the *dimension identity* $\sum_c N_{ab}^c\, d_c = d_a\, d_b$;

the *weight-free Casimir*, positive at every nonunit label, with its floor at the named least labels, its dominance growth along a row, and the drift identity; the *self-dual* θ, with the character's two-sided form cap, its square's channel list at multiplicities, dimensions and Casimirs, the base c_1, and θ's own content list; the *class data*, the vertex law's classes over the root folds with the winding floors per class; the *evaluation*, the unit-coefficient read with orthonormal characters and the coefficient carrier's evaluation identity and integration by parts; and the *presentation*, the evaluation's computation at stated factor lists. A *series* enters with one further tier: the declared residue with every field a polynomial-pair read in it at stated reach clearances, the banded families the fusion-closed interiors with the θ content list among their data, and the generic lift's discipline; a fixed member enters at the member fields alone, its reads displayed finite evaluations. The label calculus above instantiates every field, each instantiating statement the field's own derivation, and a further member's or series' instantiation is its own construction of the field data, the chain reading the fields alone: the classification's members each enter with every field a derived read at their weight tables, their positive lists, alternants, assembly, character identity, fusion rows, pairing spans and defining tables the fields' own instantiations, the count and involution fields at the member constructions (Lemma 10), and every chain derivation reads the fusion data through the fields, at every member at once.

Construction 7 (The lattice interface)**.** The carrier's region data are one interface, and every derivation reads the lattice through its fields: a set of *links* and a set of *vertices*, each link with one tail and one head, the orientation the labels' duality read; the *plaquettes*, each boundary a cyclic word of four distinct oriented links; *simplicity*, two vertices sharing at most one link; the *even-cycle read*, every cycle's length even, a two-coloring of the vertices its witness datum; the d *directions*, a partition of the links with a transverse-cut family per direction, each cut crossed by its own direction's links alone; the *translation action* on a torus window, permuting links, vertices and plaquettes and preserving directions, two shifts composing to the keys' one composite with the directions' composites one map at either order; and the signed coordinate permutations B_d, permuting the link and plaquette sets with orientation reversal dualizing labels. The fields read at two instantiation families. The region fields, the links, the vertices, the plaquettes, simplicity, and the even-cycle read with its two-coloring witness, are every carrier window's, the regions of Definition 8's directed family; the direction data with their transverse cuts read at both families. The translation action and the signed coordinate permutations are the torus family's, the windows of the momentum fibering: there a contractible cycle's wrap count is even, so its length reads even at the coordinate parity, while a winding cycle reads the side's parity and is odd at odd side, so a torus is a carrier of the interface's fields off the even-cycle read, its counts reaching the directed windows at the decimation's window-free reads (Lemma 14) over the momentum fibering's stated family, the odd tori at side five or beyond. A *label-graph isomorphism* between two supports is a bijection of their links and vertices preserving incidence, orientation and labels. The dimension enters through the direction count alone, its reads (the neighbor count $2d$, the per-link and per-vertex plaquette counts, the per-direction momentum products, the charged direction sums) polynomial in it.

Definition 8 (The carrier)**.** The sector is an indexed family with multiplicities, built from the fusion interface's fields alone (Construction 6). A *configuration* a assigns a nonunit label to each link of its *support*, finitely many links at a natural count (Construction 5); its *content* is $\sum_{\ell\in\operatorname{supp} a} C_2(a_\ell)$, positive at every configuration; a vertex of touched(a), the support's endpoints, is *occupied* where the incident labels' product, dualized on incoming links, reads an invariant count of at least one, and a is *occupied* where every touched vertex is. The window index is the occupied configurations at the region and the cutoff, and the window is the unit's line with their fibers:

$$\mathrm{Idx}(\Lambda, C) = \Big\{ a \text{ occupied} \;\Big|\; \operatorname{supp} a \subseteq \Lambda,\ \sum_{\ell \in \operatorname{supp} a} C_2(a_\ell) \le C \Big\},$$

$$\mathrm{Sect}(\Lambda, C) = \mathbb{P}\,\mathbf{1} \;\oplus \bigoplus_{a \in \mathrm{Idx}(\Lambda, C)} \mathrm{Fib}(a), \qquad \mathrm{Fib}(a) = \bigotimes_{v \in \mathrm{touched}(a)} \mathrm{inv}(v, a),$$

with $\mathrm{inv}(v, a)$ the vertex's intertwiner space, the incident labels' product's invariants at a stated independent spanning list, incoming links dualized (Lemma 3's collected reads at Definition 5's count), and $\mathrm{vmult}(v, a)$ its count, the vertex intertwiner multiplicity, a natural at every occupied vertex. The pairing is definitional as well: gram is the orthogonal direct sum of the unit line's pairing at one and the fibers' own pairings, per configuration the invariant lists' power pairing at the links' cleared block scale, one factor $[\,1 : d_{a_\ell}\,]$ per support link: positive definiteness is by construction, the power pairing a coordinate fold at positive scales. The windows are directed under $(\Lambda, C) \le (\Lambda', C')$, the inclusions $\mathrm{Sect}(\Lambda, C) \hookrightarrow \mathrm{Sect}(\Lambda', C')$ are the summand injections mapping the unit line and each fiber to themselves, compatible under composition, and the carrier is their colimit

$$\mathrm{Sect}_\infty \;=\; \varinjlim_{(\Lambda, C)} \mathrm{Sect}(\Lambda, C).$$

Proposition 2 (A window is finite-dimensional, and its dimension is a count)**.** *For a region Λ of finitely many links, $\mathrm{Idx}(\Lambda, C)$ is finite, each $\mathrm{Fib}(a)$ is finite-dimensional, the dimension is one at the unit line alone, and at an occupied index*

$$\dim \mathrm{Sect}(\Lambda, C) \;=\; 1 + \sum_{a \in \mathrm{Idx}(\Lambda, C)} \; \prod_{v \in \mathrm{touched}(a)} \mathrm{vmult}(v, a)\ ,$$

a count computed from the structure constants N_{ab}^{c}, the vertex multiplicity being $m\{a, b, c\} = N_{ab}^{\bar{c}}$ (Proposition 8).

Proof. For dominant λ the read $\langle \lambda, 2\rho \rangle$ is a fold of natural products by its shape (Lemma 7's display), so $C_2(\lambda) \ge \langle \lambda, \lambda \rangle$ and the labels of content at most C satisfy $\langle \lambda, \lambda \rangle \le C$: finitely many. At a reduced shape, at most r rows deep, the fold $\langle \lambda, 2\rho \rangle$ reads at or beyond the degree, the last place's pairs the rows' own reads, so $d_f\, Q(\lambda)$ clears $d_f\, k$ (Definition 6) and an enumeration at a stated cutoff closes on the degrees at or below it. A configuration of $\mathrm{Idx}(\Lambda, C)$ assigns to the links of its support, within Λ's finitely many, labels from that finite set, so $\mathrm{Idx}(\Lambda, C)$ is finite; and the enumeration reads the cutoff per link: a configuration's content is the fold over its links' reads, each a natural, so a partial assignment's content sits at or below every completion's and the enumeration refuses a partial beyond the cutoff with every completion refused beside it, the pruned walk and the whole product's filter one list. Each $\mathrm{vmult}(v, a)$ is a natural at an occupied configuration and the product defining $\mathrm{Fib}(a)$ has finitely many factors, so $\mathrm{Fib}(a)$ is finite-dimensional and the displayed sum is a finite sum of finite products of structure-constant reads. Both indices are needed: the content bound alone leaves a loop free to sit at every translate, all at one content, so the region is what makes the count finite. $\square$

Proposition 3 (Fusion rows lie in content balls)**.** *Every constituent of $a \otimes b$ has highest weight below the Cartan product $a + b$ in the dominance order (a constituent's top content is a content of the product, contents of a tensor product add over the factors, and a block's contents sit at or below its top (Lemma 3)), and $\hat{c}_2$ (Definition 6) is increasing there (at $\mu \preceq \lambda$ dominant, $C_2(\mu) + \sum_k \delta_k\, h_k = C_2(\lambda)$ at the dominance order's prefix gaps δ_k and the gaps h_k of the sorted* *list $\lambda+\mu+2u$ at u the unit set's sorted display (Construction 1), every summand two naturals' product), so*

$$\hat{c}_2(c) \;\le\; \hat{c}_2(a+b) \;=\; \hat{c}_2(a)+\hat{c}_2(b)+\left[\,2d_f\,\langle a,b\rangle : 2d_f^2\,\right] \qquad \textit{for every nonunit member } c \textit{ of the row,}$$

the unit member, one exactly at $b=\bar{a}$ (Proposition 8), the Casimir's kernel point, inside every content ball outright; the cross term twice the bilinear read ($C_2(a+b)=C_2(a)+C_2(b)+2\langle a,b\rangle$), $d_f\,\langle a,b\rangle$ the weight lattice's integer read, every term one form read against itself, the pair at one value across the form's scales (Definition 1; Proposition 1). So a fusion row lies in a content ball, and content balls are finite, which is what makes fuse*, Φ and the one-link step finite objects. This bound and Φ do different jobs: the bound makes the row a finite set, while Φ is the attained maximum over it, and the window index uses Φ, since a bound composed from the two contents' root sum forces a rounding whose composition depends on the order taken.*

Proposition 4 (Every element has a least window, read off its index support)**.** *An $A\in\mathrm{Sect}_\infty$ is represented by some $x\in\mathrm{Sect}(\Lambda,C)$, a unit coordinate with its family of fiber components over the index. The index support $\mathrm{ix}(A)$, the member configurations, is representative-free, because the inclusions are summand injections and map the unit line and each component to themselves; the unit coordinate is at home in every window, multiplying as the scalar it is, every window its own. An element whose index support has members reads its window data over them:*

$$\Lambda_A=\bigcup_{a\in\mathrm{ix}(A)}\mathrm{supp}\,a,\qquad C_A=\max_{a\in\mathrm{ix}(A)}\mathrm{content}(a)$$

are one value at every representative, and $\mathrm{supp}(A)=(\Lambda_A,C_A)$ is the least window whose image contains A's components. Both are counts off the index: the links the element occupies and its members' largest content. Multiplication therefore has a window bound stated on the index alone,

$$A\cdot\mathrm{Sect}(\Lambda,C)\;\subseteq\;\mathrm{Sect}\big(\Lambda_A\cup\Lambda,\;\Phi(C_A,C)\big).$$

A probe is an element of the carrier, and its window is this one.

Definition 9 (The product and the involution)**.** Sect_∞ is a unital $*$-algebra. A factor of the unit coordinate alone multiplies as its scalar, every window its own (Proposition 4); at A and u whose index supports have members, at the least windows $\mathrm{supp}(A)=(\Lambda_A,C_A)$ and $\mathrm{supp}(u)=(\Lambda_u,C_u)$ (Proposition 4), the *target window* is $\mathrm{Sect}_0:=\mathrm{Sect}\big(\Lambda_A\cup\Lambda_u,\;\Phi(C_A,C_u)\big)$, its *window list* the unit line's vector with the fibers' stated lists, the pairing's Gram of unequal-membered determinant at the lists' independence (Definition 8; Lemma 3), and the product $A\cdot u$ is the target window's element at the reads

$$\mathrm{gram}\big(A\cdot u,\;v\big)\;=\;\mathrm{Eval}\big(\bar{A}\,\bar{u}\,v\big)\qquad\text{at every window-list member } v,$$

the element the reads' solve, the adjugate column against the Gram's determinant (Definition 3), one finite ground computation per read, the right side the *unit read*: Eval reads its argument's unit coordinate in the label index (Definition 8), the invariant count's own read (Proposition 8), a finite ground computation in the structure constants alone. Its matrix elements between summands, $\mathrm{gram}(x,A\cdot y)$ for x,y window-list members, are the label-index reads of Proposition 8 on the changed edge alone, and the magnetic form is built from them. The involution is contour reversal, descending from the words' own involution, Eval one value under it.

Proposition 5 (The product reads the pairing identity, and the algebra laws are structure-constant identities)**.** *The product of Definition 9 reads the pairing identity against every element,* $\mathrm{gram}_\infty(A\cdot u,\,v)=\mathrm{Eval}(\bar{A}\,\bar{u}\,v)$ *at every* v*, at the colimit pairing* gram_∞*, two elements' read at a shared window, one value at every window; it is the target window's one element at that identity, and every window at or beyond the target window solves the identity to it; the product of two window elements sits in a window, and in the colimit commutativity, associativity and the unit are the identities*

$$N_{ab}^{c}=N_{ba}^{c},\qquad \sum_{e}N_{ab}^{e}N_{ec}^{d}=\sum_{f}N_{bc}^{f}N_{af}^{d},\qquad N_{a\mathbf{1}}^{c}=\delta_{ac}.$$

Proof. Take A and u whose index supports have members, with $\mathrm{Sect}_0=\mathrm{Sect}\big(\Lambda_A\cup\Lambda_u,\ \Phi(C_A,C_u)\big)$ the target window of Definition 9 at the least windows (Proposition 4); a factor of the unit coordinate alone multiplies as its scalar, its identity the scalar's read.

The functional reads equal members off Sect_0. Eval reads the unit coordinate, so $\mathrm{Eval}(\bar{A}\bar{u}v)$ is of equal members unless the labels of v occur among the constituents of $\bar{A}$'s against $\bar{u}$'s, dual labels at one link set and one content each (Definition 6). Those constituents sit on the links of $\Lambda_A\cup\Lambda_u$ and, by Proposition 3, at content at most $\Phi(C_A,C_u)$: they are summands of Sect_0. So $v\mapsto\mathrm{Eval}(\bar{A}\bar{u}v)$ is of equal members on every summand beyond Sect_0.

On Sect_0 *the identity is the definition's solved read.* Sect_0 is finite-dimensional (Proposition 2) and its window list is independent with its pairing Gram of unequal-membered determinant (Definition 8; Lemma 3), so the identity's reads against the list solve the one $x\in\mathrm{Sect}_0$ with $\mathrm{gram}(x,v)=\mathrm{Eval}(\bar{A}\bar{u}v)$ at every $v\in\mathrm{Sect}_0$, the adjugate column against the determinant (Lemma 23(ii); Definition 3); two stated lists spanning Sect_0 solve one element, each solution reading the identity against the other's members, the solutions' gap at equal members against every member with its self-pairing that unit at the positive pairing, so a presentation's wiring list reads the determination at its own Gram. For v in any larger window, $\mathrm{gram}(x,v)$ reads only v's Sect_0-summands, gram being the orthogonal direct sum, and both sides are of equal members off Sect_0 as above. So the product reads the identity against every v; a window at or beyond the target reads the identity at its own list to one further solved element, each reading the identity against every v, the two agreeing against every element, one element at the positive pairing on the larger window; and the product sits in Sect_0 and descends to the colimit.

The laws. Sect_∞'s elements are character polynomials and Eval reads their product's unit coefficient, so the product just constructed is the character polynomials' own, commutative and associative with the unit label as its unit. Read in the label index those three statements are the displayed identities on N (Proposition 8). The pairing and the link Laplacians are compatible with the summand injections, so they descend to Sect_∞ and restrict to every window: a pairing of two colimit elements is read at any common window, one value at every window. ☐

Lemma 14 (Matrix elements are reads at the support's reach)**.** *Let* x,y *be elements of a window with joint support* $S:=\Lambda_x\cup\Lambda_y$*. Then* $\mathrm{gram}(x,y)$ *and* $\langle x,Ey\rangle$ *read the index data on* S *alone, and the magnetic row reads only the plaquettes meeting* S*'s vertex neighborhood: at* ∂p *sharing a vertex with* S*, the entry* $\langle x,\chi_\theta(U_{\partial p})\,y\rangle$ *is the changed-edge read of Definition 9 at the labels incident to* p*; at* ∂p *off that neighborhood it factorizes,*

$$\langle x,\chi_\theta(U_{\partial p})\,y\rangle=\mathrm{Eval}(\bar{x}y)\cdot\mathrm{Eval}(\chi_\theta),\qquad \mathrm{Eval}(\chi_\theta)=\delta_{\theta\mathbf{1}}\quad(\textit{Proposition 8}).$$

Consequently a label-graph isomorphism of the supports' vertex neighborhoods transports every pairing, electric and magnetic read across windows: each entry is one polynomial-pair read in the residue, window-free once a window's region contains the neighborhood. The locality band is that *reach, the plaquette neighborhood's: a magnetic row's entries off it are the factorized reads at the unit character, the sum's unit, so every pencil row reads within one band of its support, at every window.*

Proof. gram is the orthogonal sum of the fibers' pairings at matched configurations and E scales each summand by its content, both index reads on S (Definitions 8, 10); the product $\bar{x}y$ is supported on S at bounded content (Proposition 4). For the factorization: two elements on disjoint links and vertices have Fib data at disjoint vertex sets and label data at disjoint links, so the product's index summand at (a, b) is the concatenation of the two summands, each vertex's vmult reading its own incident labels; a concatenation occupies a link exactly where a part does, so the product's unit coordinate is the unit coordinates' product, and $\mathrm{Eval}(\chi_\theta) = \delta_{\theta\mathbf{1}}$, the Kronecker delta at two labels, is of equal members, the far plaquette's term with it. The near case is Definition 9's read, a fusion computation at the labels incident to p. ☐

Construction 8 (The coefficient carrier)**.** The carrier's vertex-free companion drops the invariance constraint: per link the *coefficient family* $\bigoplus_\lambda W_\lambda \otimes W_\lambda^\vee$, one block per label (Construction 5; the blocks and duals of Lemmas 4, 8), per window the tensor product of its links' families over the window index of Definition 8, the involution the blocks' dual exchange with the unit block fixed. Eval extends as the per-link unit coefficient: a product of coefficient elements decomposes by the fusion data, and its unit coefficient is the invariant line's read on both data, the vector datum's component at the orthogonal split (Lemma 5(i); Lemma 1) against the functional datum's value on the line, the component the projection fold's read at the line's member (Lemma 3). Two one-block elements pair at the *block orthogonality*,

$$d_\lambda \, \mathrm{Eval}\big(\, \overline{u \otimes f} \ (u' \otimes f')\,\big) \ = \ \langle u, u' \rangle \, \langle f, f' \rangle \ ,$$

the product a coefficient of the dual block against the block, its unit channel one line (Proposition 8's unit read; Lemma 4(ii),(iii)) at the coevaluation's tensor, and the display the projection read at two adjugate identities: the coevaluation against a member pair reads the factors' pairing, once per datum, and its self-pairing reads the block's count against the determinants (Lemma 8(i)); two blocks' elements pair at equal members, the unit channel vacant at distinct labels (Proposition 8). The *evaluation identity* $\sum_j U_{ij} \bar{U}_{kj} = \delta_{ik}$ is the dual pairing's read: the evaluation's tensor is invariant (Lemma 8(i)'s equivariance), so its coefficient family is its unit component's constants, the Kronecker read entry by entry. gram is the orthogonal direct sum of the blocks' standard inner products at the cleared block scale $[\,1 : d_\lambda\,]$: positive definiteness is by construction at the positive scales, the gram is the evaluation's own pairing, $\mathrm{gram}(x, y) = \mathrm{Eval}(\bar{x}\, y)$, the block orthogonality collected over the summands, the multiplication walks across it at a coefficient element w, $\mathrm{gram}(w\, x,\ y) = \mathrm{gram}(x,\ \bar{w}\, y)$, the products' exchange and regrouping at Eval's one value under the involution (Definition 9; Proposition 5), and the carrier sits inside isometrically at its own gram (Definition 8): a configuration's state contracts the links' block coefficients at the vertices' invariant lists, Eval factors over the support's links at disjoint variables (Lemma 14), and two states' pairing collects the block orthogonality link by link, the links' scales against the slot pairings' product regrouped at the vertices' lists (Definition 1), the fiber gram's own display. A generator acts within each block (a matrix coefficient of λ differentiates to one), so $\mathrm{Eval} \circ X_a$ reads equal members at every argument, the unit block the constants' line, and the pairing integrates by parts, the two reads $\langle X_a x,\ y \rangle$ and $\langle x,\ X_a y \rangle$ balance partners (X_a a derivation at one value under the involution, $X_a \bar{x} = \overline{X_a x}$). Consequently every finite family of coefficient vectors' Gram is positive semidefinite (Lemma 30): a sum of squares $\sum_i \|T_i x\|^2$, each T_i a generator action or a coefficient multiplication, is a fold of squares, and where the summed operator maps the carrier to itself the read restricts to every window. The pairing's squared Cauchy–Schwarz is the two-member instance at the coordinate folds, its gap a fold of squares displayed once at the cross pairs c_{ij}, each joined to its exchange's balance partner at the memberwise swap (Definition 3),

$$\Big(\sum_i u_i v_i\Big)^2 + \sum_{i<j} \big(\, c_{ij} + \check{c}_{ji} \,\big)^2 \;=\; \Big(\sum_i u_i^2\Big)\Big(\sum_i v_i^2\Big) \qquad \text{at} \quad c_{ij} := u_i v_j \ ,$$

the cross monomials $u_i u_j v_i v_j$ entering each member once per place pair, so two vectors' squared pairing sits at or below their self-pairings' product, the comparison's witness the displayed fold; a finite family of paired slots reads it at the slots' joined coordinate list, one list.

Definition 10 (The electric and magnetic operators, and the pencil)**.** E is the electric operator, diagonal in the label index, scaling a configuration's summand by its content and reading the unit line as its kernel, so E is positive semidefinite, every content positive (Definition 8); Δ_ℓ is its per-link summand, scaling by $C_2(a_\ell)$, so $E = \sum_\ell \Delta_\ell$; M is the plaquette multiplication $\sum_p \chi_\theta(U_{\partial p})$ with $\theta = \mathrm{adj}$, the sum over the region's plaquettes, $\#p$ their count. The *pencil* is the family of form pairs $(\alpha E : \beta M)$ with H their site datum at the join

$$\alpha\, E \;=\; \beta\, M + H\big([\alpha\!:\!\beta]\big), \qquad \alpha, \beta \in \mathbb{N}$$

over the end-weight pairs of Proposition 6 with the two ends' one-member instances, an end's pair reading its one member alone, the contact end's datum the magnetic member's balance partner. On a window ε_0 is the pencil's floor and the *level gap* $\tilde{E}$ its witness, the operator gap at $\varepsilon_0\, G + \tilde{E} = H$ with $\tilde{E}$ positive semidefinite, G the window pairing's gram (Definition 8; Lemma 33, the floor the least located root); ψ denotes a window ground, a kernel point of $\tilde{E}$, every read at ψ quantifying over the grounds, and $\omega = \langle \psi, \cdot\, \psi \rangle$. The gap chain reads a coupling in its *root coordinate*: the weights $[\, q^2 : p^2 \,]$ at the pair $\tau = [\, p : q \,]$, $q\,\tau = p$ the cross-multiplied tie, a cell's reads taken at pair τ, the pair-τ rays the coordinate's own points; toward the contact end the dual coordinate weights the couplings $[\, \sigma : 1 \,]$ at pair σ. The *word sector* is $\mathrm{Sect}_\infty\, \psi$, the image of multiplication by ψ (Definition 9).

Proposition 6 (The base is a closed segment)**.** *A coupling is an end-weight pair $[\alpha\!:\!\beta]$ at $\alpha, \beta \in \mathbb{N}$, taken up to rescaling: α weights the free end's electric member E and β the contact end's magnetic member M, the coupling pencil the pair $(\alpha E : \beta M)$'s site datum at the displayed join. The base $\mathcal{C}$ is the couplings with the two ends, the one-member weightings: the free end, the electric member alone, $H = E$ there, and the contact end, the magnetic alone, H the member's balance partner. Every coupling pair is interior, both weights present by the shape of its data, and the base is the closed segment, the free and the contact end its two endpoints. A spectral quantity of the pencil homogeneous of degree one scales by one factor along a ray, so the pair of two such reads is one value on the ray, the homogeneity principle's read (Definition 1), and descends to $\mathcal{C}$.*

Proof. The pair's site datum is additive over its members and degree-one in each weight by the join's shape, so rescaling the weights by a natural u rescales the pencil by u, and scaling an operator by u scales its spectrum by u: a degree-one spectral quantity acquires that one factor, so both components of a pair of two such reads acquire it and the pair reads one value across the ray, the homogeneity principle's read, every rescaling a ground rescaling and so order-preserving by the type. The weight data up to that rescaling are the two-member rays and the two one-member classes: the segment's interior and its two ends. □

3 The statement

Write H_k for the harmonic numbers (Definition 1), r for the member's derived residue, the pairing's own fold (Construction 4; Proposition 12), the A-series' declared scalar its instance with the successor name the fundamental count $d_f = r + 1$ (Construction 1), $\mathcal{C}$ for the closed segment of Proposition 6, E_0 for the module count at the free end (Proposition 7), $\mathcal{K}$ for the invariant (Definition 11), and c_1 for the multiplicity of the adjoint in adj $\otimes$ adj, θ's coroot-support count: two at the A-series at $r \geq 2$, one at its first member and at every further member of the classification (Construction 13; Proposition 13).

Proposition 7 (The module energy is a link count)**.** *At the free end the module $\chi_\theta(U_{\partial p})$ is an eigenvector of the pencil, with $E_0 = 4$ at every member and on either carrier.*

Proof. At the free end the pencil is its electric member E alone (Definition 10; Proposition 6), and E is diagonal in the label index, scaling the summand at a by content$(a) = \sum_\ell C_2(a_\ell)$ (Definition 10). In that index $\chi_\theta(U_{\partial p})$ is the configuration assigning θ to each link of ∂p, so it is an eigenvector of E with eigenvalue its content, the fold over ∂p's links, four regardless of sharing (Construction 7): that content is $4\,C_2(\theta)$; read against $C_2(\text{adj})$ (Proposition 1) it is 4. On the coefficient carrier the module is the θ-blocks' coefficient at the boundary's four links, each link's Δ_ℓ scaling its block at $C_2(\theta)$ (a matrix coefficient of λ differentiates to one, Construction 8) with every further link's summand at the sum's unit, so the eigen-read is the four-link fold there as well, one count on either carrier. $\square$

Lemma 15 (The free end's level is the fundamental loop)**.** *At the free end the window cut of Definition 11 is a diagonal read, and*

$$\mathcal{K} \;=\; \hat{c}_2(f) \;=\; [\,r(r+2) : 2d_f^2\,] \qquad \textit{at the free end,}$$

exact and symbolic in the residue, one value at every window of the directed family (Definition 8); at a member the level is the member's floor, the least nonunit $\hat{c}_2$ of its end lists (Lemma 9), the display its A-series read, the proof's two reads at the floor's label.

Proof. At the free end the pencil is its electric member E alone, diagonal in the label index with the contents as eigenvalues (Definition 10). Every configuration's content is positive (Definition 8), so the ground is the unit's line, ψ the algebra's unit (Definition 9) and E's kernel point (Definition 10), the floor of equal members there and the level gap reading $\tilde{E} = E$, and the word sector is the window itself. The cut's pair $(\tilde{E}^2 : E_0\kappa\tilde{E})$ is then diagonal, the unit summand its kernel's and the entries $c\,\langle c : E_0\kappa\rangle$ over the configurations' contents c: the cut at κ holds exactly where every occupied configuration's content is at least $E_0\kappa$.

The floor. At an occupied configuration every touched vertex reads an invariant count of at least one (Definition 8); at a vertex meeting exactly one support link an invariant line would be an equivariant line between the unit block and a nonunit one, and an equivariant line forces one label (Proposition 8; Lemma 4(iii)), against the support's nonunit datum. So an occupied configuration's links meet each of their vertices at least twice; a finite graph of minimum degree two contains a cycle (a forest has a leaf); every cycle is even, and at least four, the graph simple (the even-cycle and simplicity reads of Construction 7); and each occupied link reads content at least the member's floor, $\hat{c}_2(f)$ the A-read (Lemma 9). Every occupied configuration's content is therefore at least four times the floor.

Attainment. The configuration assigning the floor's label ω around one plaquette's boundary is occupied: each corner reads one outgoing ω against one incoming, vmult $= N^{\mathbf{1}}_{\omega\bar{\omega}} = 1$ (Proposition 8; Definition 8, incoming links dualized), at content four times the floor exactly.

So the cut holds at every κ at or below the floor and fails beyond, the loop's entry $c\,\langle c : E_0\kappa\rangle$ on its lower side at c four times the floor: with $E_0 = 4$ (Proposition 7), the level is the member's floor, and the contents are window-free data, so the read is one diagonal computation at every window. $\square$

Definition 11 (The invariant)**.** $\mathcal{K}$ is the pencil's level read against the module count E_0 of Proposition 7, given by its window cuts: at a coupling v, a window Λ and a pair $\kappa = [\,p : q\,]$,

$$\mathcal{K}_\Lambda(v) \;\geq\; \kappa \qquad \textbf{is} \qquad \big(q\,\tilde{E}^{\,2}\big)_\Lambda : \big(E_0\,p\,\tilde{E}\big)_\Lambda \;\text{ positive semidefinite,}$$

the pair clearing its pencil to integer coefficients, a positive rescaling fixing the count (Lemma 30), with $\tilde{E}$ the level gap of Definition 10 read on the word sector (Theorem 8), and $(\cdot)_\Lambda$ the compression to the window's word sector, taken of the detector form whole: the count reads the square's compression, a compression keeping the positive-semidefinite read (Lemma 30). The display quantifies over the window grounds (Definition 10), the ground freedom's quotient read: at a ground the compression is the detector pair's pullback along word images spanning the sector, a spanning list by elimination (Definition 3), a combination reading its image vector's pair and a kernel combination reading the sum's unit, so the pulled-back pair is positive semidefinite exactly with the pair on the word sector, its entries quadratic in the ground's coordinates over the located stage (Lemma 32, the kernel's spanning list Lemma 33's), one reversal count per ground, exact by $\mathrm{LDL}^{\mathsf{T}}$ there (Lemma 30; Corollary 3), settled by integers; a ground line reads one count, the pullback quadratic's scale the homogeneity principle's read (Definition 1; Lemma 30); a kernel of one line reads its quotient at a point, and a deeper kernel indexes the display over its ground space, the window pair's read covering every member at once; and the window pair, the detector form whole, reads every ground at once, a compression keeping the read (Lemma 30), the spectral read fixed at Theorem 6. Each window cut is available at every point of the *closed* segment: a window compression is finite-dimensional (Proposition 2), so a least eigenvalue and an exact reversal count exist at either end and at every coupling ray, an end's pencil its one member's own (Proposition 6). The cut of $\mathcal{K}(v)$ is the pairs every window cut admits, the window cuts' meet over the directed windows, $\mathcal{K}(v) \geq \kappa$ that membership, the read $\mathcal{K}_\Lambda(v) \geq \kappa$ at every window; each cut, the window cuts and the meet, is downward closed, its read, the kernel point with every further level at or beyond γ, weakening as γ falls; a member pair at all is the positivity read (every pair positive by its shape); and a value, of $\mathcal{K}$ or of a window's $\mathcal{K}_\Lambda$, is its cut's located edge where one is named.

Two scalings, and which one each clause reads. Moving along a ray, $[\alpha:\beta] \mapsto [\,t\alpha : t\beta\,]$, is the freedom the pair reads through, one value on the ray (Proposition 6); rescaling the family, $E \mapsto uE$ and $M \mapsto uM$, is a second freedom, and a level of the pencil is degree one in each. $\mathcal{K}(v)$ takes E_0 at the free end against the level at v, so it is free of the second and degree one in the first: its *side* is a property of the ray. Its *value* is free of both freedoms where the two reads are taken at the two ends against each other, one family and one u common to the two; that is $\mathcal{K}$'s value at the contact end.

Theorem 4. *Let the member run over the classification's stated domain, each member once: the A-series at the declared residue $r \geq 1$ (Construction 1), and B_ℓ at $\ell \geq 2$, C_ℓ at $\ell \geq 3$, D_ℓ at $\ell \geq 4$ and the five fixed members at their weight tables (Construction 4), the index floors reading each member once with the low-index coincidences the classification's own identifications; let r be the member's derived residue, $d \in \mathbb{N}$ the direction count (Construction 7), and $[\,\alpha : \beta\,]$ the coupling's end-weight pair at $\alpha, \beta \in \mathbb{N}$, interior by the shape of its data (Proposition 6). Then:*

1. *a pair is a member of the cut of $\mathcal{K}\big([\,\alpha : \beta\,]\big)$, positive by its shape; and*

2. *at the contact end the value is the contact pair of Definition 14,*

$$(3,\ H_r)\ :$$

the channel's order count at the base c_1*, the member's* θ*-support count (Proposition 13), and the walk's pole excess on the ruler's side, the pair's equality read componentwise, a count and one cross-multiplication identity (Definition 1); and at every window whose cut names an edge the edge is attained: the compressed pair's least root, read at a word image with the two moments at equality and the first positive (Lemma 22(i),(ii)), the level the cut fixes there the mass point's own located read between the window cuts' edges and the bordered pencils' own, the mass weight's side the upper at base two and equal members at base one (Theorem 28; Theorem 12); and*

3. *every continuum datum, a finite list of supports, separations and Euclidean pairs at pair coordinates against the corner scale, is one located object: the corner cell has positive extent with the contact end interior to it, the datum's scale reads close within priced brackets at the certified rates, and the corner heights read two-sided at the cell's positive located floor, the spacing that floor's read under the unit map; and each cell's window cut's edge, where one is named, is the compressed pair's least root read at a word image at the corner scale (Theorem 22; Lemma 45; Lemma 46; Lemma 22(i),(ii)).*

Proof. Clause (i) is Theorem 13: the cells' reads at the root coordinate's rays, the counting sandwich of its clause (v) at every further end-weight ray, the cutoffs reaching the cut within Theorem 32's priced modulus at the unit electric representative. Clause (ii) is Theorem 28, the lattice point $(3, H_r)$ of the two divisor reads with the window cuts' edges read at word images at the chain's located bracket. Clause (iii) is Theorem 22: the located brackets, the scale closure, and the height floor on the corner cell, the cells' edges read at word images (Lemma 45). □

Clause (i) is a count and clause (ii) a lattice point, and $\mathcal{K}$ is the one object both read (Definition 11). Its two ingredients are each weight-free, one value at every form point (Proposition 1), and $\mathcal{K}$ is a count against a count. Clause (iii)'s located values are the family's continuum reads, the spacing among its outputs (Theorem 22). The outputs factor, the invariant residue-side and the corner outputs holding the direction and scheme dependence (Theorem 30); the fundamental-character family reads beside the adjoint's at one bracket per output (Theorem 31); and the state's space is the word sector's own (Lemma 21). Remark 1 records the correspondence between these objects and their usual names, the lattice point's reading against its rulers included.

Remark 1 (Correspondence with the usual formulation)**.** The objects above are recorded here under their standard names, so that Theorem 4 reads against the usual statement. The correspondence reads into this remark alone: every derivation above holds with the remark deleted.

1. *The theory.* $H(1,b) = E + bM$ is the lattice gauge Hamiltonian, E its electric term, a sum of link Casimirs, and M its magnetic term, a sum of plaquette characters (Definition 10). Its space is the carrier Sect_∞, a colimit of windows each finite-dimensional (Definition 8, Proposition 2); its observables are the $*$-algebra of Definition 9, whose products obey a window bound stated on the index alone (Proposition 4) and whose local algebras have $\mathfrak{A}$ as their limit (Definition 15). That bound and that limit are locality, and both are read off the index. H is a symmetric rational matrix on every window, so each level read there is linear algebra (Theorem 16). Classically the carrier's elements are the gauge-invariant polynomial functions of the link holonomies, Eval is the Haar integral of the compact group, and the pairing is its L^2

one: the compact group, its invariant mean, and its coordinate ring are one classical reading of the fusion data, which the calculus above computes exactly. The label calculus's classical identities are listed under their usual names: the blocks are the irreducible representations at highest weights, the orthogonal complement of Lemma 1 is what Weyl's integration buys as complete reducibility, the scalar endomorphisms of Lemma 4 are Schur's lemma, the alternant identity of Theorem 1 is the Weyl character formula, the alternant pair a_β reading as the determinant $\det\left(x_i^{\beta_j}\right) = \sum_w \operatorname{sgn}(w)\, x^{w\beta}$, the pair's value its members' difference and the swap grading the sign of the permutation, the full-column line of Lemma 8(ii) the determinant representation, and Corollary 1 its dimension formula, Lemma 11 the Pieri rule at one box, Lemma 12's path count f^λ the standard tableau count at the shape (the raised glyph the classical one, the unraised f the fundamental label's), Corollary 2 the alternating fusion count (Brauer, Klimyk), and Lemma 13 the Schur–Weyl invariant span. That reading extends to the local objects: the characters are the blocks' traces on holonomies, the product reading $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \cdot \operatorname{tr} B$ with $d_a = \chi_a(1)$; the adjoint character runs pointwise over $[-1, n^2 - 1]$ and the fundamental trace's modulus over $[0, n]$, the tops at $U = 1$, the form caps of Lemmas 17 and 18 read pointwise; the form line of Proposition 1 is the invariant symmetric bilinear forms on $\mathfrak{su}(n)$; the two-plaquette fiber of Construction 9 is $L^2(G \times G)^G$, the functions of the two plaquette holonomies invariant under simultaneous conjugation, the theta graph's gauge reduction; and $\mathfrak{A}$ (Definition 15) is the quasi-local C^*-algebra, the norm closure of the inductive limit of the local algebras.

2. *The vacuum.* ψ is a window ground and ε_0 its eigenvalue, so $(H - \varepsilon_0)\psi = 0$ and $H - \varepsilon_0 \succeq 0$ on $\psi^\perp$ by construction (Definition 10); on a window the ground multiplicity is a count, a divisor read (Theorem 9), and the vacuum's correlation reads sit in named window brackets, Euclidean pairs included (Theorem 20). A reader who doubts uniqueness in the limit keeps every read: the cut is stated on counting functions, every comparison a count of the pencil itself rather than a read at a chosen state (Definition 11; Theorem 19), so Theorem 4 holds at every ground.

3. *The lattice.* The carrier is stated over the interface of Construction 7, so the derivations run on its fields alone. The reading here takes $\mathbb{Z}^d$, whose plaquettes are its unit squares, at $d \geq 2$, and the plaquette chain at $d = 1$, consecutive squares sharing one link. On either, a plaquette's boundary is four links, the count Proposition 7 reads, and a link sits in the plaquettes of its direction's pairs, the per-link plaquette count the gap g at $g + 2 = 2d$ here, the per-vertex count $d\,g$, a straight cell's stencil $[\,g^2 : 2\,]$, and each at most two on the chain. d counts the space directions and time is the pencil's own, so $d = 3$ is the case of $\mathbb{R}^4$.

4. *The lattice symmetries.* E and M run over all links and all plaquettes at coefficient one, so H commutes with every symmetry of $\mathbb{Z}^d$ permuting those two sets: the translations, and the hyperoctahedral group B_d of signed coordinate permutations (Theorem 21), whose permutation factor is the coordinate exchanges and whose sign flips are the reflections. B_d is what Theorem 21 is stated over, the translations entering separately at the fibering (Lemma 39). What the continuum asks besides is $O(d)$, and Theorem 21 reads the difference at every correlation as a multiplicity: the rings coincide at degree 2, the multiplicity positive at degree 4, so the anisotropy is suppressed by two degrees in the momentum; the theorem's isotropic span reads classically as the $O(d)$-invariant ring, the first fundamental theorem for $O(d)$ the classical side's own identification.

5. *The coupling.* The pencil's parameter is the coupling. The free end's pencil is the electric

term alone and the contact end's the magnetic alone (Proposition 6), so the two ends are the strong- and weak-coupling limits.

6. *The continuum's points.* Clause (i) is stated over every interior end-weight ray, the root coordinate's rays the cells' working points (Definition 10). A reader's continuum point (a Cauchy or Dedekind real, a nested-bracket datum) reads the cut in through its own rational comparisons: the counting sandwich of Theorem 13(v) transports the rational reads to every located coupling at its stated modulus, so "at every interior coupling" holds at whatever continuum the reader brings, each point reading itself in through the rationals beside it. The chain's own points are the rational rays, and the sandwich is the whole of a further point's read.

7. *The scalar pairs.* The translation of the colon notation is a bijection and reads one way only: a scalar pair $[a : b]$ (Definition 1) is the positive rational a/b, two pairs at the cross-multiplication identity being one rational, so the stage glyph $\mathbb{P}$ reads as the positive rationals, with $\mathbb{Q}$, the field, this remark's own; under that map the displayed sum, product and order of Definition 1 are the field's own on the positive rationals, the signed readings and the zero this remark's own, a pair of equal members reading zero and an absent summand a zero term, and $H_k = \sum_{j\leq k}[1 : j]$ is the classical harmonic number $\sum_{j\leq k} 1/j$. A coupling's end-weight pair $[\alpha\,{:}\,\beta]$ (Proposition 6) is the point $[\alpha : -\beta]$ of the real projective line in the classical (E, M)-coefficient coordinates, the minus this remark's own; the scalar instance is the affine reading at positive second component, and the division is the remark's own reading of a clearing. A balance pair $\langle u\,{:}\,v\rangle$ (Definition 1) is the difference $u - v$, the subtraction this remark's own: its side is the value's sign, equal members the value zero, and a bracket of balance-pair endpoints is the signed interval its endpoints name.

8. *The names.* The document's terms of art read against their standard names, each at its defining site. The reversal count $\mathrm{rev}(A : B)$ (Lemma 30) is the negative index of inertia of the difference $A - B$, the subtraction this remark's own, and its two-splits-one-count clause is Sylvester's law of inertia. The division descent (Theorem 15) is the Euclidean algorithm, its greatest common divisor taken monic over the pair carrier and, at the integer representatives, primitive at the stated positive top with the clearing multipliers the remark's own rescaling; the cleared division at a stated top is the classical pseudo-division. The remainder lists at a monic S (Definition 2) are the quotient ring $K[x]/(S)$ read as data. The cofactor c at $Y\,c = X$ (Definition 1) is the exact quotient X/Y, computer algebra's own cofactor, and the gap g at $q + g = p$ the difference $p - q$, the remark's own operations (the classical matrix cofactors enter as the adjugate's entries, under that name). The deck relation $z^2 + 1 = wz$ (Definition 4; Theorem 11; Lemma 38) is the double cover $w = z + 1/z$, the Joukowski map, its deck involution the cover's deck transformation $z \mapsto 1/z$, and the chord $t = \zeta + \zeta^{-1} = 2\cos k$ at the classical momentum k is the Chebyshev variable of the momentum fiber (Lemma 39). The decimation (Theorem 18) is the classical block decimation, each step a Schur complement onto the head. A located element (Lemma 32) is located in Bishop's constructive sense, the bracket its datum. The d_f-ality (Lemma 42) is the classical N-ality at $N = n$, the box count's class at full columns. The beta-set (Construction 1) is the partition's classical beta-set at the positive shift, $x_i = \lambda_i + (d_f - i) + 1$ against the classical $\lambda_i + d_f - i$. A content (Construction 1) is the classical weight of the grading, 'weight' in the chain the end and form data's word; a configuration's content (Definition 8) is its total Casimir, the electric term's total. The calculus at the symbolic residue is the unitary group $\mathrm{U}(n)$'s, the determinant relations per-residue data, the presentation kernel's (Remark 2).

9. *The members.* The classification's members read as the simply connected compact simple groups at $r+1=h^\vee$, the derived residue the dual Coxeter number's read, the residue folds of Construction 4 its per-member evaluations; the title reads here. The A-series is $\mathrm{SU}(n)$ at $n:=r+1=d_f$, whose fundamental representation has dimension n and whose Lie rank is $n-1$, the declared names reading the matrix size as the fundamental count d_f and the Lie rank as the residue r (Construction 1); the member weight tables read B_ℓ, C_ℓ and D_ℓ as $\mathrm{Spin}(2\ell{+}1)$, $\mathrm{Sp}(\ell)$ and $\mathrm{Spin}(2\ell)$, and the five fixed members the exceptional groups. Theorem 4's prefix instantiates the classical statement at every simply connected compact simple group and every $d\geq 1$, each group once: the index floors read each low-index coincidence at one member, $\mathrm{Spin}(3)$ and $\mathrm{Sp}(1)$ at $\mathrm{SU}(2)$, $\mathrm{Sp}(2)$ at $\mathrm{Spin}(5)$, and $\mathrm{Spin}(6)$ at $\mathrm{SU}(4)$, with $\mathrm{Spin}(4)$ off the simple list, the identifications the classification's own.

10. *The weak-coupling comparison, and the root coordinate.* Classically the weak-coupling theory is read against lattice Maxwell theory per Lie-algebra direction: the link field is $\log U_\ell$, the magnetic energy is the abelianized flux squared, and the comparison diagonalizes in Fourier with dispersion $\varepsilon(k)^2=2|b|\,\hat{h}(k)$ at the graph Laplacian $\hat{h}$ of $\mathbb{Z}^d$, massless at $k\to 0$. That is the classical identity of the gap chain's objects: the root coordinate τ ($b=-\tau^2$, the remark's reading of Definition 10's weights) is the harmonic frequency scale, and the soft momenta the flat step reads cell by cell (Lemma 39; Theorem 19) are the sector the massless comparison leaves ungapped: the lift above the comparison is what the counts read, the corner scale its ruler (Lemma 45).

11. *The gap.* E_0 is the energy of the one-plaquette excitation at the free end, where the module is an eigenvector of the pencil (Proposition 7), and $\mathcal{K}$ is the mass of the lightest excitation above the ground read against it (Definition 11). Since $E_0=4>0$, clause (i) is the mass gap: on the word sector (Definition 10) the spectrum starts *strictly* above the vacuum. The spectrum condition itself, $H-\varepsilon_0\succeq 0$, is item 2's and holds by construction; clause (i) is the strict statement above it. Clause (ii) is its size in units of that excitation, the two ends read together, with the level met at word images window by window (Lemma 22(i),(ii)): the lightest excitation is a state the local observables create from the vacuum. A reader who wants the level as an energy restores the two scales at which $\mathcal{K}$ reads one value: it is $\mathcal{K}\,E_0\,C_2(\mathrm{adj})$, with E_0 moving with the family's scale and $C_2(\mathrm{adj})$ with the invariant form (Proposition 1), as the level does with each, which is what makes the product an energy.

12. *What clause (ii) says at the contact end.* $\mathcal{K}$ reads a value free of unit and of scale: one value at every form point (Proposition 1) and at every family scale (Definition 11, Theorem 29), a ratio of two counts, the outputs' factorization Theorem 30's, the invariant residue-side with the corner outputs holding the direction and scheme dependence. Clause (i) is strict positivity at every rational interior ray, transported to every located coupling by the sandwich (the item on the continuum's points above); clause (ii) is the lattice point $(3,H_r)$ at the contact end, its excess on the ruler's side, and $\mathcal{C}$ is closed (Proposition 6), so that endpoint is a value like any other rather than a limit approached; and the approach is the corner cell's, its extent and floor located outputs at the corner scale, the couplings' geometric mean (Lemma 45). The classical rulers are $(\ln c_1,\,1)$, the axes the usual formulation measures with; read against them the lattice point is the exact value $\mathcal{K}=c_1^3\,e^{-H_r}$ at every member, $r+1=h^\vee$ the members item's read, the base c_1 one off the A-series so the value e^{-H_r} there with the first ruler reading one, and at the A-series $8\,e^{-H_{n-1}}$ at $n\geq 3$ and e^{-1} at $n=2$ (Construction 13; Proposition 13). The first coordinate's order 2 is classically the Szegő geometric-mean read of the spectral weight, and the second the walk residue's exponent; the attainment clause reads classically as the

excitation's own state, a bound state at base two, the mass point an isolated pole below the band, and at base one the mass point at the band's edge, the mass weight's equal-membered read there (Theorem 12); item 4's $O(d)$ multiplicities are the contact end's symmetry reads.

13. *The continuum reading.* The usual formulation reads the contact end's approach through the continuum construction: the spacing a sets the units, the electric weight is $g^2/2a$ and the magnetic $1/(ag^2)$ at the usual coefficients, and the construction holds the physical gap Δ fixed while $a \to 0$ along a coupling trajectory $g \to 0$, so its premise is that the gap in lattice units, $a\Delta$, falls under every floor toward the contact end, the diverging correlation length. The corner scale maps that premise onto the chain's data exactly: the two weights' product is $1/2a^2$ at every g, one value across the couplings, so the corner height of Lemma 45 reads classically as $\sqrt{2}\, a\Delta$ at every coupling of the approach. The premise therefore reads: the gap falls below every corner-scale floor toward the contact end, the corner cell's outputs under every floor. Its negation reads: a positive floor on a positive extent, $a\Delta$ bounded below toward weak coupling. The outputs are located data per member and direction count (Lemma 45), so the formulation's continuum reading, restricted to this family, is a read of them: the corner cell's floor is positive, a located datum per member and direction count at the certificate's derived rates (Lemma 45's cell floor); clause (ii)'s pair is the contact end's own value on either outcome, and item 4's multiplicities are the reading's symmetry half.
14. *The unit map.* The classical construction chooses a trajectory $g(a)$ by hand; here the spacing is a read: at a physical gap Δ the unit is $a = h/(\sqrt{2}\,\Delta)$ at the certified corner height h (Lemma 45), one equation per scale, the whole trajectory a located function.
15. *The reconstruction data.* The Wightman reconstruction reads $(\mathfrak{A}, \omega)$: the quasi-local algebra of Definition 15 with a window ground's state, positivity definitional (Definition 8), the Hilbert space the GNS one at the word sector's own pairing (Lemma 21), the vacuum the ground; the Osterwalder–Schrader detour is the Euclidean route to what the pencil formulation holds by construction (Theorem 33), so the "at least as strong as" clause is met on the spectral side directly. The lattice correlators' certified reads are Theorem 20's displays: window brackets, clustering, and Euclidean pairs, with $e^{-u(H-\varepsilon_0)}$ this remark's reading of the Euclidean bracket families; the Wightman functions are the bracketed reads' classical reading at the moment folds' clearance (Lemma 47), the reconstruction's half there (Theorem 22), the spacing the height floor's output (Theorem 22(iii)).
16. *The spectral condition.* $\operatorname{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ is the window cut (Definition 11; Lemma 20), its edge met at a word image window by window with the weight at the level where the inclusion is read (Lemma 22); the energy–momentum decomposition is the fibering (Lemma 39); the forward cone is the cone read under the unit map (Lemma 48), the dispersion isotropic at degree two with the direction data two powers of the spacing down; and the posed $\Delta > 0$ is the corner floor's positivity read through the unit map, a located datum per member at the chain's direction count (the projection item's own reads) at the certificate's derived rates (Lemma 45's cell floor), clause (ii)'s pair the exact value at the contact end.
17. *Locality, and the net.* The regions' local algebras with the window bounds (Proposition 4; Lemma 14; Definition 15) read as the Haag–Kastler net; the algebra is commutative outright (Proposition 5), the equal-time half of microcausality, and the unequal-time half is the propagation bracket's read (Theorem 20(vi)): the lower family's reach is exact, commutator reads beyond it are the remainder's own at the certificates' width, and the classical reading's cone is the reconstruction's read at the located families (Theorem 33), riding the datum reads' closure

at the spacing output (Theorem 22(ii),(iii)); the clustering brackets are Theorem 20(iv)'s.

18. *Invariance.* Rotations: Theorem 21's count is the limit's $O(d)$ invariance, the breaking terms at two powers of the spacing under the unit map, the rate certified. Boosts: time is the pencil's own and continuous, and the degree-two dispersion is one isotropic cone with the direction data two orders down (Lemma 48); the boost covariance is the reconstruction's read at the cone (Theorem 33), riding the datum reads' closure at the spacing output (Theorem 22(ii),(iii)); translations are the lattice's, the momenta reading through the t-cells at every continuum point.

19. *Nontriviality.* The theory is interacting at every member: c_1, the cubic-coupling count, is positive (Construction 13); the connected four-point read at the theta-graph pair holds a positive located floor on a cell at the free end's side, the leading coefficient $c_1^2\,\langle S:[\,1:8\,]\rangle$ a displayed member read (Lemma 50), while a generalized free field's four-point function is its three two-point pairings' sum, Wick's read, so its connected fold reads equal members at every argument list: the floor separates the state from every generalized free field's; and clause (ii)'s value separates the family from every free field's reads.

20. *The projection.* The posed problem asks one compact simple group on $\mathbb{R}^4$, existence, and a positive gap. The statement here is instantiated at every member of the classification, every direction count, and every coupling of the closed segment, with the contact end a value rather than a limit, the constant exact where a bound was requested, and the chain constructive over the naturals with machine-checkable certificates: the posed problem is the $d = 3$, single-member, existence-only projection of Theorem 4, its clause (iii) the existence half's carrier, read through the items above, each requirement one derived read: the existence half reads Theorem 22(i)'s located brackets with the spectral floor on the corner certificate's cell, the heights two-sided at the cell's positive located floor, the datum reads' closure the reconstruction's read (Theorem 33), and the gap half is the floor's clearance at the located spectral data (Theorem 22(ii)). The emitted records read at both coordinates: the assembled chains and the corner records read at the plaquette chain's direction count, the separated-sector instance at two directions and residue two (Lemma 41), and the derivations run at every direction count with the stencil caps and the momentum sums polynomial in it (Theorem 13(iv)): the posed $d = 3$ is the derivations' instance at its stated coordinates, read as every dimension is; the emitted records key by member, their direction count the chain's, this item their statement.

21. *The representation, and the bridge to the fundamental theory.* M multiplies by the adjoint character. The bridge to the fundamental theory's coupling is Theorem 26's $b_f^{\mathrm{equiv}} = 2n\,b_X$, the plaquette characters' quadratic matching (χ_R-curvature $= d_R C_2(R)/\dim G$) generating the fundamental theory's continuum action at that ratio, a ratio of two weight-1 objects and so a count, the fundamental-character family's corner outputs beside the adjoint's at one bracket per output (Theorem 31); c_1 counts the cubic couplings of the adjoint (Construction 13).

4 Fusion data

Proposition 8 (A fusion computation is a representation computation)**.** *The fusion data are counts of the calculus: N_{ab}^c is the number of c-blocks in a product of blocks and d_a the block dimension (Lemma 5, on labels by Construction 5), $m\{a,b,c\}$ the invariant count of the triple product, the labels the label set's (Construction 5). Each is read in the label index: χ_a is the index's generator* *at a, the product of two expanding over the tensor product's blocks,*

$$\chi_a\chi_b = \sum_c N_{ab}^c\,\chi_c,$$

the coefficients the block counts (Lemma 5(iii), on labels by Construction 5); the dimension d_a is the block's count (Lemma 4(iv)); and the unit label $\mathbf{1}$ has $\chi_{\mathbf{1}} = 1$, the constants' line's.

Four reads follow, and they are what the label index runs on. Write $\mathrm{Hom}(x, y)$ *for the linear maps from x to y; then* $\mathrm{Hom}(x, y) \cong x^* \otimes y$*, the intertwiners are its invariants, and the unit label's multiplicity in any x is its invariant count. First,*

$$N_{ab}^{\mathbf{1}} = \delta_{a\bar{b}},$$

the unit read of Construction 5: the count of the unit label in the product at a and b is the full-column-power count of Lemma 8(iii), one exactly at the dual label. Second, the characters are orthonormal for the evaluation, the unit-label coefficient read $\mathrm{Eval}(\chi_c) = \delta_{c\mathbf{1}}$*, the block's invariant count, so*

$$\mathrm{Eval}\big(\chi_a\overline{\chi_b}\big) = \sum_c N_{a\bar{b}}^c\,\mathrm{Eval}(\chi_c) = N_{a\bar{b}}^{\mathbf{1}} = \delta_{ab},$$

and a product expanded in the label index therefore has unique coefficients. Third, $m\{a, b, c\} = N_{ab}^{\bar{c}}$, by $m\{a, b, c\} = \sum_d N_{ab}^d N_{dc}^{\mathbf{1}}$ with the first read; the display reads at either pairing of the associativity at the unit target (Lemma 5(iii)), $\sum_e N_{ab}^e N_{ec}^{\mathbf{1}} = \sum_f N_{bc}^f N_{af}^{\mathbf{1}}$, the first read keeping each sum at the dual label's summand alone, so $N_{ab}^{\bar{c}} = N_{bc}^{\bar{a}}$, the rotation, with the factors' exchange the transposition: the two generate the symmetry of m in its three arguments, and that is the symmetry of N at self-dual labels. Fourth, the involution's index read: the dual label's matrix is the transpose, $N_{\bar{a}\,b}^c = N_{ac}^b$: the triple count reads one value at every argument dualized, duals of blocks being blocks with the unit's count kept (Lemma 8(i)), and the third read's symmetry closes the exchange.

Lemma 16 (The adjoint character)**.** *Write f for the fundamental label, the one-box shape's, whose block is V itself at the declared count d_f (Construction 1), and $\bar{f}$ for its dual, at the residue at or beyond one. Then*

$$f \otimes \bar{f} \;=\; \theta \oplus \mathbf{1}, \qquad \textit{hence} \qquad \chi_\theta(U) + 1 \;=\; |\mathrm{tr}\,U|^2,$$

and the adjoint's contents are the letter-pair moves, one place raised and one lowered, each ordered pair of distinct places once, with the unit-monomial content at multiplicity r, the product's unit-monomial count splitting as the unit line's one with the adjoint's, $r + 1 = d_f$.

Proof. $\bar{f}$ is the class of the complement f^* (Construction 5), the one-column shape at the complement gap, r rows of one box (Lemma 8(ii)), and $\bar{f} \otimes f$ exhausts by the one-box row (Lemma 11): the dominant one-box additions of (1^r) are $(2, 1^{r-1})$, the label θ (Construction 5), and the full column (1^{d_f}), the unit label's, one block each with every further label vacant, and the factors exchange (Lemma 5(iii)): $f \otimes \bar{f} = \theta \oplus \mathbf{1}$. The product's contents are the factors' contents' sums (Construction 1): the one-box block reads each letter once and $\bar{f}$'s block each withdrawn letter once (Lemma 11's complement read), so the product reads the unit-monomial content at d_f, one per letter, and each letter-pair move once, a letter against a withdrawn one. The content refinement of Lemma 5(ii) reads the product's occupancies as the two blocks' at their counts, and the full-column line is occupied at the unit-monomial content alone (Lemma 8(ii)): θ's occupancies are the product's with the line's withdrawn, each move once and the unit-monomial content at $d_f - 1 = r$. Hence $\chi_f\chi_{\bar{f}} = \chi_\theta + 1$ (Proposition 8), and $\chi_f(U) = \mathrm{tr}\,U$ reads the display. ☐

Lemma 17 (The fusion matrix is capped by its dimension)**.** *On one link's character index, multiplication by χ_λ is the fusion matrix $N^{\mu}_{\lambda\nu}$ against the orthonormal characters (Proposition 8), row-finite on every window (Proposition 3), with count entries, symmetric at self-dual λ, and the dimension vector is its exact positive eigenvector:*

$$\sum_{\mu} N^{\mu}_{\lambda\nu}\, d_\mu \;=\; d_\lambda\, d_\nu,$$

dimensions multiplying over a tensor product (Lemma 5). Hence at self-dual λ the fusion form is capped two-sidedly at the dimension (Lemma 30),

$$N_\lambda \;\preceq\; d_\lambda \qquad \textit{with} \qquad d_\lambda + N_\lambda \;\; \textit{positive semidefinite}:$$

the two squares $(d_\nu x_\mu + d_\mu x_\nu)^2$ and $\langle d_\nu x_\mu : d_\mu x_\nu\rangle^2$, each positive by its shape, price each entry's read $2\,x_\mu x_\nu$ two-sidedly at $[\,d_\nu : d_\mu\,]x_\mu^2 + [\,d_\mu : d_\nu\,]x_\nu^2$, the comparison at its rescaling (Definition 1's homogeneity principle), the cleared weights the window's co-products, the dims' product at the pair's two factors withdrawn, every weight natural, and the eigen-identity sums the rows. The fundamental's multiplication then squares to the adjoint's: its pairing-adjoint is multiplication by $\chi_{\bar f}$, so

$$M^\dagger_{\chi_f} M_{\chi_f} \;=\; M_{1+\chi_\theta} \;\preceq\; 1 + r\,(r+2) \;=\; d_f^2,$$

at the display's $\lambda = \theta$, self-dual with $d_\theta = r\,(r+2)$.

Lemma 18 (The loop character is capped by its dimension)**.** *On every window, multiplication by a plaquette's fundamental character squares to the cap d_f^2: writing $\chi_f(U_{\partial p}) = \sum_{i,j} A_{ij}\,(U_4)_{ji}$ over a cut of the loop, $A := U_1U_2U_3$ the complementary path,*

$$\big\langle y,\; \chi_f(U_{\partial p})\, x\big\rangle^2 \;\le\; \Big(\sum_{i,j} \|\bar A_{ij}\, y\|^2\Big)\Big(\sum_{i,j} \|(U_4)_{ji}\, x\|^2\Big) \;=\; d_f^2\,\langle x, x\rangle\,\langle y, y\rangle$$

by the squared Cauchy–Schwarz of the coefficient carrier's pairing (Construction 8), every square a pairing, each sum contracting by the evaluation identity to the trace of the identity, $\mathrm{tr}(A^\dagger A) = \mathrm{tr}(U_4^\dagger U_4) = d_f$, times the pairing; at $y = \chi_f(U_{\partial p})\, x$ the display reads the comparison at the factor $\langle y, y\rangle$: at the factor occupied the order reads back (Definition 1), and at the factor of equal members the conclusion is the pairing's own side read, the two self-pairings at or above the sum's unit (Construction 8): $\langle \chi_f x, \chi_f x\rangle \le d_f^2 \langle x, x\rangle$. Hence, with $\chi_\theta + 1 = |\chi_f|^2$ (Lemma 16),

$$\chi_\theta(U_{\partial p}) + 1 \;=\; M^\dagger_{\chi_f(U_{\partial p})} M_{\chi_f(U_{\partial p})} \;\preceq\; d_f^2 \qquad \textit{as forms on every window:}$$

the left side is a square, positive semidefinite at its split, the cap d_f^2 its upper read, and the shifted term reads the character's two-sided cap at unit width below and $r\,(r+2)$ above. At a member the cap runs in the adjoint itself: over a cut of the loop at the θ-block's coefficients, $\chi_\theta(U_{\partial p}) = \sum_{i,j} A_{ij}\,(U_4)_{ji}$ with $A := U_1U_2U_3$ read in the block, the squared Cauchy–Schwarz of the coefficient carrier's pairing contracts each sum by the evaluation identity at the block to the identity's trace there, d_θ (Proposition 13's dimension fold), and the order reads back as above: the multiplication's square is capped, $M^2_{\chi_\theta(U_{\partial p})} \preceq d_\theta^2$ as forms on every window; the term reads two-sided at d_θ by one further squared Cauchy–Schwarz at a window vector against its image (Construction 8), $\langle x, \chi_\theta\, x\rangle^2$ at or below $\langle x, x\rangle\,\langle \chi_\theta x, \chi_\theta x\rangle$, at or below $d_\theta^2\,\langle x, x\rangle^2$ at the square's cap against the self-pairing's side (Construction 8), and the squared comparison reads back at the trichotomy (Definition 1), the form and its memberwise swap each at or below $d_\theta\,\langle x, x\rangle$; the A-series' shifted route is the display above.

5 The requirement, in two moments

Theorem 5 (The divisor-separation detector)**.** *The level-γ requirement is the probe family $\mu_2(A) \geq \gamma\, \mu_1(A)$ at the two moments $\mu_j(A) := \langle A\psi,\, \tilde{E}^j A\psi\rangle$ (Theorem 7's reads), and its detector pair at a spectral point λ is $(\lambda^2 : \gamma\lambda)$, the moments' spectral read: a probe's two moments read the weight fold over the located spectrum (Lemma 33), so the requirement's read at a probe is the fold of the detector pairs' at the probe's weights. The pair's members are equal exactly at the kernel point and at the edge $\lambda = \gamma$, the two neutral points, its site datum the factor read $\lambda\,\langle\lambda : \gamma\rangle$, and off them its side is λ's against γ at positive λ: weight below γ drives failure, weight beyond passes, weight at the edge is neutral, and the vacuum direction, the kernel point, is neutral as well, the vacuum's saturation untouched.*

Theorem 6 (The level condition over probes is the spectral condition)**.** *Over the word sector, the level-γ requirement (one condition per probe) is equivalent to the operator inequality*

$$\tilde{E}^2 \;\succeq\; \gamma\,\tilde{E} \quad \textit{on the word sector} \qquad (\textit{the kernel point with every further level at or beyond } \gamma).$$

Proof. ($\Leftarrow$) With the spectrum at the kernel point and at or beyond γ, every detector pair reads its upper side or equal members (Theorem 5), and every probe's weight fold with it; read at the pair's certificate, the positive-semidefinite split prices every sector vector's form through the congruence at its witness, the blocks' fold of squares (Lemma 30), and the site datum's decomposition reads the priced form as the two moments' comparison, the requirement's own display. ($\Rightarrow$) On a window the reversal count settles it: at $\operatorname{rev}(\tilde{E}^2 : \gamma\tilde{E}) \geq 1$ a lower-side pivot sits among the elimination datum's LDL^T pivots, and Theorem 14(2) reads off the elimination a pair-entried v with $\langle v, \gamma\tilde{E}\, v\rangle > \langle v, \tilde{E}^2 v\rangle$, the reversal's own read. The witness is a vector of the sector, hence $A\psi$ at some word A (Definition 10), and that word's probe fails the requirement outright, the witness's two reads its two moments (Theorem 7). $\square$

Theorem 7 (The two-moment form)**.** *The family $\mu_2(A) \geq \gamma\, \mu_1(A)$ over probes is the two-moment form of the operator inequality ($\mu_j = \langle v_A, \tilde{E}^j v_A\rangle$, both affine in the state, so a floor at the extreme states extends to their mixtures). For a multiplication probe f (a carrier element acting by its product, Proposition 5), both moments are exact, β-free polynomial-pair atoms: $[H, f] = [E, f]$, so*

$$\mu_1(f) = \omega\big(f^\dagger[E, f]\big), \qquad \mu_2(f) = \omega\big(Q_2(f)\big), \quad Q_2(f) := [E, f]^\dagger[E, f],$$

the first display Theorem 8's exact line read at the state and the second the commutator's square, and the requirement reads, per multiplication word, as one inequality between two commutator reads at the state, with the coupling entering only through the state:

$$\boxed{\omega\big(Q_2(f)\big) \;\geq\; \gamma\,\omega\big(f^\dagger[E, f]\big)}$$

(The multiplication family suffices.) Both moments depend on A only through $y = A\psi$, since ψ, $\tilde{E}$'s kernel point, gives $[H, A]\psi = \tilde{E}\,(A\psi)$ at every probe, and both sides are quadratic forms in y. A probe is an element of the carrier (Proposition 4) and the carrier's elements multiply pointwise (Proposition 5), so $A \mapsto A\psi$ is linear and its image is the word sector (Definition 10): the floor over probes and positivity of the moment form on that sector are one statement, the hopping class included.

Corollary 3 (Two moments against a divisor)**.** *Combining Theorem 7 with Theorem 6, on a window:*

$$\big(\forall A:\; \mu_2(A) \;\geq\; \gamma\,\mu_1(A)\big) \quad\Longleftrightarrow\quad \big(\tilde{E}^2\big)_\Lambda : \big(\gamma\,\tilde{E}\big)_\Lambda \;\; \textit{positive semidefinite},$$

the compression to the window's word sector taken of the detector form whole, since $\mu_2(A) = \langle v_A, \tilde{E}^2 v_A\rangle$ and $\mu_1(A) = \langle v_A, \tilde{E}\, v_A\rangle$ at $v_A = A\psi$ (Theorem 8): one reversal count on a form pair over the located stage, exact by $\mathrm{LDL}^{\mathsf{T}}$ *there (Lemma 32), whose read is the kernel point with every further level at or beyond γ. Right to left by Theorem 5's side reads, the split's transport pricing every sector vector at the site datum's decomposition (Lemma 30); left to right by the count's own witness, which Theorem 14(2) reads off the elimination.*

Lemma 19 (The sector's span read)**.** *Multiplication by a window ground is adjoint to its involution image's: at carrier elements w, x and a window ground ψ (Definition 10),*

$$\langle w,\ x \cdot \psi\rangle \ = \ \langle w \cdot \bar{\psi},\ x\rangle\ ,$$

both sides the read $\mathrm{Eval}(\bar{w}\,\psi\,x)$*, the products' one value under commutativity with the involution invariance (Definition 9; Proposition 5); and a ground's involution image is a ground: the involution keeps the pairing (*Eval *one value under the involution, Definition 9) and the pencil, E at the dual label's Casimir (Definition 6) and M at the self-dual adjoint (Lemma 16), so it keeps the level gap's kernel. The span read at a window and a ground is the product matrix's rank at the window's dimension: the matrix* $\mathrm{Eval}(\bar{v}\,w\,\bar{\psi})$ *at w over the window list and v over the product window's (Proposition 4), every entry a ground computation, the rank the elimination's read (Definition 3), one decidable read per window and ground. The word sector's window projection fills the window exactly at the span read: a window element perpendicular to every word image pairs the ground product of its involution image against every element at the sum's unit, the display's read, so the product is the sum's unit at the positive pairing (Definition 8), the matrix's kernel the perpendicular elements' own; and at the filled projection the compression to the window's word sector is the window pair whole.*

Lemma 20 (A window spectral read is the cut at that window)**.** *At a window and a pair κ, the spectral read*

$$\mathrm{spec}(H) \ \subset\ \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa,\ \infty)$$

is the window cut $\mathcal{K}_\Lambda$ of Definition 11 at κ, at every window ground at once: the split of Lemma 33 maps the level gap $\tilde{E}$ ($\varepsilon_0 G + \tilde{E} = H$, Definition 10) to a diagonal at the kernel point's read and at reads at or beyond $E_0\kappa$, the detector pair at each reads its upper side or equal members (Theorem 5, the two neutral points the ground's and the edge's), so the pair ($\tilde{E}^2 : E_0\kappa\,\tilde{E}$) is positive semidefinite on the window, and a compression keeps that read (Lemma 30): every word-sector compression reads it, the cut's own object, the detector form compressed whole; and one ground at the span read closes the identification both ways, its compression the window pair whole (Lemma 19), so that ground's instance is the spectral read outright and every ground's follows by the display above, the window cut's edge the spectral edge there. The reading back rides the gap's own floor: the detector pair's side is λ's against γ at positive λ (Theorem 5), the levels' positivity $\tilde{E}$'s positive-semidefinite certificate (Definition 10), the datum locating every root's side. At a level on its lower side the pair's read holds outright, the square sitting on its upper side from either side of its entry: the floor's certificate is the side's one source.

Theorem 8 (The moments are read at coefficient one)**.** *For a multiplication word c (its products with M at one value in either order, the algebra commutative), one exact line:*

$$E(c\psi) \ = \ c\,(\varepsilon_0 + \beta M)\,\psi + \tilde{E}\,(c\psi) \qquad \text{at the join} \quad E = \beta M + H: \qquad [E, c]\,\psi = \tilde{E}\,(c\psi),$$

so $\omega(c^\dagger[E, c]) = \langle v_c, \tilde{E}\, v_c\rangle$ exactly, on the multiplication probes' vectors: the pencil's λ is the Hamiltonian's own, at coefficient 1*, so both moments depend on A through $y = A\psi$ alone.*

Lemma 21 (The state space)**.** *The word sector is the state's own space: at a window ground ψ and carrier elements A, B,*

$$\omega(\bar{A}\,B) \;=\; \langle A\psi,\, B\psi\rangle\ ,$$

both sides the read $\mathrm{Eval}(\bar{A}\,B\,\bar{\psi}\,\psi)$*, the left at the pairing identity and the right at the involution's product read, the products' one value under commutativity joining the two (Definition 9; Proposition 5): the pairing on the word sector is the state's, and the sector is* $\mathrm{Sect}_\infty\,\psi$*, its own image (Definition 10), cyclic by construction.*

Lemma 22 (The level is attained at a word image)**.** *At a window, a window ground ψ and the located edge γ of the level condition over the word sector (Theorem 6; the window cut's own read at $\gamma = E_0\kappa$, Definition 11): (i) the edge is the compressed pair's least root; (ii) a word image reads the two moments at equality there, the first positive; (iii) at the window's spectral read at the edge the state's weight sits at the level, a divisor point.*

(i) The edge is the compressed pair's least root. The compression is finite-dimensional with pair entries (Proposition 2; Theorem 16), and $(\tilde{E})_\Lambda$, the compression of the level gap, is positive semidefinite, a compression keeping the read (Lemma 30), with the ground inside its kernel block: ψ is the unit's own word image and $\tilde{E}\,\psi$ the sum's unit (Definition 9; Definition 10). A kernel point y of $(\tilde{E})_\Lambda$ reads its self-pairing at equal members, so $\tilde{E}\,y$ is the sum's unit: $\tilde{E}$ is positive semidefinite (Definition 10) and its split reads the pairing as the fold over the blocks (Lemma 30), a fold of equal members putting the coordinates off the kernel block at the sum's unit, so y sits in that block's span and its image with it. Then $(\tilde{E}^2)_\Lambda$ reads its row at y at the sum's unit as well, the pencil symmetric with $\langle x, \tilde{E}^2 y\rangle = \langle \tilde{E}x, \tilde{E}y\rangle$ (Theorem 16): both compressions sit on the split's own columns off the kernel block, the complement list, where $(\tilde{E})_\Lambda$ is positive definite (Lemma 30). There Lemma 33 splits the pair $\big((\tilde{E}^2)_\Lambda : \gamma\,(\tilde{E})_\Lambda\big)$ over located roots, $\mathrm{count}(a)$ *the roots below a: at γ at or below the least root the roots sit at or beyond the level and the pair is positive semidefinite, the congruence's diagonal $\langle x : \varepsilon_j\rangle\, g_j$ reading its upper side or equal members there, the least root's own read the sharpened one (Theorem 14(1)), while beyond that root the count is at least one and Theorem 14(2) reads the count's own witness off the elimination. So the level condition's members are the levels at or below the least root and the located edge is that root, the window cut's own edge at $\gamma = E_0\kappa$.*

(ii) Its kernel point is a word image, and the two moments read equal members there. At the edge the site datum Z at the join $\gamma\,(\tilde{E})_\Lambda = (\tilde{E}^2)_\Lambda + Z$ has a kernel point y off the sum's unit (Lemma 33's kernel block at a located root), a vector of the word sector and hence $A\psi$ at a word A, the sector being the image of multiplication by ψ with $A \mapsto A\psi$ linear (Definition 10; Theorem 7); y sits on the complement list, where the compressed level gap is positive definite, so its first moment is positive, and the kernel read gives the second at the level's multiple,

$$\mu_2(A) \;=\; \gamma\,\mu_1(A)\ , \qquad \mu_1(A)\ \textit{positive},$$

the moments read at coefficient one on the multiplication probes' vectors (Theorem 8; Corollary 3). The level-γ requirement therefore holds at equality at that one word, the detector pair's edge its neutral point (Theorem 5), and the floor over the probes (Theorem 7) is met at a probe of its own. The edge sits off equal members with it: at equal members the second moment reads them, its display $\langle \tilde{E}\,y, \tilde{E}\,y\rangle$ putting $\tilde{E}\,y$ at the sum's unit at the positive pairing (Definition 8) and the first moment at equal members with it, against its positivity.

(iii) At the window's spectral read the weight sits at the level, a divisor point. Where the window reads $\mathrm{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + \gamma, \infty)$*, the jump comparisons' own object (Theorem 19) and the cut's identification at a ground at the span read (Lemma 20; Lemma 19), the two moments are the weight*

folds over the located spectrum (Theorem 5): the split's congruence diagonalizes the gram and the pencil at once (Lemma 33), so the level gap acts at the gaps λ_j there and its square at their squares, the weights w_j the coordinates' squares against the gram's upper-side diagonal entries, and

$$\mu_1(A) = \sum_j w_j\,\lambda_j\ , \qquad \mu_2(A) = \sum_j w_j\,\lambda_j^2\ , \qquad \big\langle\,\mu_2(A)\!:\!\gamma\,\mu_1(A)\,\big\rangle\ =\ \sum_j w_j\,\lambda_j\,\big\langle\,\lambda_j\!:\!\gamma\,\big\rangle\ .$$

Every λ_j is the kernel point's read or sits at or beyond γ, so every summand of the fold reads its upper side or equal members, while the display above reads its left side at equal members: gaps add over a fold, and a natural fold reads the sum's unit exactly where every summand does (Definition 1), so at every occupied weight λ_j is the kernel point's read or the level γ exactly, and $\mu_1(A)$ positive puts an occupied weight at γ. The level is then a located root of the window pencil, a point of its divisor at that root's multiplicity (Lemma 33; Theorem 25(i)), the word image's weight there positive, and its state is the word sector's own, the state's space (Lemma 21).

6 The window ground

Theorem 9 (The window multiplicity is a count, and the count is a divisor read)**.** *On a window the ground multiplicity* $\dim\ker\tilde{E}$ *is one integer, a count at a designation's data; it is the ground root's multiplicity, one integer per cell of the crossing divisor, reading* 1 *at the free end and over its attached cell, and moving across a located boundary only at a ground collision, the boundary's own two-sided read (Theorem 25(ii),(iii)). Where the window read is* 1*, a symmetry of the window, an orthogonal map commuting with the pencil, maps the ground line to itself at a graded read, $g\psi = \psi$ at even and $g\psi + \psi$ the sum's unit at odd, the reads of ω blind to the grading.*

Proof. The multiplicity is a count. A window compression is finite-dimensional with pair entries (Proposition 2), and $\operatorname{count}(a) = \operatorname{rev}(H\!:\!aG)$ counts its eigenvalues with multiplicity below a, exact over the pairs by $\mathrm{LDL}^{\mathsf{T}}$ (Theorem 14; the multiplicity read is Lemma 33). Theorem 15 gives a pair δ below the separation of the pencil pair's site-datum determinant's distinct located roots (Lemma 33's χ), and a designation at that width (Theorem 14) reads levels $l < h$ at a gap at most δ, the pencil pair at l positive semidefinite and $\operatorname{count}(h) \geq 1$. The bracket is narrower than the separation, so it admits the ground root alone and

$$\dim\ker\tilde{E}\ =\ \operatorname{count}(h),$$

one count at the designation's data.

The count is a divisor read. The multiplicity is the ground root's order in the pencil determinant, the split's diagonal read (Lemma 33), and Theorem 25(ii),(iii) lock it: one integer per cell of the crossing divisor, read 1 from the free end's level bracket over the attached cell (every configuration's content is positive, the unit's line the ground, Lemma 15), and one two-sided bracket read at each located boundary.

The character. At a window read of 1, a symmetry g has $Hg = gH$, so it preserves the ground eigenspace, which is the line; an orthogonal map on a line is a scalar c at $c^2 = 1$ ($g^{\mathsf{T}}g = 1$), so $c = 1$ or $c + 1$ reads equal members; and $\omega(g^{\mathsf{T}}Ag) = \langle g\psi,\, A\,g\psi\rangle = c^2\,\omega(A) = \omega(A)$: the state reads its ground twice, an even count, blind to the grading. ☐

7 The integer constants

With every pair at one value across the form's scales (Proposition 1), every constant below is literally an integer or a fusion pair: 4 (links per plaquette, the module energy); $2d$ (the neighbor count, Construction 7's direction read); the winding floor $\hat{c}_2(\omega_{c_i})$ per cell and charged direction, the charged margin's growth (Lemma 42); and the cells' entries and levels, located roots and jump distances of the decimation's own divisors, outputs rather than constants. The residue enters through one pair, $\hat{c}_2(f) = [\,r(r+2) : 2d_f^2\,]$, the minimal content, its two reads the floor's product (Lemma 9) and the clearing's count (Definition 6) at the one-value identity $r(r+2)+1 = d_f^2$, the A-series' read with the member floors the end lists' (Lemma 9); the adjoint read against it is the invariant $[\,2d_f^2 : r(r+2)\,]$, and every further residue read of the chain is a coordinate read of the cell calculus (Lemma 25(iii)). What reads $4n$, $8n$, $2n$, n in the dictionary's units (Remark 1) was that integer times the Casimir.

8 The two-plaquette object

Construction 9 (Two-plaquette fiber)**.** Fix the theta-graph of two square plaquettes sharing one link. The *invariant fiber* is the span of the permutation-presentation states in the two plaquette variables (Construction 11), with the pairing identity of Proposition 5 through the presentation Gram (§11).

Construction 10 (The two-plaquette data)**.** On the invariant fiber of Construction 9, the electric operator's link weights are $(3,1,3)$, *forced by the graph*: each plaquette has four links, one of which is shared, so three unshared each,

$$E \;=\; 3\,\Delta_U \;+\; \Delta_L \;+\; 3\,\Delta_V,$$

Δ_U, Δ_V the plaquette Laplacians, Δ_L the shared-link Casimir (simultaneous right translation, §10). Energies and the coupling are read against $C_2(\mathrm{adj})$ (Proposition 1), so $\hat{c}_2(f) = [\,r(r+2) : 2d_f^2\,]$, $\hat{c}_2(\mathrm{adj}) = 1$, while the section's displayed operators read at the trace form itself, $C_2(\mathrm{adj}) = d_f$ there, the weight clause below the conversion. The magnetic operator is χ_X per plaquette (Construction 13). Module states: $\Phi_{q_1} = \chi_{\mathrm{adj}}(U)$, $\Phi_{q_2} = \chi_{\mathrm{adj}}(V)$, on which this graph's E reads $4\,C_2(\mathrm{adj})$, the carrier's own link count $E_0 = 4$ at the weight (Proposition 7), here the weights reading $3\,d_f + 1\,d_f$ with the Δ_V read the sum's unit (Φ_{q_1} free of V). A Feshbach coefficient (Construction 12) on this graph reads

$$\big\langle \Phi_{q_2},\, M\,x \big\rangle, \qquad x \text{ the solve of Construction 12 at } M \cdots \Phi_{q_1},$$

one solve per resolvent step at the pair $(E\!:\!4d_f)$'s site datum, Q the projection off the two module states (the site datum's kernel the module space, Construction 12), the inner product the pairing identity's read (Proposition 5). The magnetic multiplications read per plaquette: $\chi_X(W)$ at a state extends its factor list at the variable's pair (Construction 11's product at the pair's presentation, Lemma 16), so a k-step image is one state per plaquette word of length k, each at its own factor list, the solves at each word's stated list and the displayed coefficient the pairing identity's fold over the words, a presentation state sitting at one factor list with the algebra's sums reading through the folds; a k-solve coefficient pairs with $C_2(\mathrm{adj})^k$ to a weight-free read, one value at every form point (Proposition 1).

9 The permutation presentation

Construction 11 (States)**.** Fix a factor list $F = (v_1, \ldots, v_m)$, each *factor* a variable $v_i \in \{U, V\}$ with its *dagger read*, undaggered or daggered. To a permutation $\pi \in S_m$ associate

$$\Phi_\pi(U,V) \;=\; \prod_{\text{cycles } (i_1\, i_2 \cdots i_k) \text{ of } \pi} \mathrm{tr}\big(M_{i_1} M_{i_2} \cdots M_{i_k}\big),$$

M_i reading v_i at an undaggered factor and $v_i^\dagger$ at a daggered. States are the Φ_π's combinations at polynomial-pair coefficients in the residue (Lemma 23). The states' product is the sites' concatenation: two generators multiply at the joined factor list, the second wiring's places shifted by the first list's count, the joined wiring the two blocks, and the coefficients multiply, the trace polynomials' own product; the two block orders read one word multiset, the blocks' exchange a letter-preserving relabeling of the joined list with the joined wiring composed at it, the product's identities one value at either order (Remark 2). Equivalently $\pi(i) = j$ records that the row index of factor i is contracted against the column index of factor j. This is the trace-polynomial reduction of the walled Brauer category: the Schur–Weyl dual presentation, in which the group enters only as the loop value d_f of the calculus.

Remark 2 (The word index, and the presentation kernel)**.** The states are indexed gauge-free by the multisets of oriented cyclic words in the four letters: Φ_π reads only its cycles' words, each up to rotation, so the positions of repeated factors are a gauge mode of the $(\pi, \text{positions})$ spelling, and a state's word coefficient is the class sum of its π-coefficients, one finite projection at every residue. Identities of character polynomials close on the word index outright, coefficient comparisons per word multiset before any pairing, holding at every residue; the Gram below enters only at the state reads. The word index's one further datum is the *kernel*: the map from word monomials to functions has one, the evaluation identity $\sum_j U_{ij}\bar{U}_{kj} = \delta_{ik}$ (Construction 8) imposing polynomial-pair relations among the Φ_π within a factor list. All function-level linear algebra is therefore performed through the pairing's Gram (§11); the null space of that Gram is the presentation kernel, computed once per factor list. Determinant (ε-tensor) relations are invisible at the symbolic residue; this is precisely the statement that the calculus computes the generic theory, and it is why $r = 1$ is its own window read while $r \geq 2$ reads generically.

10 The electric operators as group-algebra insertions

Proposition 9 (Plaquette Laplacian)**.** *The factor list's W-factors split by their dagger read (Construction 11), $W \in \{U, V\}$, and $c_f = C_2(f) = [\,r(r+2) : 2d_f\,]$ is the trace form's own read, the section's displayed operators' scale. On states of Construction 11 the insertion identity reads d_f-cleared, each sum the fold over its stated family of W-factors and the single-index sum over them all:*

$$d_f\,\Delta_W\,\pi \;+\; \sum_{\substack{i<j\\ \text{undaggered}}} \pi \;+\; \sum_{\substack{i<j\\ \text{daggered}}} \pi \;+\; d_f \sum_{\substack{i\ \text{undaggered}\\ j\ \text{daggered}}} \mathrm{C}_{rc}(i,j)\,\pi$$

$$=\; d_f\,c_f \sum_i \pi \;+\; \sum_{\substack{i\ \text{undaggered}\\ j\ \text{daggered}}} \pi \;+\; d_f \sum_{\substack{i<j\\ \text{undaggered}}} \pi \circ (i\,j) \;+\; d_f \sum_{\substack{i<j\\ \text{daggered}}} (i\,j) \circ \pi\;,$$

$$\mathrm{C}_{rc}(i,j)\,\pi = \begin{cases} d_f\,\pi, & \pi(i) = j,\\ \pi \circ (i\ k) \quad \textit{at the witness } k \textit{ at } \pi(k) = j, & \textit{else,} \end{cases}$$

the witness k at $\pi(k) = j$ off the permutation's own transpose data, the inverse permutation (Construction 1).

Derivation and verification. Δ_W, the per-variable summand of Definition 10's diagonal E, is read through the coefficient carrier's derivation action (Construction 8; Definition 7's Δ at the W-factors): each generator differentiates each W-factor once by the Leibniz rule, an undaggered factor reading its insertion at the row index and a daggered at the column index with the memberwise swap, the transpose identity's side, so Δ_W folds over the factor pairs, and each distinct pair's generator fold is the Fierz display of Definition 7, d_f-cleared. A within-factor pair reads the factor's own Casimir, the count c_f per factor; a one-sided pair reads the Fierz's transposition member, right multiplication at the undaggered pairs and left at the daggered; a mixed pair reads the memberwise-swapped fold, its transposition term joining the display's left side, the contraction C_{rc} at the wiring through π, and the closed case $\pi(i) = j$ reads the loop value d_f of the calculus (Construction 11). Every case coefficient is the Fierz display's own read; the identity instances: $\Delta_U \operatorname{tr} U = c_f \operatorname{tr} U$; the unit $\mathbf{1}$ is Δ_U's kernel point on the presentation $\mathbf{1} = [\,1 : d_f\,]\ \operatorname{tr}(UU^\dagger)$; $\Delta_U \chi_{\mathrm{adj}} = d_f\, \chi_{\mathrm{adj}}$; and the graded squares,

$$\begin{aligned} d_f\, \Delta_U\, s + 2\, s \;&=\; d_f\, (d_f + 1)\, s \quad \text{at} \quad s = (\operatorname{tr} U)^2 + \operatorname{tr} U^2, \\ d_f\, \Delta_U\, w + (d_f + 2)\, w \;&=\; d_f^2\, w \quad \text{at} \quad w = \big\langle (\operatorname{tr} U)^2 : \operatorname{tr} U^2 \big\rangle \end{aligned}$$

all cleared polynomial identities in the residue. □

Proposition 10 (Shared-link Casimir)**.** *Δ_L is the Casimir of the simultaneous right translation, Definition 7's Δ read through the joint derivation moving both variables at once, each factor of either variable differentiated once (Construction 8). Its insertion identity is Proposition 9's at the sides exchanged, the column index for an undaggered factor and the row for a daggered, all factors of both variables participating, and a factor pair joins the display at its dagger reads: a matched pair, two undaggered or two daggered, reads its transposition term on the display's right side, and a mixed pair on its left, Proposition 9's collection. The identity instances: $\Delta_L \operatorname{tr} U = c_f \operatorname{tr} U$, $\Delta_L \chi_{\mathrm{adj}}(U) = d_f\, \chi_{\mathrm{adj}}(U)$, and the cross-variable graded reads,*

$$\begin{aligned} d_f\, \Delta_L\, s + 2\, s \;&=\; d_f\, (d_f + 1)\, s \quad \textit{at} \quad s = \operatorname{tr} U\ \operatorname{tr} V + \operatorname{tr}(UV), \\ d_f\, \Delta_L\, w + (d_f + 2)\, w \;&=\; d_f^2\, w \quad \textit{at} \quad w = \big\langle\, \operatorname{tr} U\ \operatorname{tr} V : \operatorname{tr}(UV) \big\rangle \end{aligned}$$

with the module energy $E\chi_{\mathrm{adj}} = 4d_f\, \chi_{\mathrm{adj}}$, the count $E_0 = 4$ at the weight. The link translation acts through the site conjugation at g_s, $U \mapsto U\, g_s X g_s^\dagger$ at the site unitary g_s, its dagger read $g_s^\dagger g_s = 1$ the evaluation identity's (Construction 8); on the fiber invariant under that site conjugation the conjugation is absorbed, the read the proposition takes.

11 The pairing in the permutation presentation

Lemma 23 (The generic lift)**.** *A polynomial pair in the residue is a pair of ground polynomials named as Definition 1's constructed pairs are, read at every natural argument through the members' Horner reads and across representatives by the cross-multiplied identities; every entry below is one such pair. (i) A polynomial pair whose reads at infinitely many naturals are of equal members reads equal members at every argument: its first member's located roots sit inside Theorem 15's bound, finitely many. (ii) A matrix of polynomial pairs whose evaluation at infinitely many residues is the Gram of an independent list has its determinant pair off equal members at each such residue, (i) lifting the read, and every unit read is solved by the adjugate column against the determinant (Definition 3's minors), the identity $\mathsf{M}\operatorname{adj}(\mathsf{M}) = \det(\mathsf{M})\,1$ the solved witness's whole verification. (iii) A polynomial pair keeps one side beyond its members' largest located root, the leading-coefficient read at Theorem 15's leading-term bound: the settled side, one decidable read per pair. (iv) A claim whose per-residue reads are the sides and equal-member reads of finitely many polynomial pairs in the residue therefore holds at every residue beyond their computed radii once its symbolic reads hold, and the finitely many residues at or below the radii are each their own window read.*

Proposition 11 (Pairing)**.** *For states A, B on factor lists F_A, F_B, the pairing $\langle A, B\rangle = \mathrm{Eval}\big(\overline{\Phi_A}\,\Phi_B\big)$ of Proposition 5 is computed in the presentation by: (i) the involution flips daggers and transposes permutations, the transpose the inverse permutation (Construction 1); (ii) per variable, with k undaggered and k daggered factors (else the pairing reads equal members), the contraction sum*

$$\sum_{\sigma,\tau\in S_k} \mathrm{Wg}\big(\sigma\tau^{\mathsf{T}}; d_f\big) \prod_r \delta_{i_r l_{\sigma(r)}}\, \delta_{j_r k_{\tau(r)}}, \qquad \mathrm{Wg}: \quad \mathsf{G}\, w = e_{\mathrm{id}}, \quad \mathsf{G}_{\alpha\beta} = d_f^{\#\mathrm{cycles}(\alpha\beta^{\mathsf{T}})};$$

(iii) the δ-wirings close the permutation chains into loops, each loop contributing a factor d_f. The contraction sum reads the wiring's cycle words alone: the Gram's entry at two keys, each composed between one relabeling and its inverse permutation, is the two keys' own entry, the composition's cycle count the place composition's read (Construction 1), and the identity key its own image, so the adjugate column at a composed key against the one determinant reads the key's own value, the solve read across the relabeling. Every output of the computation reads at one of two tags: symbolic, one polynomial pair in the residue at its clearance radius, the factor count, read at every residue beyond the radius (Lemma 23); or direct, the residue's own contraction read through its Gram with its presentation kernel (Remark 2), at each residue at or below the radius. A symbolic read enters a residue's comparison beyond its radius alone, so a window at or below a radius reads directly, the kernel's relations read where they occupy its factor lists. At a B, C or D member the presentation runs at the defining factors' pair partitions with the member Gram (Lemma 27): a daggered factor reads its slots exchanged at the form's twist, its entry the fold of the dual pair's coefficient, the factor's own entry and the form's, $\sum_{a,b} c_{xa}\, U_{ba}\, b_{by}$ at the dual pair c and the form b, the defining table's two data (Construction 4; Lemma 27), and per variable the contraction sum runs over the partition pairs,

$$\sum_{P,Q} \mathrm{Wg}\,(P,Q) \prod_{(u,v)\in P} c_{x_u x_v} \prod_{(u,v)\in Q} c_{y_u y_v} \prod_j w_{y_j}\,, \qquad \mathrm{Wg}: \quad \mathsf{G}^{\mathrm{ser}}\, \mathrm{Wg} \;=\; 1\,,$$

the row slots at the dual pair's coefficients alone and the column slots at the pairing's weights w beside them (a matrix entry is a coordinate, so the pairing's weight enters at the paired member alone), $\mathsf{G}^{\mathrm{ser}}$ the member Gram at the loops' fold with the solve the adjugate against the determinant (Lemma 27; Lemma 23(ii)); the wirings close the chains into loops as the A-series' display's do, each loop one value: the dimension $r+3$ at B and D, the twist pair's weighted product reading the identity at the dual pair against the form, and the doubled rank $2r$ at C at the loop's parity, half the loop's edge count joined to the count of dual-pair edges read at their displayed order and of form edges read against theirs, each edge the form's multiple with the form's square reading the unit's balance partner; and an odd factor count's symbolic read equal members, the even powers the span's own carrier (Lemma 27); the two tags at the rank clearance, the factor count with the D series at its successor; a fixed member's outputs read direct outright, the finite arithmetic of its displayed member data (Construction 4).

Derivation and verification. Per variable, a product of k coefficients against k involution images is a monomial coefficient of the mixed power $V^{\otimes k} \otimes (V^{\otimes k})^{\vee}$, and Eval reads its unit component (Construction 8): the family splits orthogonally over the blocks (Lemma 1), so that component is the (a, b) entry of the invariant projection, a spanning invariant list's adjugate solve at its Gram, the membership residual's own read (Lemma 3). At $k \le d_f$ the invariants are the permutation span (Lemma 13), the projection is the span's Gram projection, and its entries are the displayed contraction sum at the Gram's solved column, Wg the adjugate column against the determinant with $\mathsf{G}\,\mathrm{Wg} = e_{\mathrm{id}}$ its whole verification: the Gram's entries are residue naturals reading $\sigma\tau^{\mathsf{T}}$ alone, the transpose the inverse permutation (Construction 1), the solve Lemma 23(ii)'s at the per-residue Grams of the independent permutation list, and the δ-wirings are the permutations' monomial entries; a product at unequal daggered and undaggered counts reads equal members there, its two degrees differing by less than a full column beyond the factor count. A scheme output at a fixed factor list is one polynomial pair in the residue agreeing with the per-residue read at every $k \le d_f$, so the identities hold as cleared polynomials in the residue (Lemma 23(i)), the generic theory of Remark 2. *The direct tag's read.* At a residue at or below the radius the read runs at the residue's own carrier. A variable's cycle word reduces at the evaluation identity (Construction 8): the daggered factors withdraw against the undaggered along the cycle, the word reads the net power, and a cycle at equal counts reads the residue's own count, a scalar factor; a state's variable datum is its *winding pair*, the net powers' occupancy family at the undaggered side against the daggered side's, the read at every wiring whose cycles each sit at one variable. The power word expands over the alternants at the residue's count: the product $p_\mu\, a_u$ is graded and regroups over the strictly decreasing exponents (Theorem 1's conclusion walk at the residue's variables), the *character read* $\chi^\lambda(\mu)$ its extraction,

$$p_\mu\, a_u \;=\; \sum_\lambda \chi^\lambda(\mu)\, a_{\lambda+u}\,,$$

one balance pair per shape of the degree at rows within the residue, so $p_\mu = \sum_\lambda \chi^\lambda(\mu)\,\mathrm{ch}_\lambda$ over those shapes, the division at a_u exact, $a_u = e\,V$ a product of monic factors (Theorem 1's assembly). The characters are orthonormal for the evaluation (Proposition 8), so two states pair at the residue at the *character fold*: per variable one shape read at both winding sides, $\sum_\lambda \chi^\lambda(\alpha)\,\chi^\lambda(\beta)$ at the matched totals, the off-diagonal shape pairs withdrawing at the orthonormality, the scalar factors' count multiplied through, and an unmatched total reading equal members. The direct member read is the at-residue Gram's own row read: at a factor list whose factors sit at one variable, a combination at coefficients occupied at the residue is the presentation kernel's member exactly where its character-fold pairing reads the sum's unit against every wiring of the list: the generators span the carrier, so rows at that unit read the self-pairing's fold at it and the pairing's definiteness closes the member, while a member's rows are the null function's own (Remark 2; Construction 8); a coefficient whose second member reaches the unit at the residue sits off the read, the occupancy its own conjunct, and a wiring at a crossing cycle reads at the factor list's own Gram, Remark 2's computation. The scheme's identity instances read at the fusion reads of Proposition 8: $\langle \operatorname{tr} U, \operatorname{tr} U\rangle = 1$, $\langle \mathbf{1}, \mathbf{1}\rangle = 1$, $\langle \chi_{\mathrm{adj}}, \chi_{\mathrm{adj}}\rangle = 1$, $\langle \mathbf{1}, \chi_{\mathrm{adj}}\rangle$ of equal members, $\mathrm{Eval}\big(|\operatorname{tr} U|^4\big) = 2$, and $\langle (\operatorname{tr} U)^2, \operatorname{tr} U^2\rangle$ of equal members; and the closed form

$$\mathrm{Wg}(\mathrm{id}; S_4) = \big[\,\langle d_f^4 + 6 : 8d_f^2\rangle \;:\; d_f^2\, g_1 g_2 g_3\,\big] \qquad \text{at the gaps } k^2 + g_k = d_f^2,$$

the gaps' existence the four-factor read's own validity ($k \le d_f$ at the independent span), is reproduced by the adjugate solve. All are cleared polynomial identities in the residue. □

12 Resolvent, module deflation, and the presentation kernel

Construction 12 (The Feshbach resolvent at polynomial pairs)**.** Given a state y on factor list F:

1. *The deflated side*: for each module presentation m in F (built as $\chi_{\mathrm{adj}}(W)$ times paired $[1\!:\!d_f]$-contractions of the other factors), the deflation y' at the join $\langle m, m\rangle\, y' + \langle m, y\rangle\, m = \langle m, m\rangle\, y$, its m-pairing reading equal members, the fold over the presentations in list order, each join at the prior deflation.

2. *The support*: the union of the π-support of y' and of the presentation kernel (Remark 2), closed under the E-insertions (Proposition 9; Proposition 10), the closure after the union (the insertions map the joined support into it).
3. *The solve*: $S\,x + k = y'$ at a presentation-kernel member k, the member's read the symbolic tag's with a window at or below the clearance radius reading its own kernel's members (Proposition 11; Remark 2), and S the site datum at the module level, the section's pair $(E\!:\!4d_f)$'s, one exact adjugate solve over the polynomial pairs, null-space augmented with the member the augmentation's read (Lemma 23(ii)), then
4. *The solution's deflation*: the solution's component off the module space, the join of step 1: E is diagonal in $\hat{c}_2$, so at the generic residue every non-module gap reads unequal members as a polynomial pair and the pair $(E\!:\!4d_f)$'s site datum has the module space as its kernel. The result is the Feshbach Q-resolvent applied to y.

The solve's error is the sum's unit *as a function*: the residual w's self-pairing reads equal members as a polynomial pair.

13 The invariant problem: the X-sector block data

The problem statement's invariant object is the walk on the category generated by the self-dual object X, and it restricts the computation to the X-generated sector of the existing calculus, with $M = \sum_i \chi_X(U_i)$.

Remark 3 (The certificate target)**.** χ_X conserves the per-cell box charge, so within the X-theory the restriction to the X-sector is a superselection fact; the sector's two data are Construction 13's, its block reads and structural identities the certificate surface (Theorem 10; Theorem 23; Theorem 24), and the bridge to the fundamental theory is Theorem 26's $2d_f$.

Lemma 24 (The adjoint square's channels)**.** *The blocks of $\theta \otimes \theta$ are, at every residue, the labels*

$$\mathbf{1}, \qquad \theta, \qquad 2\theta = (4, 2^{\,a}),$$
$$2\omega_1{+}\omega_2^* = (3, 1^{\,b}), \qquad \omega_2{+}2\omega_1^* = (3, 3, 2^{\,b}), \qquad \omega_2{+}\omega_2^* = (2, 2, 1^{\,e}),$$

named by their weights over the fundamentals and the complements, ω_j and ω_i^* (Lemma 9; Lemma 8(ii)), at the run counts a, b, e at $a + 2 = d_f$, $b + 3 = d_f$ and $e + 4 = d_f$, at multiplicities 1, c_1, 1, 1, 1, 1, with $c_1 = 2$ at $r \geq 2$ and 1 at $r = 1$; the outer pair are duals, the complement arithmetic of Construction 5 the manifest swap of the names at $\omega_j^* = \omega_{j'}$, every index pair joining to $r + 1$, and the other rows are self-dual. The list runs at every residue with each row entering at its count $m\,d$: the closed forms read the drops themselves, the dual pair's d with the factor g at $g + 1 = r$, the residue's gap from one, and the last row's the root $r = 2$, while at $r = 1$ the row list is the alternant read's, the run $4, \ldots, d_f{+}1$ of y occupied exactly at $r \geq 2$. The unit channel enters at count one on the unit line, the Casimir's kernel point, and the nonunit channels close in the residue:

label	m	d	$\hat{c}_2$
θ	c_1	$r\,(r+2)$	1
2θ	1	$[\,(r+1)^2 r\,(r+4) : 4\,]$	$[\,2r+4 : r+1\,]$
$2\omega_1{+}\omega_2^*,\ \omega_2{+}2\omega_1^*$	1	$[\,g\,r\,(r+2)(r+3) : 4\,]_{g+1=r}$	2
$\omega_2{+}\omega_2^*$	1	$[\,(r+1)^2(r+2)\,w : 4\,]_{w+2=r}$	$[\,2r : r+1\,]$

with the two closure identities

$$\sum_R m_R d_R = r^2(r+2)^2, \qquad \sum_R m_R d_R\,\hat{c}_2(R) = 2\,r^2(r+2)^2,$$

the displayed reads of the proof, the sums over all blocks with the unit channel's count one among them and its Casimir term's members equal. The module's distances close in the residue as well: against $\hat{c}_2(\theta)=1$, the gaps of the Casimir column read 1 at the unit channel and at the dual pair, $[\,g : r+1\,]$ at $\omega_2{+}\omega_2^*$ over its gap witness $g+1=r$, and $[\,r+3 : r+1\,]$ at 2θ, at the cross-multiplied witnesses $2r=(r+1)+g$ and $2r+4=(r+1)+(r+3)$: each gap is positive at its row's residues, one symbolic read at every one of them.

Proof. Corollary 2 at $a=b=\theta$: a permutation contributes where the tie $\nu_\sigma+\sigma(\theta+u)=c+u$ meets a shifted content of θ, u the unit set's sorted display, (1^{d_f}) at multiplicity r or its letter-pair moves at one each (Lemma 16), so, writing $y:=\theta+u+(1^{d_f})$, of value multiset $\{2,4,5,\dots,d_f{+}1,\,d_f{+}3\}$, a label contributes only where the strictly decreasing $c+u$ rearranges y or y with one entry raised and one lowered by 1, the result's entries distinct. Equality is $c=\theta$ at matched degree, the count 2 at $2+g=r$ against the run's adjacent value pairs, g their count: the identity's read r at the unit-monomial content on the even side against one odd transposition per adjacent value pair, the pairs sitting in the run $4,\dots,d_f{+}1$. The modifications with distinct entries are exactly five: $2\to3$ with $d_f{+}3\to d_f{+}2$ (the unit label at matched degree); $2\to1$ with $d_f{+}3\to d_f{+}4$ (2θ); $4\to3$ with $d_f{+}3\to d_f{+}4$ $(2\omega_1{+}\omega_2^*)$; $2\to1$ with $d_f{+}1\to d_f{+}2$ $(\omega_2{+}2\omega_1^*)$; $4\to3$ with $d_f{+}1\to d_f{+}2$ $(\omega_2{+}\omega_2^*)$; every other raise or lowering collides with an occupied value, the adjacent swaps returning the equality case. Each modification preserves the decreasing order, so the aligning permutation is the identity and each count is one, on the even side. At $r=2$ the run $4,\dots,d_f{+}1$ is the single value 4, so $\omega_2{+}\omega_2^*$'s two modified entries coincide: the row enters at $r\geq3$; the dual pair's modifications each move one value of the run $4,\dots,d_f{+}1$, occupied at $r\geq2$ alone, so the pair enters at $r\geq2$, $\omega_2{+}\omega_2^*$ at $r\geq3$ as above, while equality reads $r=1$.

The dimensions are Corollary 1's telescopes. At $2\theta=(4,2^{\,a})$ the factors from either end read one pair $[\,g+2 : g\,]$ over the gaps $1\leq g<r$ and telescope to $[\,(r+1)r : 2\,]$ each, the pair $(1,d_f)$ reads $[\,r+4 : r\,]$, and the middle pairs read 1: the cofactor at $4\,d_{2\theta}=(r+1)^2r\,(r+4)$. At $2\omega_1{+}\omega_{r-1}=(3,1^{\,b})$, at the gap g at $g+1=r$, the five collected reads are $[\,rg:2\,]$, $[\,r+2:g\,]$, $[\,r+3:r\,]$, g and $[\,r:2\,]$: the cofactor at $4\,d=g\,r\,(r+2)(r+3)$, the pair's second count equal, the dual's contents the balance partners (Lemma 8). At $\omega_2{+}\omega_{r-1}=(2,2,1^{\,e})$, at the gap w at $w+2=r$, the six collected reads are $[\,r-1:2\,]$, $[\,(r+1)(r+2) : r\,(r-1)\,]$, w, $[\,r\,(r+1) : w\,(r-1)\,]$, w and $[\,r-1:2\,]$: the cofactor at $4\,d=(r+1)^2(r+2)\,w$.

The Casimirs are Lemma 7's bracket at the reduced shapes: $\hat{c}_2(2\theta)=[\,2r+4 : r+1\,]$, the $m=2$ read of the tower law (Proposition 12); at $2\omega_1{+}\omega_{r-1}$, $k=d_f$, $q=r+7$, and the top row's weight $3\,\langle r+2:2\rangle$ with the run's collecting reads $\langle\lambda,2\rho\rangle=\langle 4r:2\rangle$, so the cleared join $d_f+Q+2=(r+7)+4r$ reads $Q=4\,d_f$: $\hat{c}_2=2$ at both members of the dual pair; at $\omega_2{+}\omega_{r-1}$, $k=d_f$, $q=r+5$, and the top rows' weights $2\,\langle r+2\,{:}\,2\rangle+2\,\langle r+2\,{:}\,4\rangle$ with the run's summing to equal members read $\langle\lambda,2\rho\rangle=\langle 4r\,{:}\,4\rangle$, so the cleared join $d_f+Q+4=(r+5)+4r$ reads $Q=4r$: $\hat{c}_2(\omega_2{+}\omega_{r-1})=[\,2r : r+1\,]$.

The closure identities. At $r\geq2$, with $v:=r\,(r+2)$ and $x:=d_f^2$ at the join $v+1=x$ (Lemma 16): the self-dual outer rows' counts Σ_o and the dual pair's Σ_d read the cross-added identities $2\Sigma_o+3x=x^2$ and $2\Sigma_d+5x=x^2+4$, so, with the unit's one and the θ row's $2\,v$,

$$\sum_R m_R d_R + 2x \;=\; x^2+1;$$

and the self-dual outer rows' weighted reads collect at their common factor to $x\,v$, the weights' tie $(r+4)+g=2\,d_f$ at the gap g at $g+2=r$ closing it, with the dual pair's weighted read twice its count, so

$$\sum_R m_R d_R\,\hat{c}_2(R)+4x \;=\; 2x^2+2,$$

the displayed sums v^2 and $2v^2$, the statement's two closes. At $r=1$ the surviving rows read $1+3+5=9=r^2(r+2)^2$ and $3\cdot 1+5\cdot 3=18=2\,r^2(r+2)^2$ directly. □

Lemma 25 (The window data are residue-stable on banded families)**.** *A banded family is a pair of weight words at the fundamentals' two ends: the top word P, a finite occupancy family at $\omega_1,\omega_2,\dots$ indexed from the first fundamental, and the bottom word Q, one at the complement fundamentals $\omega_1^*,\omega_2^*,\dots$ indexed from the last (Lemma 8(ii): ω_i^*'s column at the gap ℓ at $i+\ell=d_f$), with reaches a and b, the words' last keys; the member at the residue r is*

$$\lambda \;=\; \sum_{j\le a} P_j\,\omega_j \;+\; \sum_{i\le b} Q_i\,\omega_i^*\;,$$

one label per cleared residue, the reach clearance the residue at or beyond $a+b+2$, at least two keys off the words between the reaches, and the member's rows are the weight family's tail sums, $\lambda_i=\sum_{j\ge i}c_j$ at the coordinates c, each row one fold over the member keys at or beyond it. The vacuum is the unit pair, a member at every residue outright, θ is one box on each word (P at ω_1, Q at ω_1^*, the fundamental with its complement), and Lemma 24's whole list is: the families are the X-sector's labels, and the fusion-closed interior at any cutoff and depth is one word-pair list at every cleared residue.

(i) The Casimir and the dimension are single polynomial-pair reads. The words read the rows: the height m, the bottom word's coordinate total, is the row across the run between the reaches, a top row adds its tail sum, $\lambda_i=m+p_i$ at $p_i=\sum_{j\ge i}P_j$ for $i\le a$, and the rows below the run are the bottom word's prefix sums, $\sigma_t=\sum_{i\le t}Q_i$ the row at the join $\ell+t=d_f$. So $q(\lambda)$ splits into the top rows' squares, the run term $u\,m^2$ at the run count u, $a+b+u=r+1$, and the bottom rows' squares, each row a word read; $\langle\lambda,2\rho\rangle=\sum_i\lambda_i\,\langle r+2:2i\rangle$, the row weights balance pairs, splits into word reads and the run's arithmetic sum; and the degree is $k=\sum_j j\,P_j+\sum_i \ell_i\,Q_i$ at the complement columns ℓ_i, $i+\ell_i=d_f$, linear in the residue, so the degree's square against the count, $[\,k^2:d_f\,]$, is one pair term. Hence $\hat{c}_2(\lambda)$ is one polynomial pair in the residue per family (Lemma 23), its second member a product of integer shifts of the residue and its powers. The dimension is one read as well: in Corollary 1's product the run–run factors are 1, and a word-against-run factor telescopes over the position gaps g, at the joins $i+h=a$ and $i+b+g_2=d_f$,

$$\prod_{h<g\le g_2}\big[\,p_i+g\;:\;g\,\big] \;=\; \Big[\,\tbinom{p_i+g_2}{p_i}\;:\;\tbinom{p_i+h}{p_i}\,\Big]\;,$$

the top row's case at its tail sum p_i, with the bottom's its reflection, the run against σ_t at the row gap τ_{t+1}, $\tau_t=\sum_{i\ge t}Q_i$ the bottom tail sum at the join $\sigma_t+\tau_{t+1}=m$, a pair of two integer binomials polynomial in the residue each; a factor within one word is residue-free, its rows' and its positions' gaps the word's own data; and a cross factor, the top row i against the bottom row σ_t, is the pair $[\,p_i+\tau_{t+1}+g\;:\;g\,]$ at the position gap's join $i+t+g=d_f$, one factor per key pair with both members linear in the residue: d_λ is one polynomial pair in the residue per family.

(ii) The fusion rows are word reads, one row at every residue. Corollary 2 at the adjoint runs on $x:=\lambda+u$, the beta-set's sorted display, on Lemma 24's alignment: the θ-contents are the letter-pair moves at multiplicity one and the unit-monomial content at r (Lemma 16), so the diagonal reads

the unit-monomial multiplicity against one odd transposition per unit gap $x_i - x_{i+1} = 1$, and a gap reads its weight key, $x_i - x_{i+1} = c_i + 1$ at the coordinate c_i: the diagonal count is the occupied coordinates' count over the gap keys, at every shape of the width, and at a word pair one at every key off the words, beyond one at every member key. Hence, at every family off the vacuum,

$$N^{\lambda}_{\theta\lambda} \;=\; \#\operatorname{supp} P + \#\operatorname{supp} Q \;,$$

the member-key count, residue-free: 2 at every tower label $m\theta$ (the c_1 of Proposition 13), the vacuum's diagonal the defect, the Kronecker delta of Proposition 13. Off the diagonal a contribution raises one x-coordinate and lowers another by 1 with the entries kept distinct: at the run's clear keys an interior raise or lower collides with its neighbor, so the surviving moves sit within one key of the words, each target again a word pair at each reach grown by at most one, and each count an at-most-one read settled by the identity alignment of Lemma 24: a kept move's count is one exactly, the identity's own tie at the moved display with every further permutation's tie at the raised display alone, at every shape of the width, and the row is a word read, one row at every cleared residue. A depth-D fusion closure of the vacuum therefore has as members word pairs at reaches summing to at most $2D$, and its whole data list, the labels, multiplicities, Casimirs, dimensions and the network weights $w_k = [\,m_k d_k : r^2(r+2)^2\,]$, is one list of polynomial-pair reads at every residue clearing the reach sum by two, each family with its reach clearance there, the recoupling entries included through Proposition 11's generic computation, one polynomial pair per entry agreeing with the per-residue read at every residue beyond the factor count.

(iii) The residue is a cell coordinate. At fixed content cutoff, region and fiber, the window pencil's entries are the reads of (i)–(ii), so they are one list of polynomial-pair data in (t,τ) and the residue at every residue clearing the reach sum by two; the cutoff comparisons $\hat{c}_2 \le \Lambda_C$ selecting the interior are eventual-sign reads (Lemma 23(iii)); and the second members, integer shifts of the residue and its powers, keep one side there, so clearing them is a positive rescaling fixing every count (Lemma 30). The fiber pencils' entries then sit in $\mathbb{P}[t,\tau,r]$ and every count of the gap chain is Lemma 31's cell function with r a coordinate: a divisor's roots in r are located outright (Theorem 15, the leading-term bound the coarser certificate), and each root reads against the compared families' reach clearances: a comparison's residues are those clearing its families' reaches, so at every root below them the divisor keeps its leading side at every one of its residues and one symbolic count is every residue's at the cleared families; a family below its clearance reads its finitely many residues' data directly, the calculus's displays at the residue's label, one evaluation each; and Lemma 24's $r = 1, 2$ degenerations are each their own window read.

Lemma 26 (The series' window data are rank-stable)**.** *At a B, C or D member (Construction 4), a leading family is a finite weight word $W_1 \ge \cdots \ge W_a$ at positive reads on the leading keys, its member at the rank ℓ the word's own occupancy family over those keys, and its reach clearance the rank at or beyond $a+2$.*

(i) The Casimir and the dimension are single polynomial-pair reads in the rank. $q(\lambda)$ is the word's square fold; $\langle \lambda, 2\rho \rangle = \sum_i W_i\,(2\rho)_i$ folds the word against the displayed root-fold key reads, each linear in the rank; so $\hat{c}_2$ is one polynomial pair per family. The dimension is the member gap product (Corollary 1): the tail factors read one, the word-against-tail factors telescope over the position gaps to binomial pairs in the rank; the word-against-word difference factors are rank-free, and the word sum, short and long factors are single linear pairs in the rank, finitely many.

(ii) The adjoint fusion rows are word reads, one row at every cleared rank. The row runs at the alignment, Corollary 2's member ties $\nu_w + w(\lambda+\rho) = c + \rho$ over the member's Weyl list at the θ contents (Proposition 13); the shifted key $\lambda+\rho$ reads the word's values against the ρ run, consecutive values at the displayed gaps, every entry distinct, the junction gap the last value's successor, and a

word gap at the unit exactly at a repeated pair. An occupied tie's element sits below the shifted key at a natural simple fold, the image list's support read, the tie naming the fold as the two contents' gap; the kept square reads the fold against the key, twice the pairing at the fold's own square, with the key strictly dominant, every coroot pair at the member's successor; the tie collects the square against the target's own shifted key, twice the fold's pairing there the two contents' square gap, so the fold's length-weighted height sits at or below that gap at the shifted target's coroot pairs, each at or beyond one by the target's dominance, the contents' displayed lengths capping the gap at two; a simple's raise beyond one sits off the list outright, the raised content's square the raise's multiple of the letter's scaled length against the doubled ρ-dot at the raise once, the positive squares' cap at the doubled dots (Proposition 13) refusing every raise beyond one where the height cap alone admits the raise two at the length-one keys; and the square equation with the list's membership refuses every occupied solution of the finite residue beyond the one-letter reads: the surviving elements are the identity and the simple letters, each letter's fold the key's coroot multiple of its own root and the letters' images distinct at the strict dominance, one list member each. The letters read at the run: an adjacent exchange at its unit gap, and the last letter the end device, B's flip at the half value's double (the short root's read, one), C's at the long root (two), and D's last-pair exchange with both partners at the joined values (the sum root's read, one). Each count is therefore the letter fold: the identity's read at the target's own content joined against one signed read per simple letter at the raised content, the raise the key's coroot multiple of the letter's root. The evaluations close the row. The diagonal's odd fold reads the letters at the key's unit coroot pairs, the vacant coroot support, so the diagonal is the occupied support, the word's distinct-value count (Proposition 13, the tower's display its instance). An off-diagonal target is dominant $\lambda + \nu$ at a content ν, a leading word at reach grown by at most two with the sum pair at the first two tail keys the two-key growth; the coroot presentation clears the grown reach, every further key reading its displayed pair at coordinates past the reach, at D off the last key alone, whose pair reads the two tail coordinates' sum (Construction 4) and so reads the grown reach's own coordinate, occupied at the whole-growth target at the rank one beyond the boundary and vacant at every rank clearing the reach by four. Its count is an at-most-one read: one at the identity's tie alone, or the two ties evening at a named pairing, B's single-box moves against the end flip, C's tail-pair sum against the long at the exchange, and a repeated pair's moves against their own exchange; at D the difference move at the last pair, its target's last coordinate one's balance partner, is dominant at the boundary rank alone, the reach clearance's least, its count one. Every fold entry is a word read, so the row is one list at every rank clearing the family's reach, D's boundary rank joined by its last-pair member alone, and a below-clearance rank reads its finitely many labels directly, the member displays' own evaluations.

(iii) The rank is a cell coordinate. At fixed content cutoff, region and fiber, a series member's pencil entries are the reads of (i) and (ii) with the recoupling entries Proposition 11's, one list of polynomial-pair data in (t, τ) and the rank at every rank clearing the reach sums by two; the cutoff comparisons are eventual-sign reads and the divisors' rank roots read against the clearances, one symbolic count at every cleared rank with the below-clearance ranks direct (Lemma 23; Theorem 15): Lemma 25's three reads at the series' own coordinates, the A-series' the first instance.

Lemma 27 (The member pairing span)**.** *At a B, C or D member with its defining table (Construction 4), the invariants of the defining power $V^{\otimes 2k}$ at the rank at or beyond k, the D series at or beyond $k{+}1$ (the doubled-rank alternating invariant sits off the span at the tie $2\ell = 2k$), are the pair-partition span: one vector per pair partition of the $2k$ places, the partition's fold of the form's dual pairs, at the count $(2k{-}1)!!$.*

Proof. The invariant read. An invariant of the power is a vector at the unit content with every raising image the sum's unit: a unit-content top's block is the unit label's line, and a top at a further content sits at its own label (Lemma 2; Lemma 5), so the invariants are the unit-content summand's raising kernel, the two reads' conjunction. *The defining rows.* V's contents are the signed keys with the unit content at B_ℓ, and its fusion rows at a block are one box up or one box down at the cleared ranks: the alignment's form read (Proposition 13's tie analysis at the defining contents) reads the B_ℓ stay term against its correction, the unit content's one against the last letter's read at $(\lambda+\rho)(e_i^\vee) = 2\lambda_i + 2g + 1$ over the join $i + g = \ell$, one exactly at the last key off the support: equal members at every label whose support clears the last key, and the moved ties at the box targets alone; at C_ℓ and D_ℓ the signed keys sit off the root list and the unit content off the table, the rows the box moves outright. *The count.* Iterating the rows, the unit-label count of $V^{\otimes 2k}$ is the closed path fold $\langle \mathbf{1}, (U{+}D)^{2k}\,\mathbf{1}\rangle$ over the shapes at most k rows deep (the clearance; a B_ℓ stay reads at a full-depth shape, its boxes the step count's read, and the return closes beyond the walk's $2k$ steps, so the closed folds read the box moves alone), with $DU = UD + \mathrm{id}$ (Lemma 12's displayed argument at the member's shapes) and the commutator telescope $X^{m-1}U = U\,X^{m-1} + (m{-}1)\,X^{m-2}$ at $X := U{+}D$, so the fold obeys $m_m = (m{-}1)\,m_{m-2}$ from $m_0 = 1$: the double factorial, the matchings' count, and the unit-label count is the invariant count (Definition 5; Lemma 5 at table spaces). *Independence, and the Gram.* A pair partition's vector reads a monomial the further partitions miss at the rank clearing k, one key pair per part at distinct keys, so the $(2k{-}1)!!$ vectors are independent and span. The Gram is the loops' fold: two partitions' union closes into loops, each loop contracting the form's dual pairs around its cycle; at B_ℓ and D_ℓ a loop reads the dimension, $r+3$; at C_ℓ a loop reads the doubled rank at its *crossing parity*, the loop's pair count at either side with its against-order crossings along one traversal, the antisymmetric form's memberwise swap per crossing: the entry is the loops' product, $(2r)$ per loop with the parities' fold its side, one displayed traversal read per entry. $\square$

Construction 13 (The X-sector is adjoint fusion)**.** $\chi_X = \chi_{\mathrm{adj}}$ at $\chi_X + 1 = \chi_f\chi_{\bar f} = |\mathrm{tr}\,U|^2$ (Lemma 16), so the X-sector's calculus is tensoring by the adjoint, and its data are exactly two: the fusion matrix N_{adj} and the weight-free Casimir $\hat c_2$. The product expands as $\chi_{\mathrm{adj}}\chi_\lambda = \sum_\mu N^{\mu}_{\mathrm{adj},\lambda}\chi_\mu$ with unique coefficients, and the characters are orthonormal for Eval (Proposition 8), so M acts on the label index by N_{adj} itself, symmetric because $m\{a,b,c\} = N^{\bar c}_{ab}$ is symmetric and the adjoint is self-dual, and E is diagonal there with entries $4\hat c_2(\lambda)$. Every anchor of the sector is a read of that pair: $c_1 = \langle \mathrm{adj}|\chi_X|\mathrm{adj}\rangle$ is the multiplicity of the adjoint in $\mathrm{adj}\otimes\mathrm{adj}$, the $m{=}1$ alternant read of Proposition 13, 2 at $r \geq 2$ and 1 at $r = 1$, and the member-key count of θ's word pair, one box on each word, at the residues clearing the pair's reach (Lemma 25(ii)); $\langle q, M\,\mathrm{vac}\rangle$ is the multiplicity of the vacuum, 1; and $E_0 = 4$ (Proposition 7).

The channel list is Lemma 24's, one word-pair list at every residue with each channel entering at its count (Lemma 25), the two root-drops and the $r = 1$ row list that lemma's own reads, so every quantity below is a sum over the channels at their counts.

Theorem 10 (X-sector block data)**.** *With $q_i = \chi_X(U_i)$, $E_0 = 4$: $\langle q_1, Mq_1\rangle$ has the integer coefficients $\langle q_1, M\,\mathrm{vac}\rangle = 1$ and $c_1 = \langle q_1, Xq_1\rangle$, the fusion multiplicity of Construction 13 (2 at $r \geq 2$, 1 at $r = 1$); the vacuum Q-dressing reads equal members as a polynomial pair, $M\,\mathrm{vac}$ lying in the module space Q projects off (Construction 10); the module is isolated, the channel distances the Casimir gaps of Lemma 24's list. Structural identities on the constituent data: $\sum_R m_R \dim_R = r^2(r+2)^2$, the block count of the square (Lemma 5(ii) over Lemma 24's list), and $\sum_R m_R \dim_R \hat c_2(R) = 2\,r^2(r+2)^2$, cleared polynomial identities in the residue, the channel table's displayed closes (Lemma 24); the second is the uniform drift of Theorem 24 at $R = \mathrm{adj}$, where $d_{\mathrm{adj}} = r\,(r+2)$ and $\hat c_2(\mathrm{adj}) = 1$ make $d_R d_{\mathrm{adj}}(\hat c_2(R)+1)$ read $2\,r^2(r+2)^2$.*

14 The charged anchor

Proposition 12 (Charged anchor for the X-chain)**.** *Define charge as Cartan-weight density along the highest root θ. The charged tower is the $m\theta$-string, the Cartan powers of θ (Construction 5), and the residue is the pairing's own read at the highest root: the row-weight fold reads $\langle\theta, 2\rho\rangle = 2\,r$ against the pair-square fold at $2\,d_f$, the declared scalar's own value, and the tower reads*

$$\hat{c}_2(m\theta) = \big[\, m\,(m+r) : r+1 \,\big] \qquad \textit{at the derived residue} \quad r,$$

an $l(l+1)$ in shape; at $r = 1$ it is $[\,m^2 + m : 2\,]$, the integer-spin tower's weight-free read. At a member's weight table the residue is Construction 4's one fold, $\langle\theta, 2\rho\rangle = 2\,r$ at the form's read $\langle\theta,\theta\rangle = 2$, and the tower reads the member's form at its two degrees, the form additive in each argument (Construction 3), $C_2(m\theta) = m^2\,\langle\theta,\theta\rangle + m\,\langle\theta, 2\rho\rangle = 2\,m\,(m+r)$ against $C_2(\theta) = 2\,(r+1)$: the display at the member's derived residue. The linear coefficient r is fusion strictness: the abelian tower reads $C_2 = m^2$ and this one $m\,(m+r)$.

Proof. The join's constructed witness (Definition 6) reads a label at two folds, $d_f\,Q(\lambda) = \sum_{p<q}\langle\lambda_p : \lambda_q\rangle^2 + d_f\,\langle\lambda, 2\rho\rangle$, each homogeneous in the rows, degree two and degree one ($\langle mu : mv\rangle = m\,\langle u : v\rangle$, Definition 1's displayed reads); the tower label's rows are m times θ's, $m\theta = (2m,\, m^{\,c})$ at $c{+}2 = d_f$, so the tower reads off θ's two folds,

$$d_f\,Q(m\theta) \;=\; m^2 \sum_{p<q} \langle\theta_p : \theta_q\rangle^2 + m\,d_f\,\langle\theta, 2\rho\rangle\;.$$

At the reduced shape $\theta = (2, 1^{\,c})$ the square identity $d_f\,q(\theta) = k^2 + \sum_{p<q}\langle\theta_p : \theta_q\rangle^2$ (Definition 6) at $q(\theta) = r+3$ and $k = d_f$ reads the pair squares at $2\,d_f$; and the row weights $\langle r+2 : 2i\rangle$ sum to equal members over the run $2 \le i \le r$, so $\langle\theta, 2\rho\rangle = 2\,r$, the top row's read, the pairing's read the residue doubled. So $Q(m\theta) = 2m^2 + 2mr$, and against $Q(\theta) = 2d_f$, the display's own $m = 1$ read, $\hat{c}_2(m\theta) = [\,m\,(m+r) : r+1\,]$; at $r = 1$ the value is $[\,m^2 + m : 2\,]$. ☐

15 The X-tower matrix and the walk

Proposition 13 (The X-tower matrix)**.** *On the orthonormal character index the multiplication by χ_θ is the symmetric matrix of fusion counts $N^{\mu}_{\theta\lambda}$ (Proposition 8), and its compression to the tower labels $m\theta$ is the tower matrix J, fusion counts alone. Every neighbor entry is one,*

$$N^{m\theta}_{\theta,(m+1)\theta} = m\{\theta, (m{+}1)\theta, m\theta\} = N^{(m+1)\theta}_{\theta,m\theta} = 1,$$

upward the Cartan equality, $\theta + m\theta = (m{+}1)\theta$ as reduced shapes (Construction 5), downward the m-symmetry at the self-dual tower labels (Proposition 8; $(m\theta)^* = m\theta$ by the complement arithmetic of Construction 5); and the diagonal is the tower's own fusion multiplicity $N^{m\theta}_{\theta,m\theta}$, one alternant read at every m (Corollary 2 at $a = \theta$, $b = c = m\theta$). The vector $x := m\theta + u$, at u the unit set's sorted display, has coordinates $2m + r + 1$; the middle members $m + i$ at $2 \le i \le r$; and 1: pairwise distinct, and a permutation contributes only where $\sigma x = x$, the adjoint's unit-monomial content, or where the tie $\nu + \sigma x = x$ holds at a root ν, the adjoint's occupied contents (Lemma 16). At $\sigma = \mathrm{id}$ the read is r, the adjoint's multiplicity at the unit-monomial content, on the even side; a root tie forces σ to fix every coordinate off some $\{i, j\}$ (the coordinates are distinct), so $\sigma = (i\,j)$ at the two coordinates' gap one, an odd transposition at the root's multiplicity one. At $m \ge 1$ the unit gaps *sit in the run alone, $x_{k+1}+1=x_k$ at $2\le k<r$, a run occupied at $r\ge 2$, the two boundary gaps reading $m+1\ge 2$:*

$$N^{m\theta}_{\theta,m\theta}+g \;=\; r \quad (r\ge 2,\;\; g \text{ the run's unit-gap count at } g+2=r), \qquad N^{m\theta}_{\theta,m\theta} \;=\; r \;=\; 1 \quad (r=1),$$

the count 2, one value at every $m\ge 1$; at the vacuum the boundary gaps close to 1 and the two further odd transpositions even the two sides' counts, the count $N^{\mathbf{1}}_{\theta\mathbf{1}}$ the Kronecker delta at two labels (Proposition 8): the vacuum defect. So J is the free Jacobi matrix with a single boundary defect, neighbor entries one, the diagonal c_1 at every $m\ge 1$ and the vacuum's the Kronecker delta at two labels: integer data, c_1 the one parameter, the shape's reads the fusion interface's own, the Cartan strictness and the unit read at the self-dual θ (Construction 6). At a member's table the diagonal is the alignment's form read (Corollary 2's member ties at $a=\theta$, $b=c=m\theta$), the member's θ content list the root list with its balance partners at one each and the unit content at the coordinate count, the interface field's instantiation (Construction 6), derived on a block at top θ: a content's dominant image is a dominant root or the unit (the descent, Lemma 9, with the content symmetry, Lemma 6), the top's line reads one (Lemma 3), the short dominant root's read is the trace recursion's own fold there (Lemma 7), one display per two-length class,

$$(2\ell{-}2)\,m=2\,(\ell{-}1) \text{ at } B_\ell, \quad 2\,m=2 \text{ at } C_\ell, \quad 6\,m=6 \text{ at } F_4, \quad 4\,m=4 \text{ at } G_2,$$

each right side the fold of the form reads at the contents above the short dominant root, and the unit content's read is the recursion's unit fold, $m\,(r+1)$ the halved lengths' fold over the root list, at the member join $\sum_\alpha\langle\alpha,\alpha\rangle=\ell\,(r+1)$ over the positive list, one displayed fold per member ($2\ell^2{-}\ell$ at B_ℓ, $\ell^2{+}\ell$ at C_ℓ, $2\ell^2{-}2\ell$ at D_ℓ, and 8, 36, 72, 126, 240 at the fixed members' length folds): the unit content at the coordinate count. The dimension is the content list's own fold, twice the positive count with the coordinate count,

$$d_\theta=\big[\,(r+2)(r+3):2\,\big] \text{ at } B_\ell \text{ and } D_\ell, \quad r\,(2r+1) \text{ at } C_\ell, \quad r\,(r+2) \text{ at the A-series,}$$

and 14, 52, 78, 133, 248 at the fixed members, the member gap product's evaluation at θ its cross-check (Corollary 1). A tie $\nu_w+w\kappa=\kappa$ at $\kappa:=m\theta+\rho$ keeps the form's square, so $2\,\langle\kappa,\nu_w\rangle$ reads $q_0(\nu_w)$. At ν_w the unit content the element is the identity, κ dominant at every simple pair beyond the unit with its images one per element (the regular reads, Construction 4), the read the coordinate count on the even side. At ν_w a root read, a partner's case reads the doubled pairing on its lower side against the square, so ν_w is a positive β at $\kappa(\beta^\vee)=1$, the element the root's reflection alone, $w\kappa=s_\beta\kappa$ at the regular reads, odd at the transport with the fold descent reading β a letter's image; and $\kappa(\beta^\vee)=m\,\theta(\beta^\vee)+\rho(\beta^\vee)$ reads one exactly at θ's coroot pair of equal members with β simple, the root fold's pair two at the simples and beyond two at every further positive member (the fixed members' displayed reads; the series' one fold per family), each such β one odd read at the root content's one. So the diagonal joins the count z of the simples at θ's equal-membered coroot pairs to the coordinate count, $N^{m\theta}_{\theta,m\theta}+z=\ell$ at every $m\ge 1$: the value is θ's occupied coroot support, one at every member off the A-series, the displayed tables' single keys, and the A-series' the run's read above, two at $r\ge 2$ and one at $r=1$; the vacuum's ties read every simple at the pair one, the count the Kronecker delta at two labels, the defect's read at every member. With strictly positive off-tower weight: the X-tower leaks, and the family operator is the tower-compressed Jacobi pencil plus off-tower coupling, the dimension weights riding the off-tower channels alone.

16 The tower channel: chord map and divisor

Theorem 11 (The chord map and the channel's base)**.** *Pass to the dual of the walk: the spectral side, in the deck coordinate z, at the walk's level the balance pair $y=\langle y_+:y_-\rangle$, its members* *cross-added onto the two sides (Theorem 14's levels), the deck relation*

$$z^2 + y_+ z + 1 \;=\; \big(c_1 + y_-\big)\, z,$$

one display at every representative (a summand on both members fixing the value, Definition 1), degree two in z with the deck involution ι exchanging its two roots, their product the constant coefficient one and the level read jointly by both, Definition 4's relation at the chord w joining the level's members, $w + y_+ = c_1 + y_-$. The relation's two roots are equal exactly at the crossings, ι's fixed points, a double root squaring to the constant coefficient one at the display

$$(c_1 + y_-)^2 + y_+^2 \;=\; 2\, y_+\, (c_1 + y_-) + 4\ ,$$

the double root's read cross-added onto the two sides, one display at every representative (Definition 1); the chord is a member of the carrier at the lower crossing alone, and the display holds exactly at the two crossing joins $y_+ + 2 = c_1 + y_-$ and $y_+ = c_1 + y_- + 2$; the band is the segment between the two crossing levels; and the defect's root is z_* at $c_1 z_* = 1$, where the relation reads the bound-state level at the join $c_1 y_+ + 1 = c_1 y_-$.

The chord and the defect read distinct integers. The relation's two roots are the half-line walk's neighbors, the coordination count that fixes the band's width 4*, the level read jointly by the two roots. The base of the channel's value is c_1 at $c_1 z_* = 1$, the mass point's own witness. The two integers agree exactly at base two, the member-key count's two-key read (Proposition 13), and separate at base one, where the chord has two terms.*

Theorem 12 (The channel divisor)**.** *The walk's boundary entry G_0 in the deck coordinate is the renewal witness $G_0 = z + c_1 z G_0$'s series, its divisor the witness's own at the mass point z_* at $c_1 z_* = 1$, and the channel's coordinate is the order pair, the determination: the mass point a simple pole, order* 1*, the ι-symmetrized weight at order* 2*, both orders independent of the base and summing to* 3*; the base c_1 is the orders' read, the single-key member's display at base one, the mass point at the crossing.*

The divisor. The walk's resolvent reads in the deck coordinate, the level the balance pair $y = \langle y_+ : y_- \rangle$ of Theorem 11. The defect-free tail's boundary entry $m^{(L)}$ at the tail's L-site cut J_L is a solved witness of Lemma 23(ii)'s shape, the boundary entry of the adjugate column against the determinant at the pair $(J_L + y_- : y_+)$'s site datum (Lemma 30; Theorem 14's levels, the datum at $J_L + y_- = y_+ + S_L$): the boundary couples to the tail by the single entry one, the unit-triangular congruence of Lemma 30 splits the pair $(J_{L+1} + y_- : y_+)$'s datum as the boundary pivot against $(J_L + y_- : y_+)$'s, and the boundary entry solves the pivot identity,

$$m^{(L+1)}\, \big(c_1 + y_-\big) \;=\; 1 + m^{(L+1)} \big(y_+ + m^{(L)}\big), \qquad m^{(1)}\, \big(c_1 + y_-\big) \;=\; 1 + m^{(1)}\, y_+,$$

the one-site read its seed. The truncations generate one series m_1, whose fixed-point identity is the deck relation in the boundary entry, $m_1^2 + y_+ m_1 + 1 = \big(c_1 + y_-\big)\, m_1$: its two roots are the root pair, at product one, and the member at the z-multiple series is the coordinate itself, the errors e_L at $m^{(L)} = z + e_L$ obey $e_{L+1} = z^2 e_L + z\, e_L\, e_{L+1}$, so $m^{(L)}$ agrees with z beyond z-order $2L$, and $m_1(z) = z$. The defect site then reads one further pivot, its diagonal the vacuum's (Proposition 13), and the boundary read solves it at the renewal witness

$$G_0 \;=\; z + c_1\, z\, G_0,$$

the recursion's own series $\sum_k c_1^k z^{\,k+1}$ (Definition 2), coefficients naturals: its divisor is the witness's own, the least occupied degree one and the mass point z_* at $c_1 z_* = 1$, the clearing factor $\langle 1 : c_1 z\rangle$'s *root. The deck involution exchanges z_* with its ι-image c_1, the orbit's product one: z_* is the orbit's interior member, $z_*^2 \le 1$ one pair comparison. The mass point's margin is the crossing read: the relation at the mass point reads the symbol $c_1\, w = 1 + c_1^2$, its crossing pair $\langle w^2 \!:\! 4\rangle = [\,\langle c_1^2 \!:\! 1\rangle^2 : c_1^2\,]$, equal members exactly at $c_1 = 1$, and the margin $\langle 1 : z_*^2\rangle$ is the mass weight's own pair, its square the crossing pair's z_*^2-multiple (Lemma 38): $[\,9 : 4\,]$ at base two, the member-key count's two-key read (Proposition 13). Every channel datum is a read of it: the bound-state level at the join $c_1 y_+ + 1 = c_1 y_-$ (Theorem 11); the mass weight, the pair $[\,\langle c_1^2 : 1\rangle : c_1^2\,]$ at the complement $[\,1 : c_1^2\,]$, the two summing to one; and the crossing levels at their joins (Theorem 11), with c_1 Proposition 13's. The lower crossing's level reads equal members exactly at $c_1 = 2$, the infrared contact forced by the integer counting the cubic couplings, and at $c_1 = 1$ the mass point sits at the crossing, $z_* = 1$, the divisor count unchanged there, as below.*

What the channel contributes, and at which base. The divisor's data are two integer orders and one base. The mass point is a simple pole of G_0, order 1. The weight reads the clearing factor ι-symmetrized, the ι-image's cleared read and the symmetrized pair

$$z \cdot \big(\langle 1 \!:\! c_1 z\rangle \circ \iota\big) \;=\; \langle z \!:\! c_1\rangle, \qquad \big[\,\langle 1 \!:\! c_1 z\rangle\, \langle z \!:\! c_1\rangle : z\,\big],$$

whose two ι-image factors each read the base once, $\langle 1 \!:\! c_1 z\rangle$ at its leading coefficient and $\langle z \!:\! c_1\rangle$ at its root c_1, so the weight's read at the base is order 2. Both orders are independent of c_1, and the base collects at the orders' sum,

$$c_1^1 \cdot c_1^2 \;=\; c_1^3.$$

So the channel's coordinate is the integer $3 = 1 + 2$ and its base is c_1: the two orders are what the theory determines, the base what they are read at. The read is one divisor count at every position of the ι-image root, and $c_1 = 1$ places that root at the crossing, the one count throughout.

Definition 12 (The channel reads)**.** At a base c_1 the channel's derived objects (Theorem 12) are ground reads (Definition 2): the boundary entry G_0 at its renewal witness $G_0 = z + c_1 z G_0$, as the witness's two coefficient lists; the mass point z_* at $c_1 z_* = 1$; the *pole order* of G_0 at z_*, the gap of the clearing factor's root multiplicity there against the series member's; the ι-symmetrized weight's factor list $\big(\langle 1 \!:\! c_1 z\rangle,\ \langle z \!:\! c_1\rangle\big)$ over the second member z, each factor reading the base at degree one; and the *order count*, the pole order added to the factor list's count.

17 The cell certificate

Theorem 13 (Gap positivity on the interior, assembled)**.** *At every interior end-weight ray of $\mathcal{C}$ the cut of $\mathcal{K}\big([\,\alpha : \beta\,]\big)$ holds a member pair, positive by its shape: the interior is a union of located cells, the root coordinate's rays $q\,\tau = p$ read their cells directly, the counting sandwich of (v) hands every further ray its cut at the sandwich's stated width, and on each cell the cut is finitely many counts at a positive level of the cell's own.*

(i) The count has an object. On a window the compression is finite-dimensional (Proposition 2), so "the number of eigenvalues below a line" denotes something and $\operatorname{count}(a) = \operatorname{rev}(H : aG)$ *is an exact reversal count.*

(ii) The interior is covered by located cells. At any interior coupling and window, Theorem 32's modulus puts the cut's verification on a finite compression at a computable cutoff, at every coupling, and the decimation presents that compression's counts per cell of the named divisor list (Theorem 18; Lemma 41; Lemma 42): the cells over the coupling, the momenta and the residue (Lemma 31; Lemma 25(iii); Lemma 26(iii)), read at each dimension through stencil caps and momentum sums polynomial in it, cover the interior, the free cell at the free end (Lemma 15; Lemma 49), the emitted

cells of the decimation's divisor list between, and the contact cell at the other end (Lemma 44; the approach the corner cell's, Lemma 45), every boundary a located root, counts persisting at a boundary beside a positive-semidefinite cell.

(iii) Per cell the cut is a flat step, and the cutoff is priced. On each cell the flat step verifies the cut at a level κ, finitely many anchored jump comparisons, each two counts at located pairs (Theorem 19), and Theorem 32 prices the cutoff at the α=1 representative: at an admissible cutoff's order $E_{\text{low}} < \Lambda_C$ with its witness gap ς at $E_{\text{low}} + \varsigma = \Lambda_C$,

$$E_0\,\varsigma\,\mathcal{K}_\Lambda \;\le\; E_0\,\varsigma\,\mathcal{K}(v) + \beta^2\#p^2\,d_\theta^2,$$

at every cutoff: given a pair ϑ, taking $\vartheta\,\varsigma > \beta^2\#p^2\,d_\theta^2$ transports the window cuts to $\mathcal{K}(v)$ with a modulus rather than along a subsequence, at a rate the member's adjoint dimension and the region's plaquette count fix.

(iv) The level is positive per cell, and the whole read closes on finitely many cells. The cell's level is a divisor read: root collisions sit on the crossing divisor alone (Theorem 25(ii)), so on a cell the decimated head's first two jumps stay separated, the ground jump at its per-cell height, read 1 *from the free end's side (Theorem 9; Theorem 18(i)), and window-free by the decimation, off the unit fiber outright and within the named geometric brackets on it (Theorem 18(iii)); the counting functions are flat across $(\ell,\ \ell+E_0\kappa)$ at fixed pairs of the cell's certificate, each count a cell function read once (Lemma 31; Theorem 19), so a pair κ within that flat window less the brackets' width is a located datum of the cell, a content cell's the $H([\,1:\tau^2\,])$ level distance, the free and contact cells reading theirs outright (Lemma 15; Lemma 44; the contact approach at the corner scale's heights, Lemma 45) and each interior boundary one located two-sided read of the divisor, and*

$$E_0\,\mathcal{K}(v) \;\ge\; \kappa$$

on the whole cell, located point by located point, each cell read at one pair (Lemma 31). At the reach clearances of the families each count reads, one symbolic verification is every residue's, the rank divisors at their leading sides at every one of their residues off the classifications' displayed merged roots, those residues reading their tied pairs at the multiplicity, with a below-clearance family's residues read directly (Lemma 25(iii); Lemma 26(iii); the classifications of Lemma 28 and Lemma 29), and the channel-degeneration residues are each their own read. The quantifier over the dimension closes at the derivations: the fibering, the decimation, the charged grading and the flat step are derived at every d, with the stencil caps and the momentum sums polynomial in it, and the head's jump reads are counts at every window there (Theorem 9; Theorem 18(i)), so each cell's κ is a positive located datum at every dimension, the divisor's own read; a dimension's cell list is one instance of those derivations, as is a channel-degeneration residue's, residues one and two at their own channel data (Lemma 24).

(v) The reads close across each cell, at the sandwich's width. The cut's comparisons are counting-function reads, and counts obey a coupling sandwich: at couplings v, v' of a cell whose pencils' site datum S at $H(v) = H(v') + S$ is capped two-sidedly at $\rho\,G$ (Lemma 30) per decimated head: the decimated entries' derivative coefficient folds price the site datum entrywise over the pair distance, two points' values differing by at most the distance against the fold at the bound (Lemma 32's priced side read at Theorem 15's arithmetic), and at the head's order the scaled identity caps the priced datum two-sidedly where ρ clears the order's multiple of the entry price, each row's off-row magnitude fold within that multiple at the dominance comparison (Theorem 18(ii)), the cap window-free off the unit fiber and within the deck brackets on it (Theorem 18(iii); Lemma 31; against G's stored factorization), every count obeys

$$\mathrm{count}_v\langle a\!:\!\rho\rangle \;\le\; \mathrm{count}_{v'}(a) \;\le\; \mathrm{count}_v(a+\rho)$$

(Lemma 30, counts monotone under $\preceq$), so every jump bracket travels between the two couplings at width ρ: at κ below a cell's level, the level's margin names a pair distance whose sandwich width it absorbs, the pair reads hold across the cells' closed union by (ii)–(iv) with Lemma 31's boundary clause, and the sandwich transports the jump comparisons over (Theorem 19): so $\mathcal{K}([\alpha:\beta]) \geq \kappa^*$ at every interior end-weight pair, at one positive pair κ^* below the finitely many cells' located floors, the trichotomy fold over the cell list: the root coordinate's rays read their cells directly, the corner cell's rays at its own constancy (Lemma 45), every further pair ray is a located coupling between them whose margin absorbs the sandwich's width.

Lemma 28 (The rank divisors)**.** *The rank divisors of the chain's comparisons (Lemma 25(iii)) close on* 37 *records over the depth-two charge-one closure, the divisors read at $n := d_f$, the fundamental count (Construction 1): per band family the excess gap, the cleared read at $2n\,C_2(\omega_1) + \mathrm{gap}_\lambda = 2n\,C_2(\lambda)$ of Lemma 25(i)'s word data, each a product of naturals at a gap witness,*

family	gap_λ	*witness*	*clearance*
$\omega_2 + \omega_1^*$	$2n\,g$	$1 + g = n$	6
$\omega_3 + 2\omega_1^*$	$4n\,g$	$1 + g = n$	7
$\omega_3 + \omega_2^*$	$4n\,g$	$2 + g = n$	8
$2\omega_1 + \omega_1^*$	$2n\,(n+1)$		5
$\omega_1{+}\omega_2 + 2\omega_1^*$	$2n\,(2n+1)$		6
$\omega_1{+}\omega_2 + \omega_2^*$	$2n\,g$	$1 + g = 2n$	7
$3\omega_1 + 2\omega_1^*$	$4n\,(n+2)$		5
$3\omega_1 + \omega_2^*$	$4n\,(n+1)$		6

the frontier-binding comparisons, the balance pairs $\langle \mathrm{gap}_i : \mathrm{gap}_j \rangle$ over the family pairs, and the stencil cap $3\,g$ at $1 + g = n^2$, the clearance column the family's reach clearance read at n, the clearance residue's successor (Lemma 25(ii)). A comparison's ranks are those clearing its families' reaches, and the classification is the equal-members reads in the rank domain, five, every one below a compared family's clearance, one cross-multiplied evaluation each: $\omega_3{+}\omega_2^*$'s gap at $n = 2$ against its clearance 8*, and the binding pairs' $(2\omega_1{+}\omega_1^*,\ \omega_1{+}\omega_2{+}\omega_2^*)$ at $n = 2$ against* 7*, $(\omega_2{+}\omega_1^*,\ \omega_3{+}\omega_2^*)$ at $n = 3$ against* 8*, $(\omega_3{+}2\omega_1^*,\ 2\omega_1{+}\omega_1^*)$ at $n = 3$ against* 7*, and $(\omega_3{+}\omega_2^*,\ 2\omega_1{+}\omega_1^*)$ at $n = 5$ against* 8*. So every divisor keeps its leading side at every one of its ranks, every cell's count one read along the rank direction, and one symbolic verification is every rank's at the cleared families; a family below its clearance reads its finitely many ranks' data directly, the calculus's displays at the rank's label (Lemma 7; Corollary 1; Corollary 2), one evaluation each, a tied pair reading its merged level with the counts at the multiplicity (Lemma 31's boundary clause).*

Lemma 29 (The series' rank divisors)**.** *The rank divisors of the chain's comparisons at a B, C or D member (Lemma 26(iii)) close on the depth-two sector closures at the member's class group (Lemma 42(i)):* 47 *records at B_ℓ and at C_ℓ and* 83 *at D_ℓ, the divisors read at the derived residue r (Construction 4). The sectors are the unit class's, based at θ, and one per nonunit class, based at the class's end member (Lemma 42(ii)): the spinor at B_ℓ, the first fundamental at C_ℓ, and the vector with the two spinors at D_ℓ; the exchange of the last two simple letters is a relabeling of the D table, the last coordinate read to its balance partner, fixing the form and the root fold (the last key's read equal members) and exchanging the two spinor classes, each sector's base with its closure, so the spinor pair's records are one displayed list.*

The families. A sector's families are the base with an added word, the fusion row's dominant sums at the member's θ content list (Lemma 26(ii); Proposition 13), iterated twice from a charged sector's base and once from θ at the unit sector's, the vacuum's own depth-two closure; each count *is the alignment's finite signed fold, rank-free, and the displayed lists are the occupied folds. Two collision reads close the further sums, Lemma 26(ii)'s at the moved keys. At B_ℓ, a short-move tie at a target leaving the last key open pairs under the last coordinate's flip with its mixed move at the last key, ν against ν joined to e_ℓ's balance partner, one odd letter fixing the further magnitudes, the two contents at one count each: the fold reads the sum's unit, and the unit sector's short-move sums $(2,1)$ and (1) read it at $\ell \geq 3$ with $(1,1,1)$ at $\ell \geq 4$, the third's move at key three. At C_ℓ, a mixed-move tie whose target's shifted list reads the unit gap at the moved adjacent pair pairs under that pair's transposition with the long move at its second key, one odd letter: $\theta + e_2 + e_3$ and $\omega_1 + e_2 + e_3$ read the unit fold, the word $(1,1,1)$ entering at the second fusion through the difference move at $(2,1)$, and the second fusion's two further ties close under the moved pair's transposition against the long move: $(2,1) + e_3 + e_4$ reads the unit fold with its target off the sector's families outright, and $(3) + e_2 + e_3$ reads it with the word $(3,1,1)$ entering through the two-row source at $(2,1) + e_1 + e_3$. A displayed family's ties off the identity read off the content list, the two collision conditions failing at its shifted list (the moved pair's gap beyond one at the word's keys and the last key's read off the flip's, Lemma 26(ii)'s alignment fold, rank-free beyond the clearance), so its row's count is one, the identity's own read, at every rank clearing the family's reach, the added word's last key at Lemma 26's clearance.*

The excess gaps. The gap is the form's bilinear read at the member's displayed data (Construction 3's additive form; Construction 4),

$$C_2(\beta) + \mathrm{gap}_W \;=\; C_2(\beta + W)\ , \qquad \mathrm{gap}_W \;=\; \langle W, W\rangle + 2\,\langle W, \beta\rangle + \langle W, 2\rho\rangle\ ,$$

β the sector's base and W the added word, three folds over the word's occupied keys: the square fold and the doubled base fold are rank-free, the word's own form square and its doubled dot against the base's list, and the root fold's read is the word's fold against the displayed key reads, linear in the rank, so every gap is one linear read at the derived residue, displayed as a natural fold or a doubled gap at its witness, positive at every domain rank: every charged family sits strictly above its base. The unit tables, the gaps against $C_2(\theta) = 2\,(r+1)$ with the 2θ rows the tower law's $m = 2$ read (Proposition 12):

family	gap	*witness*	*clearance*
$B_\ell,\ D_\ell:\quad (2)$	4		3
$(1,1,1,1)$	$2\,g$	$g + 3 = r$	6
$(2,1,1)$	$2\,(r+1)$		5
$(2,2) = 2\theta$	$2\,(r+3)$		4
$C_\ell:\quad (1,1)$	$\langle 2r : 2r+2\rangle$	*margin* 2, *lower side*	4
$(2,2)$	$2\,r$		4
$(3,1)$	$2\,(r+1)$		4
$(4) = 2\theta$	$2\,(r+3)$		3

The charged tables, the gaps against the base's Casimir, the spinor list one table for B_ℓ and D_ℓ

with a word of odd box count at D_ℓ read at the flipped base, the class kept:

spinor word	gap	*clearance*
e_1	$r+3$	3
$e_1{+}e_2$	$2\,(r+2)$	4
$e_1{+}e_2{+}e_3$	$3\,(r+1)$	5
$e_1{+}e_2{+}e_3{+}e_4$	$4\,r$	6
$2e_1$	$2\,(r+4)$	3
$2e_1{+}e_2$	$3\,(r+3)$	4
$2e_1{+}e_2{+}e_3$	$4\,(r+2)$	5
$2e_1{+}2e_2$	$4\,(r+3)$	4

$C_\ell,\ \omega_1$	gap	*wit.*	*cl.*	D_ℓ*, vector*	gap	*wit.*	*cl.*
$(1,1,1)$	$2\,g$	$g{+}1{=}r$	5	$(1,1,1)$	$2\,g$	$g{+}1{=}r$	5
$(2,1)$	$2r+1$		4	$(2,1)$	$2\,(r+2)$		4
(3)	$2\,(r+2)$		3	(3)	$2\,(r+5)$		3
$(2,2,1)$	$4\,r$		5	$(1,1,1,1,1)$	$4\,g$	$g{+}3{=}r$	7
$(3,1,1)$	$2\,(2r+1)$		5	$(2,1,1,1)$	$2\,g$	$g{+}1{=}2r$	6
$(3,2)$	$4\,(r+1)$		4	$(2,2,1)$	$4\,(r+1)$		5
$(4,1)$	$4r+7$		4	$(3,1,1)$	$4\,(r+2)$		5
(5)	$4\,(r+3)$		3	$(3,2)$	$4\,(r+3)$		4

The records. Per sector, the family gaps and the frontier-binding comparisons, the balance pairs $\langle \mathrm{gap}_i : \mathrm{gap}_j \rangle$ over the sector's family pairs, six per unit list and twenty-eight per charged sector; and per member the stencil cap $3\,d_\theta$ at the member's dimension fold (Proposition 13), $[\,3\,(r{+}2)(r{+}3) : 2\,]$ at B_ℓ and D_ℓ and $3\,r\,(2r+1)$ at C_ℓ: the joins $10+36+1$ at B_ℓ and C_ℓ and $10+2\cdot 36+1$ at D_ℓ are the stated counts.

The classification is the equal-members reads of the displayed divisors in the rank domains, the domains the residue folds' reads (Construction 4), every even residue from two at B_ℓ, every residue from three at C_ℓ and every odd residue from five at D_ℓ: seven, each one cross-multiplied evaluation. Five sit below a compared family's clearance, the rank's data read directly: at B_2 the spinor word $e_1{+}e_2$ reads 8 with the four-key word beyond the rank's keys, and at B_3 the word $2e_1$ reads 16 so; at D_4 the unit sector's (2) and the key-four pair, the word $(1,1,1,1)$ with its flip image, read one merged level at the gap 4, and the spinor words $e_1{+}e_2{+}e_3$ and $2e_1$ read one merged level at 18; and at D_4 the vector's $(1,1,1)$ reads 8 with the five-key word beyond the rank's keys. Two sit at cleared ranks, the merged levels: the spinor words $e_1{+}e_2{+}e_3{+}e_4$ and $2e_1{+}e_2$ at D_6, the gap 36, and the vector's $(1,1,1,1,1)$ and (3) at D_7, the gap 32, each root a located rank point whose tied pair reads its merged level with the counts at the multiplicity (Lemma 31's boundary clause). So every displayed divisor keeps its leading side at every further cleared rank, one symbolic read every such rank's, a comparison's ranks those clearing its families' reaches; and a family below its clearance reads its finitely many ranks' data directly, the calculus's displays at the rank's label (Lemma 7; Corollary 1; Corollary 2), the short-move sums joining those ranks' unit closures at their own displayed folds, $(2,1)$ at 6 with (1) at $\langle 4 : 6 \rangle$, margin 2 on its lower side, at B_2 (the occupied last key at the first, the last key's own move at the second) and $(1,1,1)$ at 2 at B_3, with the last key's flip moves at D_4.

18 The count certificates

The statements below are the cell calculus's arithmetic, made finite and constructible: the counts, the located roots, and the window reads the decimation and the flat step take.

Lemma 30 (The reversal count)**.** *Over a located stage (Lemma 32), where every entry's members' equality and side are stage reads, the reversal count of a form pair, written* $\operatorname{rev}(A\!:\!B)$ *at symmetric* A*,* B *of order* N*, is the side count of the pair: the dimension of a maximal list on whose span* B *exceeds* A *at every combination off the sum's unit, the two lists of the proof its witnesses. The count is read off a pivot split of the pair's elimination datum* S*, the site structure at* $A = B + S$ *read entrywise (Definition 3): a unit congruence,* T *at its witness* T' *at* $TT' = 1$ *and* $T'T = 1$*, the two reads the statement's data,* $T^{\mathsf{T}}ST = D$ *with* D *block diagonal in nonsingular blocks of orders* 1 *and* 2 *and one kernel block, its columns'* T*-image spanning the datum's kernel; a nonsingular block reads its sides, one per unit: an order-*1 *block its entry's side, and a symmetric* 2×2 *block two side reads off its data (the determinant on its lower side reads one unit per side, the mixed block; on its upper side the identity* $S_{11}S_{22} = \det + S_{12}^2$ *puts the diagonal product on its upper side, both entries of unequal members at one shared side, which both units read), so a split reads a lower-side count, the order-*1 *blocks' sides with the* 2×2 *reads. Every symmetric* S *has a split, closed in at most* N *pivot steps, and every two splits read one lower-side count:* $\operatorname{rev}(A\!:\!B)$ *is that shared count, the split's pivot ordering certificate data, and* $\operatorname{rev} S$ *abbreviates the count at* S *its own elimination datum. A split enters cleared: over pair data the congruence reads at an integer* T*, square at the datum's order, its determinant of unequal members with the adjugate against the determinant the witness's solve (Definition 3), and the block diagonal absorbs the columns' rescaling at every side fixed, the cleared read the unit congruence's by the homogeneity principle (Definition 1), one count at either spelling. The split reads the rank with it, the kernel block's complement, one count at every split and blind to unit congruence with the reversal count: a split* (T, D) *of* S *is the split* $(T, \check{D})$ *of the memberwise swap* $\check{S}$ *(each entry's member swap) at the block sides exchanged, so the rank is the two counts' sum,* $\operatorname{rev}(S) + \operatorname{rev}(\check{S})$*. A datum is positive semidefinite where a split's nonsingular blocks read the upper side throughout, the split the statement's witness, and positive definite where those blocks' orders sum to the datum's; a form pair* $(A\!:\!B)$ *is positive semidefinite where its elimination datum is, the glyphs* $B \preceq A$ *and* $A \succeq B$ *denote that read, and* $S \succ 0$ *denotes* S *positive definite; a datum* S *is capped two-sidedly at* C *where* $S \preceq C$ *holds and the sum* $C + S$ *is positive semidefinite, two splits as the statement's witnesses. Consequently:* $\operatorname{rev}(T^{\mathsf{T}}ST) = \operatorname{rev}(S)$ *at* T *with its witness;* $\operatorname{rev}(qS) = \operatorname{rev}(S)$ *at a pair* q *above the sum's unit, a split* (T, D) *of* S *being the split* (T, qD) *of* qS *with every block's sides and determinant side fixed; a symmetric* $\mathrm{LDL}^{\mathsf{T}}$ *reads* rev *as its count of lower-side pivots, and when the leading principal minors read unequal members the pivots are their successive cofactors (*$m_j\, d_{j+1} = m_{j+1}$*), so* rev *is the minor sequence's own orientation-reversal count, the reversals along* $1, m_1, \ldots, m_N$ *(Jacobi's rule), the name's read;* $S \preceq S'$ *gives* $\operatorname{rev}(S) \geq \operatorname{rev}(S')$*; a compression's counts are at most the full form's; inertia adds at a nonsingular principal pivot,*

$$\operatorname{rev}(S) = \operatorname{rev}(P) + \operatorname{rev}(S_P) \quad \textit{at} \quad S = \begin{bmatrix} P & B \\ B^{\mathsf{T}} & Q \end{bmatrix}, \qquad B^{\mathsf{T}}C_P + S_P = Q \quad \textit{at the solve} \quad P\,C_P = B,$$

S_P the pivot's deflation, its witness C_P the adjugate column against the determinant, a pivot at stated places entering at the leading position by the congruence at the places' permutation matrix, its transpose the witness (Construction 1's transpose identity at the matrix carrier) with the count the congruence read's own, and the determinant splits with the inertia at the cleared scale: each trailing row enters at the pivot's determinant against the leading rows' combination at the solve's witness column, one scale per trailing row, the minor's scale and combination reads (Definition 3)

reading the scale's power at the deflation's order against the reduced list's determinant, the coupling's exchange withdrawn at the solve so the reduced list is block triangular with its trailing block the deflation at one further scale per row, and a block-triangular list's determinant is its diagonal blocks' product, every assignment crossing into the vacant slab reading an entry at the sum's unit: $\det P \cdot \det\big((\det P)^2 S_P\big) \;=\; (\det P)^{2o} \det S$ *at* o *the deflation's order,* $\det S \;=\; \det P \cdot \det S_P$ *its read at the off-unit scale's injectivity; two square lists' product, its entries the first's rows against the second's columns at the coordinate fold, reads its determinant at the factors' own, one joined list's two reads: at the join of the identity's memberwise swap beside the second list over the first list beside a vacant block, the trailing slab clears against the leading rows at the first list's own coefficients, one combination per row with the further summands repeated-row reads of equal members (Definition 3), onto the vacant slab beside the two lists' product, and the cleared join's determinant is the block-triangular read at the swapped identity against the product's; the slabs' exchange, one transposition per row pair (Definition 3), reads the joined list block triangular at the factors instead, and the exchange count joins the swapped identity's diagonal to an even family (Construction 1): the signs withdraw and the product's determinant is the determinants' product; inertia adds at a block diagonal outright,*

$$\operatorname{rev}(P \oplus Q) \;=\; \operatorname{rev}(P) + \operatorname{rev}(Q),$$

the addition's instance at a coupling of equal members: the two splits concatenate to a split of the block diagonal, the congruences joined at that off-block with the block lists appended, the kernel columns collected to the tail and the kernel orders added; and a symmetric matrix of unequal-membered determinant has a nonsingular principal submatrix of order at most two.

Proof. Existence. At a nonsingular principal block P of order 1 or 2, the unit-triangular congruence at upper block $\check{C}_P$, the witness's memberwise swap ($P\,C_P = B$), maps S to $P \oplus S_P$, and the recursion descends to the deflation. With every principal minor of orders 1 and 2 of equal members, the diagonal reads equal members entrywise, each order-two minor then reads S_{ij}^2, and S_{ij} with it, of equal members: a symmetric block either has such a pivot or is the kernel block, so the leftmost admissible pivot closes a split in at most N steps, the per-step congruences composing to T; in particular a symmetric matrix of unequal-membered determinant has a nonsingular principal submatrix of order at most 2.

The mixed blocks' two witnesses. At $P = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ with the determinant $\langle ac : b^2 \rangle$ on its lower side, a lower-side vector is read off the entries: e_1 at a on its lower side; $(b,\, \check{a})$ at a on its upper side, $\check{a}$ the member swap, reading $a\,\langle ac : b^2\rangle$ on the lower; and at a of equal members, where b's members differ, the vector $(w,\, b+b)$ at w the balance partner of $c+1$, its read the balance partner of $(b+b)^2$, the cross terms collecting at the constant's swap, on the lower side at b's occupied square. An upper-side vector of P is a lower-side vector of the memberwise swap $\check{P}$, the recipe read there.

The two lists of a split, and the exchange read. From a split (T, D) at lower-side count ν, the *lower list* takes, per lower-side unit, the T-image of one block-supported vector: each lower-side order-1 column, both columns of each 2×2 at the shared lower side, and each mixed block's lower-side vector. The list's ν members are independent, the strict read's own consequence (the forcing clause's read below), and S reads the lower side at every combination off the sum's unit: distinct blocks' pairings read equal members through D, so a combination reads a per-block sum, every term on its lower side or of equal members and the terms at coefficients of unequal members on the lower side. The *complement list* takes the upper-side order-1 columns, both columns of each 2×2 at the shared upper side, each mixed block's upper-side vector, and the kernel block's columns: its member count μ at $\nu + \mu = N$, the block orders' total, independent, with every S-read of a combination on its upper side or of equal members. For two splits at counts ν_1, ν_2, join the first's lower list to the second's complement list: the joint rank is one computation, and a rank defect exhibits a dependency whose read is one stage element with $x^\mathsf{T} Sx$ on its lower side and off it at once (a dependency whose lower-list part is the sum's unit sits inside the independent complement list), against a stage read's one side (Lemma 32): the joint list is independent, $\nu_1 + \mu_2 \leq N$ at $\nu_2 + \mu_2 = N$, and the exchange symmetrized reads $\nu_1 = \nu_2$. The joint-rank read at an arbitrary first list is the *forcing clause*: a list of k vectors with every S-read of a combination off the sum's unit on the lower side, joined to any split's complement list, forces $\operatorname{rev}(S) \geq k$, the strict read forcing the list's independence outright: a combination reading the sum's unit at coefficients off the unit family reads its own lower side besides, against the trichotomy. The *dual clause* runs at the complement side with the independence its own stated clause, an at-or-above read admitting a dependency at the sum's unit: an independent list of k vectors with every S-read of a combination at its upper side or equal members, joined to any split's lower list, forces $k + \operatorname{rev}(S) \leq N$.

The consequences. A split of $T^\mathsf{T} ST$ at U is a split of S at TU: the count is invariant. Splits of P and of S_P concatenate through the displayed congruence to a split of S: inertia adds, and the two forcing clauses price the identity besides: the pivot's and the deflation's lower lists, padded and lifted through the solve with every cross pairing the witness's own unit read, force one side, and the complement lists at the dual clause the other. At a coupling of equal members the lift is a padding besides, and the pricing runs without the solve: the two lower lists, each padded into its own half of the join, read the block diagonal at the two forms' sum with the cross pairings at the sum's unit, and the complement lists at the dual clause read the other side. An LDL^T is a split at order-1 blocks, the congruence by the unit-triangular witness T at $L^\mathsf{T} T = 1$, one back-substitution, onto the pivot diagonal, whose k-th leading minor is the product of the first k pivots. At $S \preceq S'$ the datum R at $S' = S + R$ is positive semidefinite, so $x^\mathsf{T} Sx + x^\mathsf{T} Rx = x^\mathsf{T} S' x$ at every x through the congruence with the R-read on its upper side or of equal members; on S''s lower list the S-read sits below the S'-read, on the lower side with it, and the forcing clause reads $\operatorname{rev}(S) \geq \operatorname{rev}(S')$. For a compression $P^\mathsf{T} SP$, the compression's lower list maps through P to an independent list with every S-read of a combination off the sum's unit on the lower side (a combination with Pc the sum's unit at c off it reads $c^\mathsf{T} P^\mathsf{T} SPc$ on its lower side and of equal members at once, against the trichotomy), and the forcing clause closes it. $\square$

Theorem 14 (The designation, and the witness)**.** *Let H, G have entries of a located stage (Lemma 32) with G positive definite, and let* $\operatorname{count}(a) := \operatorname{rev}(H : aG)$*; at a balance-pair level $\langle x : y \rangle$ (Definition 1) the count is*

$$\operatorname{count}\langle x : y \rangle \ := \ \operatorname{rev}\left(H + y\, G \ : \ x\, G\right),$$

the level's second member cross-added onto the pencil's own side, so a root on its lower side reads its bracket as one on its upper does; each count exact by symmetric LDL^T *(Lemma 30).*

1. *(The designation.) A designation of a jump of the counting function is the data (l, h, w) at $l + w = h$ with the two counts read at a gap,* $\operatorname{count}(l) + g = \operatorname{count}(h)$ *at $g \geq 1$, the width w at most the δ of Theorem 15: the bracket is narrower than the separation, so it admits one distinct root alone, at multiplicity g, and the reads are its whole verification. At the least root the lower read sharpens to the pair at l positive semidefinite, one split and one count. Designations exist at every stated width: the radius of Theorem 15 reads two starting levels, the lower below every root at the split and the upper beyond every root at the full count; from two levels at counts at a gap, the mean's count sits at a gap against one end's, the half at that gap the next bracket, the width halving, a refinement closing in the least k at $w_0 \leq 2^k \delta$ from the starting gap w_0 at $l_0 + w_0 = h_0$, each step one count; and at the least-root instance a mean at the lower end's*

count either is positive semidefinite, moving the lower level, or is singular there, the mean naming the root and one recentering at $2w' \leq \delta$ emitting the designation. The refinement runs at any stage read whose trichotomy exchanges sides along a segment, a polynomial's side read at Theorem 15's separation among them, the equality outcome naming a root outright with the recentering emitting its bracket. An emitted designation enters as certificate data, its reads re-read at every verification.

2. *(The witness u.) For any form pair $(A\,{:}\,B)$ of pair entries with* $\mathrm{rev}(A\,{:}\,B) \geq 1$*, the symmetric* $\mathrm{LDL}^{\mathsf{T}}$ *of its elimination datum has a lower-side pivot, and the corresponding column of the accumulated pair-entried transform is a vector u with $u^{\mathsf{T}}Bu > u^{\mathsf{T}}Au$, the reversal's read off the elimination. At $(H\,{:}\,aG)$ that is the certificate's witness whenever* $\mathrm{count}(a) \geq 1$*.*

Lemma 31 (The count is a cell function)**.** *Let $(H(x), G(x))$ be a symmetric pencil whose entries are polynomial in $x = (x_1, \ldots, x_m)$ over $\mathbb{P}$, with $G(x) \succ 0$ on a box X, let a be a pair, and write $D(x)$ for the pair $\big(H(x)\,{:}\,aG(x)\big)$'s site-datum determinant. Then* $\mathrm{count}(x) := \mathrm{rev}\,\big(H(x)\,{:}\,aG(x)\big)$ *is constant on every segment between stage points of X avoiding D's root locus (the stages of Lemma 32), the avoidance's certificate the proof's pivot cover: finitely many subintervals chained at shared endpoints, each with a designated principal minor of order at most two whose determinant keeps its side along the subinterval at the priced side read (Lemma 32) at a bound containing the subinterval, at order two its leading entry so as well where the determinant sits on its upper side, and beneath it the cleared deflation's own cover, the determinant's root-freeness the cover's read; the locus's points are located over D's squarefree part, the division descent's iterated divisions with the Bézout witnesses the certificate data (Theorem 15's descent at Lemma 36's squarefree read), a bracket's root count the segment count at the cleared variable (Lemma 37 at Lemma 38's clearing); hence the count is one integer on each piecewise-linearly connected cell of X off that locus; and at a stage point of the locus adjacent to a cell whose pair is positive semidefinite, every pencil root is at least a, the point's read entering as its own split (Theorem 14's certificate entry).*

Proof. Write $S(t)$ for the pair's site datum at $H = aG + S$ along the segment, entries polynomial in t, its determinant with every root off the segment; the claim is by induction on the order. At order one the count is the side's, and the entry keeps its side along each piece of the cover at the priced side read (Lemma 32), the endpoint value clearing the piece's width against the derivative's coefficient fold at a bound containing the piece. At order N: inertia adds at a nonsingular principal pivot of order at most 2, and $S(t)$, its determinant of unequal members, has one at every t (Lemma 30). The principal minors of orders 1 and 2 with a coefficient of unequal members have finitely many roots; at each root point, and at the segment's ends, the unequal-membered determinant names a minor of unequal members there, whose own roots sit off that point at a distance a bracket certifies, so the segment is covered by finitely many subintervals of pair endpoints, each with one designated minor whose members differ throughout, consecutive ones overlapping at pairs. On one subinterval with pivot $P(t)$: the pivot's inertia is constant ($\det P$ keeps one side, the order-one read; at order two, the determinant on its lower side reads one unit per side, the mixed block, and on its upper side both diagonal entries keep one shared side, their roots off the subinterval, Lemma 30); and $(\det P)^2\, S_P$ has polynomial entries, the deflation cleared by the pivot's adjugate ($P\ \mathrm{adj}\, P = \det P \cdot 1$, Definition 3's minors), the positive square fixing every count, with $\det P \cdot \det\big((\det P)^2 S_P\big) = (\det P)^{2o}\,\det S$ at o the deflation's order ($k + o = N$; Lemma 30's split), its roots off the segment, so the induction reads S_P's count constant, and $\mathrm{rev}(S) = \mathrm{rev}(P) + \mathrm{rev}(S_P)$ is one integer on the subinterval. The overlaps chain the subintervals, so the count is one integer along the segment, and cells connected by such segments read one count. At a boundary point x^* beside a positive-semidefinite cell, a root below a gives u with the read $u^{\mathsf{T}}S(x^*)u$ on its lower side; along a segment into the cell, $t \mapsto u^\mathsf{T} S(x(t))u$ is a polynomial, on its lower side at the boundary point and so on it over an initial piece its coefficient folds name, putting a lower-side direction inside the cell: against the cell's count. $\square$

Lemma 32 (The located stage)**.** *The arithmetic. A located root of a polynomial over a located stage ($\mathbb{P}$, or a field built here, every element with its bracket) is a root ε of its squarefree part S (the division descent, Theorem 15), with an isolating bracket of balance-pair endpoints (Definition 1) off S's roots, narrower than half the separation of Theorem 15, whose derivation is stage arithmetic outright, division reads, coefficient folds and bracket comparisons. The stage $K(\varepsilon)$ is $K[x]$ read at ε: at $p \in K[x]$ with g the greatest common divisor of (p, S) (Theorem 15's descent), the value $p(\varepsilon)$ reads equal members exactly where g exchanges sides on the bracket (g a factor of the squarefree S, so its one possible root there is simple, and it is ε, the bracket isolating); at $p(\varepsilon)$ of unequal members the root sits on the cofactor S_1 at $g\,S_1 = S$, coprime to p since S is squarefree, and the Bézout read $u\,p + v\,S_1 = 1$ reads at the root to the unit witness, $p(\varepsilon)\,u(\varepsilon) = 1$. Sums and products are remainder reads, the division at S, a value's unit is the Bézout read's witness, and a value's side at the root is a count read (Definition 1), the bracket the certificate naming which root, every entry an elimination read of the stage below, so the adjunction stays grounded and the recursion ends at the ground pairs' own comparisons. At $p(\varepsilon)$ of unequal members the unit witness floors the value: at U_1 the coefficient fold of u over the bracket (Theorem 15's magnitudes), the witness identity read cross-multiplied floors the margin of $p(\varepsilon)$ at $\mathrm{margin} \cdot U_1 \geq 1$; and the priced side read: at bracket endpoints inside a stated bound's segment, the bound entering at a second datum beyond the sum's unit, the radius segment among them at the leading coefficient's margin, two bracket points' values of p differ by at most the bracket's width against the derivative's coefficient fold at the bound (Theorem 15's magnitudes, the binomial identity per occupied key), so an endpoint value of p clearing that price keeps its side at every bracket point, the root's among them: the endpoint side with the cleared comparison is the side read's certificate, p's image width over the bracket priced by the fold. The construction closes on itself: $K(\varepsilon)$'s elements each have their bracket (the isolating data, their image widths priced above) and its division reads, coefficient folds, endpoint evaluations and count reads are the display's, so $K(\varepsilon)$ is a located stage and adjunction iterates; a finite root list enters as a tower, one extension per root, each extension's polynomial over the stage the earlier extensions built, the extension order certificate data: a stored tower names its order, and every read runs through the arithmetic above.*

Lemma 33 (The pencil's split)**.** *For symmetric H and $G \succ 0$ of order N over a stage, with $Z(x)$ the level pair's site datum at $x\,G = H + Z(x)$ and $\chi(x) := \det Z(x)$,*

$$\chi(x) \;=\; \det G\, \prod_{j=1}^{N} \langle x \,{:}\, \varepsilon_j \rangle, \qquad \mathrm{count}(a) \;=\; \#\{\, j : \varepsilon_j < a \,\} \quad \textit{at } a \textit{ off the roots,}$$

the factors Definition 2's linear factors at the ε_j, located roots, and a unit congruence S_χ over the stage they generate maps the pencil to diagonal form,

$$S_\chi^\mathsf{T}\, Z(x)\, S_\chi \;=\; \mathrm{diag}\big(\langle x \,{:}\, \varepsilon_j \rangle\, g_j\big), \qquad g_j \textit{ on their upper sides.}$$

Proof. Over the pairs the counting function climbs to N from the positive-semidefinite read between the radius reads of Theorem 15, and its jumps enter as designations at the separation (Theorem 14). At a located root ε, the kernel of $Z(\varepsilon)$ over $K(\varepsilon)$ has a spanning list by elimination (Lemma 32), of some dimension $m \geq 1$, G-orthogonalized within, and its G-complement W another; at z in the kernel and $w \in W$, $z^\mathsf{T} Z(x)\, w = \langle x \,{:}\, \varepsilon \rangle\, z^\mathsf{T} G w$, of equal members at the complement's defining read, so the congruence by the two lists is block diagonal, the kernel block $\langle x\!:\!\varepsilon\rangle\, G_0$ at the kernel's Gram $G_0 \succ 0$ diagonal, and

$$\chi(x) \;=\; c\,\langle x\!:\!\varepsilon\rangle^m\,\chi_W(x), \qquad c \text{ on its upper side,}$$

χ_W the deflated pencil's. $\chi_W(\varepsilon)$ is of unequal members: a kernel vector of the deflated pencil at ε has its pairings against W and against the kernel both of equal members, so it is a kernel vector of the full pencil lying inside W, and the kernel meets W at the sum's unit alone. Induction on W splits χ over located roots, each deflation's root adjoined over the stage the earlier roots built (Lemma 32), and concatenates the per-step lists to S_χ, block diagonal for G and for H at once by the pairings above; the count identity is then the congruence read (Lemma 30): at a off the roots, the pair $(H : aG)$'s elimination datum is the displayed datum's balance partner, so $\operatorname{rev}(H : aG) = \operatorname{rev}\operatorname{diag}\big(\langle\varepsilon_j : a\rangle\, g_j\big) = \#\{j : \varepsilon_j < a\}$. The completed congruence returns the factorization read: the congruence identity's determinant collects the product read at the exchanged lists (Lemma 30; Definition 3) to $\det S_\chi^2\,\chi(x)$ against the diagonal blocks' product $\prod_j \langle x\!:\!\varepsilon_j\rangle\, g_j$; the identity's linear coefficient, $S_\chi^{\mathsf{T}} G S_\chi$ at the diagonal of the scales, collects its own determinants to $\det S_\chi^2\,\det G$ against $\prod_j g_j$, and at a located root's pair data the cleared factor enters at the scale, the diagonal's entry $g_j\,(d_j x - n_j)$ at the root $[\,n_j\!:\!d_j\,]$ with the linear coefficients collecting to $\prod_j d_j\, g_j$ and the clearing $\prod_j d_j$ riding onto the factorization's first member, one value at every representative (Definition 1); and the shared factor $\det S_\chi^2$, off equal members at the unit congruence, withdraws at the products' one value key by key (Definition 2): the displayed factorization is the congruence certificate's own read. ☐

Lemma 34 (Slab pivots: the factorization, the counts, and the Green products)**.** *Let S be symmetric over a located stage with a slab split $1..L$, entries at the diagonal and neighboring slab pairs alone, $A_i := S[i,i]$, $B_i := S[i,i+1]$, and the two pivot recursions at solved witnesses, each solve the adjugate column against the determinant,*

$$X_L = A_L, \quad X_i + B_i\, R_{i+1} = A_i \quad \textit{at} \quad X_{i+1} R_{i+1} = B_i^{\mathsf{T}};$$
$$Y_1 = A_1, \quad Y_{i+1} + B_i^{\mathsf{T}}\, C_i = A_{i+1} \quad \textit{at} \quad Y_i C_i = B_i,$$

at pivots of unequal-membered determinant (a singular slab pivot re-orders within its slab by Lemma 30's principal pivots of order at most two, the count unchanged). Then: (i) the Schur complement of the tail $[i{+}1, L]$ onto $[1,i]$ maps A_i to X_i and fixes every other block: eliminating slab L touches A_{L-1} alone, the entries at $i < L-1$ against L off the band, and downward induction extends it, so one recursion datum holds the whole tail, and every pivot is symmetric: the seed is a diagonal block of the symmetric datum, and at a symmetric X_{i+1} the witness identity transposed reads $B_i = R_{i+1}^{\mathsf{T}} X_{i+1}$, so the withdrawn term $B_i R_{i+1} = R_{i+1}^{\mathsf{T}} X_{i+1} R_{i+1}$ is symmetric with X_i, the induction downward from the seed; (ii) $\operatorname{rev}(S) = \sum_i \operatorname{rev}(X_i)$ and $\det S = \prod_i \det X_i$, each elimination a unit-triangular congruence (Lemma 30); (iii) with Z_j the join's datum at $Z_j + A_j = X_j + Y_j$, read off the two recursions, the solution of $S\,u = v$ at v supported in slab j reads blockwise, at $i \le j$,

$$Z_j\, u_j = v_j, \qquad u_i = T_i\, T_{i+1} \cdots T_h\, u_j \quad \textit{at} \quad h + 1 = j\ ,$$

at the transfer factors T_k, each the memberwise swap of the witness C_k, so $Y_k\, T_k + B_k$ is the sum's unit: every Green block is a product of transfer factors and one corner solve.

Proof. (i) and (ii) are the displayed inductions and congruences. (iii): solve $Su = v$ at v supported in slab j. The rows above j read $X_{i+1} u_{i+1} + B_i^{\mathsf{T}} u_i$ at the sum's unit, the tail closing from L

downward as in (i); the rows below read $Y_i\,u_i + B_i\,u_{i+1}$ at the sum's unit at every i below j, the induction from 1 on the Y recursion; and substituting both into row j leaves $Z_j\,u_j = v_j$. The display telescopes the descent, and at $L = 2$ it is the two-slab solve outright. □

Lemma 35 (The spectator law)**.** *Let S, S' be two families as in Lemma 34 on one slab split, with equal data beyond slab w: $A'_i = A_i$ at $i > w$ and $B'_i = B_i$ at $i \geq w$. (i) One sandwich identity. The tail pivots are equal outright, $X'_i = X_i$ at $i > w$, the recursion running on equal data from an equal seed; the deviation sits in the boundary-direction pivots alone, and at $i \geq w$ the deviation datum D_i at $Y'_i = Y_i + D_i$ obeys the exact sandwich*

$$D_{i+1} \;=\; T_i^{\mathsf{T}}\, D_i\, T'_i$$

at the two families' transfer factors (Lemma 34(iii)), the two recursions' joins at equal A, B read against each other, telescoping to a product form: the deviation at depth i beyond the perturbation is D_{w+1} sandwiched by the transfer factors of Lemma 34(iii). Re-convergence of the recursion and the Green products' falloff are the one sandwich identity, stated twice. The identity reads at the reversed slab order at a stated diagonal tie besides: at two families with equal couplings below slab w, $B'_i = B_i$ at $i < w$, and the diagonals tied there at one datum, $A'_i = A_i + \delta\,1$ at $i < w$, the tail deviation E_i at $X'_i = X_i + E_i$ obeys, at $i < w$,

$$E_i \;=\; \check{R}_{i+1}^{\mathsf{T}}\, E_{i+1}\, \check{R}'_{i+1} \;+\; \delta\,1$$

at the tail witnesses' transfer factors $\check{R}_{i+1}$ and $\check{R}'_{i+1}$, the memberwise swaps of R_{i+1} and R'_{i+1} (Lemma 34(iii)): the join $X_i + B_i R_{i+1} = A_i$ read against its primed copy at the equal couplings and the tie collects the deviation to $B_i\,(R_{i+1} + \check{R}'_{i+1})$ beside the tie's datum, the witness identity $X_{i+1}R_{i+1} = B_i^{\mathsf{T}}$ transposed at the symmetric pivot ((i)) reads $B_i = R_{i+1}^{\mathsf{T}} X_{i+1}$, and the two witness identities at the pivots read $X_{i+1}(R_{i+1} + \check{R}'_{i+1}) = E_{i+1}R'_{i+1}$, so $E_i = R_{i+1}^{\mathsf{T}} E_{i+1} R'_{i+1} + \delta\,1$, the two swaps withdrawn in pairs and the tie's datum riding each step; at the tie's datum of equal members the families' data are equal below the slab, the head pivots equal outright at $i < w$ and the display the sandwich $E_i = R_{i+1}^{\mathsf{T}} E_{i+1} R'_{i+1}$; the deviation at depth i below the perturbation is E_w sandwiched down to the head with the tie's datum entering once per step, and (ii)'s caps propagate along it at the tail factors' certificates. (ii) Caps propagate geometrically. The gram's slab blocks are $G_i := G[i,i]$, the cross blocks of equal members at the index's orthogonal sum (Definition 8), each display's gram its own slab's block. At certified pairs λ_i, λ'_i with $T_i^{\mathsf{T}} G_i\, T_i \preceq \lambda_i^2\, G_{i+1}$ and $T_i'^{\mathsf{T}} G_i\, T'_i \preceq \lambda_i'^2\, G_{i+1}$ at the two families' transfer factors, a cap D_i two-sided at $c_i\, G_i$ (Lemma 30) propagates to $2\,D_{i+1}$ two-sided at $c_i\,(\lambda_i^2 + \lambda_i'^2)\, G_{i+1}$: at $u := T_i x$ and $v := T'_i x$, the cap's two splits at $u + v$ and at $u + \check{v}$, $\check{v}$ the memberwise swap, collect to $2\,u^{\mathsf{T}} D_i v$ capped two-sidedly at $c_i\,(u^{\mathsf{T}} G_i u + v^{\mathsf{T}} G_i v)$, and the certificates price the two squares. Iterating, the bracket at depth i is named and geometric. At the tied families ((i)'s reversed display) each step joins the tie's datum, $D_{i+1} = T_i^{\mathsf{T}} D_i T'_i + \delta\,1$, and at the datum's identity capped two-sidedly at $s\,G_{i+1}$ the display gains its read, $2\,D_{i+1}$ two-sided at $c_i\,(\lambda_i^2 + \lambda_i'^2)\, G_{i+1} + 2\,s\,G_{i+1}$, the two splits' own polarization at the identity's datum; iterating, the bracket is the seed's geometric read joined to the tie datum's fold over the steps' certificates, one summand per step. (iii) The bulk pivot is the deck-interior solvent. At bulk data $A_i = A$, $B_i = B$ at B of unequal-membered determinant, read in a congruence mapping G to the identity (G's factorization a stage datum, comparisons then the spectral reads of Lemma 33), the deck polynomial p, the determinant of the quadratic pair $(B^{\mathsf{T}} z^2 + B : Az)$'s site datum, obeys the deck symmetry, its coefficients equal at every index pair joining to $2m$ (A symmetric, the transpose identity exchanging B with B^{T}), the display of its deck symbol (Lemma 38), and a solution pair (Y, X) of the pivot *identities $YX = B$ and $Y + B^{\mathsf{T}}X = A$ has its solvent X satisfying $B^{\mathsf{T}}X^2 + B = AX$, and the pair's site datum factors exactly,*

$$U(z) \;=\; U_1(z)\,U_2(z) \qquad \text{at} \quad B^{\mathsf{T}}z^2 + B = A\,z + U(z), \quad zB^{\mathsf{T}} + B^{\mathsf{T}}X = A + U_1(z), \quad z = X + U_2(z)\,,$$

so X's polynomial is a factor of p (the adjugate identity reading Cayley–Hamilton). Off the band (Lemma 38's band read, the deck relation's crossing locus), $p = c\,p_{\mathrm{int}}\,p_{\mathrm{ext}}$ over a stored tower; the interior kernel $V := \ker p_{\mathrm{int}}(C)$ at the companion action C, the shift $(x, y) \mapsto (y, y')$ at its second component's witness $B^{\mathsf{T}}y' + Bx = Ay$, has a spanning list by elimination over the tower, each companion step one solved read at B^{T}, stacked $\left[\begin{smallmatrix} V_1 \\ V_2 \end{smallmatrix}\right]$, and the graph read $\det V_1$ is one tower sign. Where it reads unequal members, X_* at $X_*V_1 = V_2$ satisfies the solvent identity with polynomial p_{int} and Y_* at $Y_*X_* = B$ solves the pivot identities; and every solution whose solvent polynomial is p_{int} equals it, its graph filling the one kernel V. Three counts at a located symmetric $\hat{Y}$ and pairs $\lambda < 1$, y_0, ρ, ρ_0, at the pivot map $Y \mapsto Y^\sharp := A + B^{\mathsf{T}}T$ over the transfer factor T, the memberwise swap of the witness at $Y\,C = B$, so $Y\,T + B$ is the sum's unit (Lemma 34(iii)),

$$\hat{T}^{\mathsf{T}}\hat{T} \preceq \lambda^2, \qquad \hat{Y} \succeq y_0 + \rho, \qquad D_\circ \text{ two-sided at } \rho_0 \quad \text{at} \quad \hat{Y}^\sharp = \hat{Y} + D_\circ,$$

with $\lambda_\circ$ at $y_0\,\lambda_\circ = \lambda\,(y_0 + \rho)$, $\mu := \lambda_\circ^2 < 1$ and $\rho_0 + \mu\rho \le \rho$, make the map send the ball at deviation data D two-sided at ρ, $Y = \hat{Y} + D$, to itself and contract its deviations at modulus μ: a ball member's certificate is $\lambda_\circ$, its transfer factor priced through the floor by the identity $Y\,D_T = D\,\hat{T}$ at $\hat{T} = T + D_T$, the image deviation is (i)'s sandwich capped by (ii)'s certificates, and the center's displacement tops up to ρ. A ball member prices its form at the floor: the deviation's cap withdraws against the center's floor at the gram's read, the member's form at or beyond y_0's multiple of the gram's at every vector, so every split of a ball member reads the upper side throughout (Lemma 30). The graph read joins the validity data: where it reads unequal members, the bracket comparison D_* two-sided at ρ at $Y_* = \hat{Y} + D_*$ is one count at the tower datum, and where it holds Y_* is the ball's one solution of the pivot identities: two ball solutions read their deviation datum D through (i)'s sandwich, the two-sided cap at $c\,G$ returning at $\mu\,c\,G$ by (ii) and iterating below every pair, and a located entry below every pair is of equal members; the solvent is capped inside by the first count ($X^{\mathsf{T}}X \preceq \lambda_\circ^2$ prices every polynomial root's square below 1), so its polynomial is p_{int} and its graph fills the one kernel V. The graph read's side and the three counts are each cell data of the family's parameters (Lemma 31; Lemma 32). The scalar instance, the walk pencil's bulk data at $B = 1$: the boundary entry solves the pivot identity $m\,Y = 1$, the solvent, $p_{\mathrm{int}} = \langle z : z_* \rangle$ at the interior member z_*, and (i)'s sandwich is the error law $e_{L+1} = z^2 e_L + z\,e_L\,e_{L+1}$, contraction factor z^2, marginal exactly at the crossings, $z^2 = 1$ there.

Theorem 15 (The window is a root separation, and is explicit)**.** *For a polynomial P over $\mathbb{P}$ (Definition 2), or over a located stage, every step below one of its operations, a division, a coefficient fold or a bracket comparison: the stage's distinct roots have a computed pair separation floor and a computed radius, by ground arithmetic alone.*

The magnitude carrier. The magnitude of a balance pair is its members' larger against smaller, the sorted pair $|\langle u : v \rangle| := \langle\, \max\{u, v\} : \min\{u, v\}\,\rangle$ at the trichotomy read, one value at every pair (a summand on both members fixing it), its value at unequal members the margin, the exceeding member's gap (Definition 1). A sum's magnitude sits at or below the magnitudes' sum, and a product's magnitude is the magnitudes' product, the members' trichotomies read case by case with a memberwise swap of either datum swapping the product's members and fixing every magnitude. So a polynomial's evaluations at points of magnitude at or below a bound are capped by the coefficient fold, the coefficients' magnitudes folded against the bound's powers, one read over the occupancy family.

The squarefree part, and the radius. The division descent, iterated divisions of Definition 2, computes the greatest common divisor $c(P, P')$: the descent's last occupied remainder, monic at the rescaling by its leading coefficient (each coefficient the cofactor at the leading one), a factor of P and of P' (each remainder joins the two above it through its division, so the last is a factor of every earlier one, the two seeds among them), and greatest as named: a common factor g of the two seeds is a factor of every remainder down the descent, $P = SQ+R$ at the cofactors $g\,P_1 = P$ and $g\,S_1 = S$ reading $R = g\,(P_1 + \check{T})$ at $T = S_1Q$, the check the memberwise swap over the balance pairs, the last remainder among them. At the integer representatives the rescaling enters as data: the divisor at its primitive list with the stated positive top, the cofactors at their clearing multipliers, and the division at a stated top the cleared descent, $\ell^{\,k} P = q\,D + r$ at the top ℓ, one scale per eliminated key, the exactness the remainder's read; each remainder strips to its primitive list at its coefficients' one shared count, the walk's rows at their stated multipliers with the strip absorbed there, and every certificate reads at the cleared data. The squarefree part S is the cofactor at $c(P, P')\,S = P$, and the Bézout pair $uS + vS' = 1$ arrives with the descent at (S, S') ending at a unit: an irreducible factor q of S has some multiplicity $m \geq 1$ in $P = q^m R$, the remainders of q' (the derivative's degree drop) and of R (the exact multiplicity) at q occupied and their Bézout witnesses composing to an occupied remainder of $m\,q'R + q\,R'$ at the count m a natural, $q\,R'$'s summand the sum's unit there; at $m = 1$ the derivative is that read outright, $P' = m\,q'R + q\,R'$, and at the join $m' + 1 = m$ it is $P' = q^{m'}(m\,q'R + q\,R')$ at multiplicity exactly m': the greatest common divisor's q-part sits one power below P's, the unit at $m = 1$, so the cofactor S reads q at one, and at $S = qT$ the remainders of T and of $S' = q'T + qT'$ at q are occupied: P's roots are S's, each simple there. Every root over the stage sits in the radius segment, between B's balance partner and B at the pair $B := [\,c + \mathcal{H} : c\,]$, c the leading coefficient's margin (the top key's coefficient of unequal members, the degree's own read) and $\mathcal{H}$ the height, the largest coefficient magnitude: at a point of margin $y = 1 + g$ at or beyond B, with k the top key, the geometric telescope $g \sum_{i<k} y^{\,i} + 1 = y^{\,k}$ reads the tail's cap $\mathcal{H} \sum_{i<k} y^{\,i}$ below $c\,y^{\,k}$ at the radius' read $c\,g \geq \mathcal{H}$, so the leading term dominates and the evaluation reads unequal members on the leading term's own side, the tail joined at a magnitude below the term's.

The separation. At a root α the unit witness $v(\alpha)\,S'(\alpha) = 1$ reads at the magnitudes: V the coefficient fold of v at the radius, so the margin of $S'(\alpha)$ cross-multiplies to $\mathrm{margin} \cdot V \geq 1$*. Between two roots over the stage, at the stage gap G at $\alpha + G = \beta$, the binomial identity per occupied key at $i \geq 2$, the constant key's read equal at both points and the key-one read the gap's own join $\beta = \alpha + G$,*

$$\beta^{\,i} \;=\; \alpha^{\,i} + i\,\alpha^{\,h}\,G \;+\; G^2 \sum_{a+b+2=i} (a{+}1)\,\alpha^{\,a}\beta^{\,b} \qquad (h + 1 = i),$$

summed against the coefficients, reads $S(\beta) = S(\alpha) + G\,S'(\alpha) + G^2\,T$ at the collected tail $T := \sum_i s_i \sum_{a+b+2=i} (a{+}1)\,\alpha^a\beta^b$; the two roots' equal-membered reads leave $G\,(S'(\alpha) + G\,T)$ of equal members, so at G's unit witness $S'(\alpha)$ and $G\,T$ are balance partners, and the margin of $S'(\alpha)$ sits at or below $G\,\Lambda$ at the curvature fold $\Lambda := \sum_i \binom{i}{2}\,|s_i|\,B^{\,e}$ over the occupied keys at the joins $e + 2 = i$, $|s_i|$ the coefficient's magnitude, the splitting count $\sum_{a+b+2=i}(a{+}1) = \binom{i}{2}$ and both roots capped by the radius. So the stage's distinct roots satisfy

$$\Lambda\,V\,G \;\geq\; 1\;, \qquad 2\,\max\{\Lambda\,V,\; 1\}\;\delta \;=\; 1\;,$$

one pair comparison and one cofactor per polynomial (Definition 1), δ below the separation and at most $[\,1 : 2\,]$; the radius B is the leading-term bound. The object is a distance between roots of ground polynomials, and such a distance is a separation bound, which is explicit.

Lemma 36 (The stage splits every polynomial)**.** *Every polynomial of degree at least* 1 *over a located stage splits, over a stored tower (Lemma 32), into located linear factors and definite quadratic factors at pair data* (h, q)*, the root relation* $z^2 + q = h\,z$*, at the definiteness gap* g *at* $h^2 + g = 4q$*, the data tower elements, multiplicities by iterated divisions.*

Proof. Factorization closes over finite cofactor lists. Over $\mathbb{P}$ a polynomial reads at its primitive integer form, the pair coefficients cleared at one shared second member with the content collected, and a factor reads primitive as well, the contents splitting off (Gauss); a primitive factor of degree d is determined by its values at $d+1$ integers of occupied value, a point of equal-membered value reading its linear factor outright, and each value is one of the finitely many cofactor reads of the primitive form's value there at either side, a natural or its balance partner: the value lists are finite, each selection interpolates to one polynomial, and the division tests it, the remainder of equal members exactly at a factor (Kronecker). Over an extension $K(\varepsilon)$ the norm $\mathrm{Res}_x\big(S_\varepsilon(x), f\big)$ (Definition 3's resultant, in the variable x) maps f to the stage below; at a shift $z \mapsto z + \lambda\varepsilon$ whose norm is squarefree (the failing λ are roots of one resultant, finitely many, a scalar-pair λ off them read directly), the greatest common divisor of f with each factor of the shifted norm, the division descent's end, recovers f's factors over the extension (Trager), descending to $\mathbb{P}$. The remainder lists at an irreducible factor (Definition 2) are a discrete field, every occupied element at its Bézout unit witness, so iterating the adjunction reads a full formal root list, the *splitting data*: finite algebra over the stage, in which a root of a product is a root of one factor.

The quadratic extensions. A located positive c adjoins one extension, the remainder lists at the relation $z^2 = c$'s polynomial (Definition 2); the *adjoined root at* $z^2 = c$ is the linear monomial's class, located at the bracket of c's sorted display against 1, the two side reads the refinement's seed (Theorem 14(1)) with the equality outcome naming the root outright, so the extension is a located stage again (Lemma 32). The *located pairs* are the remainder lists at $z^2 + 1$ (Definition 2), (a, b) written $a + ib$ at i the linear monomial's class, the *key swap* the linear key's memberwise swap, an automorphism of the carrier's reads. A located pair at b of unequal members has a square root $u + iv$ on three such extensions: w the adjoined root at $z^2 = a^2 + b^2$, then over $K(w)$ u the adjoined root at $z^2 = [\,1 : 2\,]\,(w + a)$ and v the adjoined root at $z^2 = [\,1 : 2\,]\,g$ at the gap g at $a + g = w$, the order's witness (w^2 exceeds a^2 at b's occupied square, so w exceeds a and its balance partner), v's side one bracket read against b's; the tower reads $u^2 = v^2 + a$ and $(2uv)^2 = b^2$ outright, so $(u + iv)^2 = a + ib$. A quadratic over located pairs at the root relation $z^2 + q = hz$ splits over located pairs, the trichotomy of its *discriminant datum* D at $D + 4q = h^2$: at D's second component of unequal members the extensions read its square root $u + iv$, the two roots the data at $2z = h + (u{+}iv)$ and $2z + (u{+}iv) = h$; at the second component of equal members the first component D_1's trichotomy reads: on its upper side, e the adjoined root at $z^2 = D_1$ with the roots the data at $2z = h + e$ and $2z + e = h$; at equality, the double root, the datum at $2z = h$; and on its lower side, e the adjoined root at $z^2 = m$ at D_1's margin m, the roots the data at $2z = h + e\,i$ and $2z + e\,i = h$.

The Laplace descent. The claim: a stage polynomial p of degree $k \geq 1$ has a located root or a definite quadratic divisor, verified by the division. Induct on the degree's 2-adic valuation ν. At odd degree the leading term dominates beyond Theorem 15's radius, the side reads at the radius and at its balance partner differ, and a located root's designation follows at every width, Theorem 14(1)'s refinement at the entry's side reads. At positive ν, at a scalar pair t the resolvent

$$B_t(x) \;:=\; \prod_{l<l'} \big\langle\, x \;:\; z_l + z_{l'} + t\, z_l z_{l'} \,\big\rangle$$

over p's formal root list has symmetric coefficients, hence stage coefficients through the elementary reads (the lexicographic descent on symmetric monomials, one generic identity), and degree $\binom{k}{2}$ at the valuation join $\nu' + 1 = \nu$: the induction hands each t a located root or a definite divisor, either way a located pair β_t (the second component of equal members, or the adjoined root at $z^2 = [\,1 : 4\,]\, g$ at the definiteness gap g), a witness-verified root of B_t. Take the splitting data over the one tower with every β_t and i: there B_t is the displayed product and β_t, a root of B_t in a field, is one factor's own root: β_t meets a pair (l, l'). Over $\binom{k}{2} + 1$ distinct scalar pairs t two meet one pair, and there, the lesser parameter written t' at the gap w at $t' + w = t$,

$$w\, q + \beta_{t'} \;=\; \beta_t\,, \qquad h + t\, q \;=\; \beta_t$$

name the located pairs q and h, each one value by the positive gap's rescaling, the displays' own joins at the meeting pair ($\beta_t = z_l + z_{l'} + t\, z_l z_{l'}$ at both parameters), with the relation $z^2 + q = hz$'s quadratic a factor of p; the finitely many parameter pairs are each tested by the division, and the display guarantees a passing test. At a passing pair the discriminant trichotomy splits the quadratic, giving a root $a + ib$ of p with a, b tower elements. At b of equal members, a bracket read, a is a located root; at b of unequal members the key swap fixes p and maps the root to its swap image, the division of p at $z^2 + (a^2 + b^2) = 2az$ reads its remainder $Rz + T$ with $Ra + T$ and Rb of equal members at the two roots, so R and T read equal members at b's unit witness: a definite divisor. Iterated divisions close the split. □

Lemma 37 (The located sign is a count)**.** *Over a located stage K, let S be squarefree of degree N with its split over a stored tower (Lemma 36). A stage value at a root is a balance pair, its side the members' order read, the upper at the first member exceeding and the lower at the second (Definition 1); the pairing below is additive over a value's site, $H_u = H_v + H_p$ at $u = v + p$, so the weight p is the site datum and the side is its read. The carrier is S's remainder lists (Definition 2), length-N coefficient lists with the convolution's remainder as product, and M_f, the remainder multiplication at a polynomial f, is the $N \times N$ matrix whose column j is the remainder of $f\, x^j$. At a weight $p \in K[x]$ the Hermite pairing H_p is the symmetric form*

$$H_p[i, j] \;=\; \operatorname{tr} M_{x^{i+j} p}\,, \qquad \textit{the keys } i,\, j \textit{ below } N,$$

tr *the matrix trace, every entry one elimination read over K (Definition 3).*

(i) The trace adds over the split's blocks. The split's factors are coprime (S squarefree), their Bézout reads yield idempotent remainders, e^2 and $\mathrm{e}\,\mathrm{e}'$ reading e and equal members, and the CRT list is a unit congruence's base change mapping every M_f to blocks, one per factor, each the factor's own remainder multiplication; the trace reads one value at every spanning list ($\operatorname{tr} AB = \operatorname{tr} BA$*), so it adds over the blocks: a linear factor's block reads $f(\varepsilon_i)$, and a definite factor's block reads f's remainder $\alpha + \beta x$ there at $2\alpha + \beta h$, M_1 the identity and* $\operatorname{tr} M_x = h$ *at the factor's data (h, q), its companion trace and constant (Lemma 36).*

(ii) The pairing splits, and a block is one orientation read. Cross-block entries are traces of actions by $\mathrm{e}\,\mathrm{e}'$, whose remainder is the sum's unit, so the CRT list is a unit congruence of H_p onto the blocks' forms over the tower, the reversal count and the rank preserved (Lemma 30, the rank the kernel block's complement; a stage split is a tower split, so the tower's count is the stage's own). A linear block at ε_i is the one entry $p(\varepsilon_i)$ at a list member's square: at $p(\varepsilon_i)$ of unequal members it is one reversal exactly where the value's side is the lower, the reversal count at order one and the stage's own comparison; at p on the root it is the kernel block. A definite block, at its factor's data (h, q) and the definiteness gap g at $h^2 + g = 4q$ (the definite read of Lemma 36), is 2×2, and it reads one unit per side, the mixed block, exactly where its mixed entry's square exceeds its diagonal

entries' product (Lemma 30's 2×2 read); it does so exactly at p off the factor, by one additive identity of the block's traces at p's remainder $\alpha + \beta x$ there,

$$4\,\mathrm{tr}(M_{xp})^2 \;=\; 4\,\mathrm{tr}(M_p)\,\mathrm{tr}(M_{x^2p}) + \big((2\alpha + h\beta)^2 + g\,\beta^2\big)\,g,$$

one quadratic-form identity in the remainder (each side quadratic, $\mathrm{tr}\, M_{x^kp}$ linear in it), settled at the three remainders 1, x and $1+x$ by (i)'s companion rule: the excess is a square with the gap's multiple of a square, positive by its shape at a remainder off the sum's unit, and at p on the factor the block is the kernel block, every entry the sum's unit action's trace.

(iii) The bracket's pencil pairs the reads, and the counts close. At a located root ε with isolating bracket (l, l') of balance-pair endpoints off S's roots, its width w at $l + w = l'$, and $p(\varepsilon)$ of unequal members, the localized read is the form pair

$$\mathsf{B} \;:=\; \big(\,H_{x^2p} + l\,l'\,H_p \;\;:\;\; (l + l')\,H_{xp}\,\big),$$

the bracket's pencil at the three pairings, a balance-pair coefficient reading cross-added with its second member's multiple on the pencil's other side (Definition 1), its reversal count and rank its split's (Lemma 30). The three pairings admit one CRT congruence (the blocks' multiplication matrices are shared), so the pencil splits per factor. At a linear factor the datum is $p(\varepsilon_i)$ times the bracket comparison at ε_i, of $\varepsilon_i^2 + l\,l'$ against $(l + l')\,\varepsilon_i$; the sides' gap is one gap product at every position, $d\,(w + d)$ for the first side at $t = l' + d$ or $t + d = l$, and $c\,c'$ for the second at $t = l + c$, $c + c' = w$: above exactly off the closed bracket, below exactly inside. So the block at ε reads its one reversal exactly at p on its upper side at ε, and at every other located root its reversal read is p's block's own. At a definite factor the pencil's block is (ii)'s at the shifted remainder, the kernel block exactly with p's: a common factor with the bracket's own data would read the sum $g + w^2$ at the sum's unit at the endpoints' pair ($(l + l')^2$ against $4\,l\,l'$ at the width's square), or a stage root t, whose equality $4t^2 + h^2 + g = 4ht$ reads $d^2 + g$ of equal members at $2t = h + d$ or $2t + d = h$, against the summands' shapes both times; so the shift is coprime to the factor, its Bézout read mapping p's remainder to the shifted one by a unit. The bookkeeping closes on two natural identities:

$$p \text{ sits on its upper side at } \varepsilon \text{ exactly at} \quad \mathrm{rank}\, H_p + 2\,\mathrm{rev}\,\mathsf{B} \;=\; \mathrm{rank}\,\mathsf{B} + 2\,\mathrm{rev}\, H_p + 2,$$

and on its lower exactly at $\mathrm{rank}\, H_p + 2\,\mathrm{rev}\,\mathsf{B} + 2 = \mathrm{rank}\,\mathsf{B} + 2\,\mathrm{rev}\, H_p$: every other located root at p's support adds one rank to each and equal reversal reads, every definite factor off p two ranks and one reversal to each, the kernel blocks off both counts, and ε adds one rank to each with the pencil's reversal at p's upper side and H_p's at its lower. At the unit weight and any endpoint pair off S's roots the split reads the segment count: every located root inside the closed segment adds one reversal to B beyond H_1's, every root outside and every definite factor equal reversals to both, so the count is the gap at $\mathrm{rev}\, H_1 + \# = \mathrm{rev}\,\mathsf{B}$, the isolating bracket the one-root instance; at a polynomial off the squarefree site the three pairings read its distinct roots, the multiplicities entering as positive weights at the roots' moment folds, so the segment count reads the distinct roots inside, the squarefree part's own count. With the entries elimination data over K and the two counts one split each, a located sign is finitely many reads of the stage below, descending to the ground pairs' own comparisons; Jacobi's minor rule is the read's nonsingular instance (Lemma 30).

Lemma 38 (The deck symbol, and the interior factor)**.** *A polynomial p over a located stage with the deck symmetry, its coefficients equal at every index pair joining to $2m$ and its keys beyond $2m$ at the sum's unit, and its unit-monomial coefficient of unequal members is the display of its deck symbol, a stage polynomial P of degree m in the deck's spectral variable:*

$$p \;=\; \sum_{i+j=m} P_j\; z^{\,i}(z^2 + 1)^j,$$

each deck monomial its own coefficient join ($z^i(z^2+1)^j$ reading one at the top pair $i+2j$ against i, equal ends joining to $2m$), the map from P's coefficients to p's top coefficients unitriangular, the symbol one exact back-substitution, and the involution the identity on the symbol. A root pair of the deck relation $z^2+1=wz$ (Definition 4) at a located w-datum is one quadratic extension (Lemma 36), the two roots at product one, the constant coefficient, and the interior member is the one at $z^2 \le 1$, one pair comparison, strict off the crossing values, the relation's double-root values at $w^2=4$, the interior margin the crossing read's own, $\langle 1 : z^2\rangle^2 = z^2\,\langle w^2 : 4\rangle$, the relation's square at the sum $z^4+1+4z^2 = w^2z^2+2z^2$; a definite w-datum's roots enter over its pair's quadratic extensions, the interior member at the modulus comparison. The band read is P's segment count at the crossing endpoints: the bracket pencil of Lemma 37 at those endpoints reads every root of P off the closed crossing segment, reversal-count reads throughout, with the endpoint reads P's own, off equal members. At an integer display off the monic carrier the count runs at the cleared variable: the display scales to the monic at the top's magnitude, each coefficient against the magnitude's power at its key's gap below the top's predecessor with the members swapped at a top on its lower side, and the scaled display reads its roots at the display's own rescaled by the magnitude, the display's evaluation and the cleared one composing by formula at the homogeneity principle (Definition 1): the segment count reads at the crossing endpoints rescaled to the doubled magnitude. With the band read clear, over P's split at the stored tower (Lemma 36),

$$p \;=\; c\, p_{\text{int}}\, p_{\text{ext}}\,,$$

p_{int} the monic product over the located interior members, one per w-datum at its multiplicity, degree m, existence by construction, and p_{ext} the product over their ι-images, the pairing $z\,z'=1$ per extension: on the relation each symbol monomial reads $z^i(z^2+1)^j = z^m w^j$, so p is z^m against the symbol's read at w; each split factor $\langle w : w_k\rangle$ clears at z to the root pair's monic quadratic, $z\,\langle w : w_k\rangle$ reading z^2+1 against $w_k z$; so the pair quadratics' product against c, the symbol's leading coefficient and the display's unit-monomial read, is p: the degrees split evenly, m each. The definite factors are the point: they enter at their pair data, and every side is Lemma 37's count.

Theorem 16 (The pencil is a symmetric pair matrix on every window)**.** *Every level read the development takes is on a window compression, finite-dimensional with pair entries (Proposition 2): E is diagonal in the label index (Definition 10) and M is multiplication by the involution-fixed $\sum_p \chi_\theta(U_{\partial p})$ (Lemma 16) against the pairing identity of Proposition 5, hence symmetric, with* gram *positive definite by construction (Definition 8). So the pair $(\alpha E : \beta M)$'s site datum $H([\alpha : \beta])$ is symmetric there at every end-weight pair, its entries pair data at pair weights.*

19 The chain's invariant form

Theorem 17 (The fiber family is a bounded chord pencil)**.** *At the banded rows, every pencil entry within the locality band of its support (Lemma 14), and the module–pair coupling $|v|^2 = 2+t$ at the chord t of the translation, its deck relation $T^2+1=t\,T$ (Definition 4), the momentum-fiber pencil is a chord pencil of bounded degree per grade: the fiber enters each graded block through finitely many translation powers per direction, T_i^j at the direction's own translation, the power bounded by the grade's spatial support, each transpose-paired read a deck-family polynomial at the direction's chord (Definition 4): the momentum variables are the chords $t_1,\dots,t_d$, one per direction, the momentum point K their list (Lemma 39). The chain object is therefore one algebraic family over the $\big(K;[\alpha : \beta]\big)$ base at the chords, with integral chord data; its divisor records the band, the gap, and every "assembled" coefficient as reads at the ends of $\mathcal{C}$. Its fiber symbol is the family's characteristic*

Laurent polynomial $\mathcal{S}(K;k,r)$, the pair $(k:A(K))$'s site-datum determinant at the level k scalar, of finite Newton polytope by the localities above, with polynomial-pair coefficients in the residue; A is the momentum fiber of a symmetric pair pencil (Theorem 16), so a separation's block is the reversed separation's transpose, the symbol's own deck symmetry.

Lemma 39 (The window floor fibers over momentum)**.** *On a torus window of odd side L, the fibering's family the odd tori at $L \geq 5$ and the commuting and chord reads below at every odd side, the translations (Construction 7) commute with H (E and M run over all links and plaquettes at coefficient one, Definition 10): a translation T permutes the label index isometrically, an orthogonal map, its transpose the inverse permutation's matrix (Construction 1), with $T^L = 1$ along its direction. Its chord operator is $S := T + T^{\mathsf{T}}$, symmetric, commuting with H and with the gram, and T reads the deck relation at it,*

$$T^2 + 1 \;=\; S\,T,$$

the product $T^{\mathsf{T}}T = 1$ collected (Definition 4). The geometric word reads its deck symbol at the operator,

$$\sum_{a+b+1=L} T^{\,b} \;=\; T^{\,m}\,P_L(S) \qquad at \quad 2m+1 = L, \qquad P_L := 1 + \sum_{j \leq m} p_j,$$

P_L monic integer of degree m at the deck families, the partial sums keeping the top at the last family joined (Definition 4; Lemma 38), the powers matched about the middle collecting at the plus read $T^{\,j} + (T^{\mathsf{T}})^{\,j} = p_j(S)$, the root pair (T, T^{T}) of product one. Write $X := T^{\,m}P_L(S)$ for the word: at $T^L = 1$ the word is its own T-multiple, $T\,X = X$, and T^{T} against that read gives $T^{\mathsf{T}}X = X$, so $S\,X = 2\,X$, and the orthogonal $T^{\,m}$ clears at its witness $(T^{\mathsf{T}})^{\,m}$:

$$S\;P_L(S) \;=\; 2\,P_L(S).$$

Divided at the monic $\langle x:2\rangle$, P_L leaves its Horner read there, $P_L = q\,\langle x:2\rangle + L$ (Definition 2): p_j at the chord 2 reads 2 at every key, the recursion's own read from the seeds, so P_L at 2 reads $1{+}2m = L$, the word's value on the unit fiber, the fixed vectors' block: at the datum D_1 of $1{+}D_1 = T$ the product $T^{\mathsf{T}}T = 1$ expands to $1 + D_1 + D_1^{\mathsf{T}} + D_1^{\mathsf{T}}D_1 = 1$ and the chord to $S = 2 + D_1 + D_1^{\mathsf{T}}$, the two joining at $D_1^{\mathsf{T}}D_1 + S = 2$, so at $S\,v = 2\,v$ the read $\langle D_1 v, D_1 v\rangle$ is the sum's unit and $D_1 v$ with it at the positive pairing (Definition 8): membership in the kernel of the pair $(S:2)$'s site datum is the fixed read $T\,v = v$. That value is occupied at the odd side; the division display is the Bézout read, and the two factors of the split polynomial $\langle x:2\rangle\,P_L(x)$ are coprime. So the window splits orthogonally into the chord blocks: the kernel of the pair $(S:2)$'s site datum, its members the reads $S\,v = 2\,v$ at the join, and the kernels of the pair $(S:t_j)$'s site data at P_L's roots t_j, the display above reading $\langle x:2\rangle\,P_L(x)$ an annihilating polynomial of S. The roots are located and the factorization linear: S is symmetric against the positive gram (Definition 8), so its site-datum determinant splits over located roots with a unit congruence to diagonal (Lemma 33), the located linear factorization the lemma's own display, and P_L's roots are the occurring ones at that annihilation; each root is separated and bracketed by ground arithmetic (Theorem 15) and located on the stage (Lemma 32), the tower the split stores Lemma 36's. A repeated root descends at the symmetry, a datum's square read against a member being the image's pairing against itself, so the distinct roots' linear factors split the window on their own, coprime at their separation; each kernel is a spanning list by elimination (Definition 3) and the idempotents are the division displays' witnesses (Lemma 37's device, e^2 and $e\,e'$ reading e and equal members). S is transpose-closed, so the splitting is orthogonal with every complement invariant (Lemma 1), and each block is invariant for H and for the gram at the commuting. A fiber is a chord block. The count splits with the splitting, the $\mathrm{LDL}^{\mathsf{T}}$ *of a block diagonal being its blocks' at the concatenated lists (Lemma 30). The* *directions' translations commute, the action's one composition read (Construction 7), and a chord operator is the join of its direction's translation with the complementary step's, the transpose's own witness, so the chord operators commute pairwise and each maps every further direction's chord blocks into themselves; the splitting iterates direction by direction within the prior steps' blocks, each step this lemma's own at its direction's operator, and the window splits orthogonally into the joint chord blocks: the momentum point* $K := (t_1, \ldots, t_d)$ *is the chord list, one coordinate per direction, each the chord* 2 *or a located root of* P_L *along its own direction, the* K*-fiber the joint block at it, and the counts split with the joint splitting as with each step's (Lemma 30). The fiber counts reach the directed windows of Definition 8 through the decimated data, window-free off the unit fiber with the window entering through the deck brackets on it, every local read transported across windows at the label-graph isomorphism (Lemma 14). This family runs at odd side* $L \geq 5$*, where the torus's own comparisons clear its winding states: a winding support occupies at least* L *links, one at each of its direction's transverse cuts (Construction 7's cut family), each reading content at or beyond the member floor* $\hat{c}_2(f)$ *(Lemma 9), so its content is at least* $L\,\hat{c}_2(f)$ *against the plaquette level's* $4\,\hat{c}_2(f)$ *(Proposition 7; Lemma 15), the one comparison* $5 > 4$ *at every member; so the winding classes read at or beyond the level at every window of the family, and the directed windows are free of them outright (Definition 15).* ε_0 *is the least first jump over the fibers, its designation certificate data (Theorem 14), the unit's line sitting in the unit fiber, and the window cut of Definition 11 holds at level* κ *once each fiber's counting function is flat across* $(\varepsilon_0, \varepsilon_0 + E_0\kappa)$*, a fiber at the first jump reading its second jump at or beyond the top and every other fiber* k *its first,*

$$\operatorname{rev}\left(H^{(k)} : (h + E_0\kappa)\, G^{(k)}\right) \;=\; 0$$

at a pair $h \geq \varepsilon_0$ *(Theorem 14): together the reads are* $\operatorname{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ *on the window, the cut at every window ground at once (Lemma 20). On a chord block at a located root* t *of* P_L*,* S *reads the root itself and* T *the deck relation at the located chord,* $T^2 + 1 = t\,T$*; the transpose-paired reads collect at the deck families,* $T^{\,b} + (T^{\mathsf{T}})^{\,b} = p_b(t)$ *on the block, the plus read's instance at the root pair (Definition 4), so every such entry is a chord polynomial, its degree bounded by the grade's spatial support (Lemma 14). The block is doubled: its datum* A *at* $T = T^{\mathsf{T}} + A$ *squares against the band read,* $A^2 + 4 = S^2$ *at the products* $T\,T^{\mathsf{T}} = 1$ *and* $T^{\mathsf{T}}T = 1$ *collected, hence* $A^2 + 4 = t^2$ *on the block.* P_L*'s roots sit interior to the band, and the block's own pairing reads that: transposing* $T = T^{\mathsf{T}} + A$ *gives* $T^{\mathsf{T}} = T + A^{\mathsf{T}}$*, so* A*'s transpose is its memberwise swap,* $A^{\mathsf{T}} = \breve{A}$*, and at a member* v *of the block the transpose walk reads* $\langle v, A^2 v\rangle$ *as the swap of* $\langle Av, Av\rangle$*; the block read against* v *is then*

$$t^2\,\langle v, v\rangle + \langle Av, Av\rangle \;=\; 4\,\langle v, v\rangle,$$

the pair $\langle t^2 : 4\rangle$ *at its lower side or at equal members (Definition 8's positive pairing), each root block occupied at the doubling read below. At equal members* $\langle Av, Av\rangle$ *is the sum's unit at every member,* Av *with it at that pairing, so* $T = T^{\mathsf{T}}$ *on the block and* $T^2 = 1$ *at the product* $T\,T^{\mathsf{T}} = 1$*;* $T^L = 1$ *at odd* L *then reads* $T = T\,(T^2)^{\,m} = T^{\,L} = 1$ *at* $2m + 1 = L$*, the coprimality of* 2 *and* L *spelled as that word, and* $S = 2$ *on the whole block, so* t *reads* 2 *against* $S\,v = t\,v$ *there, against* t *a root of* P_L *and the two factors coprime at the division display above (*P_L *at* 2 *reading the occupied* L*). So every root of* P_L *reads* t^2 *on its lower side against* 4*, strictly interior,* A^2 *reads the pair* $\langle t^2 \!:\! 4\rangle$ *off equal members,* A*'s determinant with it, and the band root* s *at* $4s^2 + t^2 = 4$ *adjoins one quadratic extension (Lemma 36), that read cleared. An antisymmetric matrix of odd order reads its determinant at its own balance partner, the assignment fold blind to the transpose with each summand's members swapped at the odd factor count (Construction 1's swap grading): the block's order is even, the doubling's read.*

The deck families hold the doubling's own identity. At a root pair (z, z') of the relation $z^2+1 = x\,z$ at a free chord x, product one, write $W := \sum_{a+b+1=L} z^{\,b} = z^{\,m}P_L(x)$ for the word's read (Definition 4). The geometric telescope is $z\,W + 1 = W + z^{\,L}$, its square

$$z^2W^2 + 2zW + 1 \;=\; W^2 + 2z^{\,L}W + z^{2L}.$$

Join W^2+1 to both members: the relation collects the left to $x\,z\,W^2+2zW+2$ at $(z^2+1)W^2 = x\,z\,W^2$, and the plus read $z^{2L}+1 = z^{\,L}p_L(x)$ collects the right to $2W^2+2z^{\,L}W+z^{\,L}p_L(x)$. The telescope reads $2W^2+2z^{\,L}W = 2W(W+z^{\,L}) = 2zW^2+2W$ and $2zW+2 = 2(W+z^{\,L})$, so $2W$ sits on both members and clears, leaving $x\,z\,W^2+2z^{\,L} = 2zW^2+z^{\,L}p_L(x)$; at $W^2 = z^{2m}P_L(x)^2$ with $2m+1 = L$ the read $zW^2 = z^{\,L}P_L(x)^2$ puts the occupied $z^{\,L}$ on both members, and clearing it gives $x\,P_L(x)^2+2 = 2P_L(x)^2+p_L(x)$, the pair display

$$p_L(x) \;=\; \langle x\!:\!2\rangle\; P_L(x)^2 + 2,$$

one identity of the deck families, the chord free over the base and both members its polynomials, every step remainder-list arithmetic at the relation (Definition 2; Definition 4). The partial sums read the telescope additively, one join per step from the seed,

$$\langle x\!:\!2\rangle\, P_m + p_m \;=\; p_{m+1}\ ,$$

and the key list's fold collects to the symbol's own multiple,

$$\sum_{b<L} p_b \;=\; p_m\, P_L\ ,$$

one induction per side: the step joins the two new keys at the telescope, the doubling display, and the products' square $p_{m+1}\,p_{m+1} = p_{2m+2}+2$, its three collectors.

Per free translation orbit the doubling is witnessed and squeezed. Take a free orbit of the label index, its L translates $T^{\,b}w$ independent, and at a root t_j the deck-family column

$$u_j \;:=\; \sum_{b<L} p_b(t_j)\; T^{\,b}w.$$

The word's read at the root pair of the located chord t_j has $W = z^{\,m}P_L(t_j)$ the sum's unit at $P_L(t_j)$'s, so the telescope reads $z^{\,L} = 1$: the powers wrap, $p_{L+b}(t_j) = p_b(t_j)$ and $p_{L-b}(t_j) = p_b(t_j)$, with $p_L(t_j) = 2$ the displayed identity's read at the located root. Hence $S\,u_j = t_j\,u_j$: the coefficient of $T^{\,c}w$ in $S\,u_j$ is the family join $p_{c+1}(t_j)+p_{c-1}(t_j) = t_j\,p_c(t_j)$ (Definition 4), the recursion closing cyclically, its two wrapped keys the reads $p_{L-1}(t_j) = p_1(t_j)$ and $p_{L-2}(t_j) = p_2(t_j)$ against the seed $p_0 = 2$. The join is the recursion's class read at the symbol: over the remainder lists at P_L (Definition 2) the column's coordinates are the families' own classes, the chord multiplies as the free chord, and the wrapped keys read the wrap identities, $p_L = 2$ with $p_{L-b} = p_b$, one read at every side at once, a located root's reads the classes' evaluations, and a polynomial in the chord reads the column at its own multiple, the membership iterated along the Horner recursion (Definition 2). That seed against the independent translates reads u_j occupied. Its A-image is its independent partner at the pairing reads: $\langle u_j, A\,u_j\rangle$ reads its own memberwise swap at $A^{\mathsf{T}} = \breve{A}$, hence equal members, and $\langle A\,u_j, A\,u_j\rangle$ reads $\langle 4\!:\!t_j^2\rangle\,\langle u_j, u_j\rangle$ on its upper side at the strict interiority above, so $A\,u_j$ is occupied and perpendicular to u_j, the two independent members of the root block: the membership is the exchange's read, the datum and the chord two words in the one translation at the product $T^{\mathsf{T}}T = 1$, so they exchange with every polynomial in the chord across it, the datum maps each chord block *to itself, and the image row reads the moved display, $S\,A\,u_j = A\,S\,u_j = t_j\,A\,u_j$, the symbol's read at the image the member's own image. The independence reads over the remainder lists at every side at once. The column's self-pairing collects at the product families, $p_b^2 = p_{2b} + 2$, the doubled keys reading the key list once (the even keys outright, the odd keys at the upper wrap), and the key list's fold is the symbol's own multiple, the display above, so the self-pairing reads $2L$; the image's self-pairing reads $\langle 4\!:\!x^2\rangle$ against it at the relation on the column, and the pair's Gram determinant reads $4L^2\,\langle 4:x^2\rangle$. At the band's two ends the symbol divides at the linear factors: at the chord two the remainder is L, the division display above, and at the partner chord the family reads the doubled parity member key by key, 2 at the even keys and its partner at the odd, so the remainder is the unit at even depth and its partner at odd; the two cofactors' product is the determinant's witness: the band pair is the two linear factors' product's balance partner, each end's factored read is its remainder's partner at the division display, and the three partners compose, the cleared read $4L^3$ at the balance partner of the depth's parity member, occupied at every side, so the pair is independent over the remainder lists, one read per side, the located roots' reads the evaluations; and the composed witness reads the band at every stored factor besides: an occupied-length factor of the symbol dividing the band pair $\langle 4\!:\!x^2\rangle$ divides the composed constant through the identity, the division's descent at the monic factor against the constant's occupied read, so the band symbol's remainder at every stored factor is occupied, the equal-members case refused at every stored root with the block pairing's lower side closing the strict interiority above. The unit fiber reads one member of the orbit, its fixed read $T\,v = v$ forcing equal coefficients along the orbit at the elimination read (Definition 3), the orbit's own sum. The orbit's count $1 + 2m = L$ squeezes the reads to equality: each root block reads exactly two members of the orbit, the pair $\big(u_j, A\,u_j\big)$, the block at doubled multiplicity, and per fiber the classes are the free orbits at the pair base, each member against its A-image: the row structure the gradings below read off this splitting. Over the residue the squeeze is one splitting read at stated lists: the pair base enters as the column's coefficient rows, one row per key below the residue's top against its image under the doubling datum, and the orbit sum joins them to an independent list at the orbit's own count. The sum pairs every row at the sum's unit, against a coefficient row at the key's coefficient of the key list's fold, the symbol's own multiple at the residue, and against an image row through the datum's memberwise-swap transpose at the sum's own image, the fixed read's unit; a coefficient row pairs an image row at two crossed folds, the cyclic shift reading either fold as the other's exchange and the wrap's reflection returning the exchange: equal members. The coefficient rows are independent at the keys below the residue's top, the families there their own below-top displays at monic tops from the seed two, a dependency reading its coefficients at the unit key by key up the tops; an image-row dependency is the doubling datum's read at a coefficient-row combination, the combination's two translation images at one value, the orthogonality doubling the iterate's fix and the member reading the translation through the wrap, $T\,w = T^{\,2m+2}\,w = (T^2)^{\,m+1}\,w = w$ at $T^L = 1$ over the count, so the combination sits in the unit fiber's kernel and joins the sum's span while the sum pairs it at the sum's unit, its self-pairing reading that unit with the positive pairing withdrawing it; and a joined dependency's sum part reads its own square at the count with each block part its self-pairing against the whole, the positive pairing withdrawing both. The joined list's count is the orbit's own, $1 + 2m = L$, so every orbit vector reads through the splitting at the elimination read, a root-datum kernel member through the pairs alone, its sum coordinate clearing at the word's value L off the unit, the division display's read.*

The band congruence takes the block's list at pairs, each member against its A-image at the elimination's pair descent (Definition 3), the image columns rescaled at the band datum: a unit congruence off the crossings, mapping the block's entries to polynomials in t and s^2, substituting at the root's identity, every count fixed (Lemma 30) and the gram's rescaled block s^2G positive definite:

the fiber pencils' entries are one polynomial-pair list in (t, τ) over the residue at the root coordinate's magnetic weight τ^2 (Definition 10): each fiber floor is Lemma 31's count, a cell function of (t, τ), constant off the root divisor of one explicit determinant of ground data and persisting at the divisor's points beside a positive-semidefinite cell. At fixed content cutoff the entries are one polynomial-pair list in (t, τ) over the residue at every L beyond the locality band (Lemma 14), the window entering through its sampled momenta and its separation range alone. The momenta are the geometric word's symbol roots, the word identity above read at the operator: the sampled momenta are the chord 2, the unit fiber, with P_L's m located roots, one doubled block each, every root a designated bracket datum at the separation of Theorem 15.

Lemma 40 (The fiber index is its grading, and the pencil is bordered)**.** *Per fiber on the odd tori (Lemma 39) the configuration classes read two counts off the index (Proposition 4): the band components, the components of the class's support at adjacency within the locality band (Lemma 14), each with its content; and the separation profile, the band components' relative positions up to translation. The grading is total, a class being its datum, and the regions are the datum's own coordinate ranges: the head, every content below a head bound Λ_h (a parameter of the presentation, as a window's cutoff is) and every separation within the band; the tower direction, a band component's content beyond; and the slab direction, a separation beyond the locality band, the repeating shape of Lemma 34. The pencil's rows read the grading in three shapes: the diagonal, the stencil rows, and the border rows, and the tower rows repeat off the walls' collar:*

$$N_{\theta\lambda}^{\lambda+\nu} \;=\; \mathrm{mult}_\theta(\nu) \qquad \text{at every } \lambda \text{ whose consecutive gaps are at least two,}$$

one row, the adjoint's content list, the collar the labels with a consecutive gap below two; at a member's table the row reads the display at every dominant λ whose simple coroot pairs sit at or beyond the member's largest coroot magnitude, two off G_2 and three at it, the contents the member's θ list (Proposition 13), the collar the labels below that clearance.

Proof. Totality is the index's: a class's support decomposes into its band components with contents and relative positions, counts off the index (Proposition 4), and the regions are ranges of that datum. A band component below content C with L occupied links reads $L\,\hat{c}_2(f) \le C$, one pair comparison (Lemma 9), and as many within-band gaps, so its diameter is priced by C and the head is a finite index (Proposition 2).

The rows. E is diagonal at the content sum (Definition 10). A plaquette term meeting a band component's neighborhood reads its changed edge there (Lemma 14), the target's content within the fusion stencil (Proposition 3, Proposition 4) and its support within the locality band (Lemma 14), band components joining or parting within one locality-band step: a *stencil row*. A plaquette term off every band component's neighborhood factorizes (Lemma 14), creating its own loop class, one new band component, at one entry per relative position beyond the locality band, a label-graph isomorphism transporting it: a *border row*, one repeated datum, the boundary coupling of the walk's shape (§16). Every entry is one polynomial-pair list in (t, τ) over the residue beyond the locality band and the reach bound (Lemma 39; Lemma 25(iii)), at the root coordinate's magnetic weight τ^2.

The tower rows repeat. At λ with consecutive gaps at least two, every permutation but the identity drops from the graded sum (Corollary 2): a contribution at σ reads $\nu_\sigma + \sigma(\lambda + u) = c + u$ at an adjoint content ν_σ, a letter-pair move or the unit-monomial content (Lemma 16), u the unit set's sorted display, the matched-degree tie's one full column σ-fixed and absorbed into the label class (Construction 5; Lemma 24's shift the unabsorbed spelling); at gaps of $\lambda + u$ at least three the content's move sits below the entries' separations, so the join $\sigma(\lambda + u) + \nu_\sigma$ at σ off the identity keeps a reversed pair, which the strictly decreasing $c+u$ misses at every target. The identity term reads $\mathrm{mult}_\theta(\nu)$ at the tie $\lambda+\nu=c$, on the even side, the display; every target $\lambda+\nu$ is dominant there, the gaps absorbing one box. The walk's own diagonal sits on the collar and differs, 2 against r (Proposition 13): the collar is a named label set, its labels' gap list the read.

The member rows. At a member's table the deep exclusion is the coroot pairing's: a contribution at w off the identity holds an inversion list's member γ, $w\beta=\check{\gamma}$ at a positive β, and the tie reads $w(\lambda+\rho)+\nu_w=\mu+\rho$ at the target μ and an adjoint content ν_w; pairing the tie at γ's coroot moves the image across the kept form, $w^{-1}\gamma=\check{\beta}$ at the transport, and collects

$$(\lambda+\rho)(\beta^\vee)+(\mu+\rho)(\gamma^\vee) \;=\; \nu_w(\gamma^\vee)\ .$$

The right side sits at the member's largest coroot magnitude, the string's own cap (Lemma 6); the left summands read the simple pairs at the roots' natural coroot folds, so at the stated clearance the first sits beyond the magnitude and the second beyond the sum's unit, the target's pair the base's joined to one content move: the join clears the cap by two, the tie refused off the identity. The identity's term reads the display, the deep pairs absorbing each move at the stated clearance: the member rows at simple pairs at or beyond it are the θ list's own, Proposition 13's content data. ☐

Lemma 41 (The separated sector fibers over relative momentum)**.** *Per fiber on the odd tori (Lemma 39), at every $d\geq 1$:*

(i) Beyond the locality band the components factorize. At a class of $\varrho\geq 2$ band components with every separation beyond the locality band, the pencil's rows are per component: E adds the contents, a plaquette term meets at most one component's neighborhood (Lemma 14), a stencil move moves one component within its own locality band with the others spectators, and every entry is the moved component's own read at one relative position (Lemma 40). The ϱ-sector's pencil therefore agrees, beyond the collar (the classes with some separation within the locality band, a finite border block of the head, Proposition 2), with the components' tensor-sum pencil extended over every relative position,

$$H^{(\varrho)} \;=\; \sum_{a+b+1=\varrho} G^{\otimes a}\otimes H^{(1)}\otimes G^{\otimes b} \quad \textit{against} \quad G^{\otimes \varrho},$$

the fold over the factor splittings at the unit power the scalars' line, $H^{(1)}$ the single-component sector's intensive pencil at Lemma 40's index, the sector's own pencil the join of the vacuum diagonal with it, and the two pencils' site datum supported on the collar.

(ii) The relative fibering. The extended pencil's entries read the relative positions through translation differences alone, so the relative translations act on it and it block-diagonalizes over their chord operators by Lemma 39's chord splitting, run here at the relative translations: their blocks read at the chord coordinate (Lemma 39), and the relative momenta, the sampled chord values, are the chord 2 with the geometric word's symbol roots, each root a designated bracket datum at the separation of Theorem 15.

(iii) The separated bottom adds, and the sector clears the bracket. A single-component head pair positive semidefinite at the level ℓ is the form floor $H^{(1)}\succeq \ell\, G$ (Lemma 30), with its site datum S_1 at $H^{(1)}=\ell\, G+S_1$, and the tensor sum reads it ϱ times:

$$H^{(\varrho)} \;=\; \varrho\,\ell\; G^{\otimes\varrho}+\sum_{a+b+1=\varrho} G^{\otimes a}\otimes S_1\otimes G^{\otimes b}\,,$$

each summand positive semidefinite against the positive-definite tensor Gram: a product of two tensors reads factorwise, the product index's fold collecting to the factors' folds' product at the ground *identities (Definition 1), so the gram's pivot congruence at its positive pivots, the split at order-one blocks a positive-definite datum reads off its leading minors of unequal members (Lemma 30's minor sequence), and the site datum's split compose to the summand's own congruence, its image the split's blocks at the pivots' positive scales, and the image's form at every vector is the fold of the blocks' reads at the scales, on its upper side or of equal members (Lemma 30); the multiset symmetry and the window compress it, a compression keeping the split (Lemma 30). So on a bracket with top at or below $\varrho\,\ell$ the extended sector's pair is positive semidefinite at every relative fiber, the collar is a head border, and the border rows between component counts are the walk-shaped border chain: the separated sector contributes its collar to the head and its jumps sit beyond the bracket, the comparison's validity divisor the single-component level's own, located per cell (Lemma 31). At $d \geq 2$ the clearance is the slab direction's step, the relative coordinates fibered and the comparison Lemma 31's cell read, per relative fiber the single-component level's own, so the direction count enters through the fiber list alone (Lemma 39).*

Theorem 18 (The decimation: the counts live on the decimated head)**.** *Fix a fiber, its grading (Lemma 40), a head bound Λ_h, and a pair bracket $[\ell_1, \ell_2]$ of spectral levels. The pencil decimates onto its head in three located steps, each step a Schur complement (Lemma 34; Lemma 30's deflation), and the decimation is the count's own presentation.*

(i) The border direction is the walk's shape. A pencil row moves the band-component count within a named locality band, a border row creating one component and a stencil move severing or joining at most its own plaquette's arms (Lemma 40), so over that count, re-blocked along it, the pencil is a half-line chain of sectors with the vacuum class at the boundary, and the count at every level splits over the chain's pivot recursion, $\mathrm{rev} = \sum_\kappa \mathrm{rev}(X_\kappa)$ *(Lemma 34(i),(ii)): the walk's shape at matrix blocks (§16), the boundary pivot the vacuum's dressing. The split is order-free: inertia adds at every nonsingular principal pivot (Lemma 30), so each slab ordering of the elimination reads the one count, and an ordering is certificate data, its validity divisors entering the emitted list as any boundary's do. The ground jump is a divisor point of the decimated head: the level is the distance of the first two jumps of one counting function, the walk's own read, its boundary defect the vacuum's diagonal, the Kronecker delta at two labels (Proposition 13).*

(ii) The tower tails decimate on their dominance cells. A class's diagonal is intensive, the far plaquettes reading equal members on it (Lemma 14): the content sum against the local magnetic stencil, the stencil rows one capped list (Lemma 40; Lemmas 17, 18). Beyond the head bound the tail block at the bracket reads its dominance comparison: per row, the off-row entries' magnitude fold sits at or below the diagonal's entry, $f_i \leq a_{ii}$ at f_i the fold (Theorem 15's magnitude carrier), the comparison's gaps d_i at $f_i + d_i = a_{ii}$ its witnesses. Wherever it holds the peel telescopes on the comparison: at the symmetric datum (Theorem 16) a vector's form peels at the leading row, the leading square with the doubled cross fold against the trailing block's form, and each cross term joins its magnitude's squares at one square's multiple,

$$2\,a\,xy + |a|\,(x^2 + y^2) \;=\; |a|\,(x+y)^2 \quad \textit{at } a \textit{ its own magnitude, and } \; |a|\,\langle x\!:\!y\rangle^2 \; \textit{at the swap,}$$

at or beyond the sum's unit by its shape, so the leading row's magnitudes withdraw from the trailing rows' folds into their gaps and the descent closes on the order: the form sits at or beyond the gaps' fold of squares $\sum_i d_i\, x_i^2$ at every vector. Every split of the block then reads the upper side throughout, its lower list at an occupied count naming a vector pricing the form strictly below the sum's unit against the fold (Lemma 30); at occupied gaps the kernel block is vacant, a kernel column's image, off the unit family at the congruence witness's clearing, reading its form at equal members against the strict floor (Lemma 30), so the determinant sits off equal members with located brackets and the adjugate its solve (Definition 3); and the count telescopes over the pivot *recursion at the slab fold, the tail's summands each the sum's unit (Lemma 34(ii)): at the α=1 representative the comparison reads the content's growth against the stencil caps, the caps' fold linear in the stencil's term count, at most the per-vertex plaquette count's fold over the support's vertices (Construction 7; Lemmas 17, 18 the per-term caps), at the $[\sigma : 1]$ representative it is Lemma 43's read toward the contact end, and each comparison's validity is a located region of (τ, ℓ, Λ_h) at the count's coordinate, its divisor the comparison's own, the leading side in the count the one linear read. The constant-coefficient instance is the walk's boundary recursion, the deck quadratic's interior root (Theorem 12).*

(iii) The slabs decimate at the deck pivot, and the window enters through its brackets alone. Beyond the locality band the slab direction's bulk datum repeats (Lemma 40); its pivot is the deck-interior solvent of the pivot identities (Lemma 35(iii); Lemma 38), the deck orbit's interior member at $m = 1$, Theorem 12's read, valid exactly off the bulk block's own band, the crossing locus of its deck relation, and at a graph read of unequal members (Lemma 35(iii)): the decimation's validity divisors there are the tail's band and the graph read's side, located data of the bulk block alone. At $d \geq 2$ the slab step is Lemma 41's clearance, the relative coordinates fibered and the deck pivot the $d = 1$ sharpening. Two windows share every decimated datum up to their lengths, their reads differing within the named geometric brackets (Lemma 35(i),(ii)): off the unit fiber the decimated symbol is window-free outright, and on it the window enters through the brackets alone.

The decimated symbol. On the bracket, within the validity cells of (ii) and (iii), the count of the full fiber pencil is the decimated head's: a finite symmetric pencil over the located stage (Lemma 32), entries pair data of the deck coordinates and the located pivots, cell functions over (t, τ, r) (Lemma 31; Lemma 25(iii)). The spectrum on the bracket is where the decimated determinant reads equal members, Theorem 25 per cell concrete; and the cells the validity divisors cover are the decimation's own output, every boundary a located root.

The divisor list, named. The decimation's validity divisors are three families: the dominance divisors, the tower comparisons' own, one per head bound and representative ((ii)); the deck divisors, the bulk blocks' crossing loci with the graph read's side, Lemma 35(iii)'s counts ((iii)); and the jump divisors, the decimated determinant's roots at the bracket's levels (Lemma 31). The list is finite per stage, its cells partition the base, counts constant per cell and persisting at a boundary beside a positive-semidefinite cell (Lemma 31); the emitted record, the located root list with its per-cell counts and the binding read of which comparison binds at each boundary, is the decimation's displayed object, every entry one stage read of the crossing divisor (Theorem 25).

Lemma 42 (The charged cells)**.** *Fix a fiber on the odd tori (Lemma 39). The d_f-ality of a label is its chain endpoint, the fundamental or the unit label Lemma 9's descent ends at, one value at every chain: a chain's moves fix the reduced degree and each re-reduction withdraws one full column (Construction 5), so a chain to the unit label reads the label's degree at full columns outright, $|\lambda| = m\, d_f$, and one to ω_j at the two-sided join $|\lambda| + m'\, d_f = j + m\, d_f$ at naturals m, m'; two endpoints' joins against one $|\lambda|$ force one index by the sum's injectivity, a positive gap of full columns reaching d_f against the indices at most r, and the fundamentals' degrees are their indices. Two labels are one class where their degrees join to full columns, $|\lambda| + m\, d_f = |\mu| + m'\, d_f$ at naturals m, m', the endpoint the class's one fundamental or the unit; the dual's endpoint is the complement at the join $a + a' = d_f$ at nonunit endpoints and the unit at the unit's, the dual's degree joining the label's to full columns, $|\lambda^*| + |\lambda| = d_f\, \lambda_1$.*

(i) Charge is a cut-free class, and the pencil grades by it. At an occupied configuration every touched vertex reads an invariant count of at least one at the incident product, incoming links dualized (Definition 8): the unit label is a constituent, a constituent's reduced degree joins the factors' degree fold to full columns (contents of a tensor product add over the factors and a block's

contents sum to its degree, Construction 1; Construction 5), and the incoming duals read across at the join $|\lambda^*| + |\lambda| = d_f\,\lambda_1$: the vertex law, the outgoing against the incoming degree folds at the unit class,

$$\sum_{\text{out}} |a_\ell| + m\,d_f \;=\; \sum_{\text{in}} |a_\ell| + m'\,d_f \qquad \text{at naturals } m,\ m'.$$

Summing the vertex laws over the region between two parallel transverse cuts, an interior link's degree enters both folds once and the sum's injectivity reads it off: the two cuts' pairs are one class, the flux through a cut one class per direction, its endpoint the class's read. A cut transverse to direction i is crossed by direction-i links alone (Construction 7's direction reads), so the directions' classes are independent data, and the charge c is the occupancy family over the directions at the flux endpoints, a direction occupied at a nonunit class by its fundamental's index c_i, the charged directions its members; one direction at $d = 1$. At a member's table the law is class additivity: a block's occupied contents are one class (Construction 3), classes add over tensor products, the contents' sums, and a dual's class joins its label's to the unit class, so the vertex law reads at the member's class group, the classes over the root folds, finite data per member (Construction 4): the d_f-ality at the matrix units, the degree at full columns; the series' classes the contents' parity data against the root folds' spans (every even list at B_ℓ, the short family's doubled keys generating; the even-sum lists at C_ℓ, the letter-pair moves with the long family's; the even lists at key-four sums at D_ℓ, the pair moves'), two classes at B_ℓ, the spinor's odd parity against the unit's, two at C_ℓ, the first fundamental's odd sum against the unit's, and four at D_ℓ, the parity with the sum's key-four remainder, the vector's and the two spinors' with the unit's; and the fixed members' classes the adjugate rows' remainder families at the fold key, the unit's alone at G_2, F_4 and E_8 at key one, three at E_6, the key-three remainders pairing ω_1 with ω_5 and ω_3 with ω_6, and two at E_7, the key-two remainders occupied at ω_2, ω_5 and ω_7. At a member whose class group is its unit alone every configuration reads the unit class, the charged sectors are the one sector, the winding clauses read at the unit class's one instance, and the cell list's tension family drops, its root list the unit class's, every other read fixed: the one-sector read is the class fold's own derivation. E is diagonal; M fuses one link by θ, whose degree is d_f (the reduced shape $(2, 1^u)$ at $u + 2 = d_f$), fixing every link's d_f-ality; a translation permutes the index at one charge, the relabeled configuration's flux through a cut its own through the cut it came from, the cut family's grades read along the translation, one term per link along the permutation with the directions kept (Construction 7), the conservation reading the base cut's class back: the fiber pencil is block diagonal over the charges and its counts split (Lemma 30), M of the unit charge class and the vacuum line sitting there.

(ii) The winding floor, attained. Every label reads $\hat{c}_2$ at or above its endpoint's, Q falling along the chain (Lemma 9), the endpoint read $\hat{c}_2(\omega_j) = [\,j\,g_j\,(r+2) : 2d_f^2\,]$ at the complement gap $j + g_j = d_f$ (Lemma 9), its first component the $j\,g_j$ count at the $(r+2)$ weight, written $w(j) := j\,g_j$ below. The floor merges over a cut: at a family of nonunit endpoints whose degree fold sits at a nonunit class c, two members merge at their sum's trichotomy against d_f: below, to the fundamental at the sum, at the cost identity $w(a_1) + w(a_2) = w(a_1{+}a_2) + 2\,a_1a_2$; at d_f, the pair drops outright at $w(a_1) + w(a_2) = 2\,a_1a_2$, the rest still at c; and beyond, to the fundamental at the gap g at $d_f + g = a_1 + a_2$, at $w(a_1) + w(a_2) = w(g) + 2\,g_1g_2$ over the complement gaps $a_i + g_i = d_f$, the cost at or beyond $2\,g_2$ at $a_1 \le r$: every family descends to the one member c, so its content is at least $\hat{c}_2(\omega_c)$, one value with its complement's at the dual pair $c + c' = d_f$, at or above $\hat{c}_2(f)$. Two distinct endpoints' floors tie exactly at the dual pair, the sum's trichotomy against d_f their read at $j' + a = j$: below, $w(j') + a\,b = w(j)$ at the join $j + j' + b = d_f$; at d_f, the dual pair's one value; and beyond, $w(j) + a\,b' = w(j')$ at $d_f + b' = j + j'$: the gap product is the order's witness, so off the dual pairs the floors' trichotomy is one symbolic read at every residue. At a charged direction i

every direction-i cut is crossed at a nonunit net class, the cuts crossing disjoint link sets, and the directions' link sets are disjoint in turn, so the charged diagonal sits at or above the summed floor

$$L \sum_{i \text{ charged}} \hat{c}_2(\omega_{c_i}) \ \geq\ L\, \hat{c}_2(f)\ .$$

At a member's table the winding floor per class is the class's least read. A series member's is the descent's: a move is one positive root, the class kept, so a nonunit-class chain ends on its own class's end members (Lemma 9), the spinor's read at B_ℓ, the first fundamental's at C_ℓ, and the vector's or a spinor's at D_ℓ's three nonunit classes. A fixed member's is the fundamental fold's class-respecting read: a chain keeps its class and ends at a content at every coroot read at most one, whose occupied keys' classes add to its own (the remainder families of Construction 4), so at E_7 a nonunit-class end content occupies one of ω_2, ω_5, ω_7 and its Casimir clears the least of their displayed list, the minuscule member's join $[\,57 : 2\,]$, while at E_6 it occupies a fundamental of its own class or two of the further nonunit class, either read clearing the pair's shared least $[\,52 : 3\,]$, the doubled case two summands of the displayed fold (Lemma 9). The class floors add over a cut's crossing family: each nonunit-class member reads at or above its class's floor, a unit-class member's content is positive, and the floor list is subadditive at the class sums, a pair at the unit class outright, E_6's nonunit pair at the shared least against its double, and D_ℓ's spinor pair at the vector's class, twice $[\,\ell\,(r+2) : 8\,(r+1)\,]$ against $[\,r+2 : 2\,(r+1)\,]$, the cross-multiplied read ℓ against 2, a spinor's with the vector's clearing the further spinor's at the spinors' one value: a family netting class c reads content at or above the class-c floor, the merge identities above the matrix units' instance. The floor is attained by the winding strings: one straight ω_{c_i}-cycle per charged direction, at charge 1 the fundamental f along its cycle, each within-string vertex reading $N^{\mathbf{1}}_{\omega\bar{\omega}} = 1$ (Proposition 8). Two strings of distinct directions share at most one vertex, meeting exactly where their fixed transverse coordinates agree, and at a shared vertex vmult ≥ 1, *the incident product with each string's pair, whose unit constituents pair (Proposition 8; Definition 8): the union is occupied at the summed floor exactly. At the free end the charged ground is the strings', and the margin above the vacuum grows by $\sum_i \hat{c}_2(\omega_{c_i})$ per cell of length.*

(iii) At bounded excess the charged head is the dressed string, and its pencil is one window-free head at a length shift. Write the excess of a charge-c class for the witness gap of (ii)'s comparison, the summed winding floor with the excess joining to the content, the grading the charged decimation reads. The exclusions are extensive, read per cut, each cut's price direction-free, the direction count entering at the cut families' list alone, and every price is a read of the member's class floor table, (ii)'s floors with their subadditivity folds; the member price ϖ_0 is the least of the two cut price reads below, one trichotomy fold over the member's displayed class data, the d_f-ality the class read at the matrix units, each extra crossing's floor clearing the merge read, the unit-class read at $2d_f^2$ against $2\,(r+2)$ over the square's growth $d_f^2 = (r+1)^2$ and the netted pair's at $2\,r\,(r+2)$ against $2\,(r+2)$ at the occupied factor; the off-cut read below prices the deviation links at their own count, the dressing bound's read at the excess floors rather than the cut count's. A transverse cut with two or more nonunit classes among its crossing multiset's members prices an excess at or above the merge gap, (ii)'s subadditivity fold at the crossing pair against the net class's floor, positive at every nonunit pair of the member's class group: twice a class floor at a unit-class join, D_ℓ's spinor pair at the cross-multiplied ℓ against 2 with the vector's own read at a mixed pair, and E_6's shared least against its double ((ii)'s folds); at the A-series the merge gaps are the complement-gap reads, (ii)'s below and beyond costs $2\,a_1a_2$ and $2\,g_1g_2$ at the complement gaps $a_i + g_i = d_f$, each at least 2, the parts nonunit, the drop case at the dropped pair's own floors, $2\,w(a)$ at or beyond $2r$, and the unit conversion at $\hat{c}_2(\omega_j) = [\,j\,g_j\,(r+2) : 2d_f^2\,]$ at $j + g_j = d_f$ (Lemma 9), each at or above $[\,r+2 : d_f^2\,]$, the instance. A cut crossed beyond its net class prices the excess at its extra crossings' own floors:

an extra crossing at the unit class and a nonunit label reads $\hat{c}_2$ at or above the unit class's least nonunit read, the chain keeping the class and ending at the unit through a dominant root's content or on a unit-class end content (Lemma 9; (ii)): θ's read one at the A-series, at D_ℓ, and at E_6 and E_7 (the unit-class fundamentals' reads at or beyond θ's, the Casimir lists' trichotomies, and a two-key end content at or beyond the nonunit pair's doubled least, the display's cross reads positive), the vector's $[\,r+2:2\,(r+1)\,]$ at B_ℓ and the short dominant root's $[\,r:r+1\,]$ at C_ℓ; and a crossing family netting to the unit class holds two members at their classes' floors, at or above twice the least class floor, the A-series' fundamental pair $2\,\hat{c}_2(f)$ the instance. And an occupied link off every charged direction's cuts reads the member's floor outright (Lemma 9), the A-series' $\hat{c}_2(f)$.

The head classification. So at excess below Λ_h the cuts crossed beyond their one net crossing number at most L_0, the least natural at $\varpi_0\,L_0 \geq \Lambda_h$, one cross-multiplied read at the member price, the A-series' at $(r+2)\,L_0 \geq \Lambda_h\,d_f^2$, a two-string class with every cut so and exiting the head at $L > L_0$ outright, and beyond that threshold the head's classes are the dressed string: the winding strings of (ii), one straight ω_{c_i}-cycle per charged direction sharing at most one vertex per pair, deviating and dressing within the bound the excess floors above set, each crossing one more localized dressing, the far local components tensor factors beyond the collar (Lemma 41). The rows read at the dressing's neighborhood (Lemma 14), the string's far links at one repeated per-link datum, and within the fiber the classes are the dressings' translation orbits at the pair base, one pair per orbit (Lemma 39), every entry a finite stencil sum at translation powers: beyond L_0 and the locality band (Lemma 14) the off-diagonal entries and the excess reads are one polynomial-pair list in (t,τ) over the residue, the label-graph isomorphism of dressed neighborhoods transporting them along the string and across windows. The diagonal splits off the one extensive read, uniformly over the head: off the dressings' neighborhoods a head class's cells are label-graph isomorphic to the straight strings' own, so the content and the magnetic stencil reads repeat cell by cell (Lemma 14) and each dressing's deviation is finitely many cells of the intensive head. At a far cell the magnetic read is off-diagonal in the label index: a straight cell meets a plaquette at one link at most, consecutive links collinear, the term writes θ on the free links, three at a link-sharing plaquette and four at a vertex-sharing one, and Eval *reads the unit coordinate, so every index component of the term's product sits off the class's own configuration, the far terms factorizing at* $\mathrm{Eval}(\chi_\theta)$*, the Kronecker delta at two labels (Lemma 14; Proposition 8). So the strings' per-cell diagonal at the α=1 representative is the electric read alone,*

$$\varpi \;:=\; \sum_{i\ \text{charged}} \hat{c}_2(\omega_{c_i})\ ,$$

window-free and coupling-free, the charged head pencil is

$$H_c \;=\; L\,\varpi\,G \;+\; H_{\text{int}},$$

H_{int} the window-free intensive head at the stencil rows, and a charged count at a level λ is H_{int}'s count at the shift cross-added onto the level (Theorem 14's balance-pair levels), the two pencils one matrix at matched levels: the length enters the charged head through the shift, the sampled momenta and the separation range alone. The magnetic datum at a far cell is its dressing fold, the tower decimation's per-cell Schur read at the cell's stencil and the bracket's level, one adjugate solve per pivot (Theorem 18(ii)), each atom read at the entry's square against its gap, $[\,e^2:g\,]$, the pivot's cofactor at the coupling's two reads, entering the dressed diagonal at the squared magnetic weight, $[\,p^4\,:\,q^2\,]$ on the ray at $[\,q^2\,:\,p^2\,]$ and τ^4 at the α=1 representative: its atoms are the stencil's changed-edge entries (Proposition 8) at their content gaps, a link-sharing plaquette at the row targets μ of $\omega_{c_i}\otimes\theta$, one entry per word target (Lemma 25(ii); Lemma 26(ii) at the further series), at the content pair $3+\langle\hat{c}_2(\mu):\hat{c}_2(\omega_{c_i})\rangle$, and a vertex-sharing plaquette at its own loop, the

plaquette content 4 *(Proposition 7), its entry the vertex's own word read. A counting function is monotone in its level, so the flat step's charged comparison rides the shift against the unit fiber's anchor through the tension, the balance pair of the string's dressed per-cell floor against the unit class's,*

$$\Big\langle\, \varpi + \sum_{i\ \mathrm{charged}} D_i^{\mathrm{vac}} \,:\, \sum_{i\ \mathrm{charged}} D_i^{\mathrm{str}} \,\Big\rangle,$$

D_i^{str} *the direction-i cell's weighted dressing fold and* D_i^{vac} *the unit class's at the matched slab, its entries one at the plaquette content* 4 *(Construction 13; Proposition 7), the far plaquettes' folds entering both members equally, a summand on both members fixing the value (Definition 1), so the pair reads at the stencil neighborhood alone, a located datum within Lemma 35(ii)'s brackets: at tension off its lower side on the cell the least-length read extends to every larger length, and the tension's root is one more located divisor of the list, the free end reading the electric member alone, both folds at the end's one-member read, the value* $\sum_i \hat{c}_2(\omega_{c_i})$ *among the integer constants. The folds' atoms are residue reads, the entries word reads and the gaps the distance classifications (Lemma 25(ii); Lemma 24; Lemma 9; the member gaps the words' bilinear reads, Lemma 26(ii)), so the tension's divisor takes the residue as a coordinate with its roots against the compared families' clearances (Lemma 25(iii); Lemma 26(iii)). At the end weights the ray's factors read the pair whole,* q^2 *at the electric member and* $[\,p^4 : q^2\,]$ *at the two folds, at* $q\,\tau = p$ *(Lemma 43's seam identity): the tension's side is a ray datum. The folds' position lists read the cell's stencil, the link-sharing and vertex-sharing plaquettes at the interface's plaquette counts (Construction 7), both members linear in the term count, so the tension's divisor takes it as one further coordinate, a linear read with its root located per charge vector and count.*

(iv) The charged decimation, and the cell. The grading and the decimation run per charge: every pencil row preserves the charge ((i)), so each diagonal, stencil and border row of Lemma 40 is a row of one charge sector, read there; beyond L_0 *the charged index is the dressed string of (iii), one dressed* ω_{c_i}*-string per charged direction, its entries the window-free list there, with the far local components reading by the vacuum sector's own rows (Lemma 41); the tower and slab decimations are Theorem 18(ii),(iii), with (ii)'s dominance comparison run against the charged diagonal's floor (ii), the head bound a parameter of the presentation. The charged counting function on a bracket is therefore the charged decimated head's, a cell function of* (t, τ, r) *with located boundaries (Lemma 31; Lemma 25(iii); Lemma 26(iii)), the comparison's validity divisor its own, and the charged cell's entry that divisor's first positive root, an output (Lemma 33). The head's level list, its multiplicities and its block structure are the reads of that head pencil, pinned per cell.*

Theorem 19 (The cut is a flat step of the decimated counting function)**.** *On a coupling cell, within the decimation's validity cells (Theorem 18), the window cut of Definition 11 at level* κ *is reached by located jump comparisons on the decimated symbols: per fiber and per charge (the* d_f*-ality class of Lemma 42(i),* M *of the unit charge class), the counting function's first jump sits at or above* $\varepsilon_0 + E_0\kappa$*, and on the vacuum's own sector its second jump does, the first the window's* ε_0*, the boundary pivot's own root (Theorem 18(i)). Each comparison is two counts at located pairs: the jumps are count data on the decimated head, cell functions of* (t, τ, r) *with located boundaries (Lemma 31; Theorem 18), the momenta read cell by cell at one bracket pair each (Lemma 39), the charged sectors' counts the reads of Lemma 42, and the contact pair's value the* X*-sector's read (Definition 14; Remark 3).*

The comparisons reach the cut. They read $\mathrm{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ *on the window, the cut at that window at every ground (Lemma 20). Window-freeness is the decimation's, off the unit fiber outright and within the deck brackets on it (Theorem 18(iii)). Per cell the verification is a finite list of anchored jump brackets, one pair of counts each: the cells' reads are the gap* *chain's data, each level naming its representative and transported across the seam at the factors of Lemma 43's identity.*

Theorem 20 (The ground reads are bracketed)**.** *Fix a window at a coupling with the ground bracket of Theorem 9 and the slab split of Lemma 34 along the decimation's direction (Theorem 18), within the decimation's validity cells and at the spectator certificates λ_i (Lemma 35(ii)).*

(i) A ground read is a stage read. The ground ψ has a spanning list by elimination over the ground root's stage (Lemma 33), and at an observable A the read enters as the pair of the two pairings, $\langle\psi, A\,\psi\rangle$ against the gram's $\langle\psi, G\,\psi\rangle$, its value the cofactor (Definition 1), Definition 10's ω at the unit gram; each pairing is one stage read, its entries Lemma 14's at the supports.

(ii) The tail rides the transfer factors. At the ground root the kernel rows of the slab split close from the far end, $X_{i+1}\,\psi_{i+1} + B_i^{\mathsf{T}}\psi_i$ the sum's unit per row (Lemma 34(iii)'s row read), the tail pivots of unequal-membered determinant on the validity cells, the ground's root the decimated head's own (Theorem 18(i)); so a depth-j component is a j-fold transfer image and its weight is capped geometrically at the gram's slab blocks (Lemma 35(ii)),

$$\langle\psi_{i+1},\, G_{i+1}\,\psi_{i+1}\rangle \;\le\; \lambda_i^2\,\langle\psi_i,\, G_i\,\psi_i\rangle,$$

the certificates' product pricing the tail's whole weight beyond any depth.

(iii) The ground transport, and the window sandwich. The transport display: at two pencils on one head at the unit gram, their site datum S at $\tilde{P} = P + S$, the later ground $\tilde{\psi}$ at a root drift δ and the datum capped two-sidedly off the drift, S' at $w\,G$ (Lemma 30) at $S = \delta\,G + S'$, with the earlier ground simple at the flat window's clearance γ', the later ground's part φ off the earlier line obeys

$$\gamma'^{\,2}\,\langle\varphi, G\,\varphi\rangle \;\le\; w^2\,\langle\tilde{\psi}, G\,\tilde{\psi}\rangle, \qquad \gamma'\,w_\omega \;\le\; 4\,c_A\,w,$$

w_ω the width of the one bracket holding the two grounds' reads of an observable at the head cap c_A: the first display is the pencil identity read at φ with the cap's polarization (Theorem 32's display), and the second collects the reads' pairing members, three off-line terms each polarized at the cofactor pair $[\,w : \gamma'\,]$, a read the cofactor at its gram's weight ((i); Definition 1), and at w beyond γ' the head cap alone reads the width at twice itself, below the display's right side. Two windows enter at their own roots' site data at one shared clearing: below depth b the couplings are equal and the diagonals tied at the roots' drift, each window's own data beyond (Lemma 35(i)'s tie), and the display instantiates at the head decimated onto the depths at or below an observable's depth j: the two heads differ at the drift along the leading diagonal and at the depth-j pivot, the pivot's deviation the tied sandwich of Lemma 35(i) from depth b down to j, capped by (ii)'s certificates with the tie's datum entering once per step over the gap g at $j + g = b$, with the grounds' weights beyond the head priced by this theorem's (ii); the drift sits within the jump bracket the decimation transports across windows (Theorem 18(iii)), and the clearance is the cell's second-jump separation (Theorem 15; Theorem 25(ii)): a ground read of a supported observable is window-free within named brackets, geometric in the gap at $j + g = b$, the brackets' data the spectator certificates, cell data of the family's parameters (Lemma 35(iii)).

(iv) The separated reads cluster. At observables A, B of supports separated by j slabs, at the unit gram, the connected read is one pairing of two off-line vectors: with $\hat{B}\psi$ the connected datum at the join $B\psi = \omega(B)\,\psi + \hat{B}\psi$, off the ground line by its construction,

$$\omega(AB)\,\langle\psi, G\,\psi\rangle = \omega(A)\,\omega(B)\,\langle\psi, G\,\psi\rangle + \langle\hat{A}\psi,\, G\,\hat{B}\psi\rangle,$$

one rearrangement at the gram-symmetric multiplications; at the dual representative, the pair $(\sigma E\!:\!M)$'s site datum $P(\sigma)$ (Lemma 43), the off-line vector solves the sourced pencil,

$$\tilde{E}\,\hat{B}\psi = \sigma\,[\,E, B\,]\,\psi \qquad \text{at the level gap}\;\; \varepsilon_0\,G + \tilde{E} = P(\sigma),$$

the magnetic part commuting (the algebra commutative, Proposition 5) and the source supported at B's reach (Lemma 14); so $\hat{B}\psi$'s weight at A's side rides the Green products through the j separating slabs at the certificates (Lemma 34(iii); Lemma 35(ii)) against the clearance γ': A's support is the split's leading slabs at their joined width, the weight's head part the cross-cleared sum over them priced by the certificates' fold at the head's side (this theorem's (ii) at the leading slabs), and the ride anchored at the source slab's own block weight with the crossing certificates pricing the j separating steps; and the connected pairing is that weight at A's head cap, the cap's polarization closing it (Theorem 32's display): the two reads $\omega(AB)$ and $\omega(A)\,\omega(B)$ read one value within the bracket the crossing certificates and the clearance price, geometric in j.

(v) The Euclidean brackets. At a pair u, the level gap $\tilde{E}$ (Definition 10) with its located gaps γ_j at $\varepsilon_0 + \gamma_j = \varepsilon_j$ (Lemma 33), $\varkappa$ the largest gap, and a natural N beyond $u\,\varkappa$, the lower and upper members are the N-fold reads of two solved witnesses,

$$L_N := \ell^N \quad \text{at} \quad N\,G\,\ell + u\,\tilde{E} = N\,G, \qquad U_N := \upsilon^N \quad \text{at} \quad \big(N\,G + u\,\tilde{E}\big)\,\upsilon = N\,G,$$

each solve one Green solve at a located pair (Lemma 34(iii)), the upper's datum $N\,G + u\,\tilde{E}$ positive definite ($G \succ 0$ with $\tilde{E}$'s split, Lemma 30); the split maps both to diagonal reads at the gaps (Lemma 33), the lower's $[\,g_\gamma : N\,]$ at the witnessed gap $u\gamma + g_\gamma = N$ and the upper's $[\,N : N + u\gamma\,]$, where the cleared product identity and the cleared Bernoulli display close the bracket,

$$g_\gamma\,\big(N + u\gamma\big) + u^2\gamma^2 \;=\; N^2, \qquad g^{\,N} + N\,a\,b^{N'} \;\geq\; b^{\,N} \;\geq\; g^{\,N} + N\,a\,g^{N'} \quad (a + g = b,\;\; N' + 1 = N),$$

so the members' gap is at most $[\,u^2\varkappa^2 : N\,]$ at every spectral point, each member's read between the two at every larger N (the refinement comparisons, the Bernoulli display again), and a read $\omega(A \cdot B)$ at the pair of families is priced by the weighted polarization at the two probes' gram reads (Theorem 32's display), the spectral width the factor: the Euclidean read at u is the datum the nested brackets locate, one located object per pair u, the reader's continuum point reading itself in through the pair brackets beside it. The m-point reads iterate the displays, one bracket per separation and one family per pair, every entry a stage read by (i); and the iteration is capped, the growth cap: an m-point bracketed read's magnitude (Theorem 15's carrier) sits at or below the probes' head caps' fold against the separations' certificate factors, one factor per separation, each pair step the weighted polarization at the probe's head cap against the prior fold's gram weight (Theorem 32's display) and each separation riding the Green products at the certificates (Lemma 34(iii); Lemma 35(ii)), every constant a located pair, the head caps the compressions' top roots (Lemma 33) and the factors the certificates' own.

(v′) The decomposition display. At a convex splitting of the state into translation-invariant states, $\omega = \sum_i w_i\,\omega_i$ at pair weights folding to one, each component's reads agree with the state's within the connected reads' widths: at the translate fold S_n, the n-translate average of a probe A, each component reads its own mean square below the fold's, $\omega_i(\bar{S}_n S_n)$ at or beyond $\omega_i(\bar{A})\,\omega_i(A)$ by the invariance with the pairing's positivity, while the state's read collects the pair reads, $\omega(\bar{S}_n S_n)$ within the clustering brackets of $\omega(\bar{A})\,\omega(A)$ at the mean's own weight and the off-diagonal widths' geometric fold (the display of (iv)): the weighted square fold $\sum_i w_i\,\langle\,\omega_i(A) : \omega(A)\,\rangle^2$ sits below the widths' read at every n, one located bracket per probe.

(vi) The propagation bracket. The gram is the fibers' block diagonal (Definition 8) and $\tilde{E}$ is banded at the locality band (Lemma 14; Lemma 40's rows), so the witness e at $G\,e = \tilde{E}$ is banded, ℓ reads the join $N\,\ell + u\,e = N$ at the unit's N-fold on the right, L_N is a polynomial in the banded e, and its reach is N bands, while U_N sits within (v)'s width of it and a Green solve's beyond-band blocks are the transfer products at the certificates (Lemma 34(iii); Lemma 35(ii)). An operator reading a region's index data alone commutes with one reading a disjoint region's, the fibers' factors

disjoint (Lemma 14; Lemma 41(i)); so at an involution-fixed observable A reading the leading slabs alone and its evolved probe $U_N A U_N$, the probe reads within the reach at one bracket, the beyond-reach remainder's form reads capped by the width and the crossing certificates' geometric read, and every ordering difference against an involution-fixed observable beyond the reach is the remainder's own read: the commutator reads at separation beyond N bands are priced by the bracket alone, the certificates cell data of the family's parameters (Lemma 35(iii)).

(vii) The scale sandwich. On the corner cell, at scales $\eta' < \eta$ (two equal scales reading one pencil outright) the pencils' site datum is $g\,E$ at the squares' gap g, the order's witness at $\eta'^2 + g = \eta^2$, one positive-semidefinite form, and the transport prices at the grounds' own electric weights rather than at a head cap. The drift is a Rayleigh read at the earlier ground,

$$\varepsilon_0(\eta') \;\le\; \varepsilon_0(\eta) \;\le\; \varepsilon_0(\eta') + g\,e', \qquad e'\,\langle\psi', G\,\psi'\rangle = \langle\psi', E\,\psi'\rangle,$$

the lower side counts monotone (Lemma 30), the upper the earlier ground as trial pair; and the off-line weight reads at the later ground's electric square weight,

$$\gamma'^2\,\langle\varphi, G\,\varphi\rangle \;\le\; 2\,g^2\,\langle\tilde\psi, E^2\tilde\psi\rangle + 2\,\delta^2\,\langle\tilde\psi, G\,\tilde\psi\rangle,$$

δ the drift, the transport display's pairing step at the two-term square (the pencil identity read at φ, (iii)); and the electric square weight closes on one finite sum over the head's shells at (ii)'s caps,

$$\langle\tilde\psi, E^2\tilde\psi\rangle \;\le\; \Big(\sum_i\,(4\,C_i)^2\prod_{j<i}\lambda_j^2\Big)\,\langle\tilde\psi_0, G\,\tilde\psi_0\rangle,$$

C_i the i-th shell's Casimir bound (the electric read at most $4\,C_i$ there), $\tilde\psi_0$ the head-shell component, the shells finitely many at the stated cutoff and each summand one comparison at the certificates: two scales' reads of one observable sit in one bracket at the priced width, (iii)'s read display at the priced cap, the weights the grounds' own reads. A support's site count at a scale is the scale count against the corner scale, the count of the scale's multiples at or below the support's separation, two comparisons its read, and on the corner disconjugacy certificate's cell the floor holds at every tail scale: the clearance is located there, the site counts stabilize beyond a scale, finitely many scale counts changing at all, and a datum's scale reads close within the stated widths, their scale sum one cofactor read at the upper family's rate.

Theorem 21 (Euclidean restoration is a count)**.** *Take the carrier at the lattice interface's fields and a region mapped to itself by the hyperoctahedral group B_d of signed coordinate permutations (Construction 7).*

The symmetry passes to the fiber symbol. $E = \sum_\ell \Delta_\ell$ and $M = \sum_p \chi_\theta(U_{\partial p})$ run over all links and all plaquettes at coefficient one (Definition 10), and B_d permutes those two sets. A signed permutation reverses link orientations, which dualizes labels, so E is fixed because $C_2(\bar\lambda) = C_2(\lambda)$ (Definition 6) and M is fixed because $\chi_{\bar\theta} = \chi_\theta$ at the self-dual adjoint, a reversed boundary reading the dual label's character. Hence $H([\alpha\!:\!\beta])$ commutes with the action. A lattice symmetry g permutes the label index isometrically and maps each direction's translation to the permuted direction's own, so it maps chord blocks onto chord blocks (Lemma 39): the direction permutation permutes the block index $(t_1,\ldots,t_d)$, and an orientation flip reads T at $T^{\mathsf T}$, the chord operator $S = T + T^{\mathsf T}$ fixed with it. So the K-fiber's pencil and the gK-fiber's are congruent at the relabeling, the pencil's rows and columns read along one permutation of the window index, an orthogonal map; the site-datum determinant reads one value across it, the assignment fold reindexed along the relabeling: each assignment pairs with its relabeled own at the entries' product kept, the row and column exchanges *composing at an even join (Definition 3's exchanged reads at Construction 1' descent), and their characteristic polynomials agree, coefficient by coefficient:*

$$\mathcal{S}(gK;k,r) \;=\; \mathcal{S}(K;k,r)$$

with $\mathcal{S}$ the fiber symbol of Theorem 17. Every coefficient of $\mathcal{S}$ in k is therefore one value at every relabeling of the chord list: a flip fixes each chord at the operator's own read, so B_d reads the chord list through its direction permutations alone, and the coefficients are the permutation-invariant chord polynomials, one coefficient identity per congruence; their momentum reading is even in each coordinate, the deck relation's two roots exchanging at a flip with the chord fixed (Definition 4). The derivation runs unchanged at every number of arguments. A lattice symmetry permutes the label index and maps the fibers isometrically, an orthogonal map commuting with H, so at a window whose multiplicity reads 1 *(Theorem 9) it reads the ground line by a graded read the reads of ω are blind to: ω is B_d-invariant, and hence every read $\omega(A_1 \cdots A_m)$ is B_d-invariant, jointly in the positions and the orientations B_d permutes. The momentum transform of the read family is its translate family read at monomials: per argument the translate key list $a^{(r)}$, one key per direction below the side, the read $\omega\big(T^{a^{(1)}} A_1 \cdots T^{a^{(m)}} A_m\big)$ at the composite shifts, and the transform the occupancy family of those reads at the per-argument exponent lists, the translate keys' own matched lists (Definition 1), the momenta $K^{(1)}, \ldots, K^{(m)}$ the exponent coordinates, one list per argument; a lattice symmetry moves a translate read to the g-moved keys' own, the direction permutation permuting each argument's keys and a flip reading a key to its balance partner at the wrap (Lemma 39's $T^L = 1$), so the transform is B_d-invariant with the reads, the moves the coefficient identities.*

What the continuum asks is $O(d)$, and the difference is a multiplicity. The momenta enter as formal coordinates at the transform's own carrier, one matched list per argument, and two moves settle every degree: a reflection, one coordinate K_i read to its balance partner across all m momenta at once, sends every monomial at an odd count of that coordinate to its balance partner, the coefficient's members then equal, and a transposition of coordinates equalizes what survives.

1. *Degree* 2*: the rings coincide, at every m and every d. The reflection leaves only $\sum_i K_i^{(r)} K_i^{(s)}$, and the transposition fixes its coefficient, so the degree-*2 *B_d-invariants are spanned by the pairwise inner products $K^{(r)} \cdot K^{(s)}$, each of which is $O(d)$-invariant already. The leading term is isotropic because the invariant ring at that degree is the inner products' span; the inclusion B_d-inv $\subseteq O(d)$-inv at that degree is the whole of what the coincidence asks, and the two moves above exhibit it.*

2. *Degree* 4*: breaking enters. The isotropic span at a degree is the pairwise inner products' polynomial span, item 1's ring at its products, and the breaking multiplicity is the independent invariants' count beyond it, the classical $O(d)$-invariant ring the isotropic span's reading in the dictionary alone: the derivations read the two spans and their counts. The reflection leaves the coordinate indices paired, so the space includes diagonal quartics alongside the inner products' products, and the count reads one of them, $p_4 = \sum_i \big(K_i^{(1)}\big)^4$, every coordinate total four. At $m = 1$ every invariant is a cleared combination of p_2^2 and p_4 over the balance pairs (Definition 1), the clearing datum two at the paired monomial's splitting count, $K_i^2 K_j^2$ read at its two factor orders, and the two are independent, a combination at the sum's unit reading each coefficient at the quartic and the paired monomial in turn: multiplicity* 1 *at the join $1 + 1 = 2$, at every $d \geq 2$, with p_4 off the isotropic span at the paired coefficient. At every m the quartic p_4 is an invariant off the isotropic span at $d \geq 2$, its two reads at the first argument's own monomials, so the multiplicity is positive there.*

3. *The rate is the gap between those degrees, uniformly in m. An odd degree's every invariant reads equal members at every monomial, the coordinate totals joining to the degree with one*

member odd; so the leading term sits at degree 2 and the first breaking invariant at degree 4, the breaking suppressed by two degrees in the momentum, the gap at $2 + 2 = 4$, at every such family and at every d. What varies with m is the multiplicity at degree 4; the rate stays that gap.

Degree-2 isotropy therefore holds at every order and at every member, since the statement is about which forms exist rather than about which are small, and the ring at degree 2 is the inner products' span. Restoration is a multiplicity read, and the degree-two rings' coincidence is exactly what the continuum asks.

Theorem 22 (The continuum reads)**.** *Fix the member and the direction count. A continuum datum is a finite list of supports, separations and Euclidean pairs stated against the corner scale at pair coordinates (Lemma 45; Theorem 20(v)), its observables the algebra's at those supports (Definition 9).*

(i) The data are bracketed, and the spectral heights read two-sided on the certificate's cell. At every scale of the corner cell the datum's read sits in the stated brackets (Theorem 20(i)–(vi)), the support counts read as the scale counts against the corner scale (Theorem 20(vii)), and two scales' reads sit in the scale sandwich's priced brackets (Theorem 20(vii)). On the corner disconjugacy certificate's cell (Lemma 46, the decimated chains' at the block clauses) the floor locates the clearance at every tail scale, the ground-level sandwich sums to one cofactor read at the upper family's rate (Lemma 45), the heights read two-sided at the certificate's rates, the site counts stabilizing with the floor, and a datum's scale reads close on the moment folds' clearance cell (Lemma 47).

(ii) The bracketed reads cohere. Every finite Gram of bracketed reads is positive semidefinite within its brackets, the pairing definitional at every scale (Definition 8; Lemma 30); reads at disjoint supports read one value at either order, and an ordering difference beyond the reach is the remainder's own read (Theorem 20(vi)); separated reads pair to their product within the crossing certificates' brackets (Theorem 20(iv)); the chain's own heights are two-sided located data at the cell's positive located floor, two-sided at the certificate's rates on its cell, so the spectral data sit at the ground with the floor's clearance in the corner scale there (Lemma 45's cell floor; Lemma 20); and the momentum reads are isotropic at degree two with the direction data two degrees down, the multiplicities scale-free (Lemma 48; Theorem 21).

(iii) The spacing is an output. The corner height reads two-sided at the certificate's rates on its cell, at the cell's positive located floor, one output bracket (Lemma 45's cell floor).

20 The network constituents

Theorem 23 (Network reduction and the two-vertex coupling law)**.** *(i) A three-link path labeled R is one edge labeled R (the convolution identity), so the theta graph reduces to the three-edge Θ-graph and the state space is $\Theta(R_1, k, R_2; \alpha, \beta)$-networks with E diagonal, eigenvalue $3C_2(R_1)+C_2(k)+3C_2(R_2)$. (ii) Couplings between sectors differing on the shared link need the label-index reads of Proposition 8 on the shared edge alone: the module–pair weights are*

$$w_k \;=\; \big[\, m_k\, d_k : r^2(r+2)^2 \,\big], \qquad \sum_k m_k d_k = r^2(r+2)^2$$

(the block count over Lemma 24's list, Lemma 5(ii)),

with d_k and m_k the dimension and the multiplicity of Proposition 8 at the channel k, so the sum is the completeness of $\mathrm{adj} \otimes \mathrm{adj}$*, whose dimension is $r^2(r+2)^2$; the weights are the channel table's* *reads (Lemma 24),*

$$w \;=\; \Big([\,1 : r^2(r+2)^2\,],\; [\,c_1 : r(r+2)\,],\; [\,(r+1)^2 r(r+4) : 4r^2(r+2)^2\,],$$
$$[\,g\,(r+3) : 4r(r+2)\,]_{g+1=r},\; [\,g\,(r+3) : 4r(r+2)\,]_{g+1=r},\; [\,(r+1)^2(r+2)\,w' : 4r^2(r+2)^2\,]_{w'+2=r} \Big)$$

over the channel list, the leading-term reads in the residue: the first two weights' second members exceed in degree, and the four channel weights read $[\,1:4\,]$ *at the top (Theorem 15's leading terms). (iii) The weights are the data. The slot-two self-energy assembled from these weights and the network energies is* $\sum_k w_k\,[\,q_k : 2q_k + p_k\,]$ *at* $\hat c_2(k) = [\,p_k : q_k\,]$*, the factor's gap the module pair's network eigenvalue against the module count,* $3 + \hat c_2(k) + 3$ *against* $E_0 = 4$ *at* $\hat c_2(\mathrm{adj}) = 1$ *((i)'s eigenvalue; Proposition 7), the resolvent factor a constructed pair. The recoupling constituent's content is the finite tetrahedral list: the within-pair middle matrix* $\langle(\mathrm{adj}, k', \mathrm{adj})|\chi_X(U_1)|(\mathrm{adj}, k, \mathrm{adj})\rangle$ *at the middle index the channel table's occupied Casimir values, the dual pair's shared value one joint block at the summed weight, computable per entry by the* E_{link} *interpolant, the projection onto one Casimir value a polynomial in* E_{link} *(Construction 2's Lagrange read) entered cleared at the node differences' product, one value across the clearing scale (Definition 1's homogeneity principle), at the channel distances' occupied gaps (Theorem 10).*

21 The drift and the divisor

Theorem 24 (The uniform drift)**.** *At every label* R *the adjoint row reads* $\sum_k m_k d_k \hat c_2(k) = d_R d_{\mathrm{adj}}(\hat c_2(R) + 1)$*, the fold over the row's channels at the multiplicities against the dimensions: the dimension-weighted fusion walk's Casimir drift is exactly* $+1$*, uniformly in* R *(*$\sum_k w_k \hat c_2(k) = 2$*, the channel table's second closure identity, Lemma 24). With strictness (the unit label is the one label of dimension one: a nonunit reduced shape reads an occupied head row at its positive degree against the withdrawn full column's vacant last row, the rows weakly descending (Construction 1), so a consecutive row gap* $h \geq 1$ *sits at some* $\lambda_{i+1} + h = \lambda_i$*, its gap factor* $h+1$ *against the unit display's one with every further factor at or beyond its left partner (Corollary 1's display), so* $d_\lambda \geq 2$*), the closure datum is a constant rather than a decaying one.*

Proof. The member tier. Trace the Casimir over $W_R \otimes W_\theta$ at the trace read (Lemma 3), one value at every orthogonal spanning list. At an exhaustion the blocks' lists join to one orthogonal spanning list, the blocks pairwise perpendicular (Lemma 5(i),(ii)), and the fold splits per block at the cleared products, each block's diagonal pairings the scalar's multiples of the self-pairings (Lemma 4(ii); Lemma 7): the trace reads $\sum_k m_k d_k\, C_2(k)$, one summand per channel at its count. The tensor's operators read by the Leibniz rule and the Casimir's fold collects at the factors (Construction 3),

$$C_\otimes \;=\; C_R \otimes 1 + 1 \otimes C_\theta + 2\sum_\alpha [\,\langle\alpha,\alpha\rangle : 2\,]\,\big(E_\alpha \otimes F_\alpha + F_\alpha \otimes E_\alpha\big) + Z\;,$$

each root term's two orders splitting at the four arrangements with the one-sided pairs collecting to the factors' own folds, and Z the diagonal fold's cross read, the form's additivity at the joint contents, $q(\mu+\nu) = q(\mu) + q(\nu) + 2\,\langle\mu,\nu\rangle$ per content pair. The trace factorizes over the tensor list, two tensors pairing at the factors' pairings' product (Lemma 5(iii)), so each term's trace is the factor traces' product: the two one-sided terms read $C_2(R)\,d_R\,d_\theta$ and $d_R\,C_2(\theta)\,d_\theta$; a root term's factor trace is a fold of cross-content pairings, each of equal members at the content sum's orthogonality (Construction 3), so the root terms' traces read equal members; and Z's trace is the double fold $2\sum_{\mu,\nu} \mathrm{mult}_R(\mu)\,\mathrm{mult}_\theta(\nu)\,\langle\mu,\nu\rangle$, the form's bilinearity collecting it to $2\,\langle F_R, F_\theta\rangle$ at the content folds

$F_X := \sum_\mu \mathrm{mult}_X(\mu)\,\mu$. The θ-block's contents are simple folds, the root list's members with the unit content (Proposition 13's content list) at each root's fold over the simple list (Construction 3), so the pairing collects at the halved products to F_R's coroot pairs, $2\,\langle F_R, \alpha_i\rangle = F_R(\alpha_i^\vee)\,\langle \alpha_i, \alpha_i\rangle$, and each reads equal members: the multiplicities are one value at every reflection (Lemma 6), a content joined to its s_i-image reads the coroot pair with its balance partner, and a fixed content's pair is its own partner. So

$$\sum_k m_k d_k\, C_2(k) \;=\; d_R\, d_\theta \left(C_2(R) + C_2(\theta)\right) \,,$$

and the display clears at $C_2(\theta) = 2\,(r+1)$ (Proposition 12), one value across the clearing (Definition 1's homogeneity principle): $\sum_k m_k d_k \hat{c}_2(k) = d_R d_\theta(\hat{c}_2(R) + 1)$.

The label calculus at the residue. At the matrix units the constant family holds every consecutive coroot pair at equal members, so the content folds read the constant family and the derivation runs at the alternant display's recursion (Theorem 1), the display at every content list of the block's width with a key at the count's unit reading the moved folds' one value. Sum the display over the recursion family's keys: the moved folds reindex at the move's own round trip, per place pair $a < b$ and per occupied target κ the sources at the shifts $1 \le l \le \kappa_a$, one source per shift at the reverse move, the second letter's weight reading the target's own $\kappa_b + l$ and the shifted first letter's $\kappa_a + l$; the two reindexed folds share the shift fold $\sum_{l \le \kappa_a} l$, a summand on both members fixing the value (Definition 1), and the summed display reads

$$q(\lambda + u)\, d_\lambda + 2 \sum_\kappa \mathrm{mult}(\kappa) \sum_{a<b} \kappa_a \kappa_b \;=\; \sum_\kappa \mathrm{mult}(\kappa)\, q(\kappa + u) + 2 \sum_\kappa \mathrm{mult}(\kappa) \sum_{a<b} \kappa_a^2 \,.$$

The swap symmetry (Lemma 6's read at the matrix units) reads each place fold place-free, the first moment M_1, the square fold M_2 and the product fold M_{11} one value per place and place pair with $d_f\, M_1 = k_\lambda\, d_\lambda$ at the contents' degree read, so the summed display collects to $q(\lambda{+}u)\, d_\lambda {+} 2\, p\, M_{11} = d_f\, M_2 + 2\, U_1\, M_1 + q(u)\, d_\lambda + 2\, p\, M_2$ at p the place pairs' count and U_1 the unit display's total. The fused carrier's family splits twice, over the channels at the exhaustion's content refinement (Lemma 5(ii)) and over the factors at the contents' addition (Construction 1), so each moment fold reads both ways, the channels' counted fold against the factors' split at the further factor's count,

$$M_1(F) = d_\theta\, M_1(R) + d_R\, M_1(\theta) \,, \qquad M_2(F) = d_\theta\, M_2(R) + 2\, M_1(R)\, M_1(\theta) + d_R\, M_2(\theta) \,,$$

with $M_{11}(F)$ at M_2's shape, the cross read $2\, M_1(R)\, M_1(\theta)$ shared: the collected display at the fused carrier, read against the two factors' own collected displays at the further factors' counts, withdraws the quadratic folds (a summand on both members fixing the value) and leaves

$$\sum_c N^c\, q(c + u)\, d_c + q(u)\, d_R d_\theta \;=\; \big(q(R + u) + q(\theta + u)\big)\, d_R d_\theta + 2\, d_f\, M_1(R)\, M_1(\theta) \,,$$

whose d_f-multiple reads the moments at the degree folds, $d_f\, M_1(R) = k_R\, d_R$ and $d_f\, M_1(\theta) = d_f\, d_\theta$ at θ's degree d_f. The square identity (Definition 6) with the row read $\langle \lambda, 2\rho\rangle {+} (d_f{+}1)\, k_\lambda = 2 \sum_i \lambda_i u_i$ (Theorem 1's collection at Lemma 7's fold) reads $d_f\, q(\mu{+}u) = d_f Q(\mu) {+} k_\mu^2 {+} d_f (d_f{+}1)\, k_\mu {+} d_f\, q(u)$ at every shape; every channel shares the degree $k_c = k_R + d_f$, the contents adding over the factors at θ's degree, so the dimension identity $\sum_c N^c d_c = d_R d_\theta$ (Lemma 5(iii)) withdraws the shared reads and the display closes at $Q(\theta) = 2 d_f$ (Lemma 7):

$$\sum_c N^c\, d_c\, d_f Q(c) \;=\; d_R d_\theta \left(d_f Q(R) + 2 d_f^2\right) \,,$$

the display's $\hat{c}_2$-cleared read at $[\, d_f Q : 2 d_f^2\,]$ (Definition 6). $\square$

Theorem 25 (The divisor identification, and the crossing divisor)**.** *(i) The identification. The spectrum is the root locus of the exact fiber symbol (Theorem 17): on a window a level is a pencil root exactly where the determinant reads equal members, the factorization read of Lemma 33 at the gram's determinant off the sum's unit, and the Feshbach determinant reads equal members with it, one Schur factor of the whole at a complement of unequal-membered determinant (Lemma 30), at every coupling outright, each root a located divisor point.*

(ii) The crossing divisor. For a symmetric stage pencil of the chain with $G \succ 0$, a window compression or a fiber pencil, the entries polynomial in the base coordinates over the stage's field (Lemma 39's band congruence), write $\chi(x)$ for the level pair's site-datum determinant (Lemma 33's χ) and take the squarefree split $\chi = c\prod_j S_j^j$ over the base's polynomial pairs, the division descent's iterated greatest common divisors, the equality test cross-multiplied through the iterated division, each S_j taken primitive over the base's polynomial ring with the contents collected into c, itself a base polynomial (Gauss): the split is an identity of base polynomials and specializes at every stage point. The crossing read is

$$D \;:=\; \prod_j \operatorname{Res}_x\big(S_j, S_j'\big) \;\cdot \prod_{i<j} \operatorname{Res}_x\big(S_i, S_j\big)\;,$$

the resultants Definition 3's at the variable x, a base element off the sum's unit: a factor of equal members reads an occupied cross pair on a squarefree or coprime pair, against its Bézout read $up + vq = 1$ (Definition 3). At a base point off D's root locus the specialized split keeps the generic structure: the leading x-coefficients multiply to $\det G$ *there, on its upper side by the split, so every one reads unequal members and the degrees hold; each S_j stays squarefree and the pairs stay coprime, the resultants' own reads there; so every root of χ at that point is a root of exactly one S_j, simple in it, of multiplicity j in χ, and distinct roots stay distinct: root collisions sit on D's located root locus alone, the jump divisors of the decimation's list; a decimated head reads its counts as the fiber pencil's on its bracket, so its jumps inherit the confinement.*

(iii) The ground's read. On a cell with pairs $l < h$, the pencil pair at l positive semidefinite and the counting function flat at h across the cell, the ground multiplicity is $\operatorname{count}(h)$*, both reads cell functions read once (Lemma 31); by (ii) it moves across a located boundary only where a ground collision sits on the boundary, the boundary's own two-sided bracket counts. On the vacuum sector's pencil at the free end's side it reads* 1*: a level bracket of pairs (l, h) isolating the free ground, the unit's line against content $4\hat{c}_2(f)$ (Lemma 15), reads the pair at l positive semidefinite and* $\operatorname{count}(h) = 1$ *at the free end, both reads constant over the attached cell (Lemma 31), so the least root sits in the bracket at multiplicity* 1 *there.*

Theorem 26 (The scheme bridge is a count)**.**

$$\big[\, d_{\mathrm{adj}}\, C_2(\mathrm{adj}) \;:\; d_f\, C_2(f) \,\big] \;=\; 2\, d_f$$

*exactly, the pair of two weight-*1 *objects, one value at every form point (Proposition 1), and the read is a count: the f-flux bridge's component at the contact end is $b_f^{\mathrm{equiv}} = 2d_f\, b_X$, one rescaling of the coupling coordinate.*

Proof. $d_{\mathrm{adj}} = r\,(r+2)$ with $\hat{c}_2(\mathrm{adj}) = 1$ (Lemma 16; Definition 6), and the fundamental count d_f with $\hat{c}_2(f) = [\, r(r+2) : 2d_f^2\,]$ (Lemma 7), so the pair reads the cross-multiplied natural identity $r(r+2) \cdot 2d_f^2 = 2\, d_f \cdot d_f \cdot r\,(r+2)$; the $r = 2$ evaluation is 6. ◻

Definition 13 (The ruler excess)**.** At the residue r the *ruler excess* is the polynomial

$$A(u) \;:=\; \sum_{j=1}^{r} [\,1 : j\,]\, u^j$$

(Definition 2), the one polynomial supported from key one with $A' + u^r = 1 + u\,A'$, r the pairing's residue.

Theorem 27 (The walk residue)**.** *The tower pairing is forced. At the residue r: for the principal adjoint tower (the $k\theta$-string, θ the highest root, $\hat{c}_2(k) = [\,k(k+r) : r+1\,]$), the Casimir $C_2(m\theta) = m\,(m+r)$ of Proposition 12 splits into a square part and a row-weight part, and the pairing is the second over the whole:*

$$\varphi_m = \big[\,m\,r : m\,(m+r)\,\big] = [\,r : m+r\,], \qquad \varphi_m + \psi_m = 1 \quad at \quad \psi_m = [\,m : m+r\,]\ ,$$

*two weight-*1 *objects read as one pair, one value at every form point (Proposition 1), every datum forced, ψ_m the pairing's complement, the one-value read. Its generating object is*

$$\Psi(u) := \sum_{m\geq 1} [\,1 : m\,]\, \psi_m\, u^m = \sum_{m\geq 1} [\,1 : m+r\,]\, u^m\ ,$$

and the shifted series $F(u) := u^r\Psi(u) = \sum_{j\geq r+1}[\,1 : j\,]\, u^j$ satisfies the recursion $F' = u^r + u\,F'$, coefficientwise one read per degree: its divisor pair reads the least occupied degree r and the pole order 1 *at u=1, both naturals. Against the harmonic ruler $R(u) := \sum_{m\geq 1}[\,1 : m\,]\, u^m$, the ruler at the recursion $R' = 1 + u\,R'$, its pole order* 1 *as well, the pole excess is one polynomial, coefficientwise exact and finitely supported:*

$$R(u) \;=\; F(u) + A(u)\ , \qquad A(u) \;=\; \sum_{j=1}^{r} [\,1 : j\,]\, u^j \quad (\textit{Definition 13})\ ,$$
$$A' + u^r = 1 + u\,A', \quad \textit{the support from key one}\ ,$$

the one polynomial with the geometric derivative, closing $F' + A' = R'$ term by term, a divisor-characterized object. The value's second coordinate is its read at $u = 1$, the polynomial evaluation

$$A(1) \;=\; H_r = \sum_{j=1}^{r} [\,1 : j\,]\ ,$$

one pair per residue, the ruler's excess over the walk; and the coefficientwise record at the Ψ-level is the finite telescope, exact at every M,

$$\sum_{m=1}^{M} [\,r : m(m+r)\,] + \sum_{j=M+1}^{M+r} [\,1 : j\,] \;=\; H_r\ , \qquad [\,1 : m\,] = [\,1 : m+r\,] + [\,r : m(m+r)\,]\ ,$$

*the tail itself an r-term expression. One object, one pole, one read: φ fixes Ψ, Ψ's divisor fixes F', and the ruler's excess is the polynomial A, the construction settling the coordinate outright. The exponent is an integer at $r = 1$ alone; the pairing's own pole residue r and the weight-*1 *tower sum $[\,r+1 : r\,]H_r$ agree with it there and separate from $r = 2$, so the residue-one instance leaves them together, and the construction rather than the instance is what selects.*

Definition 14 (The contact pair)**.** At the contact end the decimated head's counts are the walk's (Lemma 44: the contact end's reads are the fusion matrix's own), and two divisors read them: the channel's, the boundary resolvent G_0 with its ι-symmetrized weight (Theorem 12), and the tower's, the generating object Ψ against the harmonic ruler (Theorem 27). The *contact pair* is the two divisors' reads: the first coordinate is the channel's order count at the base c_1 (Definition 12); the second is the ruler's excess over the walk read at one, $A(1)$ (Definition 13).

Theorem 28 (The constant's assembly)**.** *The contact pair of Definition 14 reads factor by factor, and there are two: the walk exponent H_r on the ruler's side, the polynomial excess $A(1)$ (Theorem 27) and coupling-free; and the channel's bit count, read off the divisor of G_0 (Theorem 12). The pencil's other degree-one read is the module count E_0 at the free end against the level at the contact end, and $\mathcal{K}$ is the pair of those two reads (Definition 11, Theorem 29). What makes the two divisor factors below a read of it is Theorem 25: the Feshbach determinant reads equal members exactly on the spectrum, so the level Definition 11's cut fixes is a point of the fiber symbol's divisor, and the coordinates below are that point's. That level reads two-sidedly at the walk's own divisor: at the contact end the head's counts are the fusion matrix's, the walk's divisor data (Lemma 44; Proposition 13), a window cut the chain's compression at that window and the sector's within it, a compression of a compression (Lemma 30); the truncation pair's two counts bracket the chain's at every level below the seed floor at every cutoff with an agreeing pair a located count of the chain itself, the scale key's cutoff clearing every window level below the seed floor at once (Lemma 45), so the window cuts' edges sit at or beyond the chain's and the bordered pencils' at or below, counts monotone under the comparisons (Lemma 30), the level the mass point's own located read between them; the mass weight is the pair $[\,\langle c_1^2 : 1\rangle : c_1^2\,]$ against its complement $[\,1 : c_1^2\,]$, the two summing to one, its side the upper at base two and equal members at base one, the mass point at the crossing there, the divisor count one value throughout (Theorem 12); and each window cut's edge, where one is named, is the compressed pair's least root, read at a word image with the two moments at equality and the first positive, the state the word sector's own (Lemma 22(i),(ii); Lemma 21). So the pair's two coordinates read:*

$$\big(3,\ H_r\big),$$

and both are divisor reads. The second is the walk's pole excess against the harmonic ruler, the polynomial read $A(1) = H_r$ on the ruler's side, off F''s divisor pair, the least occupied degree r and the pole order 1 at u=1 (Theorem 27). The first is the channel's order count: the boundary entry at its renewal witness $G_0 = z + c_1 z G_0$ reads its divisor as the witness's own, the mass point at $c_1 z_* = 1$ a simple pole, order 1, and the ι-symmetrized weight reading the base at order 2 (Theorem 12); the two orders against the base give $c_1^1 \cdot c_1^2 = c_1^3$.

The pair $(3, H_r)$ is what the theory determines: two orders, both naturals, summing to 3, and an exact pair, the first coordinate read at the base c_1, a fusion multiplicity moving with the residue (Construction 13). The two reads are of two divisors, the channel's (G_0, Theorem 12) and the tower's (Ψ, Theorem 27), and the coordinates read them alone.

Theorem 29 (The scaling form, and the constant as a pair of reads)**.** *The pencil is degree-one homogeneous in the end weights, $H([\,u\alpha : u\beta\,]) = u\,H([\alpha : \beta])$, so a rescaling multiplies every spectral read of the family by one u; the pair of two degree-one reads therefore reads one value at that scale, at whichever points of the base the two are taken. The overall scale of H is the one freedom the base is already taken up to (Proposition 6), so the statement is written on the end-weight pair alone: the ray is the only variable.*

Each end has its read, and the invariant is their pair. At the free end, where the pencil is E, the module is an eigenvector with the link count E_0 (Proposition 7); at the contact end the value *is the level. Both are degree-one, so by the paragraph above the pair descends, and that pair is $\mathcal{K}$ (Definition 11). It descends jointly: the two reads enter symmetrically, each a read rather than the other's measuring scale. Both are weight-free, one value at every form point (Proposition 1), and $\mathcal{K}$ is a count against a count.*

Consequences. (1) $\mathcal{K}$ is dimension-free, both of its coordinates divisor reads: the pole excess built from the Casimir, the order count from the channel divisor, each read d-free. (2) The scheme bridge $b_f^{\text{equiv}} = 2d_f\, b_X$ (Theorem 26) rescales the coupling coordinate, acting on the base by $[\alpha : \beta] \mapsto [\,\alpha : \lambda\beta\,]$, whose fixed points are exactly the two ends, so $\mathcal{K}$ is invariant under it: the bridge locates a coupling in the fundamental theory (Remark 3).

Theorem 30 (The decomposition)**.** *The chain's named outputs factor. The invariant's two coordinates are residue-side divisor reads, one value at every direction count and every family scale: the order count off the channel's divisor and the pole excess off the tower's (Theorem 28; Theorem 29), the per-member table the pair $(3, H_r)$ at the member's derived residue with the base the θ-support count (Proposition 12; Proposition 13), two at the A-series at $r \geq 2$, one at its first member and at every further member. The corner outputs, the extent, the floor and the spacing, hold the whole direction and scheme dependence: each is a located datum per member and direction count (Lemma 45), moving with the family's scale as the level does, while the invariant reads one value across the directions and the scales, the dimension-freeness Theorem 29's first consequence.*

Theorem 31 (The two-action comparison)**.** *At the fundamental-character family, the pencil at the magnetic member $\sum_p \big(\chi_f(U_{\partial p}) + \chi_{\bar f}(U_{\partial p})\big)$, the chain's corner machinery runs at the swapped member: the sum is symmetric at the dual pair, each term capped as a form at the fundamental's own square (Lemma 18); the fibering, the truncation caps and the momentum reads are the derivations' own instances there, while the member moves the ality by one, so the charged grading reads at its own class data and the comparison runs at the outputs both families define, the extent, the floor and the spacing at the unit-class head, located data per member and direction count. The invariant is fixed under the bridge, the rescaling $b_f^{\text{equiv}} = 2d_f\, b_X$ of the coupling coordinate with the two ends its fixed points (Theorem 26; Theorem 29), so the two families' invariants read one pair, and each corner output's comparison is one bracket at the two families' certificates, finite and decidable.*

Theorem 32 (Truncation control)**.** *At the coupling $[\alpha : \beta]$ the truncation at Casimir cutoff Λ_C is a projection Π, the statement homogeneous on the ray, and the off-block coupling is the magnetic form's cap. M is multiplication by $\sum_p \chi_\theta(U_{\partial p})$ (Definition 10), each plaquette term capped as a form, two-sided at the adjoint dimension (Lemma 18, the member cap; the A-series' shifted square at unit width below and $r\,(r+2)$ above), so the sum is capped two-sidedly and with it the pencil's magnetic part (Lemma 30),*

$$\beta M \;\preceq\; W \;:=\; \beta\,\#p\, d_\theta \qquad \textit{with} \qquad W + \beta M \;\; \textit{positive semidefinite},$$

the adjoint dimension times the region's plaquette count at the magnetic weight, $d_\theta = r\,(r+2)$ the A-series read (Proposition 13's dimension fold), independent of the cutoff; E being diagonal reads its off-block part at the sum's unit ($\Pi E\, \Pi'$ at the complement projection Π' at $\Pi + \Pi' = 1$), so the off-block coupling of H is the magnetic one alone, under the one form cap W. The cap's compression (Lemma 30) bounds the magnetic part inside the removed block, whose electric diagonal sits at or above $\alpha\Lambda_C$, so that block's floor f reads the cross-added comparison $W + f \geq \alpha\Lambda_C$; writing E_{low} for the higher of the two window levels the window cut of Definition 11 reads plus the cap W, an admissible cutoff reads $E_{\text{low}} < \alpha\Lambda_C$, and at its witness gap ς,

$$E_{\text{low}} + \varsigma \;=\; \alpha\Lambda_C,$$

the removed block's floor sits at least ς above each level. A compression lowers each count below a line (Lemma 30), raising each low level; and the cap's two splits, read at the vector $Wx + \varsigma y$ and at its partner $Wx + \varsigma\,\check{y}$, $\check{y}$ the memberwise swap of y, x in the head and y in the removed block, collect to the polarization: the sum $2\langle x,\, \beta M\, y\rangle + c_W\langle x, x\rangle + \varsigma\langle y, y\rangle$ reads on its upper side or of equal members, at the pair weights W, ς and the shift cap c_W, the cofactor at $\varsigma\, c_W = W^2$ (Definition 1, ς the admissible cutoffs' own gap), so the shifted split reads $H + c_W \oplus \varsigma \;\succeq\; H_{11} \oplus H_{22}$ with the removed block's floor above every level, counts are monotone under $\succeq$ (Lemma 30), and each level's downward shift is at most c_W; the level $E_0\mathcal{K}$ is the two jump reads' gap, so the two cuts' values bracket each other at the priced width,

$$E_0\,\varsigma\,\mathcal{K} \;\le\; E_0\,\varsigma\,\mathcal{K}_\Lambda + \beta^2\,\#p^2\,d_\theta^2, \qquad E_0\,\varsigma\,\mathcal{K}_\Lambda \;\le\; E_0\,\varsigma\,\mathcal{K} + \beta^2\,\#p^2\,d_\theta^2,$$

degree matched on the ray as the level is, the two reads' bracket priced by one pair comparison each way at every α, the member's adjoint dimension ($r^2(r+2)^2$ at the A-series) and the region's plaquette count its data: the truncation is certifiable at every window size with exact atoms, and the window cuts approach $\mathcal{K}$ with a modulus rather than along a subsequence. The gap chain reads the modulus at the α=1 representative, and the display is homogeneous on the ray (Proposition 6), every representative's read one rescaling of it.

Lemma 43 (The dual truncation)**.** *At the coupling $[\,\sigma : 1\,]$ at a pair σ, the cap, the floor and the compression of Theorem 32 read with one substitution, the two terms' roles exchanged in the bound's inputs: the off-block coupling and the removed block's magnetic part are capped as forms by a W of the fusion data (the fusion form's two-sided dimension cap on a label index, Lemma 17; per plaquette term over its stencil at the member's d_θ, Lemma 18 with Lemma 14), W linear in the stencil's term count, at most the per-vertex plaquette count's fold over the support (Construction 7), the removed block's electric diagonal sits at or above $\sigma\Lambda_C$, and each level shifts downward under the compression by at most the shift cap c_W, the cofactor at $\varsigma\, c_W = W^2$ (Theorem 32), at the order $\lambda + W < \sigma\Lambda_C$'s witness gap ς, $\lambda + W + \varsigma = \sigma\Lambda_C$. A count at level λ is therefore pinned on the finite head whenever the bracket $[\lambda,\ \lambda + c_W]$ is clear of the head's divisor (Lemma 31, the level as the parameter): every count of the dual pencil, the pair $(\sigma E\!:\!M)$'s site datum $H([\,\sigma : 1\,])$, lives on a finite head pencil, at every cutoff with*

$$\sigma\,\Lambda_C \;>\; \lambda + W + c_\vartheta \qquad \text{at} \qquad \vartheta\, c_\vartheta = W^2,$$

one pair comparison at a clearance pair ϑ, the admissible set an output whose floor rises toward the contact end. The transfer is one-sided: the compression lowers counts and raises levels, each level's downward shift capped at c_W (Theorem 32; Lemma 30), so between two head jumps the lower comparison reads at its equality, the head ground at or above the full one, and the clearance rides the upper jump alone: the admissible levels are those beyond the ground bracket's top whose sum with c_W sits below the second jump's bottom, the measured gap spent on one side.

The count beyond every cutoff. The display defines the dual pencil's count at λ over the unbounded content range: at a level whose bracket $[\lambda,\ \lambda + c_W]$ is clear of the head's divisor, it is the head count at any cutoff beyond the display's bound, and two admissible cutoffs read one integer: the smaller head is the larger head's own truncation (the projections nest), the display runs on the larger head at the one cap W (a compression of a cap is a cap, Lemma 30) with the removed block's floor at least ς, and the clear bracket pins the count both times.

The seam identity. The ray at end weights $[\,q^2 : p^2\,]$ names one integer pencil pair, $(q^2E : p^2M)$, with its site datum S at the join $q^2E = p^2M + S$, and the two coordinates' representatives are the datum's positive rescalings,

$$q^2\,H\big([\,1 : \tau^2\,]\big) \;=\; S \;=\; p^2\,H\big([\,\sigma : 1\,]\big) \qquad \text{at} \qquad q\,\tau = p, \quad p^2\sigma = q^2,$$

so every subspace read agrees at rescaled levels: a rescaling-fixed direct summand reads at rescaled levels, the fibers among them (Lemma 39), and on each,

$$\operatorname{count}_{[1:\tau^2]}(\lambda) \;=\; \operatorname{count}_S(q^2\lambda) \;=\; \operatorname{count}_{[\sigma:1]}(\sigma\lambda)$$

at every level, the reversal count blind to the positive factors (Lemma 30), brackets, jumps and extents rescaled by the one factor with their levels. A stored level therefore names its representative, and a seam comparison is two reads of the one integer pencil at matched levels, the factors the display's.

Lemma 44 (The near-contact cell)**.** *The dual pencil, the pair $(\sigma E : M)$'s site datum $P(\sigma)$, is affine in σ, and on a fiber head (Lemma 43) its entries are one polynomial-pair list in (t,σ) over the residue beyond the locality band (Lemma 39; Lemma 14), the window entering through its sampled momenta and its separation range alone, the residue a further coordinate beyond the reach bound (Lemma 25(iii); Lemma 26(iii) at the further series). Every count of the dual chain is therefore Lemma 31's cell function over (t,σ,r) with located divisors: the contact cell is the contact end's, its σ-extent σ_1 the divisor's first positive root, an output of Lemma 25(iii)'s reads, the residue direction the classifications' (Lemma 28; Lemma 29 at the further series), the divisors' roots against the families' clearances; anchors at pair σ are window reads; and $r = 1$ enters as a residue read like any other, $c_1 = 1$ flowing through the channel data (Lemma 24).*

The truncation pair. On the X-sector cell's chain (the vacuum's adjoint closure, one label per index entry at the electric read $4\hat{c}_2$, Construction 13) the chain's count at a level is bracketed by two finite divisors. The first is the plain truncation, a compression, its counts at or below the chain's (Lemma 30). The second is the frontier-bordered pencil: the head with one auxiliary row per frontier label (a head label whose adjoint fusion leaves the cutoff, Proposition 3), coupling one and auxiliary diagonal the pair $[\,\langle 4\sigma c_+ : d_\theta + \lambda\rangle : \varkappa_B\,]$, its first member the seed floor $F := \langle 4\sigma c_+ : d_\theta\rangle$ at the level on the second member: c_+ is the least Casimir beyond the cutoff among the frontier's targets, and F floors the whole excluded block: an excluded label of the closure is nonunit at the unit class (the X-sector's own, Lemma 42(i)), so it moves, one move within the adjoint stencil descending the Casimir (Lemma 9's moves), the moved pair an occupied row read, the fundamental against its complement collecting the target's count through the associativity at the two one-box rows, the full-column summand reading the sum's unit at the moved target and one summand occupied, the withdrawn box's row read against the added one's (Lemma 11; Lemma 16; Construction 5; Lemma 5(iii); Lemma 8(ii)), and the moves chain at the strict descent (Lemma 9), each step's source a member of its target's adjoint row, the chain's first head target inside the degree window at the cleared read's own bound (Proposition 2): every excluded label reads a frontier target at or below its own Casimir, c_+ at or below the whole block; d_θ is the magnetic form cap, kept by every principal block (Lemma 17); and the border cap $\varkappa_B$ is the product of the coupling's largest row and column sums, $BB^{\mathsf{T}} \preceq \varkappa_B$ on the frontier support, the two-sided Schur read. Between any two cutoffs the excluded block sits at or above F (σ times its Casimir floor against the magnetic cap), so at every level below F eliminating it puts the larger head's read at or above the bordered pencil's own Schur read, the auxiliary pivots positive there and inertia additive (Lemma 30): at every level below F the chain's count sits between the pair's two counts at every cutoff, and a level whose two counts agree is a located count of the chain itself, the count beyond every cutoff (Lemma 43). At $r = 2$ the label lattice is two-row, a reduced shape at most r rows deep (Construction 5), and the head decimates by Theorem 18(ii) at the dual representative, the deep rows geometric by Lemma 35(i),(ii). A shell ordering of the elimination is certificate data (Theorem 18(i)), its validity divisors (the seed's dominance root and the envelope's own positivity ceiling) entering the emitted list as any boundary's do; the deep tower rows repeat off the collar (Lemma 40), the decimation's constant-stencil direction.

The contact end is interior to its cell. At the contact end the head pencil is the magnetic member's balance partner (Proposition 6), its level pair's datum the balance partner of $M + \lambda G$, and the counts are fusion-matrix reads, the walk's own divisor data (Theorem 12; Proposition 13): at a level off that divisor the cell's determinant reads unequal members at the contact end, so the contact end sits inside the contact cell and its counts are the cell's, direct reads; at a level on the divisor, Lemma 31's boundary clause extends the positive-semidefinite cells into the closed endpoint.

Lemma 45 (The corner pencil)**.** *Toward the contact end the chain reads in the corner coordinate: the ray $[\,\eta^2 : 1\,]$ at the scale pair η, its levels at the corner scale, a level the product $\eta\,\eta'$ at a height η', itself a balance pair (Definition 1). The corner data are the dual pencil's at the substitution $\sigma = \eta^2$, $\lambda = \eta\,\eta'$ (Lemma 43), and the corner count is the reversal count of the form pair*

$$\operatorname{rev}\big(\,\eta^2 E + \eta\,\eta'_- G \;:\; M + \eta\,\eta'_+ G\,\big) \qquad \textit{at} \quad \eta' = \langle\,\eta'_+ \!:\! \eta'_-\,\rangle,$$

the height's members cross-added onto the pair's two sides (Lemma 30; Theorem 14), the display cleared by the scale's square at the seam identity's integer pencil (Lemma 43), every entry's weight a natural product of the scale's members and the height's. The corner's shifted scaling: at the magnetic form cap d_θ (Lemma 17) the shifted pencil $Q(\sigma) := \sigma E + S_\theta$, at S_θ the pair $(d_\theta\, G\!:\!M)$'s site datum, is a sum of two positive-semidefinite forms (the electric diagonal's; the cap's, Lemma 17), and at scales $\sigma' < \sigma$ (two equal scales reading one pencil outright) with the cofactor ρ at $\rho\,\sigma = \sigma'$,

$$\rho\, Q(\sigma) \;\preceq\; Q(\sigma') \;\preceq\; Q(\sigma), \qquad \rho\,\mu_j(\sigma) \;\le\; \mu_j(\sigma') \;\le\; \mu_j(\sigma),$$

μ_j the shifted roots, $Q(\sigma')$ the sum $\rho\, Q(\sigma) + w\, S_\theta$ at the gap w, the scales' order's witness at $\rho + w = 1$, read against its second summand positive semidefinite, the root reads the comparison's counts (Lemma 30), cutoff-free and window-free, a form comparison compressing. The ground's electric weight rides the contact drift: at the dual representative the ground identity reads

$$\begin{gathered} \sigma\,\langle\psi, E\,\psi\rangle \;=\; \varepsilon_0(\sigma)\,\langle\psi, G\,\psi\rangle + \langle\psi, M\,\psi\rangle, \\ \sigma\,\langle\psi, E\,\psi\rangle \;\le\; g\,\langle\psi, G\,\psi\rangle \quad \textit{at the drift } g \textit{ at } \;\varepsilon_0(\sigma) + m_+ = g, \end{gathered}$$

the magnetic read at or below the top root m_+ of $(M\!:\!G)$ on the head, the contact end's ground level the top root's balance partner (Lemma 33; the contact end's reads Lemma 44's), the order witnessed by the counts monotone in the scale (Lemma 30): the electric weight is priced by two located roots, the drift its own cap, and the scale sandwich's electric weights read it at the grounds' own data (Theorem 20(vii)). The drift obeys a positive-rate cap, the upper family at the contact end: on the X-sector chain's head at the scale key, at a natural N and a pair ξ at the gap w at $\xi + w = 1$, the two comparisons at N (the weighted power $2\,d_\theta A^2 (2N{+}1)^{2p}\,\xi^{4N}$ at or below w at the count reads A, p, and the grade reach $c_r\,(4N^2+1)$ inside the scale key's cutoff) certificate data. The grade key is the highest root's coroot pair, $G(\lambda) := \lambda(\theta^\vee)$, the interface's root coroot pair at θ (Construction 3), one natural per label: at the fundamental fold $\lambda = \sum_i m_i\,\omega_i$ the key is the dual fold's read, $G = \sum_i m_i\, c_i^\vee$ at the halved summands $2\,c_i^\vee = c_i\,\langle\alpha_i, \alpha_i\rangle$ of the residue fold, naturals at one or beyond joining to $\sum_i c_i^\vee = r$ (Construction 4), so the coefficient sum sits at or below G, the vacuum reads the key at the sum's unit, and ρ reads it at r, the simple coroot reads one each. Three reads price the family at the key. The move cap: a θ content is the unit content or a root with its balance partner, the block's list at top θ (Construction 6's content list; Lemma 16; Proposition 13), each content at or below the top (Lemma 3), the join $\nu + \sum_i d_i\,\alpha_i = \theta$ at natural d_i reading $\langle\nu,\theta\rangle \le \langle\theta,\theta\rangle = 2$ against θ's dominant reads (Lemma 2(ii)) at the halved products, with *the balance partner's read the further side: $\nu(\theta^\vee)$ sits at magnitude at most two, and the key moves by at most two under adjoint fusion. The Casimir cap: the fundamental display (Lemma 9) reads*

$$C_2(\lambda) \;=\; \sum_{i,j} m_i\, m_j\, \langle \omega_i, \omega_j\rangle + \sum_i m_i\, \langle \omega_i, 2\rho\rangle \;,$$

every summand capped at the member's fundamental reads against the coefficient sums, each at or below G: at a fixed member the reads are the adjugate rows' displayed folds (Construction 4), and at a series member the rows' magnitudes sit at or below the key (the sorted coordinates: the key telescopes over the coroot presentation's consecutive gaps to the top pair's sum with the last coordinate withdrawn at B_ℓ, the top coordinate at C_ℓ, and the top pair's sum with the next-to-last coordinate withdrawn at D_ℓ, each withdrawn coordinate at or below the kept pair's second member at the sorted order, the last magnitude below its neighbor at D_ℓ), the coordinate squares and the displayed key reads folding to the cap: $\hat{c}_2(\lambda)$ sits below $c_r\,(G^2\!+\!1)$ at one cleared member pair c_r. The counts: the labels at one key value are the dual fold's solutions, at most $(G\!+\!1)$ per simple key, and the member gap product's factors read at or below the θ factor, $\mathrm{dot}(\lambda\!+\!\rho,\alpha) \le \mathrm{dot}(\lambda\!+\!\rho,\theta)$ at the join's read against $\lambda\!+\!\rho$'s coroot pairs with $(\lambda\!+\!\rho)(\theta^\vee) = G + r$, so the per-key label and dimension counts sit below $A\,(G\!+\!1)^p$ at member reads c_r, A, p (Corollary 1; Lemma 25(i) with Lemma 26(i) at the further series, the fixed members' reads the adjugate rows'). The dimension-weighted family is

$$\phi_\xi \;:=\; \sum_{G(\lambda)\le 2N} d_\lambda\, \xi^{G(\lambda)}\, \chi_\lambda,$$

the sum over the chain's labels at the grade key, the characters orthonormal (Proposition 8) and the vacuum's term reading one; at the A-series the key reads the reduced shape's top row, the pair $\langle \lambda_1 : \lambda_{d_f}\rangle$ at the reduced shape's one-member site with the full column's read of equal members (Construction 5), one value per label. The move cap prices the magnetic read termwise at the dimension eigen-identity $\sum_\mu N^{\mu}_{\theta\nu}\, d_\mu = d_\theta\, d_\nu$ (Lemma 17), every summand a natural at pair weights with the key's weight falling as the key rises:

$$\langle \phi_\xi,\, M\,\phi_\xi\rangle + B \;\ge\; \xi^2\, d_\theta\, \langle \phi_\xi, \phi_\xi\rangle,$$

B the top two key values' boundary term at the cap d_θ, the two keys' part of the pairing priced at the counts' squares, below w times the vacuum's term at the weighted power comparison. The electric read prices at the key's caps, $\hat{c}_2(\lambda)$ below $c_r\,(G^2+1)$ with the per-key label and dimension counts below $A\,(G\!+\!1)^p$, the pairing's per-key weight the fold of the dimensions' squares at or below the count's own square $A^2(G\!+\!1)^{2p}$, and the weighted geometric sums close on two displayed reads. The power cap is one rising-product read,

$$(L+1)^{\,j} \;\le\; (L+1)(L+2)\cdots(L+j)\;,$$

every factor $L+i$ at or beyond $L+1$; and the graded sums at the rising-product weights, $S_j(M) := \sum_{L\le M}(L+1)\cdots(L+j)\,x^L$ at $x+y=1$, descend by Pascal's identity at the rising products, $(L+1)\cdots(L+j+1) = L\cdots(L+j) + (j+1)\,(L+1)\cdots(L+j)$, the exact displays

$$y\,S_{j+1}(M) + (M+1)\cdots(M+j+1)\,x^{M+1} \;=\; (j+1)\,S_j(M)\;, \qquad y\,S_0(M) + x^{M+1} \;=\; 1\;,$$

so $y^{\,j+1}S_j(M) \le j!$ at every M. At $x = \xi^2$ the gap reads $y = w\,(1\!+\!\xi)$ at or beyond w, the vacuum's unit term floors the pairing, and

$$w^{\,2p+3}\,\langle \phi_\xi,\, E\,\phi_\xi\rangle \;\le\; 4\,c_r\,A^2\,(2p+2)!\;\langle \phi_\xi, \phi_\xi\rangle \;:$$

the rate's data are $q = 2p+3$ and $C_E = 4\,c_r\,A^2\,(2p+2)!$, member reads. The Rayleigh read at ϕ_ξ then closes the drift's rate against the head's top root m_+ at its cap d_θ (Lemma 17): at the scale's w, the bracket $w^{\,q+1} \le \sigma \le (2w)^{\,q+1}$ the certificate's own read, one power comparison,

$$g(\sigma)^{\,q+1} \;\le\; C_g^{\,q+1}\,\sigma,$$

one cross-multiplied comparison per scale at the collected read $C_g = 2^{\,q+1}C_E + 2\,d_\theta + 1$, the join read at the count carrier: the Rayleigh read puts the count occupied at the assembled margin's unit shift, the scale's electric power at the bracket's upper side with the boundary read over the cap's weight; the head's first occupied level $\varepsilon_0(\sigma)$ reads the count vacant at its own level and at every level below; the monotone count read (Lemma 30) holds the vacant level strictly below the occupied one, the two levels' difference site certified positive semidefinite at the unit gram's scalar copy with the equal-level reading refused at the vacant certificate, so $\varepsilon_0(\sigma)$ sits at or below the assembled margin; the top root's cap prices m_+ at d_θ (Lemma 17); and the drift $\varepsilon_0(\sigma) + m_+ = g$ obeys the margin comparison, the power comparison's own drift datum, a located C_g per member and direction count: the contact drift falls at a certified positive rate, the halving scales' ground-level widths of the scale sandwich then geometric over the tail at the contact-drift display's weights, their sum one cofactor read (Theorem 20(vii)); and the tail's count read is the corner disconjugacy certificate's (Lemma 46). On a fiber head the entries are Lemma 44's list at the substitution, one polynomial-pair list in (t, η, η') over the residue beyond the locality band, so every corner count is Lemma 31's cell function over the (t, η, η') base with located boundaries, and the corner divisor is Theorem 25's crossing read at that base. The corner cell is the contact end's, and its data are two located outputs per member and direction count, each with its leading-term radius (Theorem 15; Lemma 25(iii); Construction 7's direction reads): the extent η_1, the corner divisor's first positive root in the scale over the cell, and the floor κ, the flat step's height extent on it (Theorem 19). The ray coordinate reads the outputs back at squares: the ray $[\,a : b\,]$ sits in the extent exactly at $a\,d_1^2 \;\le\; n_1^2\,b$ at $\eta_1 = [\,n_1 \!:\! d_1\,]$, and a level reads through its height, the cofactor at $\eta\,\eta' = \lambda$ (Definition 1), the height against the floor one pair comparison: the level's square against the weight product, the two end weights' own scale.

The boundary families. The corner cell's boundaries are the located roots of four families over the (t, η, η') base, each family one of the chain's comparisons read at the substitution and entering the list as any boundary does (Theorem 18, the emitted record), the corner divisor's own root list in the scale beside them, the extent its first positive member: the jump family, the pencil determinant's roots at the bracket's heights, the decimation's own jump divisors at the substitution (Theorem 18); the frontier family, the decimated envelope's dominance comparison at its seed, the frontier Casimir's η^2-weight against the tail weight, a height at which the two envelope counts agree a located count of the chain itself (Lemma 44's decimated presentation; Lemma 43), so the cutoffs enter through the envelope pair alone, and the scale keys its cutoff: $\Lambda(\eta)$ is the least natural at $4\,\eta^2\,\Lambda \ge d_\theta + \lambda_+ + \vartheta$, read over the window's top level λ_+ at a clearance pair ϑ, one comparison per scale, the seed floor $F = \langle 4\sigma c_+ \!:\! d_\theta\rangle$ then clearing every window level by ϑ at once ($c_+ \ge \Lambda$, the frontier's least Casimir beyond the cutoff), so the envelope pair brackets the chain's count at every scale of the tail together, the per-scale cutoff quantifier riding the one key read, the key's comparison its own witness at every member, d_θ the member's dimension fold with the A-series' $r\,(r+2)$ its read (Proposition 13), and the agreement heights' divisors the jump family's own (Lemma 43); the deck family, the bulk blocks' crossing loci with the graph read's side and the spectator caps' three counts, each a cell datum of the corner base (Lemma 35(iii); Lemma 38), the $r = 1$ reads among them, and the contraction cap read on the cell at the boundary root's isolation datum: the cell is a component of the divisor list's complement, the located root's bracket its margin, and the cap's margin squares to the crossing read, $\langle 1 \!:\! z^2\rangle^2 = z^2\,\langle w^2 \!:\! 4\rangle$ (Lemma 38), the walk's mass point at the *crossing pair* $[\,9:4\,]$ *at base two (Theorem 12); and the tension family, the charged sectors' tension, the winding shift's dressed floor against the unit class's dressing folds, and the separated sectors' clearance at the height floor (Lemma 42(iii),(iv); Lemma 41(iii)), one root list per charge vector. Every window read sits in the list's cells: the momenta are sampled in* t *(Lemma 39), the slabs read within the spectator brackets (Theorem 18(iii)), the cutoffs within the envelope pair, and the sectors at their families; so the extent and the floor are outputs of the one list, and the cell's* t*-section at the outputs is the whole range exactly where the list's* t*-roots sit off it, itself a located read. The rank direction inherits: the substitution fixes every entry's residue dependence, so the families' divisors are the chain's own read at corner base points, the deck base the member-key count, one value at every cleared residue (Proposition 13; Lemma 25(ii)).*

The cell floor. The flat step's height extent on the corner cell is a positive located datum of the cell: the ground root is simple over the cell (Theorem 9; Theorem 25(iii)), the second jump is separated at Theorem 15's floor, and the counting functions are flat between them at the certificate's two lines (Lemma 46). The heights read two-sided on the cell: the lower side the floor's clearance, the upper the truncation's second root against the bordered pencil's ground, the compression's counts at or below the chain's and the bordered pencil's at or above (Lemma 30; the truncation pair, Lemma 44), the two root bounds exchanging; an extent interval closes at its mixed window, the ground's read at the interval's top with the second's at its bottom, counts monotone in the coordinate (Lemma 30), so the window is flat across the interval and the height clears the interval's floor there, per member and direction count. Each cell reads its edges at word images: the flat window's upper end is the cell's second root, a point of its own divisor (Lemma 33; Theorem 25(i)), and the window cut's edge there, where one is named, is the compressed pair's least root, read at a word image with the two moments at equality and the first positive, the state the word sector's own (Lemma 22(i),(ii)), one word image per window read on the cell at the corner scale. The extent's positivity is the contact end's interiority read, the σ_1 *brackets Lemma 44's own outputs, and the floor's is the certificate's, the rates derived and the gap a cut member's read, positive by its shape (Lemma 46(v),(vi)), the rank direction at the key's bracket and the families' clearances (Lemma 25(iii)).*

Lemma 46 (The corner disconjugacy certificate)**.** *On a band-one head at the dual representative, at a scale pair* η *and a probe level cross-added in the diagonal (Theorem 14's spelling), the count below the probe is the pivot sequence's lower-side count (Lemma 30), and the read at most one is settled by four displayed comparisons at the corner scale on the depth axis: the entries are polynomials in the scale and the product* $m\,\eta$ *(the corner scale's multiples, a pair read each), and every comparison below is a two-variable polynomial read of those data.*

(i) The pivot comparison. At a positive list v *on a segment with the termwise read*

$$v_m + v_{m+2} \;\leq\; a_{m+1}\, v_{m+1}$$

and the entry read $p_{m_0} v_{m_0} \geq v_{m_0+1}$*, every later pivot of the segment clears its entry read,* $p_m\, v_m \geq v_{m+1}$*: the pivot identity* $p_{m+1}\, p_m + 1 = a_{m+1}\, p_m$ *with the two reads* $a_{m+1}\, v_{m+1} \geq v_m + v_{m+2}$ *and* $p_m\, v_m \geq v_{m+1}$ *drives the induction, one cross-multiplied step per depth.*

(ii) The segment supersolutions are stated polynomial profiles. A segment's list enters as one polynomial V *in the scale and the depth's two pair reads* $u = m\,\eta$ *and* $s = m^2\eta$*, read per depth,* $v_m = V(\eta,\, u,\, s)$*; the two pair reads join at the relation* $u^2 = \eta\, s$*, the products' one read at the substitution, so* V *reads at* u*-degree at most one, its datum two slabs of scale polynomials at the square-scale key, the scale-free slab and the* u*-slab, a product's* u*-square term re-entering the scale-free slab at the relation's read: the termwise comparison of (i) is then one polynomial read in the depth and the scale, its positivity on the segment closed by the leading-term radius in the depth (Theorem 15) with the finitely many depths below the radius each one read, and the well's data*

enter at the square scale, where the probe line $g(\eta)$ meets the band read 2 at a stated pair rate and the comparison's polynomials read the scale through the pair reads alone. The profiles enter as stated polynomial data at the recurrence's comparison: at the probe line's rate q the diagonal reads $a_m = 2 + \eta\, W$ at the well read, the balance pair

$$W \;=\; \big\langle\, [\,4 : r+1\,]\,(s + r\,u) \;:\; q\,\big\rangle,$$

its first member the tower's own electric read, $[\,4 : r+1\,]\,(s + r\,u) = 4\,\eta\,\hat{c}_2(m\theta)$ at the substitution (Proposition 12), the neighbor fold has one scale factor, $v_m + v_{m+2} = 2v_{m+1} + \eta\, D[V]$ at the displayed action

$$m^{\,j} + (m{+}2)^{\,j} \;=\; 2\,(m{+}1)^{\,j} + 2 \sum_{i\geq 1,\ 2i\leq j} \binom{j}{2i}\,(m{+}1)^{\,j-2i}$$

read per monomial of the pair coordinates, so (i)'s comparison is the polynomial read $D[V] \leq W\,V$; a stated-degree profile is a supersolution exactly where the remainder $W\,V_N - D[V_N]$ and the profile read positive on the segment, two located divisor reads at the stated degree, the degree certificate data held at its statement. The constant-floor instance is the walk's own list, the recurrence $U_j + U_{j+2} = g_i\, U_{j+1}$ at the boundary's unit seed $U_1 = 1$ read from the boundary (the truncation data of Theorem 12's recursion, integer-polynomial in the floor); seams glue at one entry read each.

(iii) The crossing, and the rebound. One located depth reads its pivot on the lower side; the following pivot exceeds its diagonal, the identity at a lower-side predecessor read on the exceeding side (the rebound read), and seeds the later segments' entry reads at the diagonal itself. A depth whose minor reads equal members re-orders within its slab at Lemma 30's principal pivots of order at most two (Lemma 34), the count read unchanged across the splice: the deep end takes the order-one kernel block, and an interior depth its order-two block with the following minor read through the spliced pair.

(iv) The dominance tail. At depths whose diagonal clears 2 by a stated margin the constant list is the supersolution, and the pivots stay clear to the head's end; and at a stated allowance, a pair, the geometric list is one as well, each member the prior member's multiple at one pair of the scale's data, the pair's gap the allowance's read of the depth step at the seam depth: the neighbor sum joins the arriving member's double at the gap's square against the withdrawn member, so the margin's clearing of the squared gap's doubled price, the gap at most the pair's half, reads the termwise comparison at every deeper depth, the dominance comparison's own growth, and the constant list is the vacant allowance's instance.

(v) The ground witness, and the lower line as the family's own read. The witness family is $x_m = s\,\langle s_2 : s\rangle$ at the square-scale pair $s_2 = N^2\eta$, the endpoint depth N the least natural beyond one at $40\,N^4\eta^2 \geq 363\,(r+1)$, one key read per scale, the comparison the rate's two top terms' equality below; the family's read at the endpoint is equal members and beyond it the sum's unit. The form read telescopes at unit bonds, the boundary bond's square x_1^2 the displayed first tail term,

$$\sum_{m\geq 1} a_m x_m^2 \;=\; 2\sum_{m\geq 1} x_m x_{m+1} + x_1^2 + \sum_{m\geq 1} \langle x_{m+1} : x_m\rangle^2 + \eta \sum_{m\geq 1} W_{-}\, x_m^2\ ,$$

W_{-} the lower line's well read, the bond pair per depth the displayed square $(2u{+}\eta)^2\,\langle s_2 : 2s{+}2u{+}\eta\rangle^2$, and each sum is one closed read of the depth's powers at the endpoint (the power sums' binomial identities), the closed reads' top coefficients at the endpoint's powers $[\,8 : 315\,]$ at the family's square, degree nine, $[\,44 : 105\,]$ at the difference sum, degree seven, and $[\,4 : r+1\,][\,8 : 693\,]$ at the well sum's square part, degree eleven. The lower rate is the family's own cleared read,

$$q_{-}\,\eta \sum_{m\geq 1} x_m^2 \;=\; [\,4 : r+1\,]\,\eta^2 \sum_{m\geq 1} (m^2 + r\,m)\,x_m^2 + \sum_{m\geq 1} \langle x_{m+1} : x_m\rangle^2 + x_1^2\ ,$$

one pair per scale and residue, the cofactor at the closed reads (Definition 1), its two top terms $[\,33:2\,]$ *at the cofactor against* ηN^2 *and* $[\,20:11\,(r+1)\,]$ *at the factor* ηN^2*, the tops' cofactor reads, whose equality is the key's comparison,* $40\,N^4\eta^2 = 363\,(r+1)$*: at the line* $2+q_-\eta$ *the form read's two sides read equal members at the family, a positive-definite read excluded, so the least root sits at or below the line (Lemma 30; Lemma 33), the family its own witness. The second root prices from above by the pair read: at a cap line and a second stated family* x' *of the witness's shape, the span's compressed pencil is one* 2×2 *read whose entries are power-sum reads of the two lists, and a compression's counts sit at or below the full form's (Lemma 30), so a compressed count of two at the cap line places two chain roots below it: the second root below the cap, the heights two-sided on the cell.*

(vi) The probe lines, and the floor. The lower rate is the witness family's own read ((v)); an upper rate is a multiple, $q_+ = \mu\,q_-$ *at a pair* $\mu = [\,a:b\,]$ *beyond one, and the multiple cut is the pairs at the count's read, at most one below the line* $2+\mu\,q_-\eta$*: the counting function is monotone in its level (Lemma 30), so the cut is downward closed toward one, and membership at a member is settled by (i)–(iv)'s reads at its line, a member's reads its whole verification. The ground sits at or below the lower line by (v), so the counting function is flat from the lower line to a member's, and the rates' gap,* o *at* $q_- + o = \mu\,q_-$*, is the read* $o = [\,g:b\,]\,q_-$ *at the member's gap* g *at* $b+g=a$*, positive by its shape: the gap clears* $o\,\eta$ *at every tail scale and the height's square clears* o^2*, one cross-multiplication per scale, the floor per member and direction count at every pair of the cut. The residue direction is a coordinate: the rates are reads of the key's power sums, so the comparisons' polynomials take the residue as a coordinate at the key's bracket (Lemma 25(iii)), their roots read against the families' clearances, one symbolic read at every residue at the cleared families, and the channel-degeneration residues read the walk's own displayed degenerations, the mass point at the crossing among them (Theorem 12; Lemma 24).*

(vii) The block chain. On a decimated head's block chain (Theorem 18(i); Lemma 34) the certificate runs at block data, the band-one clauses its scalar instance: the count splits over the depth pivots (Lemma 34(ii)); a segment's pivots stay positive at the ball certificate of Lemma 35(iii) read at per-depth stated centers, one matrix polynomial in the scale and the depth's pair reads, the three counts cleared by the centers' determinants to polynomial comparisons and read as cell functions over the corner base (Lemma 31), the center defect the drift's slot; the crossing is one block's split at one lower-side unit with its witness list certificate data (Lemma 30), and the rebound is the split's rank-one read, the following pivot at or beyond its diagonal less the positive part's transfer, one congruence identity and one comparison; the dominance tail reads at block diagonal dominance; and the ground witness and the cap pair read lift through the compression unchanged ((v); Lemma 30). The collar's finitely many blocks read exactly, the bonds enter through their squares alone, and the certificate's data gain the center family with its pairs and the crossing split.

The certificate's data are one stated pair rate with every profile comparison read at it and the scale's cleared ceiling with every tail scale read under it, each datum its reads' validity divisor's located edge in its own direction, the designation's inner endpoint (Theorem 14), the count's line read at every rate at or below the stated one, the cut members' lines among them at their multiples of the lower rate, the transfer the diagonal's monotonicity in the rate, the rate entering the well read on its pair's second member, one cross-multiplied comparison per depth, the segment list with each boundary the cross-multiplied pair comparison of $m^2\eta$ *against a stated pair and with consecutive boxes overlapping at the two further depths the termwise comparison reads, a depth inside one box's top stepping to a square scale inside the next box's, the profiles' stated degrees, the boundary seed's data, the three clearings of the boundary value, of the boundary comparison's object, the shifted diagonal's multiple of the profile against the rate's multiple of the shifted profile, and of the termwise comparison's scale strip, each floored past its origin by the ceiling's priced fold, the deep* *coefficients' fold against the ceiling's second member's powers at the stated degree, the floor at one power beyond the fold's and the squared floors' square at one beyond twice it, the origins vacant at the graded device's scale-free vacancy: the depth read's origin is the profile's own, the comparison's origins read the band clearing's multiples of vacant depth-read origins, the termwise comparison's origin is vacant at every profile and rate, the two origin slabs' withdrawal, and the strip's origin is the affine depth reads' second difference, the depth read's scale coefficient at a vacant slab the first inner's own joined to the depth's multiple of the u-slab's origin; the slope allowance, a pair, with the dominance margin and the tail's three scale-free comparisons, the margin located at the final device's floor, one cross-multiplied read, the margin against the allowance's squared read at the geometric tail's handover cap, and the gap's half cap at the ceiling, and per segment and seam the box devices' finitely many polynomial comparisons in the scale and the scaled depth on stated boxes, their reads located divisor data, the lower and upper rates and the witness pair (v)'s derived reads at the endpoint key (Theorem 15; Lemma 31). A box device prices a profile's depth reads from below on its box at the scale's cleared ceiling, two naturals c_0, d_0 with the scale's bound $\eta\, d_0 \leq c_0$ read cross-multiplied: the depth's read splits as the scale-free part, the scale-free slab's origin coefficients at the square scale, joined to the scaled remainder, and the remainder's cap follows the profile's grading one order down: its origin coefficients, the scale-free slab's scale coefficients with the u-slab's origins, read on the box at the chained priced side read, the scale coefficients above their floor's balance partner and the u-slab's origins at their squared price with the square scale the box's own coordinate, each comparison at its own cut list and its floor at or beyond the sum's unit, while its deeper coefficients price at the ceiling's fold through the box top's cleared read into the floors, the fold's power the stated degree's successor at the depth shift's second clearing read once more at the scale's bound, the floor at one power beyond the fold's and the squared floors' square at one beyond twice it, the squared fold's floor at or beyond the sum's unit, so the read clears the sum's unit wherever the scale-free part clears the floors' join at the chained priced side read with the remainder inside it; the box's data are the composite endpoints with the cut lists, the priced side reads' bound, the box top's cleared read, the floors and the shape clearing. The split follows the profile's grading: a profile occupied at the scale-free order prices there, and a profile vacant there prices at the scale order, the depth's leading part the depth multiple of the u-slab's origin reads joined to the scale-free slab's scale-key reads, one read per depth and linear in it. The depth family collects at two chains: every depth's square clears its square scale at the ceiling's cross-multiplied bound, and at origin reads above the sum's unit on the box a deeper multiple only raises the read, so the family's clearing of the tails' join reduces to one line, the scale-key read's above the join; piece by piece along one cut list the piece keeps the line's upper side, or the squared comparison's, the square scale's multiple of the origin reads' square at the ceiling's cross-multiplied read above the squared gap of the join to the scale-key read, whose read returns the depth multiple past the gap at the trichotomy of squares; the origin reads' positivity is the second chain, a vacant u-slab reading the first chain alone: the two cut lists, each piece at its one comparison's kept side, the device's further data at the leading part. The well's oscillation reads at the corner scale, one read at every scale, so one segment list runs the whole tail, and the one certificate settles the count at most one at every depth and every tail scale at once. The entry reads chain across the segments to the crossing seam: an entry read crosses a seam at the crossed products' read, the outgoing profile's shift against the incoming profile joined at the swap to the incoming's shift against the outgoing, positive on the seam's box, the products' injectivity reading the crossed entry back at positive profile values; the crossing seam is the two chains' break at the count comparison alone, the rebound seeding the later chain's entry reads; the first lower-side pivot sits past it with the later boxes' reads running from below it, and the count read closes beyond it at (iii)'s rebound seed and (iv)'s dominance tail at the final seam's own read, the last profile's shift joined to the allowance's multiple of the step read of the profile clearing the* *profile on the seam's box, the flat window's lower end the ground bracket's own (Theorem 9).*

Lemma 47 (The moment folds)**.** *On a band-one head at the dual representative, at the ground's recurrence $b_m\psi_m + b_{m+1}\psi_{m+2} = A_{m+1}\psi_{m+1}$ (the eigen-identity's row read, the diagonal at the ground root cross-added), the ground enters every datum read through two moment streams,*

$$\rho[P] \;:=\; \sum_m \psi_m^2\, P(m)\ , \qquad c[P] \;:=\; \sum_m \psi_m\, \psi_{m+1}\, P(m)\ ,$$

P over the polynomial weights, and the streams solve the recurrence's own folds, exact at every scale:

(i) The fold identities. Folding the recurrence against $\psi_m P(m)$ reads

$$c\big[\, b\, P\,\big] + c\big[\, b\, P^+\,\big] \;=\; \rho\big[\, A\, P\,\big] \qquad (P^+(m) := P(m{+}1))\ ,$$

and the fold against $\psi_{m+1}P(m)$ reads a second identity of the recurrence at the shifted weights, its second-bond term reduced by one further fold of the recurrence: every correlator beyond the two streams re-enters them, so the family closes on ρ and c alone, one identity per weight, each a display of the recurrence at a polynomial weight. A commutator's ground read is of equal members outright (a commutator of two symmetric operators' pairing at a ground reads equal members), which is why the folds rather than commutators close the system.

(ii) The graded system, and its clearance. At the corner the weights read in the pair coordinates ($u = m\eta$, $s = m^2\eta$): the diagonal A_m reads σE_m at weight-degree two, so each fold relates the streams' degree-k reads to degree-($k{+}2$) at one scale factor, and the identities at weights through a stated degree, with the eigenvalue's bracket (the certificate's two lines, Lemma 46(vi)) and the unit read $\rho[1]$ at the gram, are one linear system for the truncated moment vector over the located stage, its entries polynomial in the scale and the rates: the solve is the elimination's (Definition 3), and the system's determinant clearance is one located divisor read over the corner base (Lemma 31), the validity cell the certificate's architecture.

(iii) The truncation cap. Beyond the stated degree the streams' tails read in the dominance region, where the ground's components ride the transfer factors (Theorem 20(ii)): the beyond-degree weight is priced by the certificates' geometric caps against the turning read, one comparison at the stated degree, the degree held at its statement.

(iv) The scale comparison. Two scales' systems share their shape with coefficients differing by displayed polynomial reads of the scale, so on the clearance cell the two moment vectors differ by the elimination's cofactor reads at those differences, one width per moment, and the widths take the scale factor: the streams' reads close across the scales on the clearance cell.

(v) The datum assembly. A single-region datum read is a stream read at banded weights (Lemma 14), so its scale closure is (iv)'s on the clearance cell; separated and Euclidean data assemble from single-region reads at the crossing certificates (Theorem 20(iv)–(vi)), the certificates cell data (Lemma 31), so the assembled reads close where the parts do. The block chains read the streams at block weights through the folds of their own recurrence, Lemma 34(iii)'s row read.

Lemma 48 (The cone read)**.** *The jump family's momentum structure at the corner is isotropic at degree two: on the decimation's validity cells the decimated determinant's root locus is the fiber symbol's own, one Schur factor at a complement of unequal-membered determinant (Theorem 25(i)); the fiber symbol's coefficients are permutation-invariant polynomials of the chord list at every coupling of the approach (Theorem 21, at Lemma 45's base), their momentum reading even in each coordinate at the deck relation's exchanged roots (Definition 4; Theorem 21); the degree-two invariants at that evenness are the standard square's span (Theorem 21, degree 2 at the rings'* *coincidence); and the direction data enter at degree four, the breaking multiplicities positive there. So the jump family's loci at the corner read the isotropic square at degree two with one located coefficient per root, the root reads Lemma 32's at Theorem 15's separations, and the direction reads two momentum degrees down.*

Lemma 49 (The free cell)**.** *The pencil at the α=1 representative, the pair $(E\!:\!\tau^2M)$'s site datum $H([\,1:\tau^2\,])$, is affine in τ^2, and on a fiber head (Theorem 32) its entries are one polynomial-pair list in (t,τ) over the residue beyond the locality band (Lemma 39; Lemma 14), the window entering through its sampled momenta and its separation range alone, the residue a further coordinate beyond the reach bound (Lemma 25(iii); Lemma 26(iii) at the further series): every count near the free end is Lemma 31's cell function over (t,τ,r) with located divisors, and the free cell is the free end's, its τ-extent the divisor's first positive root, an output, the contact cell's construction at this end (Lemma 44).*

The free end is interior to its cell. At the free end the pencil is its electric member E, diagonal at the free reads: the decimated symbol's jumps are the occupied contents and the level is the member floor label's loop read, one diagonal computation at every window (Lemma 15). At a level off the content list the cell's determinant reads unequal members at the free end, so the free end sits inside its cell and its counts are the cell's, direct reads; at a level on the list, Lemma 31's boundary clause extends the positive-semidefinite cells into the closed endpoint. The free-end reading of Definition 11 is those reads.

Lemma 50 (The four-point floor)**.** *Fix two plaquettes sharing one link, the theta graph (Construction 9), $q_i := \chi_\theta(U_{\partial p_i})$ the pair's modules, and the ray $[\,1:\tau^2\,]$ at a window ground ψ beyond the graph's locality band (Lemma 14). The connected four-point read at the pair, each field twice, is the moments' partition fold at $\mu_{ab} := \omega(q_1^a q_2^b)$, one term per partition of the four fields at the blocks' moment product, the weight $j!$ at the join $j+1=b$ over the block count b, the one-block term at weight one, and the side the count's parity,*

$$\varkappa_4 := \left\langle\, \mu_{22} + 2\mu_{10}^2\mu_{02} + 2\mu_{01}^2\mu_{20} + 8\mu_{10}\mu_{01}\mu_{11} \;:\; 2\mu_{10}\mu_{12} + 2\mu_{01}\mu_{21} + \mu_{20}\mu_{02} + 2\mu_{11}^2 + 6\mu_{10}^2\mu_{01}^2 \,\right\rangle .$$

At every member the read holds a positive located floor on a cell at the free end's side: at the pair fold $S := \sum_k w_k\,[\,1:6+\hat c_2(k)\,]$ over the θ-square channel list (Construction 6's field; Lemma 24 the A instance) at the module–pair weights w_k, the pair networks' channel norms $[\,m_kd_k : d_\theta^2\,]$ (Theorem 23(i),(ii), the displayed weights the A instance; Lemma 5(ii)), the read sits within the priced tail's bracket of the jet's value

$$\hat\varkappa_4 \;=\; \tau^4\,c_1^2\,\big\langle\, S : [\,1:8\,] \,\big\rangle \;,$$

whose pair's side is its upper at every member, the margin at or beyond $[\,c_1 : 42\,d_\theta^3\,]$.

The magnetic jet. The ground's jet at the free end is the displayed second-order solve (Construction 12's shape): $\hat\psi := 1+\tau^2\psi_1+\tau^4\psi_2$ at $\psi_1 = [\,1:4\,]\sum_p q_p$, each module an eigenvector at the plaquette content four (Proposition 7), and ψ_2 the solve at $M\psi_1$'s components off the unit line: a single-plaquette channel $\chi_k(p)$ at the content $4\,\hat c_2(k)$, an adjacent pair's product at its channel split $q_pq_{p'} = \sum_k F_k$ over the shared link's channels, the contents $6+\hat c_2(k)$ and the weights $\langle F_k,F_k\rangle = w_k$ (Theorem 23(i),(ii)), and a distant product at eight, each component's solve the content's cofactor.

The moments' displayed orders. On the theta graph the evaluation splits at the shared link and the outer word: the shared read pairs the two loops' channels by the evaluation identity and the outer word reads the unit label (Theorem 23(i); Proposition 11), so $\mathrm{Eval}\big(f(U)\,g(V)\big)$ is the two unit coefficients' product and the graph's free moments are the per-plaquette reads $m_a := \mathrm{Eval}(\chi_\theta^a)$, with $m_0 = 1$, m_1 of equal members at the nonunit θ, $m_2 = 1$, $m_3 = c_1$ and m_4 the square's pairing

(Proposition 8). A far insertion factorizes at its own links (Lemma 14): a plaquette off the pair holds a link off the union, the union's four-cycles the two boundaries (the incidence count, the boundary the plaquette's own word), so the jet moments read the pair's insertions alone, the far diagonal reads entering the pairing and the gram equally, a summand on both members fixing the value (Definition 1): with $T_a(k)$ the k-count of the a-th power ($T_0(k)$ the unit read, $T_1(k)$ the θ read, $T_2(k) = N^k_{\theta\theta}$, $T_2(\theta) = c_1$), the jet moments read

$$\hat{\mu}_{ab} \; := \; m_a m_b + \tau^2 \,[\,1:2\,]\,\big(m_{a+1}m_b + m_a m_{b+1}\big) + \tau^4 D_{ab} \;,$$
$$D_{ab} + [\,1:8\,]\, m_a m_b = [\,1:16\,]\big(m_{a+2}m_b + 2m_{a+1}m_{b+1} + m_a m_{b+2}\big)$$
$$+ [\,1:8\,]\sum_{k\neq \mathbf{1}} N^k_{\theta\theta}\,\big[\,T_a(k)\, m_b + m_a\, T_b(k) : \hat{c}_2(k)\,\big] + T_a(\theta)\, T_b(\theta)\, S \;,$$

D_{ab} the join's datum over the balance pairs, the first fold the pair insertions' square reads, the second the single-plaquette channels' solves, and the third the pair networks': a product state's pairing against the pair networks reads at the θ-labeled free links alone, the gram's configuration split (Definition 8), so $\langle q_1^a q_2^b,\, F_k\rangle = T_a(\theta)\, T_b(\theta)\, w_k$.

The collection. The partition fold collects the displayed orders: the τ^2 reads enter the fold's two members equally (m_3m_2-type terms on each side), the single-channel folds and the m_4 reads enter D_{22} and twice D_{20} equally, and the τ^2-cross products collect against the square terms, so the fold closes on the pair-network read against the distant product's, $\hat{\varkappa}_4 = \tau^4 c_1^2\,\langle S : [\,1:8\,]\rangle$, the jet moments' partition fold at the displayed orders: the pair solve at the shared channels' contents against the two modules' first-order reads at eight.

The side. The channel list's two closure identities, $\sum_k w_k = 1$ and $\sum_k w_k\, \hat{c}_2(k) = 2$, the square's exhaustion count and the uniform drift (Lemma 5(ii); Theorem 24; Theorem 10's displays the A instances), read $\sum_k w_k\,(6 + \hat{c}_2(k)) = 8$: the product of S with that fold, against the weights' squared fold $(\sum_k w_k)^2 = 1$, collects per channel pair, each pair's cleared read the square identity $(6+\hat{c}_2(k))^2 + (6+\hat{c}_2(l))^2 = \langle \hat{c}_2(k){:}\hat{c}_2(l)\rangle^2 + 2\,(6+\hat{c}_2(k))\,(6+\hat{c}_2(l))$, one display per pair, so

$$8\,S \;=\; 1 + \sum_{k<l} w_k\, w_l\,\big\langle \hat{c}_2(k) : \hat{c}_2(l)\big\rangle^2\,\big[\,1 : (6+\hat{c}_2(k))(6+\hat{c}_2(l))\,\big] \;,$$

every summand a square at pair weights, positive by its shape at distinct channel Casimirs, and the fold at or beyond its unit against θ term, the unit channel at the kernel point and weight $[\,1:d_\theta^2\,]$ against the θ channel at one and weight $[\,c_1 : d_\theta\,]$ (Proposition 8; Proposition 13's dimension fold): $8S$ exceeds one at the margin $[\,c_1 : 42\,d_\theta^3\,]$ or beyond, at every member.

The tail, and the cell. The jet's identity is exact at the displayed join,

$$E\,\hat{\psi} + \tau^4\,[\,\#p:4\,]\,\mathbf{1} + \tau^6 M\psi_2 \;=\; \tau^2 M\,\hat{\psi} \;,$$

$E\psi_1 = M\mathbf{1}$ outright (the modules' fold, its unit coordinate the sum's unit) with $E\psi_2$ against $M\psi_1$ the join at $[\,\#p:4\,]\,\mathbf{1}$, the unit coordinate of $M\psi_1$ at $m_2 = 1$ with every cross product's at m_1 of equal members, so the jet solves the pencil to the τ^6 read: at $\mathbf{1}$ read as $\hat{\psi}$ against $\tau^2\psi_1 + \tau^4\psi_2$, the pencil's site datum H at the ray reads

$$H\hat{\psi} \;=\; \hat{\varepsilon}\,\hat{\psi} + \rho \;, \qquad \rho := \tau^6\big(\,[\,\#p:4\,]\,\psi_1 + (M\check{\psi}_2)\,\big) + \tau^8\,[\,\#p:4\,]\,\psi_2 \;,$$

the jet level $\hat{\varepsilon}$ the balance partner of $\tau^4[\,\#p:4\,]$ and ρ the residual, the check the memberwise swap. The window ground ψ at its level ε_0, the level gap $\tilde{E}$ (Definition 10) and the clearance γ', every further root at or beyond it on the free cell (Lemma 15; Lemma 49), prices the jet's part φ off *the ground's line by the transport display's pairing step (Theorem 20(iii)): $\gamma'\langle\varphi,\varphi\rangle \le \langle\varphi,\tilde E\varphi\rangle = \langle\varphi,\tilde E\hat\psi\rangle$, the ground in the gap's kernel, and $\tilde E\hat\psi$ reads $\langle\hat\varepsilon\,{:}\,\varepsilon_0\rangle\,\hat\psi+\rho$ at the identity, so*

$$\big(\gamma'+\langle\varepsilon_0\,{:}\,\hat\varepsilon\rangle\big)\,\langle\varphi,\varphi\rangle \;\le\; \langle\varphi,\rho\rangle\ ;$$

the drift $\langle\varepsilon_0:\hat\varepsilon\rangle$ is floored at the magnetic member's cap $W := \#p\,d_\theta$ (Lemma 18 per term; Theorem 32's cap at the ray's magnetic weight): the ground's level reads $\langle\psi,E\psi\rangle$ against $\tau^2\langle\psi,M\psi\rangle$ over $\langle\psi,\psi\rangle$, the electric read on its upper side or equal members and the magnetic read capped, so ε_0 sits at or beyond the balance partner of τ^2W while $\hat\varepsilon$ sits at or below the sum's unit, and the pairing's squared Cauchy–Schwarz (Construction 8) reads

$$\langle\gamma'\,{:}\,\tau^2W\rangle^2\,\langle\varphi,\varphi\rangle \;\le\; \langle\rho,\rho\rangle \qquad \text{at}\quad \tau^2W<\gamma'\ .$$

The residual's weight is the caps' fold. The magnetic member's square reads $\langle Mx,Mx\rangle \le \#p\sum_p\langle\chi_p x,\chi_p x\rangle \le W^2\langle x,x\rangle$, the count's fold of squares, $\#p\sum_p a_p^2 = (\sum_p a_p)^2+\sum_{p<q}\langle a_p:a_q\rangle^2$ (Definition 1), at the per-term square cap (Lemma 18); the first-order weight is $\langle\psi_1,\psi_1\rangle = [\,\#p:16\,]$ ($\langle q_p,q_p\rangle = m_2$ with the cross pairings at m_1's square); and the solve's weight is floored at the free end's level $\ell := 4\,\hat c_2(f)$ (Lemma 15): every component of ψ_2 sits at content at or beyond ℓ (the single-plaquette channels at $4\hat c_2(k)$ over the member floor, Lemma 9, the pair sectors at $6+\hat c_2(k)$, the distant products at eight), so $\ell\,\langle\psi_2,\psi_2\rangle \le \langle\psi_2,E\psi_2\rangle = \langle\psi_2,M\psi_1\rangle$ (ψ_2 off the unit line) and the squared Cauchy–Schwarz reads $\ell^2\langle\psi_2,\psi_2\rangle \le W^2[\,\#p:16\,]$, the cofactor Z at $\ell^2Z = W^2[\,\#p:16\,]$ its cap. The squares' fold twice, $(a+b)^2\le 2a^2+2b^2$, then reads

$$\langle\rho,\rho\rangle \;\le\; \tau^{12}\,Q(\tau)\ , \qquad Q(\tau) := 4\,[\,\#p:4\,]^2[\,\#p:16\,]+4\,W^2Z+2\,\tau^4[\,\#p:4\,]^2Z\ .$$

The moments' brackets read at two states, ω the ground's and the jet state $\hat\omega := \langle\hat\psi,\cdot\,\hat\psi\rangle$ against $\langle\hat\psi,\hat\psi\rangle$, at the observable $A = q_1^aq_2^b$'s cap $c_A := d_\theta^{a+b}$, the loop cap's square iterated over the word, $\langle\chi_\theta x,\chi_\theta x\rangle \le d_\theta^2\langle x,x\rangle$ per letter, with the form two-sided at the count at every vector (Lemma 18), both states' reads at or below the cap. Each moment's bracket is the transport display's width read at the jet as the later ground, one polarization each (Theorem 20(iii)): at the pair w at $w\,\langle\gamma'\,{:}\,\tau^2W\rangle = \gamma'\,\tau^6K(\tau)$, $K(\tau) := K_0+\tau^2K_1$ over the root caps $K_0^2 \ge 4[\,\#p:4\,]^2[\,\#p:16\,]+4W^2Z$ and $K_1^2 \ge 2[\,\#p:4\,]^2Z$ (the pair $[\,x+1:2\,]$ clears every x's square at $(x+1)^2 = 4x+\langle x\,{:}\,1\rangle^2$), the display's first read holds, ${\gamma'}^2\langle\varphi,\varphi\rangle \le w^2\langle\hat\psi,\hat\psi\rangle$ at $\langle\hat\psi,\hat\psi\rangle\ge 1$ (ψ_1 and ψ_2 off the unit line), and the width reads

$$\gamma'\,\big|\,\omega(A)-\hat\omega(A)\,\big| \;\le\; 4\,c_A\,w\ .$$

The jet state's moment is the pairing at the full jet, P_A against G at $P_A := \sum_{i,j}\tau^{2(i+j)}\langle\psi_i,A\psi_j\rangle$ over $\psi_0 := \mathbf{1}$ and $G := P_{\mathbf{1}}$, degree eight, while the displayed $\hat\mu_{ab}$ reads P_A against G through τ^4; the datum N_A at $P_A = \hat\mu_{ab}\,G+N_A$ sits at the keys three to six, and $|\hat\omega(A)-\hat\mu_{ab}(\tau)| \le |N_A(\tau)|$ at $G\ge 1$. Its keys are capped at the weights: P_A's key three, $2\langle\psi_1,A\psi_2\rangle$, sits at or below $c_A([\,\#p:16\,]+Z)$ in magnitude (the polarization at weight one) and its key four, $\langle\psi_2,A\psi_2\rangle$, at or below c_AZ; G's key two is $[\,\#p:16\,]$, its key three, $2\langle\psi_1,\psi_2\rangle$, sits at or below $[\,\#p:16\,]+Z$ and its key four at or below Z; so at $\hat\mu_{ab}$'s three displayed coefficients $\hat\mu_0,\hat\mu_1,\hat\mu_2$ the magnitude fold reads

$$|N_A(\tau)| \;\le\; \tau^6\,T_A(\tau)\ , \qquad T_A := n_3+\tau^2n_4+\tau^4n_5+\tau^6n_6$$

at the key caps $n_3 := (c_A+|\hat\mu_0|)([\,\#p:16\,]+Z)+|\hat\mu_1|\,[\,\#p:16\,]$, $n_4 := (c_A+|\hat\mu_0|)\,Z+|\hat\mu_1|([\,\#p:16\,]+Z)+|\hat\mu_2|\,[\,\#p:16\,]$, $n_5 := |\hat\mu_1|\,Z+|\hat\mu_2|([\,\#p:16\,]+Z)$ and $n_6 := |\hat\mu_2|\,Z$. The displayed moments sit at or below the caps below their own comparisons' first located roots: per index pair the magnitude fold $|\hat\mu_0|+\tau^2|\hat\mu_1|+\tau^4|\hat\mu_2|$ reads against c_A, at the free end $|\hat\mu_0| = m_am_b$, the unit or *the sum's unit against d_θ at three or beyond (Proposition 13's dimension fold), so the comparison holds there and its first located root joins the cell's list (Theorem 15). The partition fold's bracket telescopes factor by factor: at a term's moment product, one factor moved from ω's read to $\hat\mu$'s at a time, $|\prod_i \mu_i - \prod_i \hat\mu_i| \le \sum_i \delta_i \prod_{j\ne i} c_j$ at the moments' brackets δ_i and the caps, the ω factors' the states' reads and the $\hat\mu$ factors' the displayed comparisons' below the cell's list; every term has field count four, so a factor's move at (a_i, b_i) costs $\delta_i\, d_\theta^{4-a_i-b_i}$ at $\delta_i \le [\, 4c_i w : \gamma' \,] + \tau^6 T_i$, and the fold's factor count at the block-count weights is 75, one per moment factor of every term (the one-block term one, the seven two-block terms two each, the six three-block terms three each at weight two, the four-block term four at weight six), so*

$$\big|\varkappa_4(\omega) - \varkappa_4(\hat\mu(\tau))\big| \;\le\; 75\, d_\theta^4 \Big([\, 4w : \gamma' \,] + \tau^6 \sum_{(a,b)} T_{q_1^a q_2^b} \Big) \;=:\; \tau^6 B(\tau)$$

at $d_\theta^{4-a-b} T \le d_\theta^4 T$; and the fold at the displayed orders reads $\varkappa_4(\hat\mu(\tau)) = \tau^4 c_1^2 \langle S\!:\![\,1:8\,]\rangle + \tau^6 R(\tau)$, R the fold's keys three to eight at the displayed orders, the collection its keys through two. So the read sits within the tail's bracket of the jet's value,

$$\big|\varkappa_4 - \tau^4 c_1^2 \langle S\!:\![\,1:8\,]\rangle\big| \;\le\; \tau^6\, \mathrm{tail}(\tau) \;, \qquad \mathrm{tail} := B + |R| \;,$$

$|R|$ the magnitude fold of R's coefficients, and two windows' reads sit within the spectator brackets (Theorem 20(iii)), the anchor window the graph's band neighborhood. The cell is the τ-segment below the cell list's least located root, the displayed comparisons' roots joined to the comparison's (Theorem 15), the comparison the leading read against twice the tail's fold at the drift's clearing,

$$\langle \gamma'\!:\!\tau^2 W\rangle\, c_1^2 \langle S\!:\![\,1:8\,]\rangle \qquad \textit{against} \qquad 2\,\tau^2 \Big(75\, d_\theta^4 \big(4K(\tau) + \langle \gamma'\!:\!\tau^2 W\rangle \textstyle\sum T \big) + \langle \gamma'\!:\!\tau^2 W\rangle\, |R|(\tau) \Big) \;,$$

one polynomial in τ, positive at the free end (the side, $8S$ above one) and on its lower side at $\tau^2 W = \gamma'$ (the left member of equal members against an occupied right), so the comparison's first root sits below γ' against W and the cell inside the drift's clearance outright: on it $\varkappa_4$ reads its upper side at the margin $[\,1:2\,]\, \tau^4 c_1^2\, \langle S\!:\![\,1:8\,]\rangle$ or beyond, at every window beyond the band, the floor's data located member reads.

22 The reconstruction

Theorem 33 (The located reconstruction)**.** *The passage from the Euclidean bracket families to the relativistic ones is a theorem over the located stage. Its inputs are the bracketed ground reads (Theorem 20), the restoration counts with the cone read (Theorem 21; Lemma 48), and the pairing's definitional positivity (Definition 8); its outputs, per member and direction count at every window of the corner cell:*

(i) The spectral data are located divisor data. Per fiber the energy roots with their weights are the split's own (Lemma 33), the Euclidean families read as the weight folds over them (Theorem 20(v)), and on the clearance cell the moment folds' solve returns the truncated moment vector from the streams' reads (Lemma 47), the located divisor read its clearance: the divisor data read the Euclidean families as their weight folds, one elimination per stream.

(ii) The relativistic bracket families are those data at the cone's coordinates. Per probe pair the family is the weight fold over the located roots at the fiber's chord, its covariance identity reads the restoration counts' (the momentum reads isotropic at degree two with the direction data two degrees down, Lemma 48), and the families are window-free within the spectator brackets (Theorem 20(iii)), the growth caps their m-point folds' reads (Theorem 20(v)).

(iii) The state reads close the families. The pairing's positivity is the families' own Gram read at every finite probe list, the cut is their spectrum condition (Lemma 20), the word sector is the state's space, cyclic (Lemma 21), and every read holds at every window ground at once: each output is a located bracket read of the inputs, one solve and one count per datum.

23 The quasi-local algebra

Definition 15 (The quasi-local algebra)**.** $\mathfrak{A}$ is the carrier of the infinite lattice with its filtration by least windows: every element has one (Proposition 4), so $\mathfrak{A}$ is the union of the regions' local algebras. A winding operator exists only on the finite torus: its index support wraps a cycle of the torus, while every configuration of the infinite lattice's index has finite support (Definition 8), so winding operators lie outside every local algebra and outside $\mathfrak{A}$ with them.